Observers Are All You Need

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Abstract

Observer-Patch Holography (OPH) models observers as finite patches that retain records, compare overlap readbacks, and repair disagreement. Convergent repair produces protected public normal forms. On a declared signed-record branch, the normalized response metric has an exact rank-three quotient and an abstract continuous Euclidean completion. Authenticated read-from provenance generates finite informational posets; modular screen geometry supplies Lorentz-frame kinematics and a conditional Einstein construction.

Complete reversible port response and endogenous observer transport force the local Standard Model gauge Lie algebra. Under a specified matrix-current and fermionic matter contract, anomaly balance and tensor descent determine the charges and maximal faithful matter image $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$. A supplied charged-scalar/Maxwell action with positive mass squared and nonnegative quartic coupling has a nonlinear real zero-current sector whose trajectories converge at first order to smooth Neumann continuum solutions under uniform refinement and Ritz initialization. On the fixed volume mesh, its full reduced interacting kinetic metric is complete. The declared Laplace–Beltrami Hamiltonian has a unique self-adjoint closure, a neutral Hilbert sector and normalized strong-domain Gaussian states. Exact software readback reconstructs coupled field histories, and their ordered configurations determine a Jacobi model clock at supplied energy.

Declared local screen-cell maps have interval-certified fixed points. A frozen primitive-port branch fixes three linked dispersion coefficients and a degree-six icosahedral angular shape.

Keywords: observer patch holography; quantum foundations; emergent spacetime; fixed-point consensus; Standard Model structure; quantum gravity

What This Paper Contributes

This synthesis develops the observer interpretation, continuum and Hilbert-space constructions from one supplied matter action, and two quantitative closures. The local equation

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}$$

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is the stronger result: its fixed-point uniqueness schema and outward-rounded interval certificates give one root for each declared map on the full analytic domain. The direct global proposal $N = \log M_0(\mathfrak{U}_N)$ asks a trial universe to reproduce its logarithmic correctable-record capacity. The finite consensus and capacity algebra is established separately. A source-derived packet realizes the fixed-cutoff construction at $D = 24$. A bounded target-clean counterfamily has incompatible zero sets under shared base, positivity, and carrier controls. It does not establish universal all-rung membership in the complete A1–A3 capacity-source contract or an executable-to-Lean bridge. Direct N is not evaluable on that incomplete source antecedent, and the stronger source-class verdict does not follow. The source-derived physical hadronic transport for P, completion of the source antecedent and selection of one physical capacity for N, the universe-level carrier attachment, the horizon/record identification, and the common screen/electroweak load carrier are consumed as premises and are not constructed here. Numerical comparisons and falsifiers are stated in their owning sections.

A supplied charged-scalar/Maxwell action with positive mass squared and nonnegative quartic coupling admits first-order convergence of its real-sector trajectories to smooth Neumann solutions under uniform refinement and Ritz initialization (Theorem 3.14). On the fixed mesh, the interacting kinetic metric is complete, and the declared Laplace–Beltrami Hamiltonian has a unique self-adjoint closure (Theorem 3.17). Exact software restoration supports numerical coupled histories, whose changing configurations give a nonturning Jacobi model clock (Propositions 3.9 and 3.10). The geometry, matter law and couplings are supplied; physical calibration requires separate evidence.

OPH combines familiar mathematics under one operational constraint. Finite observers compare the records visible on overlaps, repair mismatches by declared recovery moves, and accept as public the normal form that survives those comparisons. The bounded observer patch is the primitive formal object. Its physical size and carrier realization are branch data. Spacetime, fields, and particles enter through additional physical readout maps. A machine-checked receipt makes all seven clauses theorem-level data at one fixed regulator. Its proper-overlap witness uses two operational observers whose local records differ before typed restriction, agree on a nonzero common section accessible to both, and fail after a controlled corner mutation. No cross-regulator naturality, source realization, or higher-overlap coherence is supplied. A companion common-origin theorem traces each visible event's support region, public record observable, and Lorentz chart coordinate to one event packet (Section 2.2).

The declared twelve-port repair mean supports a conditional carrier construction. The normalized long-response kernel selects an intrinsic rank-three projector in port counting space. The limiting Gram form has a three-dimensional radical on the real six-control extension. The signed cumulative port-record/load module has no nonzero metric kernel. Its source readout is exactly onto $\mathbb{Z}^6$, and the conservative seam currents span the even-sum submodule D_6. Both modules embed densely in the quotient under the pullback response metric and complete to an abstract continuous local Euclidean translation carrier. The ordinary six-dimensional lattice metric on D_6 is not used. The limit precedes the real quotient and integer-module completion; finite-step kernels retain full rank on the signed sector. The construction is canonical up to port-label-preserving isometry and chooses no Cartesian axes. Cumulative record addition acts simply transitively on the exact record carrier and by isometries on its completion. Thirteen finite covariance/isometry table identities in that proper-action packet use Lean's `native_decide`, so those subreceipts trust the native compiler and runtime in addition to the kernel and are not kernel-only proofs. Under explicit A2-natural feasibility and objective premises plus an A3 unique-minimizer premise, the internal seam weights are forced to $1/60$. This gives an exact internal Markov average and the translation-invariant Dirichlet generator $L = I - P$. Plane waves diagonalize L, its carré-du-champ is a positive sum of squared seam differences, and response-metric normalization gives the complete thirty-edge cosine character. The source theorem fixes this internal record action under its named selection premises. It does not identify the action with a physical field, position, clock, or propagation law. A1–A3 do not by themselves provide a proved selection of the repair mean, the commutative signed-record module, or the Gram readback topology. No overlap/refinement gluing, physical scale, or identification of record translations with physical space is supplied.

The consensus paper proves the same-source finite normal-form result. The presentation-invariant treatment in Ref. [14] separates hidden implementation choices from quotient-visible data. It does not identify carriers with different visible port incidence, topology, record processes, or repair laws. The spacetime and Einstein paper owns the controlled modular route to Lorentz and Einstein structure. The complete twelve-port response in A1 and the endogenous overlap holonomy in A2 force the abstract local compact Lie algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. This result does not assume an ambient continuous gauge group or a matrix current. The distinct transportable-sector/Tannaka route is conditional. Under the explicit inverse-port contract, incidence and target-blind inverse-port readback independently derive $R = -J$, without reconstructing the current generators or their bracket. Under the gauge paper's conditional matrix-current and rank-15 matter contracts, anomaly balance and tensor descent give the Standard Model charge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. Source selection of the matrix current, matter action, and physical global quotient, together with laboratory identification of the finite current and flux sectors, physical family attachment, and equality with the Tannaka current, enters the chain as a named premise set. The particle paper contains the downstream hierarchy, mass, and flavor calculations; the screen paper contains the finite carrier and record model. The common-domain Einstein tower and the physical closure packets required for P and N are consumed premises, not constructions of this paper.

The exact word algebra of the registered recurrence is $\operatorname{span}\{I, A, A^2, A^3\}$. It is commutative and four-dimensional, so it does not supply the twelve generators, their bracket, or the nonidentity proper rechartings required by the physical current premise. This bounded result does not cover order-sensitive port perturbations. The target-free diagonal phase lift has first-order rank twelve and an abelian bracket. Adding the connected adjacency tangent generates $\mathfrak{u}(12)$, with derived rank 143. The required current therefore needs a non-diagonal source response with derived rank eleven. Other port actions are not covered by this control. For the canonical oriented carrier, the complete target-free A_5-equivariant alternating-bracket space over $\mathbb{Q}$ has dimension fourteen and an exact rational basis. This is a search-space theorem. It does not select a Jacobi bracket, a compact current, source histories, or holonomy. The complete Jacobi condition has exact quadratic coefficient-row rank 38 and an 11+27 rowspace decomposition. The compact real locus of the resulting fourteen-parameter variety is exactly three families, two of them mirror $\mathfrak{su}(3) \oplus \mathfrak{so}(3)$-type cells, conditional on three named textbook lemmas (Section 3); the noncompact components away from the analyzed channel supports remain unclassified [24, 25].

This paper supplies the shared interpretation and closure language. Its claims are typed: finite theorem, controlled scaling result, conditional physical branch, numerical diagnostic, and interpretation are not interchangeable. A counterexample to a load-bearing theorem, a nonunique or target-leaking closure, failure of the physical readout premises, or a carrier-dependent change in the observable quotient falsifies the corresponding OPH claim. The axiom-forced local gauge Lie algebra, conditional matter image, controlled Lorentzian limit, and two dimensionless closure equations define the integrated claim. The typed premise sets above delimit the boundary between this theorem chain and full physical closure.

One typed construction and its source assumptions

The finite carrier, accepted repair, support geometry, gravity, abstract compact current, and conditional matter belong to one typed construction. Accepted repair produces the quotient-visible normal form. A1's complete twelve-port response and A2's endogenous holonomy force the abstract local Standard Model gauge Lie algebra. A declared matrix current and matter packet provide conditional realizations, while the source producer derives only the inverse-port response. Each composition uses the objects and premises displayed here.

Three meanings of screen

The word "screen" is used for three related objects that must not be identified without a receipt.

1. The *local carrier boundary* is the twelve-port oriented interface of one Echosahedral carrier on the declared branch. Its incidence has $(V, E, F) = (12, 30, 20)$.
2. The *federation screen* is the routed system of interfaces, records, repairs, and checkpoints of many carriers at finite cutoff.
3. The *support screen* is the observer-facing geometric chart. On the spherical branch it is the refined conformal S^2 used for caps, collars, modular flow, and Lorentz reconstruction.

Local icosahedral incidence does not determine the topology of the federation nerve. A federation of identical local carriers can be routed as a path, a cycle, a higher-genus complex, or a spherical complex. The map from routed carriers to a support-visible spherical nerve is therefore a physical bridge, not a change of notation.

Structure-sensitive, presentation-invariant physics

OPH is not neutral under arbitrary changes of substrate. It is invariant under changes of presentation that preserve the complete observer-visible carrier signature. On the Echosahedral branch that signature contains

$$\mathcal{C}_{i,r} = (\mathcal{A}_{i,r}, \rho_{i,r}, P_{i,r}, I^{\mathrm{or}}_{i,r}, \mathcal{R}_{i,r}, \mathcal{U}_{i,r}, \mathsf{Chk}_{i,r}, \mathsf{Resp}_{i,r}, c_{sr}),$$

where $P_{i,r}$ is the port set, $I^{\mathrm{or}}_{i,r}$ is oriented incidence, $\mathcal{R}_{i,r}$ is the record algebra, $\mathcal{U}_{i,r}$ is the repair or feedback interface, $\mathsf{Resp}_{i,r}$ is the visible response law, and c_{sr} is the refinement lineage. Hidden coordinates, port names, worker partitions, materials, and wiring presentations are silent when an isomorphism preserves this whole tuple and its error model. A change in port number, incidence, orientation, accessible algebra, response, repair law, clock, or refinement lineage need not be silent. A cube and an icosahedron are therefore different carrier contracts even when both are built from the same material.

A carrier body is not automatically an observer. It realizes an observer only when it supplies bounded access, self-readback, durable records, record-conditioned feedback, boundary prediction against controls, and checkpoint continuation. One carrier may pass that test. A connected subfederation may pass it instead. No theorem fixes primitive observer size by counting carrier bodies.

The common finite computation

At cutoff r, source-bound carrier data are routed into an observer-patch federation. Accepted repair then acts on the physical quotient:

$$\begin{aligned}\mathsf{SourceCarrierTower}_r &\xrightarrow{\text{realize/route}} \mathsf{ObserverFederation}_r\\ &\xrightarrow{\pi_r} \mathsf{PhysicalQuotient}_r \xrightarrow{\mathrm{Rep}_r} \mathsf{PublicNormalForm}_r.\end{aligned}$$

The last arrow is the consensus result only under semantic-dependency-complete transactions, coherent union-collar payloads, repair completeness, local diamonds, protected records, and the stated endpoint conditions. A collection of oscillators with equal frequency does not supply those clauses.

Physical phase locking can instantiate one synchronization layer. For a routed edge $e = ((i,a),(j,b))$, a source-produced phase record may certify frequency entrainment and a stable relative phase,

$$\dot{\theta}_{i,a} - \dot{\theta}_{j,b} \longrightarrow 0, \qquad d_{S^1}(\theta_{i,a} - \theta_{j,b}, \delta_e) \le \varepsilon_e.$$

That certificate becomes a consensus parent only when the phase record fixes a commensurability map for the exposed packets and is tied to the accepted repair ledger, semantic records, an independently calibrated clock, and the confluence premises. Phase locking can synchronize an interface. It does not by itself make the interface an observer, settle semantic disagreement, or produce physical time.

Two projections of one source

The public normal form has two separately typed projections:

$$\begin{aligned}\mathsf{AuthenticatedSemanticHistory}_r &\longrightarrow \big(\mathsf{FiniteInformationalCauset}_r, \mathsf{CanonicalSourceHeight}_r\big),\\ \mathsf{ExactPortResponse}_r &\longrightarrow \big(\mathsf{RankThreeCarrier}_r,\ \mathsf{UnitDirections}_r \simeq S^2 \simeq \mathsf{FutureNullRays}\big),\\ \mathsf{PublicNormalForm}_r &\xrightarrow{\text{carrier-to-support}} \big(\mathsf{Support}_{S^2,r}, \mathsf{FiniteCapBWCertificate}_r\big),\\ \left.\begin{matrix}\mathsf{FiniteCapBWCertificate}_r\\ \mathsf{CompatibleModularStateTower}_r\end{matrix}\right\}_{\text{same tower}} &\longrightarrow \mathsf{BW/KMS}_r \longrightarrow \mathsf{Lorentz/H^3Kinematics}_r.\end{aligned}$$

$$
\begin{aligned}
&\mathsf{FiniteStrictPoset}_r \longrightarrow \mathsf{ExactAuthenticatedCausetLog}_r,\\
&\left.\begin{array}{r}\mathsf{FiniteInformationalCauset}_r\\ \mathsf{CanonicalSourceHeight}_r\\ \mathsf{RankThreeCarrier}_r\end{array}\right\} \longrightarrow \mathsf{AmbientLorentzCarrier}_{1+3,r},\\
&\left.\begin{array}{r}\mathsf{FiniteInformationalCauset}_r\\ \mathsf{AmbientLorentzCarrier}_{1+3,r}\end{array}\right\} \longrightarrow \mathsf{AuxiliarySeparatedForwardPlacement}_r,\\
&\left.\begin{array}{r}\mathsf{AmbientLorentzCarrier}_{1+3,r}\\ \mathsf{SuppliedSpatialReadback} + \mathsf{EdgeSpeed}_r\end{array}\right\} \longrightarrow \mathsf{ForwardConePlacement}_r,\\
&\mathsf{ForwardConePlacement}_r \xrightarrow{\substack{\text{equal-height separation,}\\ \text{incomparable spacelike}}} \mathsf{FaithfulFiniteCausalPlacement}_r,\\
&\mathsf{FaithfulFiniteCausalPlacement}_r \xrightarrow{\substack{\text{physical signals+operational clocks,}\\ \text{source refinement+count-volume,}\\ \text{dimension+manifoldlikeness,}\\ \text{topology+uniqueness}}} \mathsf{EffectiveSpacetime}_{3+1},\\
&\left.\begin{array}{r}\mathsf{NineSourceDirectionBalances}_r\\ \mathsf{SymmetricFields} + \mathsf{Ward/Bianchi}_r\\ \mathsf{AlgebraicCoupling} + \mathsf{Steps} + \mathsf{Connectedness}_r\end{array}\right\} \longrightarrow \mathsf{FiniteEinsteinForm}_r,\\
&\left.\begin{array}{c}\mathsf{EffectiveSpacetime}_{3+1}\\ \mathsf{FiniteEinsteinForm}_r\\ \mathsf{curvature} + \mathsf{physical\ stress}\\ \mathsf{vacuum} + 8\pi\mathsf{G\ scale}\\ \mathsf{small\ ball} + \mathsf{smooth\ convergence\ control}\end{array}\right\} \xrightarrow{\text{physical promotion}} \mathsf{SmoothPhysicalEinstein}_{3+1}.\\
&\left.\begin{array}{c}\mathsf{CompleteResponse}_{12}\\ \mathsf{EndogenousHolonomy}_{A_5}\end{array}\right\} \longrightarrow \mathfrak{u}(1)\oplus\mathfrak{su}(2)\oplus\mathfrak{su}(3)\\
&\qquad \xrightarrow{\text{conditional matrix and matter packet}} \mathsf{MaximalFaithfulMatterImage}\\
&\qquad \dashrightarrow \mathsf{PhysicalGauge/QFT}.
\end{aligned}
$$

The dashed arrow denotes source and laboratory maps not constructed here.

The finite carrier-to-support leg is constructed from one oriented icosahedral incidence nerve: twelve carrier charts, thirty seam algebras, and twenty nonvacuous triple restrictions. The same bound artifact supplies confluent seam repairs, an operational observer receipt, and a refinement-natural oriented S^2 support limit. The separate geometric 2π-KMS comparison and FiniteCapBWCertificate are not outputs of that finite bridge. The state tower, common-comparison maps, compatible state/vector data, modular controls, and cofinal modulus form an independent compatible modular-state tower. The BW theorem consumes both inputs on the same refinement tower. Independently, authenticated read-from provenance supplies the finite event carrier and generated order. Every finite strict partial order also compiles exactly into an abstract authenticated log; the supplied relation and unthreaded snapshots make this grammar expressivity, not OPH dynamics or physical selection. Its canonical source height is zero at roots and one plus the maximum direct-parent height otherwise, equals the attained longest authenticated-parent-chain length, and strictly increases on generated ancestry. The exact source Gram quotient supplies a positive rank-three carrier. Its direct sum with an independent real axis,

$$W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}, \qquad Q(t,x) = t^2 - g_{\mathrm{src}}(x,x),$$

is a four-dimensional ambient target carrier with Lorentz inertia $(1,3)$, and source-unit directions are exactly its future-null rays. Canonical source height enters only through a scaled event placement. This finite construction removes the free event order, supplied placement rank, rank-four chart, and fitted signature. Once the support leg produces a conformal S^2, the independent classical identity $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ gives the same celestial Lorentz cone and the three-dimensional observer-frame fiber $H^3 = \mathrm{SO}^+(3,1)/\,\mathrm{SO}(3)$.

Every finite event log admits an auxiliary injective one-way realization: enumerate its events along one source axis and take a height scale larger than the enumeration diameter. Distinct same-height events are then spacelike. The enumeration is arbitrary and generally creates unsupported cone comparisons, so it supplies neither source selection nor faithful physical coordinates. A supplied event-local spatial readback and edge speed bound send generated precedence into the future cone. Equal-height spatial separation and a strict spacelike inequality for every increasing-height pair unsupported by ancestry derive converse cone support and exact two-way order–cone equivalence. Event separation follows from antisymmetry, and the two-way equivalence gives exact interval preservation. Physical spacetime requires one source-selected refinement family preserving order, placement, and source directions, with dense and isotropic physical links on S^2, calibrated $\#I/\rho \to \mathrm{Vol}(I)$, independent dimension and manifoldlikeness tests, stable thickened-antichain topology, and convergence to a unique distinguishing Lorentzian limit. Event count is not public capacity N. The current source-history family proves exact induced-prefix custody from 24 to 384 events, but its width remains two and the prescribed single-frame source-port placement of the 24-event control is not injective or order-reflecting. That ansatz stipulates common-frame port anchors and consumed-record barycentres; it is an exact custody family and a placement falsification, not a spacetime-density refinement or a no-go for other source-selected placements.

All 30 Hasse links of that frozen source order nevertheless carry source-native carrier-local port labels and normalized null-ray representatives in a hidden reference presentation. Used-port ranks are $2, 1, 2$; pooled rank is non-invariant under independent carrier relabeling, so no local rank-three cone or frame gluing follows. A separate four-seed nested Poisson $3+1$ control recompiles Hasse links and full orders exactly through declared-parent-list-free provenance. Coordinate-map injectivity, sampled-interval membership, and the fixed-volume Poisson count channel are independently rechecked, jointly forming a construction-seeded causal-set faithful-embedding control, while only carrier and order have induced nested restrictions and event material is regulator-specific. These establish link-decoration and grammar compatibility, not OPH selection, manifoldlikeness, or a continuum. The full current receipts and standard-library checkers are archived under `evidence/source_causal_history_family/` and `evidence/causet_likeness/`; the exact local-domain inputs and receipts consumed by the comparison are archived under `evidence/local_domain/`.

On the same finite event type, the source-order Einstein theorem replaces all-null balance by nine supplied balances on fixed algebraic source-direction representatives and, with symmetric fields, an algebraic coupling, four step maps, Ward/Bianchi identities, and connectedness, derives the all-null and Einstein-form tensor relations. The directions are algebraic rather than observed signals; the informational order does not select a field, step, or balance. The matrix fields and steps remain supplied. Reading them as smooth curvature and physical stress requires a same-family tensor-curvature reconstruction converging to the smooth Einstein tensor, or the separately stated continuum small-ball/null-balance identification, together with same-source stress, Ward/Bianchi, coupling, scale, and remainder data on the physical continuum family. Scalar-curvature convergence is only a diagnostic. The separate `composedEinsteinBranch` returns the conditional Einstein-form composition under those typed inputs; it does not supply them. Any admitted continuum spacetime is an emergent effective description, not a fundamental object inserted into the finite carrier.

The second projection begins with an exact finite result on the certified Echosahedral lineage. The twelve-port module decomposes as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

The declared integer counting and normalized central-readback cost realization gives the twelve unit lines and exact gap. An append-only signed-event machine generates those integer loads. Conservative whole-unit seam repairs preserve total load and strictly decrease $V(N) = \sum_i N_i^2$; minimum move count is natural under every carrier rotation and the declared refinements. Divisibility of total load by twelve is necessary for consensus, while an explicit eighteen-move path settles the declared full-pile packet. A half-unit display rescales event values and the repair threshold together, so it is a units convention on the same move graph. Oriented incidence independently gives the antipodal pairing, proper A_5 action, and rank-three Gram frame, and determines the antipode J. Complete reversible response and endogenous overlap transport force the abstract local Standard Model gauge Lie algebra. Under the explicit inverse-port contract, the signed central involutive responses are exactly $\pm J$, with common sign conventional. Conditional also on the matrix current and rank-15 matter contract with its unique charge-conjugate projector pair, anomaly and tensor-descent certificates give the hypercharge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The target-blind producer derives the inverse-port response, without selecting the matrix current. Separate band premises select the rank-three response. Tensoring it with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ action come from that table. A distinct local-domain receipt checks the declared tensor-identity operator and conditional gap inheritance without source-selecting the matter action or transporting the twelve-port Spin packet. Laboratory identification of the finite matter carriers, current, and line sectors, exclusion of extra light sectors, physical matter-pole and continuum family identification, scalar attachment and dynamics, and quantum-field construction require separate maps.

The compact sector-category/Tannaka route conditionally reconstructs a compact group by a logically independent route. Physical family identification and extra-sector exclusion require separate identifications; they do not enter the contract-conditional finite gauge implication. Physical family identification requires an additional map on the charged-lepton and quantitative selector/gap branches. Physical unification requires a source-bound commuting square identifying its reconstructed compact group with the group acting through the Echosahedral current response:

$$\begin{array}{ccc} \mathsf{A5PortResponse}_r & \longrightarrow & G_r^{\mathrm{screen}} \\ \downarrow & & \downarrow\simeq \\ \mathsf{TransportableSectorCategory}_r & \longrightarrow & G_r^{\mathrm{DR}}. \end{array}$$

On those premises the abstract Lie-type agreement is exact. The construction supplies neither the physical vertical maps nor an identification of the two group actions. In the same way, the rank-three response band and declared generation table form a conditional complex rank-45 candidate, while three physical generations require matter-pole, continuum, seam-selection, persistence, and complement-complete refinement receipts. The rank-three result depends on the complete-band and cost-order premises; it is not a consequence of the icosahedral graph alone.

Finite controls and scope

The finite A_5 evaluator control has 60 reachable correctable public records on $\mathcal{H}_k = \ell^2(A_5) \otimes \mathbb{C}^k$:

$$M_0 = 60, \qquad D_{\mathrm{raw}} = 60k, \qquad \Delta_{\mathrm{raw}} = 60(k-1).$$

Raw equality occurs only at $k = 1$. Publicly inert multiplicity makes D_{raw} implementation-dependent, so the result is an evaluator control rather than physical capacity closure.

The unified claim has a precise scope. Consensus and geometry/gravity are composable branches of one source-bound self-reading carrier tower. The finite gauge branch uses the same carrier architecture, and A1–A2 force its abstract local Standard Model gauge Lie algebra. The matrix current and matter action are conditional realizations and are not joined to that tower by a common source construction. Local icosahedral incidence by itself constrains only the carrier module. The physical maps that turn these constraints into one inhabited universe are additional assumptions. Matching dimensions or symmetry labels does not supply them.

Canonical Observer Language

An abstract observer patch is the algebraic object

$$\mathsf{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \mathrm{Chk}_i)\,.$$

Here $\mathcal{A}_i$ is the finite or regulated accessible algebra, ρ_i the accessible state, $\mathcal{R}_i$ the exact or approximate record algebra, $\mathcal{I}_e$ the overlap-visible interface algebra, $\pi_{i,e}$ the visible restriction, $\mathcal{U}_i$ the allowed update and repair instruments, and Chk_i the checkpoint data used for continuation.

A support patch, such as a cap $P_i \subset S^2$ or a causal diamond, is a geometric chart of the abstract patch on a geometric branch. A carrier patch is a physical or digital implementation that realizes the same visible interface and record statistics within a declared error. Echosahedral hardware is therefore a reference carrier architecture for a fixed-cutoff implementation surface. The physical claim lives at the quotient of visible records, interface statistics, repair maps, and checkpoint continuation.

The quotient removes presentation data such as coordinates, labels, worker layout, and hidden ancillas. Quotient-visible port number, oriented incidence, topology, and response maps are physical branch data when the observer can recover them. Implementation invariance therefore does not imply carrier neutrality.

The observer patch is therefore the formal access structure used by the proof. A human observer, a horizon cap, and a hardware module can realize or chart it, but none of those examples is the definition. Physical predictions attach to quotient data: visible interface statistics, record readout, repair maps, and checkpoint continuation. A carrier implementation is admissible only when those data are invariant within its declared error model.

Definition 0.1 (Semantic observer history)**.** *For an observer token O, the observer-readable history is a labeled informational poset*

$$\mathsf{H}_O = (E_O, \preceq_O, \ell_O).$$

The elements of E_O are semantic record events, $\preceq_O$ is informational dependence, and ℓ_O is the observer-visible event label. The strict part of the order is the transitive closure of certified read-from parenthood, and $\preceq_O$ is its reflexive closure. The strict part is machine-checked as the least strict extension of the direct parent edges. A direct edge requires the child to certify the resource, the parent to write it, the child's pre-commit snapshot to name that parent as writer, and the pre-commit value to equal the parent's post-commit value; a raw writer label alone creates no edge. One adjacent independent swap preserves the authenticated direct-parent relation and its closure only under the fresh-ID and duplicate-free hypotheses on both executions; the general swap-chain theorem preserves only the raw writer-citation relation. Executor stutters are removed before the commit list is formed. Event identity is defined from the canonical semantic payload, observer token, visible footprint, and *semantic dependency parents. It is not derived from worker ID, repair iteration, queue position, retry counter, timestamp, or packet latency metadata. This definition is not a theorem that arbitrary schedule rewrites preserve labels or payloads. Latency is execution gauge only when it does not alter informational dependence, delivered data, timeout-sensitive operations, or operational clock records.*

This finite semantic history is causal-set-like in a precise, limited sense. Lean proves that reflexive authenticated ancestry is a partial order and that every interval is finite at each finite cutoff, so the abstract order-theoretic causal-set axioms are exact. Standard causal set theory uses a locally finite partial order as a discrete proto-causal structure [1, 2]. Finiteness supplies local finiteness here, while the proved relation is informational read-from dependence. Identifying it with physical causality first requires an exact causal-order embedding and manifoldlikeness. A causal-set faithful embedding additionally requires the appropriate density uniformity; recovering a continuum also requires dimension and a count-to-volume rule [3, 4, 5]. Stable homology, for example, is necessary rather than sufficient evidence for manifoldlikeness [6]. Under the appropriate distinguishing hypotheses, causal order fixes the conformal geometry and volume fixes the conformal factor [12, 13]. The Lorentz-invariant causal-set approximation uses Poisson sprinkling [11]; an OPH commit log is not a Poisson sprinkling, and its event count is not a spacetime volume. No physical embedding, volume law, four-dimensional manifold limit, or causal-set Hauptvermutung is a consequence of Definition 0.1. The conditional event branch therefore keeps four inputs distinct: authenticated order and its exact longest-parent-chain source height, the rank-three local carrier, celestial/Lorentz frame-and-cone data, and a calibrated physical event-count measure distinct from public capacity N. Their eventual source-selected composition does not constitute spacetime; refinement, physical cone/order faithfulness, manifoldlikeness, dimension, and count-to-volume receipts are required for an emergent continuum. The exact algebraic bridge identifies the unit directions of the rank-three source Gram quotient with S^2 and with future-null-ray labels. It does not identify them with a physical sky or signals. A $d = 4$ effective reading requires the physically faithful images of authenticated local read-from links to cover those directions densely and isotropically, since a Lorentzian d-geometry has null-direction space S^{d-2}. That coverage is not supplied. The count calibration additionally requires $\#I/\rho \to \mathrm{Vol}(I)$. Refinement must preserve the order, directions, and count density while soldering rank-three neighborhoods with $|B_{\mathrm{car}}(r)| \asymp r^3$ over a scaling window. This belongs to compatibility of the four inputs rather than adding a fifth one. Neither the local rank-three band nor the seeded global carrier wiring fixes event dimension.

The machine-checked composition derives the finite ambient Lorentz target carrier rather than receiving it through a supplied rank-four chart. Authenticated semantic provenance fixes the event carrier and generated order. Its canonical source height is computed from authenticated parenthood—zero at roots and one plus the maximum direct-parent height otherwise—and is proved strictly increasing on generated ancestry; it is exactly the longest authenticated parent-chain length, and an attaining chain exists. Every finite strict partial order also compiles exactly into an abstract authenticated semantic log. The generic authenticated-parent relation is the whole supplied order rather than its Hasse reduction alone. This shows grammar expressivity, while supplying neither the order's OPH dynamics nor one threaded execution. Independently, the exact rank-three source Gram quotient supplies the spatial carrier, and its direct sum with a real axis is a four-dimensional ambient target carrier with Lorentz inertia $(1, 3)$, not an intrinsic dimension estimator for the finite poset, with source-unit directions exactly the future-null rays. Source height enters only when a positive scale and spatial readback define an event placement. A finite enumeration along one source axis with a scale larger than its diameter gives every finite log an explicit injective one-way placement. It is auxiliary and not order-reflecting. For a physical-candidate placement, the edge speed bound proves one-way cone compatibility. Equal-height spatial separation and strict spacelikeness of unsupported increasing-height pairs imply converse support and exact two-way agreement. Antisymmetry then derives injectivity, so the placement is an exact finite causal order embedding; the two-way equivalence preserves every source interval on the placed image. The formal Boolean diamond inhabits this interface with null parent edges and spacelike independent branches, proving that the construction admits an exact non-chain finite witness. It is not a physical-continuum witness. The same-event-type Einstein theorem works in a separately supplied $3+1$ tensor interface and uses nine algebraic source directions plus supplied symmetric fields, an algebraic coupling, four step maps, Ward/Bianchi identities, and connectedness to derive all-null balance and the Einstein-form tensor identity. No provenance link selects the fields or identifies them with physical curvature and stress.

The continuum claim is therefore narrower and stronger: one physical source-selected refinement must pass exact two-way order/cone agreement, count–volume calibration, independent dimension and manifoldlikeness tests, stable topology, and Lorentzian-limit uniqueness, with convergent stress data and either tensor-curvature convergence or the independent continuum small-ball/null-balance identification on that same family. Scalar-curvature convergence alone is diagnostic. Exact replay of the current $4, 8, 16, 32, 64$-round histories proves exact induced-prefix custody from 24 to 384 events, but width remains two and ordering fraction rises from 0.8841 to 0.9922. One prescribed single-frame source-port placement of the 24-event control is noninjective and not order-reflecting. It stipulates one shared port frame and arithmetic consumed-record barycentres, so these are positive custody results and a useful ansatz falsification, not a spacetime-density limit or a no-go for other source-selected placements [24]. All 30 Hasse links of that same 24-event order are authenticated direct reads and carry source-native carrier-local port labels plus normalized null-ray representatives in a hidden reference presentation. The used-port ranks are $2, 1, 2$, and their pooled rank is non-invariant under independent carrier relabeling. This is an exact decoration result, not a local rank-three cone, frame gluing, a global placement, or physical causality. The full embedded-history receipt, compact projection, and local checker are `evidence/source_causal_history_family/source_causal_history_family_receipt.json`, `evidence/source_causal_history_family/source_causal_history_family_publication_projection.json`, and `evidence/source_causal_history_family/verify_source_causal_history_family_projection.py`. The separate post-hoc exploratory causet-likeness comparison, frozen after evaluation for exact replay, compares the current 2,304-event local-domain cut with Poisson-sprinkled $3+1$ Minkowski, de Sitter flat-patch, and nonmanifold controls. None of its 736 qualifying intervals of size 32–72 lies in the frozen exploratory four-dimensional ordering-fraction band. This is a current-cutoff dissimilarity result for the captured event semantics, not manifoldlikeness and not a no-go for a different physical event carrier and density refinement. The complete mirror and checker are `evidence/causet_likeness/causet_likeness_receipt.json` and `evidence/causet_likeness/verify_causet_likeness_receipt.py`. The exact underlying local-domain package and its archive checker are under `evidence/local_domain/`. The receipt also gives a positive grammar control. At 12 levels from four nested Poisson-sprinkled Minkowski $3+1$ families, declared-parent-list-free provenance reconstructs the seeded Hasse links and full orders exactly. Coordinate-map injectivity, sampled-interval membership, and the fixed-volume Poisson count channel are independently rechecked; together they form a construction-seeded causal-set faithful-embedding control. Only carrier/order restrictions are induced, event material is regulator-specific, and the declared three-sigma count checks pass. Mean Myrheim–Meyer diagnostic estimates at target means $64, 128, 256$ are $4.02824, 3.84595, 3.90346$ [8]. This proves compatibility of the provenance grammar with a $3+1$ causet carrier and order, not OPH selection, refinement, similarity, manifoldlikeness, dimension, volume, or a continuum.

Definition 0.2 (Observer registry groupoid and namespaces)**.** *Observer identities live in a registry groupoid, not in a flat UUID table. Objects are observer tokens*

$$\mathrm{RID}(O) = (\mathrm{kind}(O), [\mathrm{birth}(O)]_{\mathsf{H}}, \mathrm{lineage}(O)),$$

where $\mathrm{kind}(O)$ *lies in the disjoint namespace*

$$\{\textit{patch observer},\ \textit{cap observer},\ \textit{future observer}\}.$$

The birth event is a semantic event class in H*, and lineage is carried by explicit continuation, split, and merge arrows. Local registries on overlapping presentations must preserve observer kind, birth event, lineage arrows, visible observer structure, and checkpoint cuts. The global registry is the descent groupoid or colimit of those local registries when the cocycle and trivial-monodromy checks pass. A split creates child tokens with parent links; it does not silently assign one identity to multiple descendants.*

Definition 0.3 (Operational observer clock)**.** *The modular parameter of an observer-facing cap pair is not, by itself, an arbitrary operational clock reading. A clock claim supplies an observer-accessible instrument*

$$\Theta_O : \mathrm{Chk}_O(D) \to \mathrm{Prob}(\mathbb{R})$$

and a declared affine calibration law. For a corresponding observer O'*, the required finite or exact certificate has the form*

$$\Theta_{O'}(\Phi_{O*}\rho) \approx (A_{a,b})_* \Theta_O(\rho), \qquad A_{a,b}(\tau) = a\tau + b,$$

with an explicit residual bound from state mismatch, calibration error, and record-readout error. Execution counters are provenance; observer record order, modular parameter, and clock uncertainty are separate semantic or operational fields.

Remark 0.4 (Observer-clock naturality boundary)**.** *This paper proves observer-first quotient and readback claims only where the cited theorem surface supplies the required data. Scheduler-independent observer history and clock naturality require a history-augmented quotient relation with history-coherent diamonds, registry descent, state-preserving observer-algebra extraction, support-cap chart naturality, and the operational clock instrument above. A public evidence bundle can falsify the claim by showing counter-dependent semantic event keys, duplicate or namespace-colliding registry identities, broken registry cocycles, non-isomorphic semantic history DAGs across executors, or clock residuals exceeding the declared bound. Implementation field names alone are not the proof.*

The formal time-and-order ledger enforces this boundary at the type level. Universe closure, repair execution order, observer record order, modular parameter, worldline realization, clock readout, proper time, and an optional global time function are eight separate structures. In the committed source environment, the complete matrix of 56 ordered transitive coercions between distinct ledger layers fails to elaborate. Interpretations pass through an explicitly named realization map without a function coercion. More strongly, composition with any strictly increasing real function preserves the bare clock interface. A three-record control sends readings (0,1,2) to (0,1,8) and proves that the resulting clock is not related to the original by any positive affine map. Record order therefore supplies order data and an order-compatible scalar readout, while fixing neither affine structure, origin, nor physical unit.

The bounded observer-time continuation keeps the missing receipt visible. If a supplied event atlas and record map satisfy an affine coordinate law along a supplied future-unit timelike direction, every precedence has future-timelike displacement in every overlapping chart. On this calibrated branch the positive clock increment along that same supplied affine history is additive, and its square equals the chart-invariant Lorentz quadratic interval. One shared public event cannot determine affine rate and origin. Two ordered shared events determine the unique positive-affine interpolation of their four readings. At a third shared event, affine consistency is equivalent to an exact cross-multiplication equation. Distinction of the event and both readings from the two anchors makes this a nondegenerate check with no new fit parameter. Calling it held out requires a separate predesignation and custody protocol. The affine and cubic controls synthesize separate atlases from their clocks; they pass and fail this algebraic check without constructing two physical clocks on one calibrated geometry. A separate literal three-record, two-chart example proves the geometric interface is inhabited. These are bounded conditional algebraic theorems with finite controls: the source atlas, affine unit-speed law, shared-event correspondences, refinement transport, physical instrument, common physical clock geometry, predesignation protocol, three-clock network, and SI unit are not constructed, and no global time function or modular-to-clock identity follows.

The companion formal tower interface packages the next structural layer in one object: finite observer and record fibres, observer-indexed record orders, private matrix algebras, commutative public subalgebras, selected states and generators, and functorial refinement maps with explicit compatibility laws. An exact constant adaptor packages one existing projective partition and certified state using a discrete order and zero generator. This is an inhabited packaging witness, not a nonconstant source-derived tower, repair endpoint, causal net, physical evolution, geometry, clock, or continuum limit.

This definition does not fix the primitive observer size. The axioms require finite or regulated access at fixed cutoff and permit connected support regions with different finite or regulated algebras. A primitive observer may therefore be a variable finite federation of screen cells. The stronger claim that elementary carriers are isomorphic fixed-capacity cells would require a separate branch: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlap maps, refinement preservation of the cell type, and a proof identifying one cell or a fixed block of cells with the primitive carrier. One Echosahedron is therefore a candidate primitive carrier on the homogeneous branch. It is not the definition of an observer patch.

Failure Tests

OPH is wrong as stated if a load-bearing theorem admits a counterexample under its declared assumptions, if the controlled modular/Einstein branch cannot be instantiated under physically realizable conditions, if the exact rank-three screen-band theorem fails under its named premises, if a claimed physical family or sector attachment is contradicted inside its stated class, if a quantitative closure is nonunique or target-leaking, or if a physical implementation shows branch-essential dependence on hidden carrier presentation after visible quotient data are held fixed. Concrete falsifiers include failure of a displayed Maxwell, pure-Yang–Mills, or pure-Einstein quadratic kernel to have its stated transverse or transverse-traceless (TT) classical mode under all of that theorem's action/background/phase hypotheses; failure of a claimed quantum pole after its full quantization and spectral receipt is supplied; an observed low-energy X/Y-type $(3, 2, \pm 5/6)$ gauge generator on the realized product branch; a fourth light matter generation, which would falsify a claimed three-family attachment if that attachment were supplied, while leaving the finite rank-three band theorem and the three-to-five window as separate statements; nonunique $P_\star$ or N_{CRC} closure; a failed fixed-cap entropy-to-Einstein tensor upgrade; or carrier-dependent observables after the visible interface is held fixed. A failed continuation does not erase the recovered core unless that continuation is one of the stated parents of the core claim.

Contents

1 Introduction

Observer-Patch Holography approaches unification through the records that bounded observers can share. Each patch has local state, an accessible boundary, readback and repair operations. Overlap agreement constrains its public observables. The attraction is the reach of this same architecture: protected records, a constrained gauge Lie type, and conditional geometric and matter constructions can be connected by explicit mathematical maps. On the supplied charged-scalar/Maxwell branch, one action supports controlled real continuum trajectories and a fixed-mesh interacting Hilbert space.

The description is quantum-algebraic: patches carry algebras and states, record events use the trace/Born rule on that declared representation, and the gravity branch uses generalized entropy. An observer screen is an operational access cut; the shared spherical support is a separate geometric structure. Every additional representation, source law and physical realization map counts among the inputs to the corresponding result. The theory-of-everything proposal is that one common realization carries these constructions and reproduces observation. Its scientific value is assessed by the structures forced under those inputs and by independently testable consequences.

The computational interpretation has a sharp boundary. A conventional simulation evolves surrogate universe states,

$$U_t \longrightarrow U_{t+1} \longrightarrow U_{t+2} \longrightarrow \cdots .$$

OPH uses a fixed-point computation, but the equation alone is not the definition of simulation. In the fundamental description there is no global timeline on which the contents of spacetime are updated. The readback-and-repair operator acts on observer-readable world candidates and selects a stable solution

$$\mathcal{T}(W) = W.$$

The companion consensus paper [17] defines an OPH simulation by five independent clauses: recovery-derived endogenous update, nontrivial quotient-readable records, strict overlap-repair descent with a schedule-independent normal form, elimination of a proper candidate basin in the selected boundary/sector fiber, and implementation/clock closure. Its simulation theorem proves that the selected finite OPH packet meets those clauses under the stated boundary and record hypotheses and then derives the fixed-point equation. The same theorem gives an identity-map/constant-functional counterexample, so an ordinary equilibrium is insufficient as a simulation.

The local readout of the selected normal form supplies an internal record history, not by itself a physical clock. Operational time requires an observer-readable transition process, event correspondence, and affine calibration; it is not the external update parameter. The calibration import is typed one level further in the Lean corpus: the 2019 SI anchors, the cesium hyperfine frequency, the speed of light, and the Planck constant, enter as exact rational literals with a unique inhabitant, so the dictionary itself carries no measurement uncertainty, and the entire empirical content of the import is one declared tick duration. Under a declared tick the internal step count, angular rate, and energy gap convert exactly to laboratory seconds, hertz, and joules, the conversions compose affinely with the committed clock algebra, and two distinct tick declarations give distinct laboratory frequencies for the same internal rate, so the calibration is an import and never a derivation. In OPH, "simulation" means this certified observer-facing self-consistency structure, not a bare fixed point and not a frame renderer. The executable OPH-FPE simulator and its finite-run receipt bundles are a separate public evidence surface for finite OPH experiments [24]; they do not replace the theorem-level definition above. One public large-run archive records a level-six icosahedral diagnostic with 81,920 patch rows, 122,880 routed seams, and 2,048 materialized patch-observer neighborhoods. Conditional on one immutable, source-bound, distinct integer authority per patch, each inconsistent seam has a unique winning endpoint. All sixteen shuffled replays remove the 102,415 initial mismatches and agree on one authority-bound terminal hash. A standalone verifier reconstructs the exact terminal state and hashes from the archived primitive arrays without importing simulator code. The authority source is not selected by the present axioms or shown compatible across refinement maps, and the neighborhood records cause no later port writes. A separate eight-carrier diagnostic supplies that literal mechanism: 96 probe/read/write transactions restore perturbed port coordinates from full committed integer records, with ablation, record-counterfactual, A_5, idempotence, and commutation checks. The feedback component remains separate from the large run and has no physical realization.

The effective bulk is reconstructed from compatible screen data. Locality comes from the way collars control recovery across patch boundaries. Gauge freedom is the freedom to change the local description without changing any shared observable. Records are the stable parts of a patch that can be checked by other patches. Inside the declared tensor category, matter is the charge and excitation content that survives the certified transport and fusion tests. Completeness of the physical light sector is a separate source-grammar and laboratory attachment premise.

The first major geometric result is Lorentz-frame kinematics. On the declared support-visible branch, modular flow on screen caps becomes geometric, and the compact cap-normal theorem identifies a sky direction with $q(\Omega) = (1, \Omega)$, represents an oriented round cap by $n_C = (\cot\alpha, \csc\alpha\, \mathbf{c})$, proves $\eta(n_C, q(\Omega)) = 0$ exactly on the cap boundary with the interior/exterior sign rule, and gives $n_{gC} = \Lambda_g n_C$. Future unit observer frames then form $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$, exactly three-dimensional. A cap determines an H^3 plane/half-space, not an event position or populated bulk. On the conditional frame-response branch, quotient-visible record tokens whose calibrated modular cap responses factor through frame-local H^3 data admit exact frame identification and stable finite-noise frame estimates. The finite frame radius is $R_H[(L/\alpha)\varepsilon + (2/\alpha)\sigma]$ for exact net minimization, with a positive residual gap required before a unique finite frame value is reported. Event location requires a source-selected spatial readback and the order-faithful placement conditions of the source-causal continuum branch. Generalized entropy at fixed cap then gives a tensor first-variation relation in the large-scale regime. The absolute Einstein equation additionally uses one source-derived common-domain tower with uniform asymptotics, universal coupling, a vacuum reference, and independent scale readouts; that tower enters as a premise and is not constructed here. The proposed cosmological capacity relation is $\Lambda_{\mathrm{CRC}} = 3\pi/(GN_{\mathrm{CRC}})$. The Gibbons–Hawking entropy and the Planck-2018 late-time de Sitter benchmark give a concrete normalization [57, 59]: with $R_{\mathrm{dS}} \simeq 1.66 \times 10^{26}\,\mathrm{m}$ and $\ell_P \simeq 1.616 \times 10^{-35}\,\mathrm{m}$, the bare horizon ratio is $N_{\mathrm{patch}} = (R_{\mathrm{dS}}/\ell_P)^2 \simeq 1.05 \times 10^{122}$, the entropy capacity is $N_{\mathrm{scr}} = S_{\mathrm{dS}} = A_{\mathrm{dS}}/(4\ell_P^2) = \pi N_{\mathrm{patch}} \simeq 3.31 \times 10^{122}$, and $\Lambda\ell_P^2 \simeq 2.85 \times 10^{-122}$. OPH uses N_{scr} for the Gibbons–Hawking entropy capacity and N_{patch} for the bare horizon area ratio. The proposed input-free closure begins at finite cutoff with a frozen carrier dimension D, the unclosed terminal quotient fiber, local and interface record atoms, compatible public global sections, endogenous reachability, a frozen publicness policy, and the globally coupled semantic checkpoint kernels. Their compound confusability graph G_q gives the exact finite readback

$$M_0(q) = \alpha(G_q).$$

This correctable-code capacity includes joint checkpoint compatibility and confusability. A cyclic checkpoint can preserve every label while fixing only constant functions, and equal local marginals can hide different joint capacities. The first producer is the set

$$\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

A scalar exists only when the whole nonempty terminal fiber agrees. A faithful capacity-carrier representation gives $M_\varepsilon \le D$; exact equality means a complete rank-one public record basis. If the scalar exact map is total, monotone, and deflationary on a declared finite chain, top-down iteration reaches its greatest fixed point. The universe-level cosmic record-closure equation and its stable finite closure condition are

$$N = \log M_0(\mathfrak{U}_N), \qquad \mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star.$$

The finite public-section, correctable-code, boundary, scalarization, order, extension, refinement, and sewing implications can be proved from the stated data. A source-derived packet realizes the fixed-cutoff construction at $D = 24$ inside its declared finite source category. An executable all-rung counterfamily shares base agreement, positivity, the carrier bound, and finite incidence, action, projection, composition, extension, and sewing controls. Its reversible, copy-collapse, two-class, and hidden-spectator completions have different exact slack-zero sets. Lean proves the corresponding all-rung arithmetic result. This establishes nonidentifiability for the bounded completion class. The complete A1–A3 source contract has not been shown to contain those completions at every rung, and no executable-to-Lean bridge establishes that membership. Direct N is not evaluable on the incomplete capacity source antecedent; the stronger source-class verdict does not follow. A positive universe-level result requires a complete source antecedent, one physical zero, and proof that both sides of the strange loop read the same invariant quantity. Its slack would be

$$s(D) = \log D - \log M_0(D)$$

with one physical zero.

The declared finite candidate family covers the seventeen-row menu for the executed capacity families, two declared P maps, one direct fixed-cutoff control, the common-load baseline, reserve maps, RC-LOAD, the existing construction aggregate, and five hierarchy packets. Fifteen of the thirty entries have certified unique fixed points. Across these thirty entries, none supplies target-independent selection, a constructed same-quantity identification, and a complete source return map together. The input catalogue, nuisance quotient, coordinate and scheme catalogue, and candidate grammar are not frozen, so no entry qualifies as a source-only candidate in this declared family. Both declared P maps have certified unique stable roots while their selection, same-quantity construction, and complete return maps are unproved. The four multiplicative CAP-P maps fix whichever positive normalization seed is supplied and therefore do not select that seed; the two additive rows have no positive fixed point. The conditional RC-LOAD equation has positive discriminant, maps its declared macroscopic domain into itself, is contractive there, has one macroscopic fixed point, and excludes its microscopic root from that domain. Its pricing law, same-quantity identity, and target-independent selection are premises. The audit excludes laboratory, cosmological, and absolute-scale attachment from its qualification rule. It neither exhausts other source laws nor opens an observational comparison.

Two further physical identifications would be required afterward: the correctable-record carrier must be identified with the de Sitter horizon record for the de Sitter area law, and the screen load must be identified with the electroweak load by a positive, unital, refinement-natural map for the electroweak bridge. An independently produced operational scale is a commuting-square test of the code capacity, not its definition. The electroweak bridge supplies a mathematical candidate of $N_0 = 3.5321315434\ldots \times 10^{122}$ on the source-forward branch only after the common-load receipt. The weighted Planck base-ΛCDM read-off is $3.3129270981\ldots \times 10^{122}$; the uncorrected bridge candidate is about 6.6 percent above it. The separately premised finite-presence and Poisson carriers give $3.2920978773\ldots \times 10^{122}$ and $3.3000722254\ldots \times 10^{122}$. Exact positive composition countermodels select neither action from the finite source. Their comparisons are retrospective. Four manifest-authenticated official DESI DR2 BAO+CMB base-ΛCDM chains, transformed sample by sample, instead give

$$\Lambda\ell_P^2 = (2.96770 \pm 0.03978) \times 10^{-122},$$

with weighted 2.5–97.5 percent interval $[2.88997, 3.04530] \times 10^{-122}$ [27, 25]. The transform retains the sampled H_0–Ω_Λ covariance. It is a retrospective, target-informed ΛCDM posterior display, not an OPH likelihood, theory uncertainty, or predictive pull. The physical statement additionally requires the source-derived record carrier and common-load premises. Thus neither the physical fixed-point identity nor the cosmological display $\Lambda_{\mathrm{CRC}} = 3\pi/(GN_{\mathrm{CRC}})$ is established by the bridge calculation.

The second result is a finite Standard Model recognition package. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence expresses J as a polynomial in adjacency. A target-blind impulse and port readback derive $R = -J$, whose relative sector pattern on $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3'} \oplus \mathbf{5}$ is exact. Under the conditional matrix current and declared fermionic Spin category, the anomaly and tensor-descent calculations give the maximal faithful matter image

$$\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)/\mathbb{Z}_6.$$

This conditional matter conclusion uses the stated premises alone. The matter theorem fixes exact hypercharge and a three-color carrier. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The physical matrix current, matter action, and global quotient need source receipts. No laboratory interpretation of them is claimed. Separately, the CKM and weak-sector conditions give $3 \le N_g \le 5$. The target-free matter reduct does not fix the count. The packet also gives integer charge for color singlets and, on the declared product branch, the absence of the simple-GUT X/Y gauge channel; scalar mediators, dimension-six baryon violation, and general proton stability stay open. Inside the screen an exact selection is certified: among single complete faithful in-window multiplicity objects of the coefficient space, the operational seam-cost order $5 - \sqrt{5} < 6 < 5 + \sqrt{5}$ has the rank-three band as unique strict minimizer under two named premises: realization of one complete multiplicity object inside the screen, and comparison by the operational cost order. Tensoring this response band with the declared fifteen-state table gives a conditional rank-45 candidate. The declared finite simulator realizes the band comparison. Its per-band adjacency channels have the frame triplet as strict minimizer, and the response resolvent of the declared Laplacian generator has four pole clusters at the band costs. The rank-three frame residue sits at the lowest positive generator frequency. The channel is unitary, so every mode preserves its norm and this statement supplies no relaxation rate. The nondegenerate chirality grading and diagonal $\mathbb{Z}_6$ action are properties of the generation table. A separate 8,662-node local-domain receipt checks the declared operator $D_\sigma \otimes I_{45}$ and conditional gap inheritance. The source does not select this matter action, and no certified source, domain, or transport bridge joins the twelve-port Spin packet to the local operator domain. Matter-pole identification, the continuum Spin/locality limit, physical seam selection, refinement persistence, and laboratory attachment are separate premises; external copy completions stay grammar-indistinguishable, and the canonical rank-three screen band is a family-fiber candidate conditional on those receipts. On a declared trace-balanced block carrier $V = C \oplus W$, with dimensions $3+2$ and hypercharges $(-1/3, 1/2)$, the exact exterior package $\Lambda^2 V \oplus \Lambda^4 V$ branches as one fifteen-state Standard Model generation, supports the three one-Higgs invariant lines, cancels all five listed perturbative gauge and mixed anomalies, and has weak-doublet multiplicity $3 + 1 = 4$ with even Witten parity. This is an axiom-forced Lie-type theorem and a conditional finite representation theorem. Laboratory identification of its current and finite flux sectors, physical line-spectrum attachment, equality with the independently reconstructed Tannaka current, extra-light-sector exclusion, physical interpretation of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, scalar dynamics, genuine 1PI realization, and continuum QFT are separate premises. The subsequent QFT landing is a typed graph: exact finite quantization and formal perturbative quantization are parallel descendants of the finite local action, while the nonperturbative continuum requires its own observable tower; the conditional implications are explicit, and the OPH-native producers enter as premises. The selected exterior package is not the full even Clifford module, because $\Lambda^0 V$ is omitted. On separately stated Maxwell, perturbative/deconfined Yang–Mills, and pure-Einstein action/phase branches, the quadratic reductions have the corresponding classical null carrier modes. Photon, gluon, or graviton particle claims require the additional physical-Hilbert-space and spectral-pole receipts.

The finite exterior statement has a direct algebraic realization. The basis of $\Lambda(\mathbb{C}^3 \oplus \mathbb{C}^2)$ consists of the 32 subsets of five coordinate modes. Removing the vacuum and top line leaves the ten color/weak bidegrees in the component table, with exact charge, parity, conjugation, square-zero, and anticommutation identities. A supplied map from those rows into nonzero sectors of a projective record partition, together with supplied central-weight labels, preserves the diagonal $\mathbb{Z}_6$ weight-kernel arithmetic. A separate supplied selection mask is forced to one of the two parity rows. No group action on the partition projectors or physical attachment of that mask follows. Source selection of the maps, physical matter dynamics, continuum spin–statistics, and laboratory charge remain separate.

The third result is quantitative and sits outside the recovered core. Its primary equation defines a fixed point between a declared outer detuning and a declared electromagnetic endpoint map:

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}, \qquad A_T(P) = \alpha_{\mathrm{em}}^{-1}(0; P).$$

The outer coordinate is the normalized detuning $(P-\varphi)/\sqrt{\pi}$. The inner coordinate is the Thomson-limit coupling returned at the same trial input. The contraction-based theorem gives existence and uniqueness on a closed interval. Outward-rounded interval certificates verify the concrete hypotheses, enclose one root for each declared map, and exclude a second root across the full analytic domain. No measured value is inserted into either solve. Physical identification requires a source-derived same-scheme hadronic transport, target-independent selection of the applicable map, and a typed bridge proving that its two coordinates read the same physical cell.

The secondary global extension uses the exact finite correctable-record capacity $M_0(q) = \alpha(G_q)$ and the universe-level equation $N = \log M_0(\mathfrak{U}_N)$. Its finite public-section, carrier-bound, scalarization, order, and refinement implications are conditional theorems. The fixed-$D = 24$ checkpoint packet and whole-fiber scalar readback are exact inside their declared finite source category. A bounded all-rung counterfamily has incompatible zero sets without carrying the full observer packet across capacity. Universal all-rung membership in the complete A1–A3 source contract and its executable-to-Lean bridge are unproved. Direct N is not evaluable on that incomplete antecedent. A complete source antecedent with one physical finite-size slack zero, the universe-level carrier attachment, the horizon identification, and the common screen/electroweak carrier are premises this construction consumes and does not supply. The weighted Planck base-ΛCDM capacity $3.3129270981\ldots \times 10^{122}$ and the conditional source-forward electroweak value $3.5321315434\ldots \times 10^{122}$ are comparison coordinates. Their 6.6-percent difference does not enter the construction of the direct capacity map. The finite-presence and Poisson values require separate carriers, and their comparisons are retrospective.

The local pixel coordinate and independent scale certificate are

$$P := \frac{a_{\mathrm{cell}}}{\ell_\star^2}, \qquad \gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star = \frac{3\pi}{\ell_\star^2}.$$

If the two declared-map outputs acquire their missing physical identifications, $P_\star$ and $N_{\rm CRC}$ fix $\Lambda_\star\ell_\star^2 = 3\pi/N_{\rm CRC}$ and $\Lambda_\star a_{\rm cell} = 3\pi P_\star/N_{\rm CRC}$. They fix the dimensionless geometry. The selected scale certificate supplies the SI scale product $B_\star = \Lambda_\star N_{\rm CRC} = 3\pi/\ell_\star^2$. The selected scale certificate does not construct the cosmic readback map or remove the uncorrected 6.6-percent central-value mismatch. Under those same identifications, the pixel $P_\star$ together with the Λ-located working capacity $N_\Lambda = 3.31 \times 10^{122}$ fixes a canonical geometric pixel area and a total geometric cell count on an equal-area screen chart,

$$K_{\rm cell} = \frac{A_{\rm screen}}{a_{\rm cell}} = \frac{4N_\Lambda}{P_\star} \simeq 8.12 \times 10^{122}.$$

This count is a measured-side comparison display and does not enter either closure. At the conditional bridge capacity the same chart count reads 8.66×10^{122}. The result supplies a geometric chart count. It does not choose a primitive observer capacity. The quantity $P_\star/4 \simeq 0.408$ nats, or about 0.588 bits, should not be read as $\log\dim\mathcal{H}_{\rm cell}$ for an autonomous tensor factor. OPH capacity lives on shared cuts, edge centers, and constrained overlap data; adjacent geometric pixels are not assumed to factor into independent Hilbert components. No fixed primitive observer algebra follows from $P_\star$ and $N_{\rm CRC}$ alone. The no-G burden of the scale proof record is the clock hierarchy. The first load-bearing object must be a strict source-root proof for $\alpha_U(P_\star)$; the source-check branch is only a witness. After that, the cesium gap $\varepsilon_{\rm Cs} \sim 10^{-33}$ must be emitted by source-side electroweak transmutation, QCD/hadronic data, flavor and nuclear data, and a cesium hyperfine spectral readout, with no dependency path from measured gravity or any calibrated electroweak or gravity scale. The public α_U record is a one-dimensional comparison-branch fixed-point certificate, evaluated at the CODATA-located comparison pixel $P_C \simeq 1.6309682094$:

$$\begin{aligned}\alpha_U(P_C) &= 0.041124336195630495,\\ \frac{v}{E_\star} &= 2.0199803239725553 \times 10^{-17},\end{aligned}$$

and the Krawczyk image of $[0.041123336195630494,\ 0.041125336195630496]$ lies strictly inside that interval. The source-side diagnostic branch excludes the public Thomson endpoint upstream. Its certified forward pixel is $P_{\rm fwd} = 1.630972095858897\ldots$. That source branch is a witness, not a full endpoint proof; it does not supply the strict interval root for the Ward-projected self-map. The dimensionless formula is the strongest clean hierarchy statement. It does not supply an independently physical $E_\star$, a mass in GeV, or a complex propagator pole. The full SI gravity row requires the full set of cesium-clock source records. The QCD and hadronic weakpoint is narrow. The low-energy Thomson endpoint and the cesium-clock display require confined-quark and nuclear source payloads that the first-principles trunk in this paper does not emit. A source-only Thomson endpoint requires a target-independent choice of the physical map, a typed same-quantity bridge, and a source-derived hadronic spectral backend: source QCD parameters, a quotient ensemble, finite Euclidean vacuum transfer, Ward-normalized electromagnetic currents, the two-current spectral marginal, same-scheme remainder, systematics, and no-target-leak receipts.

Four fine-structure coordinates must be kept separate. The source/root analysis gives the undressed declared-map witness $\alpha_{\rm root}^{-1} = 136.994835177413\ldots$ (interval-certified unique fixed point of that numerical map, with enclosure width 7.2×10^{-24}). The declared map is incomplete: no derivation identifies this root with a physical fine-structure endpoint. Adding the finite-screen unified gauge-width contribution at the CODATA-derived comparison pixel gives the mixed-provenance, no-hadron diagnostic

$$A^{\rm fp}_{\alpha_U} = 137.0359595136\ldots.$$

The certified self-consistent gauge-width fixed point is $\alpha^{-1} = 137.035660136946577\ldots$. The mixed diagnostic $137.0359595136\ldots$ combines the inner value at P_{fwd} with α_U at the CODATA-derived comparison pixel, is not a fixed point of any single declared map, and is excluded from the physical output set. An executed empirical closure, using external $e^+e^- \to$ hadrons spectral data, records

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z) = 0.027609 \pm 0.000112, \qquad \alpha^{-1}_{\mathrm{emp}}(0) = 136.3827548175$$

with interval $[136.3670480603, 136.3984651934]$. The measured endpoint value 137.035999177(21) for $\alpha^{-1}(0)$ lies outside that interval. The resulting same-scheme anchor correction is $[0.6198609041, 0.6505569679]$ inverse-alpha units; the standard on-shell reference deficit 0.631 and the exact closure value 0.6379 both lie inside it, and their difference is the scheme term carried by the anchor-bridge premise. This empirical payload tests the endpoint map and exposes the missing contribution; it does not supply a source-only OPH hadron law. The two-current spectral measure covers the running-α/HVP marginal. HLbL and rare-decay rows require higher-point and transition spectral sectors from the same source law. For that reason the compact quantitative stress test is the hierarchy row above: it avoids the public Thomson endpoint, G, Λ, W/Z, Higgs mass, hadronic payloads, and the cesium clock.

The same hierarchy exponent has a local/global capacity form:

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N^{\mathrm{EW}}_{\mathrm{CRC}}}{\pi}\right)^{-P_\star/12}, \qquad N^{\mathrm{EW}}_{\mathrm{CRC}} = \pi \exp\left[\frac{6\pi}{P_\star \alpha_U(P_\star)}\right].$$

This is the electroweak bridge capacity, about 3.53235×10^{122} on the declared comparison branch at P_C; the interval-certified value at the forward pixel P_{fwd} is $N_0 = 3.5321315434 \times 10^{122}$. Identifying it with the cosmic capacity requires the source-derived correctable-public-record producer and the common screen/electroweak load-carrier identification. The weighted Planck base-ΛCDM capacity $3.3129270981\ldots \times 10^{122}$ is the comparison coordinate; the source-forward bridge value is about 6.6 percent above it. The conditional finite-presence and Poisson candidates are $3.2920978773\ldots \times 10^{122}$ and $3.3000722254\ldots \times 10^{122}$, respectively. Their physical attachments are distinct, and exact countermodels select neither one from the finite source. Their comparisons are retrospective. The factor 12 has a branch-specific geometric reading. On the declared echosahedral carrier lineage, twelve equal-trace primitive central port atoms are part of the finite architecture. The named counting realization additionally declares the integer defect fiber $\sum_p q_p = 12$ and the normalized central-readback Hilbert–Schmidt cost $H(q) = \sum_p q_p^2$. The identity $H(q) = 12 + \sum_p (q_p - 1)^2$ uniquely selects one unit per port with exact gap two. An append-only signed-event machine realizes the integer load: writes contribute $+1$, retractions contribute -1, and each port readback sums its atomic events. A conservative repair transfers one whole unit across a seam with mismatch $d \geq 2$. It preserves total load and changes $V(q) = \sum_p q_p^2$ by $-2(d-1)$, so every repair path terminates. Divisibility of the total load by twelve is necessary for consensus and is not sufficient on its own; the declared full-pile packet has an explicit eighteen-move settling witness. Minimum move count is natural under all sixty carrier rotations and the declared refinements. The half-unit display rescales the event values and repair threshold together, leaving the event graph and move-count cost unchanged. It is a units convention rather than a competing half-event mechanism. The positive readback scale is a units convention. For the separately declared uniform twelve-port reference, the local Kullback–Leibler Taylor coefficient is $6I$ and its Hessian, equivalently the Fisher matrix, is $12I$. This infinitesimal curvature does not determine the exact discrete cost. Oriented $(12, 30, 20)$ incidence then derives the unique graph-distance three antipode, six axes, proper automorphism group A_5, and the canonical rank-three icosahedral Gram frame. Edge-center collars expose the defects as central ports. Reversible write/check orientation gives an oriented 24-slot screen register. Independently, the product adjoint has count $m_{\mathrm{rep}} = 2(8+3+1) = 24$. Their equality is bookkeeping and supplies no load or clock. The exterior matter package gives the exact four-copy weak multiplicity; attaching that multiplicity to the physical screen load is an explicit hypothesis.

The oriented coefficient module is

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

Let $\mathfrak{g} = D(P_{12}) \subseteq \mathfrak{u}(H)$ be the complete faithful unitary response supplied by A1. Then $\dim \mathfrak{g} = 12$, and equivariance gives $\dim \mathfrak{g}^{A_5} = 1$. A2 makes the proper recharting action internal to this same response, so its centre is fixed pointwise and has dimension at most one. If the centre vanished, compact classification would leave only $\mathfrak{su}(2)^4$. Each simple factor has A_5-fixed dimension zero or three under an internal action, contradicting the one-dimensional fixed space. The centre therefore has dimension one, while the semisimple part has dimension eleven. The unique compact simple-dimension split is $11 = 3 + 8$, which forces

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

No ambient continuous gauge group, factor dimensions, or matrix current enter this argument.

A conditional matrix witness uses the six antipodal axes. Its exact frame maps split the even and odd modes, while the outward faces orient the complementary triplet. Pulling the block commutator back through

$$\Theta(b + d_G + d_W) = \left(\kappa_E\left(\tfrac{1}{2} U d_G\right) + i\Phi(b), \kappa_W(d_W)\right)$$

constructs

$$(P_{12}, [\ ,\ ]_\Theta) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2)$$

with center $\mathbf{1}$, derived dimension eleven, and a noncentral $\mathbf{5}$. This bracket acts on coefficient fluctuations; the central record projectors commute. Its four nonzero equivariant coefficients give an injective realization with a positive Hilbert–Schmidt pullback, A_5 covariance, inner action, and algebraic refinement naturality.

Separately, incidence expresses the antipode J as a polynomial in adjacency. Under the explicit inverse-port contract, target-blind port readback derives $R = -J$. The simulator producer executes the maximal-distance impulse/readback protocol, and the port-current certificate recomputes the incidence polynomial and sector signs independently. It does not reconstruct D, the matrix generators, their bracket, or the same-current holonomy. Source realization of the abstract response, physical carrier attachment, laboratory identification with measured Standard Model currents, and the intertwiner to the independently reconstructed Tannaka current are separate premises.

Anomaly freedom of the declared fifteen-state module forces the determinant balance, and primitive integrality fixes the charge pair $\pm(-1/3, +1/2)$; the common sign is charge conjugation. This completes the conditional map to $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$, whose connected group is

$$S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

The exhaustive tensor-kernel computation and Smith invariants independently fix the same $\mathbb{Z}_6$ kernel on those tensors and compute its conditional character/cocharacter arithmetic. The determinant balance, the measured non-split Spin lift, the central embedding, and the refinement descent are receipt-checked; the rank-15 matter object is selected by the exhaustive 1024-subset anomaly scan with the fermionic-parity grading as an output, and the declared deck and six-axis relations reproduce the $\mathbb{Z}_6$ descent. They do not source-select the complete character lattice or physical quotient. Hypercharge, chiral content, and this tensor quotient come from the stated premises within the chain. Scalar existence and multiplicity, four-dimensional instanton sectors and theta periodicity, laboratory current identification, physical matter-pole attachment of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, and continuum QFT are separate premises.

The positive twelve-term sum-to-twelve theorem establishes the all-one split inside its declared integer counting fiber. The finite algebra separately establishes the abstract matrix commutator, dimension $9 + 3$, a matrix noncentrality witness, the six-axis lattice quotient, and the sixfold central-axis arithmetic. On the declared echosahedral lineage, the normalized central-readback Hilbert–Schmidt cost establishes the strict gap, while oriented incidence establishes inverse pairing, proper A_5 action, the rank-three port frame, and refinement/relabeling naturality. A derivation of the integer fiber and physical discrete cost from the complete three-axiom schema is not claimed. This result also does not derive the echosahedral carrier from arbitrary OPH data. The noncentral current lift, trace-balanced group, Spin/deck descent, and one-generation matter calculation are exact conditional on the response and matter contracts. Physical source binding, laboratory current identification, equality with the Tannaka current, scalar multiplicity, physical three-family attachment, and continuum QFT are separate premises. The face-C_3 representation theorem gives a three-dimensional minimal extension; physical family attachment is a separate premise. See Refs. [15, 16].

The Standard Model gauge paper packages this quantitative surface as a dependency-audited compression test. A row counts in the accidental-hit estimate only when its declared source map has no dependency path from the measured target or a calibrated proxy. If p_i bounds the conditional accidental-hit probability for row i after the preceding accepted rows, then

$$P_{\mathrm{acc}} \le \prod_i p_i.$$

Only rows whose source maps have no dependency path from a measured target or calibrated proxy enter the accidental-hit bound. A compression statement must include the declared settings and basins. The proof load is the fixed-point maps, uniqueness certificates, source records, and falsifiers.

The declared map for P places an outer detuning coordinate and an inner electromagnetic coordinate on one trial domain. The outer coordinate sits above the exact self-similar entropy balance point $\varphi = (1 + \sqrt{5})/2$. The inner coordinate is returned by the electromagnetic map. Each certified map has one mathematical root on its declared interval. The measured fine-structure constant tests the resulting endpoint and does not enter the solve. Interpreting a root as the local physical closure coordinate requires the unconstructed same-scheme hadronic transport, target-independent map selection, and the same-quantity bridge.

The named P branches organize the electroweak carrier chart, the low-energy electromagnetic endpoint, the conditional Higgs/top surface, the rejected reciprocal-ray and register-Clebsch quark candidates and the six-coordinate physical quark interface, a target-informed weighted-cycle neutrino comparison candidate, and the local cell/edge consistency used by the gravity readout. The hierarchy theorem fixes $v/E_\star$. An independent physical $E_\star$ and the separate pole receipt govern any mass in GeV. The gravity normalization itself comes from $\ell_\star^2$, and P cancels in the Newton area-law readout. The quantitative branches distinguish the electroweak chart, conditional law, and inverse target adapter; the target-informed charged-lepton candidate; the NuFIT 6.1 rejection of the neutrino candidate; its tautological declared-basis transport and missing physical charged basis; and the hadron construction [35].

The screen-microphysics construction makes patches, overlaps, edge observables, records, repair maps, and observer checkpoints explicit at finite cutoff through a federated patch-carrier architecture. This regulated observer-and-record surface carries the basic consistency machinery directly.

The three axioms.

Axiom 1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. The axiom does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

Axiom 2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. The axiom constrains accepted shared data only, and the domain it quantifies over is supplied by the Axiom 1 interfaces rather than by the axiom itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\operatorname{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\operatorname{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, Axiom 2 supplies a closed path γ_a and one projective implementer $[U_a] \in \operatorname{PU}(H_{r,i})$ satisfying

$$\operatorname{Ad}_{[U_a]} D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0 ,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. Axiom 2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

Axiom 3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by Axioms 1 and 2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \underset{\rho \in \mathcal{K}_r}{\arg\min} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. The axiom selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal basis, including its constraint map and non-implications, is given in the canonical three-axiom reference.

The closure principle. The axioms describe the observer screen. The global closure principle states that the simulating and the simulated description are one system. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity.

Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point on its declared analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as diagnostics of the declared source maps.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the three-axiom source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\text{pres}} = N_0\left(1 - \frac{P}{24}\right), \qquad \ln\frac{N_{\text{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\text{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122}. The full weighted Planck base-ΛCDM chain gives the comparison coordinate 3.3129×10^{122}. The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

The gauge consequence shows how the axioms cooperate. Carrier incidence gives the twelve-dimensional A_5-module and its one fixed line. A1 supplies a faithful compact response that is complete under reversible composition. A2 makes the proper recharting action internal to that same response. Geometry alone leaves many equivariant laws, and the response conditions without the internal action leave a centreless compact alternative. Together they exclude $\mathfrak{su}(2)^4$, whose fixed dimension under such an action is divisible by three. The centre must therefore be one-dimensional, leaving the unique semisimple split $11 = 3 + 8$. The resulting local algebra is $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$.

The descent repair law serves a different purpose. Given termination and the local diamond, it turns compatible local corrections into a schedule-independent public record. Simulator runs can test candidate repair laws and their relation to the carrier response. The Lie-type theorem itself uses the complete response and internal transport clauses. Source tomography, its matrix realization, matter representation, physical global quotient, and laboratory attachment are separate.

For the declared mean $T = I - (5I - A)/60$, let P_3, P_5, P_3' denote the three nonconstant spectral projectors, of ranks $(3, 5, 3)$, ordered from the slow to the fast response. The centered complete port-probe kernel is

$$C_n = QT^{2n}Q = r_{\text{slow}}^{2n} P_3 + (9/10)^{2n} P_5 + r_{\text{fast}}^{2n} P_3', \qquad r_{\text{slow}} = \frac{55 + \sqrt{5}}{60}, \quad r_{\text{fast}} = \frac{55 - \sqrt{5}}{60}.$$

The strict order $0 < r_{\text{fast}} < 9/10 < r_{\text{slow}} < 1$ makes the normalized kernel converge to P_3. Unit-diagonal normalization gives the exact twelve-port Gram form $G = 4P_3$. Intrinsically, $H = \operatorname{ran} P_3$ and $v_p = 2P_3 e_p$ satisfy $\langle v_p, v_q \rangle = G_{pq}$. Equal port weights fix these angles; strictly positive unequal weights preserve the leading rank-three range and generally change its Gram form. The conditional signed cumulative port-record/load module injects densely and completes to H only after the normalized limit has produced the degenerate form and its real radical has been quotiented. The exact source readout is onto $\mathbb{Z}^6$, while the thirty conservative seam boundaries have image D_6 with Smith invariants $(1, 1, 1, 1, 1, 2)$. Their pullback response metric has the same completion. Record addition supplies homogeneous internal translations. Ordered histories, repair costs, and exact record distance remain separate data, and the finite state-dependent repair kernel does not descend to position alone.

Finite measures follow the same pattern. Counting and trace give exact measures on the carrier algebra. Agreement and repair decide whether those measures survive on the public quotient without schedule dependence. A physical readout must preserve the resulting measure before it can be identified with a laboratory or spacetime quantity.

Collar recovery, generalized-entropy structure, and sector completions are named premises at the results that consume them. Quantum mechanics and quantum field theory are effective descriptions carried by the observer-patch architecture.

The quantitative selection chain. Self-reading, overlap consensus, modular geometry, transportable charge, entropic gravity, and stable records define the structural branch. They do not prove that the branch is inhabited, force the icosahedral screen-to-Standard-Model attachment, or construct the global capacity packet. The two closure equations test independently produced local and global readback maps. If both equations hold with the declared selection rules, the branch admits at most one coordinate pair (P, N). This statement does not prove that exactly one physical universe exists. The local uniqueness certificate gives existence and uniqueness on its certified interval, and the domain-global certificate bounds $\sup |g'| < 1$ on every piece of a 256-piece subdivision of the declared numerical domain ($\alpha^{-1} \in [100, 200]$, both readout maps, empty exceptional set), so each declared readout map has exactly one fixed point on that domain. This certificate does not derive a physical endpoint relation for the incomplete declared map. Public global sections, zero-error graph capacity, approximate stability, the carrier bound, whole-fiber scalarization, greatest-fixed-point order theory, and exact refinement stabilization are finite implication theorems. A source-derived fixed-cutoff simulator packet at $D = 24$ verifies the finite evaluator contract inside its declared source category. A bounded capacity-indexed counterfamily shares base agreement, positivity, the carrier bound, and the executable finite controls while giving different exact zero sets. This proves nonidentifiability for the bounded completion class. The complete A1–A3 source contract has not been shown to contain those completions at every rung, and no executable-to-Lean bridge establishes that membership. Direct N is not evaluable on its incomplete source antecedent, and the stronger source-class verdict does not follow. A physical universe-level map requires a complete antecedent and one physical zero, together with the universe-level carrier attachment. The de Sitter and electroweak consequences then require the horizon–record identification and the common screen/electroweak load-carrier identification, respectively. The operational resolution experiment is an independent test, not the producer. The certified value $3.5321315434 \times 10^{122}$, evaluated at the forward pixel P_{fwd}, is a solution of the declared bridge equation, not of the unconstructed universe-level public-record producer; it is about 6.6 percent above the weighted Planck base-ΛCDM coordinate $3.3129270981 \ldots \times 10^{122}$. The separately premised finite-presence and Poisson expressions give retrospective residuals of -0.63 and -0.39 percent. Exact neutral and multiplicative global-capacity completions share the same local survival datum and satisfy positive disconnected-cut composition and finite cut-count regrouping, but they disagree after one cut. The finite source therefore selects neither expression as a global action, and its horizon branch has no capacity object to attach. The two electromagnetic endpoint residuals have no valid target-blind score: the exposed target conflates total and residual coordinates, and no active decision threshold exists. A valid test requires a complete method fixed before genuinely withheld data are examined, or an independently verified clean-room producer.

The synthesis surface gives the operational picture first, then places the technical proof machinery after the reader has the construction in hand.

2 Main Results

The OPH papers support the following unified picture.

2.1 Theorem and source-object map

The proof architecture separates three distinct jobs. A paper theorem proves that stated finite or analytic premises imply a conclusion. A simulator or source construction produces the premises on a declared refinement tower. A receipt records enough public data for an independent verifier to check that the produced object satisfies them. A successful finite run does not replace an implication proof, and an implication proof does not construct its physical source object.

The receipt discipline has an executed bounded source run with a locally frozen, hash-pinned contract said to have been written before execution. The contract and result first enter the public Git history together, so this is not an independently timestamped preregistration. Its extracted counts and class labels are mirrored as kernel-decided Lean literals, with each collapse of the truncated data stated as a theorem instead of repaired by synthetic data. The paired state and transition artifacts do not supply one common reference through either of the two audited direct mechanisms: the exact spectral obstruction excludes a mixing-mode-retaining linear intertwiner into the idempotent heat bath, and the denominator obstruction excludes a deterministic empirical pushforward. Stochastic, nonlinear, reverse-direction, dilated, and enriched-source routes are not excluded by these theorems and require a separately sourced common-reference, collar, and refinement construction. The run is therefore reproducible, mechanism-scoped negative source evidence rather than an end-to-end common-object realization, and its integers carry no physical claim [25].

Mathematical domain	**Result**	**Required source object and boundary**
Consensus protocol	Semantic-dependency-complete transactions with one coherent canonical union-collar aggregate per conflict component satisfy the local diamond. Descent, this diamond, and repair completeness give a schedule-independent quotient normal form. Coherent recovery squares and pentagons give parenthesization-independent collar gluing. At one finite regulator, equality of complete public readbacks defines a literal kernel quotient equivalent to the range of realized signatures. Finite termination, terminal completion, confluence, fixed-point completeness, and repair-output and enabledness congruence imply one public endpoint for every completed schedule from every representative of one public class.	The engine supplies functional supports, read/write closures, aggregate payloads, protected and history data, repair completeness, and concrete finite peaks. Atomicity without semantic closure fails on a two-bit protected-observable countermodel; pairwise collar compatibility without coherent union data fails on parity and GHZ countermodels. The endpoint theorem assumes its confluence, completion, and quotient-congruence packet. No exhibited source repair family supplies the whole packet. Terminal completion is a finite postcondition rather than scheduler fairness, and the result supplies no cross-regulator endpoint or physical world.

Mathematical domain	**Result**	**Required source object and boundary**
Compact relativity and gravity	Authenticated semantic provenance fixes the finite event carrier and generated order, from which the exact longest authenticated-parent-chain height is computed. Every finite strict partial order compiles exactly into an abstract authenticated log; its generic parent relation is the whole supplied order rather than its Hasse reduction alone. Every finite log has an auxiliary injective one-way placement in the source carrier. Independently, the exact rank-three source Gram quotient gives a positive spatial carrier whose direct sum with a real axis is four-dimensional with quadratic form $t^2 - g_{\mathrm{src}}(x,x)$ of inertia $(1,3)$, and source-unit directions are exactly future-null rays, hence form S^2. A positive `timeScale` multiplies source height only in a supplied event placement; an enumeration-dependent injective forward-causal placement exists for every finite log. For a source-selected candidate, the edge speed bound gives one-way cone compatibility. Equal-height separation and spacelike unsupported increasing-height pairs imply exact two-way order–cone equivalence. Antisymmetry derives injectivity, while the two-way equivalence preserves every source interval on the placed image. Cap-normal reconstruction independently yields the same Lorentz group and the observer-frame fiber H^3. On the same event type but in a separately supplied $3+1$ tensor interface, nine algebraic source directions plus supplied symmetric fields, an algebraic coupling, four step maps, Ward/Bianchi identities, and connectedness imply the finite Einstein-form tensor identity. No theorem identifies tensor-coordinate differences with those of the	Canonical source height is not an operational clock, and the source quotient is not physical position. A physical continuum requires one refinement family with exact two-way order/cone agreement, dense and isotropic link directions, count–volume calibration, independent dimension/manifoldlikeness tests, stable topology, Lorentzian-limit uniqueness, and convergent curvature and stress data, where the Einstein reading specifically requires tensor-curvature convergence or the independent continuum small-ball/null-balance identification; scalar curvature alone is diagnostic. The current 24–384-event source-history family proves exact induced-prefix custody but has width two; its prescribed single-frame 24-event placement is noninjective and not order-reflecting. The mass shell, proper-time, and Einstein readings retain their separately stated physical calibration and field-identification premises.

Mathematical domain	Result	Required source object and boundary
Finite local action domain	One target-clean source capture gives an exact finite informational event complex whose ancestry is regenerated from authenticated read-after-write provenance. The capture also gives six observer neighborhoods with induced affine wiring transitions, an exact seam-sign layer, and typed scalar, chiral, and gauge sections. The declared unit-counting realization has a sign-twisted local derivative. Exact deterministic-section evaluations test its adjoint, kinetic, covariance, gluing, refinement, and boundary relations, while the signed-graph theorem fixes its kernel. Separate band premises select the rank-three response, and its tensor product with the declared fifteen-state table is a conditional rank-45 candidate. The table carries the chirality and diagonal $\mathbb{Z}_6$ action. This local receipt checks the separately declared operator $D_\sigma \otimes I_{45}$ and conditional inheritance of the signed seam operator's positive gap. It does not source-select the matter action or bridge the twelve-port Spin packet to this domain.	The fitted spectral columns are retained only as local operator coordinates, not as evidence for spacetime dimension or Lorentz signature. A faithful source-selected causal placement, continuum topology, Stiefel–Whitney identification, matter-pole identification, physical seam selection, laboratory attachment, and continuum operator limits require separate inputs.

Mathematical domain	Result	Required source object and boundary
Screen microphysics	Exact finite coarse graining sends modular potentials through log-sum-exp. A faithful equilibrium state leaves the reversible generator underdetermined. Rate affinities, the weighted graph Hodge split, entropy production, and a convention-fixed finite KMS jump theorem give the equilibrium operator layer. The $A_5 \times \mathbb{Z}_2$ register has an eight-parameter self-adjoint commutant, its twelve-port commutant is generated by icosahedral adjacency, and it admits no equivariant directed 24-cycle. A source-derived 24-channel realization carries the edge modular grading exactly when its channel-closure residual vanishes; with A_5 and reversal covariance, the compressed spectrum has sign-paired $1, 3, 3, 5$ degeneracies, while the four gap values are source-rate data. On the certified echosahedral lineage, the declared integer counting and normalized readback-cost realization supplies the twelve unit lines. Local port incidence independently derives inverse pairing, proper A_5 action, and the rank-three frame.	The source supplies coarse states or channels, conductances or rates, KMS frequency operators, positive rate matrices, the oriented channel map and its provenance, quotient lumpability, cycle-affinity, channel-closure, commutant, and refinement receipts. Register symmetry classifies finite operators only. It supplies neither a clock nor a physical gauge-current lift. Local icosahedral incidence does not produce the global federation nerve or its S^2 support receipts. Physical phase locking is a candidate implementation bridge to overlap repair; no theorem identifies it with transactional confluence, modular flow, or an observer clock.

Mathematical domain	Result	Required source object and boundary
Finite event algebra	The span of a projective record partition is a commutative star subalgebra exactly star-isomorphic to complex functions on its nonzero projector labels. A common linear isometry can copy two distinct sharp states from one normalized blank only when they are orthogonal. Partition pinching and partition averaging have explicit normalized Kraus families and preserve trace. The two-scale public relaxation and Poissonized repair families obey exact semigroup laws, and on bounded endomorphisms of a real Banach space the latter is exactly $\exp[t\gamma(E-I)] = E + e^{-\gamma t}(I-E)$. Positive unital maps of the active-record function algebra are exactly real row-stochastic kernels. Every star automorphism of the public function algebra is uniquely pullback by a label permutation, so every pointwise-continuous real-parameter group of arbitrary public star automorphisms is trivial. On one finite full private matrix block, every star automorphism is unitarily inner, while a fixed self-adjoint Hamiltonian gives a unitary real-parameter flow obeying the von Neumann equation. For nonselective projective pinching, $S(\mathcal{E}\rho) - S(\rho) = D(\rho\\|\mathcal{E}\rho) \geq 0$; this identity is separate from a normalized selective Lüders branch. Arbitrary partition pinching is exactly the uniform random-unitary average over all 2^k independently signed block reflections. A complementary support boundary is exact: the totalized matrix logarithm sends every projection to zero, so its raw relative-entropy formula assigns zero to an orthogonal pure-state pair whose support inclusion fails.	The projective partition and input state are finite source data. The full mixed-state no-broadcasting implication enters through an explicit adapter. The finite complete-positivity predicate is defined in a separate module that proves the averaging, pinching, and nonnegative-parameter relaxation channels completely positive and trace preserving; the packet's own formal claims are Kraus normalization, Kraus form, and trace preservation. The operator-exponential theorem supplies no source rate or physical clock. The public automorphism classification is complete only for the exact finite active-record function algebra and reaches the matrix public algebra through the equivalence with its matrix realization. The private theorem covers one simple full block; the converse derivation of a coherent Hamiltonian from a pointwise-continuous automorphism group on that block is Theorem 2.2. The entropy identity is an analytic matrix proof outside the Lean export. The sign-average identity and totalized-log countermodel are Lean-checked, but the package supplies no support-aware extended entropy, pinching Pythagoras, spectral majorization, or maximum-entropy chain. No physical rate or clock is attached.

Mathematical domain	Result	Required source object and boundary
Finite Born-frame boundary and the effect-valuation representation	The twelve declared central port atoms form one classical context with an eleven-dimensional weight simplex. A distinct qubit spinor adapter has six disjoint antipodal binary contexts whose additive weights have affine dimension six; the trace-one Hermitian/Born slice is three-dimensional and obeys three exact golden-ratio relations. Representation there is unique when it exists, but exact weights show that neither existence nor positivity follows. On the full celestial/Bloch sphere, the continuous weight $F(n) = (1 + n_z^3)/2$ is normalized on antipodes yet non-affine. The positive bridge is a theorem: every valuation on the effects of a finite matrix algebra that is nonnegative, normalized, and additive on coexisting effects equals $E \mapsto \mathrm{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, with no continuity axiom. Finite receipts locate the missing hypothesis rather than deriving it: sharp binary webs admit the cube valuation as a non-Born extension, and one unsharp three-effect context excludes that particular extension. The same finite battery nevertheless admits the planar cubic response $F_y(n) = (1 + n_y^3)/2$, which agrees with the maximally mixed Born weight on every displayed axis while remaining non-affine away from them. The real S_3 algebraic web attached to source-realized gauge labels is also blind to the Pauli-Y direction: two distinct complex states give equal weights on every declared projector candidate. The missing algebraic direction is nevertheless exact. If P is the declared diagonal projector and Q its noncommuting image under the declared two-dimensional representation of a source-realized gauge label, then	The spinor projector family is a declared mathematical adapter, not a source-produced public instrument. The celestial countermodel has a sharp role: it proves that projector webs alone underdetermine the weight, and the planar control proves that one finite unsharp battery does not force a global valuation. Positive closure requires a source-realized complex-tomographically-complete effect/instrument web, an operational coarse-graining or equivalence theorem deriving noncontextual additivity, and only then preregistered frequencies as validation. Frequencies alone do not prove the universal valuation law. On the committed static effect/count fixture this boundary is machine-checked in exact form. Every committed effect has trace one, so a sum of two committed effects is an effect precisely when it resolves the identity, and any assignment sending the sure effect to one and normalized in each context is additive on every coexistent sum formed within the committed effect set. A cubic deformation of the run Born functional, shaped to vanish on the committed effect set, reproduces every generated expected frequency, stays in $[0, 1]$ on all effects, obeys the complement rule, and is additive on every committed coexistent sum, yet it fails additivity at an explicit effect pair with values $35/64$ against $143/256$ and is the Born functional of no state. Conversely, full cross-context additivity together with the static fit values forces the Born functional of the declared matrix uniquely. The phase lift is in the complex operator algebra generated by the source-attached algebraic pair, but this is a

Mathematical domain	Result	Required source object and boundary
Finite regional observer net: bounded interface	A proof-carrying enrichment of one finite consensus tower packages a finite region order, isotony, exact commutation on declared-disjoint regions, covariant refinement, compatible regional expectations, and idempotent local repair. A declared repair that fixes remote observables gives exact Heisenberg nondisturbance for every matrix state. Compatible local sections have a unique glue only on nonempty finite subregion families carrying the explicit restriction and uniqueness receipts. The API name `FiniteCover` does not include a joint-coverage law. A separately declared basis split and Kraus completeness give a generic reindexed marginal identity; no theorem identifies the regional algebras with its matrix factors.	Contravariant restriction maps and their retraction of isotony are strong extra premises, not consequences of an ordinary isotonic net. The rich-fibre packet supplies a stronger conditional construction on the second preregistered fresh-seed bounded run: four target-blind greedy-disjoint observers realize split-fibre counts $3, 3, 4, 4$ on pairwise-disjoint twenty-node windows (the pilot run failed its gates and is closed negatively), and the mirrored net carries a genuinely noncommutative block algebra with an embedded two-by-two designated-fibre matrix corner at every observer region, kernel-backed window disjointness, elementwise commutation across distinct windows, the character-and-block restriction laws, the computed four-window coverage of the top algebra, and the drop-one control showing every window is load-bearing. The enrichment rule, region lattice, restriction system, base-point state, and identity repair are declared readings rather than source-produced objects. The matrix corner is not a regional tensor-factor or `TensorSplitReceipt` attachment. This is a conditional algebraic packet, not the required source-attached regional-factor theorem.

Mathematical domain	Result	Required source object and boundary
Finite regional observer net: carrier join and time slices	For each of four retained observers, counted transition operators together with diagonal field projectors generate the full matrix algebra on the realized alphabet. The fully support-disjoint retained pair 86/247 defines a positive rational correlated state with exact counted marginals, recovered by a conditional-expectation diamond whose regional pair erases the correlation, so coverage cannot imply unique reconstruction. A typed hinge-fibered join places both committed carriers in one object: the 182-dimensional 86/88 and 169-dimensional 86/247 diagonal layers embed injectively into a 338-dimensional pushout over the 13-dimensional shared observer-86 hinge, the embedded images intersect exactly in the hinge, both transported counted states restrict to the same committed observer-86 occupation law, and the committed anchoring and walk-step transports descend to one join evolution intertwined by both embeddings. On the corrected one-step evolution interface, every net-compatible slice family over the committed walk evolution is constant, a one-step region change forces the slot exchange, and a slot-alternating inhabitant exists whose evolved data generate the next regional algebra at every step with a provably non-step-invariant region family.	The join lives at the diagonal record layer with declared fiber-uniform sections: no star-algebra amalgamation, order structure on the quotient, net morphism, tower morphism, or uniqueness claim accompanies it. The two-observers-as-factors reading remains a declared postprocessor, and no nonconstant source tower is supplied; neither a source realization nor a scoped no-go is supplied. The exchange factor of the slot-alternating inhabitant is a constructed internal symmetry, not a source transition. Source production and physical attachment of positive quantum channels, spacetime causality, physical time, the source content of the registered time-slice premise, continuum field theory, sectors, and laboratory attachment are not supplied.

Mathematical domain	Result	Required source object and boundary
Boltzmann transport	Convergence of the discrete generators to the null Hamiltonian vector field, vanishing carré du champ, tightness, and uniqueness of the limiting ODE imply deterministic Liouville transport. A limiting collision generator gives Boltzmann transport; nonzero limiting quadratic variation gives a diffusive limit.	The simulator supplies quotient-lumpable transition kernels, physical time steps, first and second conditional moments, mass-shell and boundary defects, and schedule/refinement-natural convergence. Path confluence alone supplies none of these transport limits.
Particle and gauge sectors	Complete reversible response and endogenous overlap transport force the abstract local Standard Model gauge Lie algebra from the twelve-port module. The declared charged-double-triplet matrices give an exact full-rank conditional witness. The simulator producer derives the inverse-port response constraints, not the matrix generators or their bracket.	The source supplies no ordered current tomography, same-current holonomy, laboratory identification with Standard Model currents, or intertwiner to the independently reconstructed Tannaka current; the identification consumes these as premises. A register adjacency or modular generator cannot be relabeled as this current.
Yang–Mills gap	For the source-defined finite collar family, the finite-type Dobrushin/minorization theorem gives one positive projection/transfer-gap modulus uniform in location, boundary condition, system size, and the declared cofinal refinement tower. Removing locality or the stated mixing control admits finite countermodels with a collapsing gap.	The physical compact-gauge branch must produce the admissible collar types, transition matrices, uniform constants, refinement maps, reflection-positive vacuum, and continuum regularity packet. The finite theorem closes the repair-gap implication. The Clay-facing continuum construction is conditional on that source receipt.

This map fixes the mathematical type of each result. The cited specialist papers give the detailed definitions and proofs. A finite operator, receipt, or simulation trace retains its type when used in another argument.

1. **Observers and records become physical objects.** Finite patches, shared overlaps, record algebras, repair maps, and observer checkpoints are represented on the screen as part of the construction.

2. **The spacetime implication remains conditional.** On the controlled Einstein-branch geometry readout, the finite cap-normal Bisognano–Wichmann (BW) certificate (support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and independently normalized

geometric 2π-KMS (Kubo–Martin–Schwinger) convergence), together with an independent mixed Gelfand–Naimark–Segal common-comparison premise on the same tower, makes the support-visible cap modular automorphism geometric and supplies Lorentz kinematics. Recoverable generalized entropy together with the null modular bridge, bounded-interval transport, fixed-volume small-ball area variation, admissible fixed-cap stationarity, and timelike tensor upgrade then supplies the Einstein relation on one source-derived common-domain tower with certified tails and independent physical identifications. The finite artifacts test algebra, evaluator logic, manifests, deletion rules, and synthetic controls. They do not certify such a tower; the tower enters the Einstein relation as a premise. Bare finite consensus supplies only the upstream quotient-normal-form input. The screen-capacity branch proposes the dimensionless cosmological-constant relation below. The identification of a cosmic record-closure value with the electroweak bridge requires the common screen/electroweak load-carrier identification. A fixed-cutoff source-derived simulator packet exists at $D = 24$. A bounded capacity-indexed counterfamily has different exact fixed sets under shared base, positivity, carrier, and executable finite controls. The exact verdict applies to this bounded completion class. No all-rung membership in a complete A1–A3 capacity-source contract or executable-to-Lean bridge is supplied. Direct N is not evaluable on the incomplete source antecedent:

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star, \qquad \Lambda_{\mathrm{CRC}}\ell_\star^2 = \frac{3\pi}{N_{\mathrm{CRC}}}, \qquad \Lambda_{\mathrm{CRC}} = \frac{3\pi}{GN_{\mathrm{CRC}}}.$$

The last display uses the selected scale certificate $G_{\mathrm{geom}} = \ell_\star^2$. The Einstein-branch antecedent is instrumented end to end: a typed common-domain source tower with hash-bound provenance and cross-source splice rejection, and fail-closed instruments with adversarial negative controls for geometric modular normalization, GNS cyclicity and modular intersections, the informational-ancestry cone candidate, and same-source stress/coupling, together with semantic countermodels proving each receipt family load-bearing [24]. Two clauses are theorem-grade: coupling universality holds with zero spread for every icosahedrally equivariant source law, and generator positivity holds by construction on the declared law family. A separate target-clean capture inhabits the finite local action domain: 2,304 informational record events, six observer neighborhoods with exact induced affine wiring transitions, and an observer-visible seam complex with 8,662 carriers, 11,816 seams, and 38 frustrated triangles. Two coordinate routes through the same $\mathbb{F}_2$ rank implementation agree with the exact rank identity and give lift-ambiguity rank 3,117. The section, operator, boundary, and replay certificates are exact [24]. The finite object inhabits the typed domain without establishing the source-causal continuum or Einstein antecedent. The source-forward bridge formula has the certified mathematical value $3.5321315434\ldots \times 10^{122}$; the weighted Planck base-ΛCDM comparison coordinate is $3.3129270981\ldots \times 10^{122}$. The bridge value is about 6.6 percent above it. On the declared one-class branch, physical class selection and a scalar-weighted presence-survival attachment would give $3.2920978773\ldots \times 10^{122}$; the separate exponential mean-count branch would give $3.3000722254\ldots \times 10^{122}$. Exact positive composition countermodels select neither action on global capacity. The named-law branch is not evaluable on this source class, and both comparisons are retrospective. A uniqueness certificate for the bridge formula does not establish the unique zero of the public-record slack, select one physical capacity, or construct its universe-level carrier.

3. **The finite gauge chain forces the abstract local Standard Model gauge Lie algebra.** Complete reversible response and endogenous overlap transport supply the load-bearing premises. Under the explicit inverse-port contract, incidence and target-blind port

readback derive the signed response. Under the conditional matrix current and matter representation, anomaly-forced determinant balance and exhaustive tensor descent fix exact hypercharge, a three-color carrier, a common $\mathbb{Z}_6$ kernel, and the maximal faithful matter image $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$. The cover and its three nontrivial central quotients carry the same local tensors. The six-axis menu matches the $\mathbb{Z}_6$ quotient after its relations are declared, so the physical global form is not source-selected. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. Within this gauge chain, physical family interpretation and extra-sector exclusion require separate identifications; charged-lepton and quantitative branches retain their separately stated declared premises. The same product structure gives the structural absence of the simple-GUT X/Y gauge channel. General proton stability is not implied. Conditional on the declared one-generation matter table and baryon and lepton labels, an exact dimension-six census admits $QQQL$, $QQ\,u_R e_R$, $QLu_R d_R$, and $u_R u_R d_R e_R$. Their Wilson coefficients, physical emission, decay amplitudes, and lifetimes are not supplied. On the ordinary electromagnetic branch, the unbroken $\mathrm{U}(1)_Q$ factor reduces the compact-gauge curvature to $F_Q = dA_Q$; after separately assuming the low-energy Maxwell action and its nonzero kinetic term, variation gives $dF_Q = 0$ and $d{*}F_Q = g_Q^2 {*}J_Q$, i.e. Maxwell's equations after canonical electromagnetic normalization. The rank-three screen band and declared fifteen-state generation give a conditional rank-45 tensor candidate. The table carries its chirality and diagonal $\mathbb{Z}_6$ action. On a separate finite local domain, a receipt checks the declared operator $D_\sigma \otimes I_{45}$ and conditional inheritance of the signed seam operator's positive gap. The source does not select this action, and the local domain is not certified as the domain of the twelve-port Spin packet. Interpretation as three physical families requires matter-pole identification, a continuum Spin/locality limit, physical seam selection, refinement persistence, and laboratory-current receipts. A target-clean finite local-domain capture separately supplies scalar, chiral, and gauge section spaces and a sign-twisted kinetic Hamiltonian. The exact kinetic identity makes the Hamiltonian positive semidefinite, and the frustrated seam layer makes its kernel trivial. Its dimensionless gap is therefore strictly positive. A sparse numerical refinement gives $0.1175367\ldots$ with a relative residual below 1.5×10^{-14}. This number is not a certified gap bound, a selected physical transition, or an SI energy. Physical fiber selection and clock attachment are separate [24]. The transportable-sector/Tannaka route is a separate conditional compact-group classification. The target-blind producer derives $R = -J$, without deriving the matrix current. Physical matter typing, identification of the finite current with the Tannaka current, and laboratory-current attachment are separate premises.

4. **The measured endpoint tests the local closure.** The comparison pixel $P_C \simeq 1.6309682094$ is inferred from the measured fine-structure constant solely for comparison: $\alpha^{-1}(0) = 137.035999177(21)$, $\alpha(0) \simeq 0.00729735256433$. The forward model with the source anchor and Ward-projected electromagnetic transport measures the loop-closure residual: 3.0×10^{-4} relative on the source chain and 2.5×10^{-6} relative for the certified numerical gauge-width fixed point $\alpha^{-1} = 137.035660136946577\ldots$, a certified fixed point of the declared map. Physical promotion additionally needs source-derived Ward-projected transport, target-independent map selection, and the typed same-quantity bridge. The mixed-provenance diagnostic $137.0359595136\ldots$ is a fixed point of no single declared map and is excluded from the physical output set. The separate root $136.994835177413\ldots$ is an interval-certified output of an incomplete declared map; no derivation relates it to the physical fine-structure endpoint. The comparison distinguishes the source-side diagnostic from the endpoint residual.

5. **Edge sectors also admit two-dimensional Yang–Mills and worldsheet-style effective**

descriptions. The heat-kernel edge partition function reorganizes the same boundary data into the two-dimensional Yang–Mills form, and on the declared large-edge branch this yields a controlled worldsheet effective description. This 2D bridge sits beside the Standard Model gauge paper's controlled compact-gauge repair-gap theorem. The string-vacuum selector uses this bridge only as the effective edge-language input; critical worldsheet closure, Bouchard-Donagi cohomology, operator-safety realization, threshold matching, and moduli locking are separate premises. The rank certificate gives proxy rank zero on the specified incomplete slice because the executable proxy consumes no BD value, and it supplies no physical slice or Jacobian; a certified completion that cuts the physical slice to five or fewer dimensions stays open. The Bouchard–Donagi construction is therefore a structural benchmark rather than a selected string witness.

6. **The compact-gauge branch carries the 4D Euclidean Yang–Mills form and isolates the gap theorem.** On the declared controlled compact-gauge branch, with the four-dimensional scaling chart, reflection-positive ordinary vacuum, no additional gauge-invariant relevant dimension-four pure-gauge operator on that branch besides curvature squared, repair completeness, and the Standard Model gauge paper's explicit finite cylinder system in force, OPH proves projective weak-* / GNS extraction of a support-visible cylinder cluster family. Identification of that family with the Euclidean Yang–Mills action additionally uses the renormalized four-dimensional regularity/universality receipt. Weighted observation-fiber resampling is the finite conditional-expectation projector, with the noncircular matrix-recognition criterion of Ref. [14]. A concrete active repair law is identified with that projector only after its independently extracted transition matrix passes the support, equal-fiber-row, and weighted detailed-balance receipt. Identification of Euclidean transfer with the repair generator, vacuum persistence, and passage of a uniform finite-stage gap use the separate finite ground-state-transform, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts. The checked 244-type Ising collar table is an exact calibration of that finite receipt format, not evidence for the physical compact-gauge source family. Only on that certified branch is $\Delta_{\mathrm{YM}} = \Delta_{\mathrm{rep}}$. Weak-* extraction alone is not the Clay-facing axiomatic construction.

2.2 The observerhood receipt and the common-origin bridge

The observer named in the title has a theorem-level bounded receipt. Over any instance of the companion formal tower interface, its base structure packages clauses 1 through 6 and the own-observer half of clause 7 as typed data: a bounded accessible star subalgebra of the private algebra containing the observer's own public record algebra; a self-readback map sending accessible elements into that public algebra and fixing each of its elements; durability, with every record element annihilated by the observer's declared generator; record-conditioned control maps preserving the accessible interface; a prediction map whose output never strictly precedes its input in the observer's own record order; label and prediction families commuting with the declared refinement maps; and a declared readout equal to the trace pairing of each record element against the observer's own state, with every record element fixed by every declared control. Readback idempotence and refinement stability of the readout follow. An explicit witness inhabits the receipt away from the constant adaptor: private algebra $M_2(\mathbb{C})$, a nonzero generator given by the scaled commutator with $\mathrm{diag}(1, 2)$, which annihilates every diagonal record element while moving the raising matrix unit, control by conjugation with the involution $\mathrm{diag}(1, -1)$, three chain-ordered records, and a nonconstant non-identity prediction map. Four negative controls (an erasing readback, an identity generator, a resetting prediction, and a constant misreading) each violate one attained field on the same data. The base structure is operational and finite: it selects no unique observer over a given tower, attaches no physical instrument, makes no consciousness claim, and supplies no source realization of the tower data. A separate fixed-regulator mixin closes the cross-observer half on an access cut. It requires two distinct labels, exact agreement of both accessible interfaces with the cut, equality of every committed record after the two typed restrictions to the owner-region meet, equality of own readouts, and a nonzero common restriction. The common section is thereby accessible to both observers. Its proper-meet witness uses two distinct diagonal owner algebras, a scalar bottom algebra, and corner-character restrictions: observer-indexed records differ before restriction and agree nontrivially only after restriction. A one-corner mutation makes the overlap receipt false. The operational overlap-event consumer takes two operational observers owning two charts in which one event is visible and proves equality of the two owner-region restrictions of its packet record together with two-sided self-readback. Thus all seven operational clauses are inhabited at one finite cut. Cross-regulator naturality, triple-overlap coherence, and source realization are not supplied [25].

An access cut on the finite regional observer net above assigns each observer one declared support region and one accessible star subalgebra squeezed between the observer's public record algebra and the regional algebra of that region, stable under the declared restrictions below it; the observer-local algebra at a region is the meet of the regional algebra with the accessible algebra, and the net receipts transport to this meet. A finite event world over the supplied Lorentz soldering binds one packet p_e, with packet carrier fixed independently of the quotient state, to each event e, with declared chart ownership. The common-origin theorem states that for every active visible chart/event pair (i, e),

$$\operatorname{supp}(\mathsf{p}_e) \preceq R_{O(i)}, \qquad \operatorname{rec}(\mathsf{p}_e) \in \mathcal{L}_{O(i)}\big(\operatorname{supp}(\mathsf{p}_e)\big) \cap \mathcal{P}_{O(i)}, \qquad x_i(e) = \operatorname{geo}_i(\mathsf{p}_e),$$

where $O(i)$ is the declared owner of chart i, $R_{O(i)}$ its observer region, $\mathcal{L}_{O(i)}$ its observer-local algebra, and $\mathcal{P}_{O(i)}$ its public record algebra: one event packet determines the event support region below the owner's region, the public record observable in the meet of the observer-local algebra at that support with the owner's public algebra, and the Lorentz chart coordinate as a readout of the same packet. A selected coarse-to-fine refinement transports all three conjuncts. The witness chain is strict, with the public algebra strictly inside the accessible algebra strictly inside the private algebra, and the witness records separate packets. Charts and observers are distinct types, ownership is declared data, and no physical region, coverage, instrument, spacetime, causal cone, event actualization, or continuum object is claimed [25].

Coexistence of regional operator algebras on this net interface has an exact boundary.

Theorem 2.1 (Scalar-character obstruction and character-block coexistence)**.** *For every $n \geq 2$ there is no unital complex-algebra homomorphism from the full matrix algebra $M_n(\mathbb{C})$ into the scalars, and no unital complex-algebra homomorphism into the scalars accepts an anticommuting pair of units. On this net interface, at any regulator of nonzero dimension, a region whose local algebra consists of scalars admits above it no region whose local algebra contains a unital copy of $M_n(\mathbb{C})$ with $n \geq 2$: the declared star-homomorphic restriction retraction would compose with the copy to a scalar-valued unital homomorphism on a full matrix algebra. At any such regulator, a region lying above a region with commutative local algebra carries no anticommuting pair of units: the declared restriction would send the pair to commuting units whose product equals its own negation. No full matrix factor of dimension at least two sits above a scalar overlap region, and the standard tensor-factor local-algebra picture with trivial overlap is impossible on this interface. The boundary is inhabited: over a dimension-four private algebra, the block algebras of matrices* $\operatorname{diag}(a, a, A)$ *and* $\operatorname{diag}(B, b, b)$ *with arbitrary two-by-two blocks A and B are each noncommutative,* *commute elementwise, neither contains the other, and each retracts onto the scalar overlap algebra through the character evaluating its one-dimensional corner block.*

Proof. The matrix ring $M_n(\mathbb{C})$ is simple, so the kernel of a unital homomorphism into the scalars is trivial and the homomorphism is injective, while the off-diagonal matrix unit is a nonzero element whose square is zero and whose scalar image would be a nonzero complex number whose square is zero. For the anticommuting form, a unital homomorphism into the scalars sends units to nonzero scalars, and applying it to the anticommutation relation forces the product of the two image scalars to vanish. The interface statements compose these arguments with the declared restriction into the scalar or commutative region. The inhabitant is finite linear algebra over one declared regulator with identity repair maps; the general matrix obstruction, the interface compositions, the interface laws, both noncommutativity witnesses, the mutual noncontainment, and the corner-character retractions are checked in Lean [25]. □

The declared finite C^*-algebra/retraction interface is therefore compatible with noncommutative regional coexistence through the displayed character-block witness. The obstruction excludes the full-matrix and anticommuting presentations stated in the theorem; it does not classify every possible coexistence presentation. The declared overlap algebra is a proper subalgebra of the set intersection of the two block algebras, and genuine coverage semantics or regional-factor receipts for a source-produced net are not supplied.

Dynamics on the private block carries a matching rigidity statement.

Theorem 2.2 (Finite Stone converse on one full private block). *Let α_t, $t \in \mathbb{R}$, be star automorphisms of one full finite matrix block $M_n(\mathbb{C})$ with $\alpha_0 = \mathrm{id}$, $\alpha_{s+t} = \alpha_s \circ \alpha_t$, and every orbit map $t \mapsto \alpha_t(x)$ continuous. Then there is a self-adjoint $H \in M_n(\mathbb{C})$ with*

$$\alpha_t(x) = e^{-itH}\, x\, e^{itH} \qquad \text{for all real } t \text{ and all } x \in M_n(\mathbb{C});$$

the conjugating family $U_t = e^{-itH}$ is unitary with $U_0 = 1$ and $U_{s+t} = U_s U_t$, and H is unique up to one additive real scalar multiple of the identity.

Proof. Continuity upgrades to differentiability at parameter zero by interval averaging: the group acts on the matrix-unit basis as a norm-continuous matrix family whose short-interval average is invertible, and the fundamental theorem of calculus recovers the family from its primitive. The derivative at zero is a star-compatible Leibniz endomorphism. Every complex-linear Leibniz endomorphism of the block is an inner derivation, with an explicit matrix-unit witness proved from scratch; star compatibility symmetrizes the witness into one self-adjoint generator; two generators with one commutator action differ by a real scalar multiple of the identity; and the group differential equation identifies the flow with conjugation by the coherent unitary family. Every step is checked in Lean [25]. □

On the finite private block, a pointwise-continuous symmetry flow of the observables therefore has no choice except Hamiltonian form; the additive energy offset is the only freedom. The theorem extends to the general finite private algebra: every unital star subalgebra of a finite complex matrix algebra, containing the ambient identity, is star-isomorphic to a finite direct sum of full matrix blocks, and on that block model a pointwise-continuous star-automorphism group fixes every block projection and is blockwise conjugation by the unitary family of one time-independent self-adjoint Hamiltonian per block. Block fixing is derived rather than assumed; a single automorphism may swap isomorphic blocks, and the swap is realized on the square two-block algebra. The parameter is a real number; no physical clock, source-selected Hamiltonian, or physical time identification is claimed.

The measurement and dynamics surfaces share one object. A composed frame-duality theorem (`schroedinger_frame_duality`) consumes the registered record representation, operational effect additivity, and one supplied pointwise-continuous flow through a single typed bundle and proves that the unique Busch–Gleason state of the Heisenberg-shifted registered frame is exactly the propagator conjugation of the unique state, at every flow parameter: the Schrödinger picture of the Born frame is forced by the Heisenberg action of the rigid generator. The flow stays a supplied premise and its parameter is not physical time; what the theorem removes is any freedom to represent the measurement statistics and the dynamics on different state objects.

Six receipts of the formal stack share one typed record. A common-world architecture record carries one declared positive mass, one state frame, one flow frame, one projective partition with state, one declared Maxwell evolution bundle on the twelve-port local carrier, and one declared electroweak premise bundle whose base instantiates the Standard-Model structure premises. For every inhabitant, the kinematic mass-shell, internal-clock, free-evolution, and causal-order receipts hold of the same mass, frame, flow frame, partition, and state on one Hermitian Lorentz module, with the shared fields identified definitionally, and four join theorems place the four-momentum, the clock worldline, and every nonnegative flow displacement inside the cone order that carries the finite causal nets, with the four-momentum future-causal from the causal net's bottom vertex. Those nets and their Lorentz-module cone order are declared kinematic data, not the source-derived informational read-from order. On the screen side, the temporal Maxwell receipt and the electroweak-breaking receipt hold of the same record, and the pinned port-dual measure of the record's structure premises evaluates the Maxwell Gauss load on the one twelve-port index type, with every weight equal to $1/12$. An explicit instance inhabits the record (`Lean/Geometry/CommonWorldKinematicsWitness.lean`). On the base record the two islands are joined only by conjunction: no typed dictionary sends ports, seams, or faces to points or cones of the module, the identification of the Maxwell step index with the kinematic proper time is a declared dictionary carried by the instrumented extension below, and no common variational principle spans both carriers; clock calibration and carrier identification are not supplied. Gravitation, thermodynamics, matter dynamics, quantization, and observer readout appear in no field of the record, so the record does not establish the common-world target.

One typed dictionary joins the two islands beyond conjunction. The rank-three source Gram quotient of the six-axis port adapter, read from the registered twelve-port incidence table, maps onto the rest fiber of the record's declared frame by a linear equivalence composed from committed isometries, and the pullback of the rest metric through that dictionary is exactly the screen quotient Gram at every declared frame, with the standard-frame case equal to the committed candidate readback and with every integer record control carried to the committed screen chain up to one committed normalization radius. An extended record carries the dictionary and the metric clause as fields; an explicit inhabitant extends the committed instance, and a composed receipt proves the eight island packets, the bridge clause with its oriented-chart commuting square, and the transported inverse-square strength read on the screen Gram, of one record simultaneously (`Lean/Geometry/CommonWorldIslandBridge.lean`). Dropping the metric clause readmits a mismatched pair: the same base witness extended by the dictionary scaled by two is well typed and violates the clause at an exhibited unit class. The dictionary is a candidate local readout between two mathematical objects and identifies nothing physical; the shared clock, the common action, common dynamics, the map from ports, seams, and faces to the module, and the physical spacetime attachment are not supplied.

Four further typed joins extend the bridged record. A step-scaled Maxwell bundle sits beside the declared unit-step bundle on the same twelve-port carrier and shares its initial seam data and seam current; the unit-step law is the unit instance of the scaled law, every certified Courant constant is at least the committed eigenvalue 5, so every certified step lies below one, and the receipt carries the explicit unbounded unit-step mode beside the uniform bound on the scaled fields. The phase instrument joins on the two-dimensional record algebra: its diagonal-context channel is exactly the committed pinching of the diagonal partition, which refines the record's one-block partition as every partition does, and the swap-twisted instrument violates that identity at the run state. A declared positive step duration δ identifies evolution step n with the proper-time parameter $n\delta$ on the record's clock worldline: the clock at step n is $e^{-imn\delta}$, it repeats after N steps exactly when $mN\delta \in 2\pi\mathbb{Z}$, successive step events are causally ordered, and a zero or negative step collapses the steps or breaks the causal clause. A repair law whose reference is the diagonal of the instrument's preparation carries a horizon record, and the Einstein-branch register at that law states the first-law clause on the simplex tangent space (`Lean/Geometry/CommonWorldInstrumentJoin.lean`). A formal same-index product readout packages the two islands: a single function of the step index produces the pair of the scaled staggered-form value of the record's certified bundle and the clock worldline event at proper time $n\delta$, with the index recovered from the joined object, successive joined events causally ordered, the record's step below the sharp certificate, and, for zero seam current, the joined value constant along every causally ordered index pair while the electric and magnetic energies stay within the committed multiples of the joined value at index zero. These identities are inherited independently from the event and field projections; no theorem relates the field value to the event beyond the shared index. On the committed inhabitant the joined value is constant while the clock does not repeat with period one step, and the product fixes no clock calibration (`Lean/Geometry/CommonWorldMaxwellClockJoin.lean`). A chosen quadratic clock functional can be added formally to the Maxwell functional: conditional on its declared endpoints, the worldline is the unique stationary path in the committed Lorentz form, whose exact expansion carries the action of the variation as the entire quadratic remainder, and the direct-sum action, the committed scaled Maxwell window action plus the worldline action on the shared window, is stationary in its Maxwell sector exactly at the scaled Ampère and Gauss clauses and in its worldline sector exactly at the record's clock line, with the committed configuration stationary in both sectors. The sum is proved a direct sum, partial variations independent in both directions, and because the form is Lorentzian a displaced spatial history strictly beats the stationary worldline, so stationarity is proved without any minimality claim (`Lean/Geometry/CommonWorldJointAction.lean`). This zero-coupling sum is a regression surface, not a source-selected physical common action; its displayed relative sector coefficient is likewise declared and cannot be fixed by the decoupled stationarity equations. The step dictionary is declared and identifies no physical time. For the carrier map itself a candidate exists: twelve exact golden-ratio-ring rays embed the ports into the spatial slice of the committed Lorentz module, reproduce the committed Gram table up to one squared-length normalization, respect the seam, face, and antipode incidence in both directions, carry the full sixty-element committed action through two exact rotation matrices and a sixty-word generation certificate, act compatibly with positive-factor barycentric refinement on nondegenerate mesh points, and land spacelike for the committed determinant form; the canonical assignment and its negation both satisfy every clause, so the selection is declared rather than forced, and the embedding carries no scale, causal content, or observer readout (`Lean/Geometry/ScreenCarrierMapCandidate.lean`). The candidate is dynamics compatible: a constrained transport structure, whose two generator inhabitants are kernel-verified row by row against the committed seam and face tables, intertwines the scaled Ampère update, the Gauss constraint with pulled-back charge, the staggered energy form, and the committed gauge transformation exactly and uniformly in the step, so no calibration is selected; through the committed word certificate every row of the sixty-element action is the port component of such a transport, the generators act geometrically through the two exact rotation matrices, and transport commutes with barycentric refinement along both routes. A declared interaction candidate couples the worldline increment to the seam electric field through the embedded seam directions: it is gauge invariant at every relative normalization, its zero value recovers the decoupled direct sum as the regression limit, and an explicit pulse configuration separates normalizations, so the coupling constant is declared rather than forced (`Lean/Geometry/CarrierDynamicsCompatibility.lean`). At a declared nonzero step h, varying the coupled action in both sectors shows what the candidate couples: the field equations are the committed Ampère and Gauss clauses at augmented sources, and the induced sources are polarization-type, a step difference of the worldline seam current and a neutral bound load, so the candidate carries no monopole charge and its force on the worldline is the coupling times the step difference of the embedded field, with the exact worldline equation $2m\,\Delta^2 x = \kappa\,\Delta E$. One declared identification of the induced Gauss-law load at a port with a declared value q fixes the normalization uniquely, $\kappa = -qh/(12 - 4\varphi)$ on a seam crossing, with the constant computed exactly from the golden-ratio seam data; the identification is step-local and increment-proportional, only κ/h is fixed, and the minimal-coupling shape pairing the potential itself with the increment is the live monopole alternative (`Lean/Geometry/ChargeFixedInteraction.lean`). The dynamical content of the mass–energy identity is located in the same setting as a conditional theorem: under a declared composite shape in which the rest parameter plus the modular energy of the cap first law multiplies the committed worldline functional, stationarity is exactly $(m + E)\,\Delta^2 x$ equal to the impulse of a declared potential, so internal energy enters the inertial coefficient with slope one, while a second declared shape entering the same ledger additively has coefficient m, agrees with the first at zero energy and on the committed worldline, and is separated from it by an explicit parabola history; the slope with which internal energy enters inertia is selected by the shape, both shapes are declared enrichments by the realized-history Legendre non-identifiability, and nothing derives the identity (`Lean/Geometry/InternalEnergyInertia.lean`).

The monopole route named there is built as a declared point charge hopping along the ports of the thirty-seam complex, minimally coupled to the seam and port potentials with the source weights of the committed window action. At a declared nonzero step h, its induced load has total q at every step and its induced current is $-(q/h)$ on the hopped seam, oriented so that the committed continuity equation holds identically; the coupled action equals the window action at the augmented sources with no endpoint term, so the field equations are the committed scaled Ampère update and Gauss constraint at the augmented sources. For a general source pair the gauge change of the source pairing is the gauge function paired against the continuity residual plus an endpoint term, and invariance under endpoint-vanishing gauge functions is equivalent to continuity on the interior steps. Along the Ampère evolution the staggered energy moves by $(q/2)(E_n(e) + E_{n+1}(e))$ on a forward hop across e and is constant at rest; no energy of the charge and no equation of motion for the path are defined. At the endpoints of a crossed seam the E-paired induced load is $-(\kappa/h)(12 - 4\varphi)$ times the step difference of the unit hopping load, and equals the hopping step difference exactly at $\kappa = -qh/(12 - 4\varphi)$; at the exhibited seam a port away from the endpoints carries a nonzero E-paired load while the hopping difference vanishes, so the polarization is the step difference of the monopole at the seam endpoints only. At nonzero h the charge is unconstrained by the field sector, and the path, charge, nonzero step, and coupling shape are declared (`Lean/Geometry/PortChargeMinimalCoupling.lean`).

The monopole route joins the clock action through a declared class of worldlines. A seam-step worldline is a declared start port and a declared step sequence, each step a rest or a signed seam of the carrier map candidate; it generates a Lorentz-module path with time coordinate τn for a declared unit τ and spatial coordinate the cumulative sum of the signed seam vectors, and a hopping port path. The seam vector is the ray difference of its endpoints in $\mathbb{Z}[\varphi]^3$, and at every step the ray of the occupied port minus the ray of the start equals the spatial coordinate exactly, so the injective ray map reads the port off the path. At every crossing step and a declared nonzero field step h, the E-paired induced load at the two endpoints of the stepped seam equals $-(\kappa/h)(12-4\varphi)$ times the step difference of the unit hopping load, and equals the hopping step difference of charge q at $\kappa = -qh/(12-4\varphi)$; the worldline seam current is the $\mathbb{Z}[\varphi]$ pairing of each seam vector with the stepped vector, ± 4 at the stepped seam and at its antipodal seam (the two carry one vector) and nonzero at a neighbouring seam, and the hopping current is $-(q/h)$ times its projection onto the stepped seam divided by four. At nonzero h, the transported action, the monopole coupled action plus the clock action of the generated path, has field-sector stationarity equal to the committed Ampère and Gauss equations at the transported sources; the clock action is $\sum(\tau^2-4)$ over crossings plus $\sum \tau^2$ over rests, timelike exactly when $\tau^2 > 4$. Distinct units give distinct paths with one hopping path and one source family, and at nonzero h the unit is unconstrained by the field sector (`Lean/Geometry/WorldlineHopTransport.lean`).

The transported charge has a discrete balance. Varying the transported action over the declared class of closed two-step variations, the replacement of one step pair by an admissible pair with the same endpoint, gives an exact action difference: the clock part is the original step norms minus the varied ones (rest 0, crossing 4, the unit cancels), the interaction part is q times the potential pairing $h\varphi_n(u) - \langle \mathbf{1}_{u\to v}, A_{n+1}\rangle$ summed along the alternative route minus the original one, and the window action cancels. Exchanging a forward crossing of seam e with an adjacent rest moves the action by $q\,h\,E_{j+1}(e)$ at zero clock cost, so, for $q \neq 0$ and $h \neq 0$, the exchange is stationary exactly when the scaled electric seam field vanishes at the delayed step; replacing two rests by a round trip across e moves it by $-8 - q\,h\,E_{j+1}(e)$, so at zero field rest is neither stationary nor a local minimum in this class, the port form of stationarity without minimality. For static potentials the interaction difference is q times the port-potential difference at the intermediate port plus q times the discrete circulation of A around the four-leg loop, and a round trip on one seam has zero circulation. The class, the potentials, and the shapes are declared (`Lean/Geometry/TransportedChargeForceLaw.lean`).

The closed carrier also fixes the neutrality condition rather than hiding it in the field solver. A hopping load and current obey the discrete continuity equation identically; the sourced scaled Ampère and Gauss equations admit a history exactly when the initial load is neutral and continuity holds, so one isolated hopping charge is compatible only at zero charge. The viable neutral branch is explicit. Put charges q and $-q$ on two declared seam-step worldlines and form one action from the source-free Maxwell window, both minimal couplings, and both clock actions. Its field variations are equivalent, for $h \neq 0$ and endpoint-fixed seam-potential variations, to the interior Ampère equations at $m < N$ and Gauss equations at $n < N+1$ for the pair's own sources. The action is invariant under gauge functions vanishing at the window endpoints. Replacing either path over the same closed two-step class, with both fields and the other path fixed, gives an exact off-shell clock-plus-interaction action difference, not by itself an equation of motion. At $q = 1$ and $h = 1/2$, crossings of seams 0 and 29 recover an explicit nonzero-current, Coulomb-started field-stationary inhabitant. It fails the declared closed-two-step exchange-stationarity condition in both path slots: for every $N \geq 1$, delaying either the positive seam-0 crossing or the negative seam-29 crossing while fixing the fields and the other path changes the same action by $5/12$ for every pair of clock-unit values within the fixed quadratic clock-action form. The two results are charge conjugates; applying both canonical exchanges simultaneously changes the action by $5/6$. These are only the displayed discrete exchanges, not continuum first-variation claims or a no-go for other paths, fields, or action shapes. The counterexample is specific to that Coulomb start. On the window $N = 1$, a second exact rational temporal-gauge history at the same $q = 1$, $h = 1/2$, and the same crossing pair satisfies the sourced Ampère and Gauss equations. The four possible intermediate ports for each closed two-step path replacement are exhausted exactly. Endpoint routes have zero action difference, and the two triangular detours have clock difference -4 cancelled by interaction difference $+4$. Thus one finite action is stationary in both field slots and unchanged under every in-window closed two-step replacement in either path slot. The seam currents are -2 and $+2$ on seams 0 and 29, and the initial loads are $+1$ and -1 at the corresponding ports. At the separately declared clock unit $\tau = 3$, both crossings are timelike in the carrier's Lorentz-module metric. The charge, paths, step, rational history, clock unit, coupling shape, and variation class remain declared; the theorem establishes existence without uniqueness or source selection. It proves no minimum, stability, or dynamic selection. No laboratory-time calibration or continuum limit is claimed (`Lean/Screen/SeamChargeContinuity.lean`, `Lean/Screen/NeutralPairCoupledAction.lean`, `Lean/Screen/NeutralPairJointStationaryWitness.lean`).

Inside the rest-diluted class the law reads as a position law. A general one-step replacement with shifted later ports has the exact difference of the one-step clock term plus q times the pairing difference at the step plus q times the sum of the later pairing differences at the shifted ports, reducing to the one-step term under declared shift invariance and to the closed two-step difference when the route rejoins at the second step. Within a block of k rests and one crossing, moving the crossing by one position changes the action by $q\,h\,E(e)$ at the delayed index (minus that at the crossing index for an advance) with zero clock cost, so, for $q \neq 0$ and $h \neq 0$, the crossing position is stationary exactly when the seam field vanishes at the reachable indices, and under a static nonzero field it is never stationary and drifts with the sign of qhE. The block momentum $2m((k+1)\tau, s_e)$ is unchanged by every in-block move, which swaps the per-step momenta. Along the respective scaled Ampère evolutions sourced by the original and delayed hopping currents, and with the original field static across the crossing step, the delay difference equals h times the hop's field-energy transfer $(q/2)(E_j + E_{j+1})(e)$ (`Lean/Geometry/TimelikeClassForceLaw.lean`).

The same class meets the field sector's stability window. For nonzero h, uniform boundedness of the electric seam energy for every zero-current datum holds exactly when $h^2(3+\sqrt{5}) < 4$; inside that strict window the magnetic face energy is also uniformly bounded, while equality and larger steps admit an explicit zero-current solution with unbounded electric seam energy. No bound on the gauge potential A follows. A crossing step is timelike exactly when the unit satisfies $\tau^2 > 4$. Under the declared identification of the worldline unit with the field step ($\tau = h$) no step lies in both, so inside the strict energy window every per-step crossing is spacelike and a timelike per-step crossing sits above the threshold with an unbounded electric-energy solution; the two thresholds differ by the factor $2\varphi^2 = 3 + \sqrt{5}$. A block of k rests and one crossing is timelike exactly when $2 < (k+1)\tau$, which inside the window forces $\sqrt{2}\,\varphi < k+1$, at least two rests per crossing, with the two-rest block timelike exactly for $2/3 < h$; the block speed $2/((k+1)\tau)$ is below one exactly on timelike blocks and its supremum over the window is one, approached and never attained. Without the identification the thresholds are independent ($\tau = 3$, $h = 1/2$ gives timelike crossings with uniformly bounded electric seam and magnetic face energies), and the corpus joined witnesses are not identified. The identification is declared and nothing selects it (`Lean/Geometry/SeamStepSpeedLimit.lean`).

Which rule the hosted internal process follows along such a worldline is compared exactly. Index accrual, one internal step per shared step index at the declared unit, is the per-step member of the internal-action family; proper-length accrual is the declared proper-time principle. Over a window with c crossings the two differ by exactly $E\,c\,(\tau - \sqrt{\tau^2 - 4})$. For $E \neq 0$ this difference is nonzero on every window with a crossing at every timelike unit and is zero exactly on resting windows; the midpoint refinement of a worldline with a crossing is the generated path of no seam-step worldline. For $E \neq 0$, index accrual fails refinement invariance for every nonzero unit while proper-length accrual passes, so the source chain advancing per shared index is the rule that refinement invariance excludes on every moving worldline with nonzero ledger. The uniform worldline with one crossing per $k+1$ steps has proper length $(k\tau+\sqrt{\tau^2-4})/((k+1)\tau)$ times that of a resting worldline over the same index count, below one for $\tau>2$, tending to one in k, $\sqrt{5}/3$ at $\tau=3$, $k=0$; under the two declared readings of one internal unit the recurrence clock's mean return 61511/7155 reads as that many shared steps on every worldline, or as the inverse factor times it on the uniform one. Neither rule is selected by the source or the join (`Lean/Geometry/SourceClockRateAlongWorldlines.lean`).

The length rule has a construction. Clock the hosted process by the accumulated proper length in units of a declared internal unit u: the clocked index $c(n)$ is monotone with each increment in $\{0,1\}$ for $\tau\le u$, equals the shared index on the resting worldline at $u=\tau$, and on the uniform worldline with k rests per crossing obeys $n\,d\tau/u-1<c(n)\le n\,d\tau/u$ on full periods with d the dilation factor, so $c(n)/n\to d\tau/u$, exactly d at $u=\tau$. The clocked return count of the recurrence chain equals the chain's return count at $c(n)$, every per-chain-step quantity converts with the same bound, and the clocked index is invariant under midpoint refinement by construction. No constant step duration of the joined architecture reproduces the accumulated proper length on a moving uniform worldline with at least one rest between crossings; the all-crossing case is matched by the constant duration $\sqrt{\tau^2-4}$, so the clocked chain is a declared enrichment of the join, two internal units give two clocked indices, and nothing selects the unit from the source (`Lean/Geometry/ProperLengthClockedChain.lean`).

The slope with which internal energy enters inertia is then selected by two declared principles, each with an exact theorem. On the momentum map, let the composite four-momentum range over the declared family $P_\lambda(m,E;u)=(m+\lambda E)\,u+(1-\lambda)E\,e_0$, with u on the future unit hyperboloid and e_0 the rest direction; every member has scalar coordinate $m+E$ at the standard frame. Under an oriented Lorentz map L the family transports as $P_\lambda(Lu)=L\,P_\lambda(u)+(1-\lambda)E\,(e_0-Le_0)$, so P_λ is frame covariant exactly when $\lambda=1$ or $E=0$, the converse witnessed by an explicit boost with parameters $(5/4,3/4)$; the Lorentz square is $(m+E)^2+2(1-\lambda)E(m+\lambda E)(u^0-1)$, so the shell identity at the rest energy holds at every frame exactly when $(1-\lambda)E(m+\lambda E)=0$. The selection is kinematic, on the momentum map, and the slope-one member is the composite momentum of the inertia precursor with no factor (`Lean/Geometry/CompositeMomentumCovariance.lean`). On the action, the declared midpoint refinement of a worldline leaves the proper length invariant, doubles the step count, and halves the quadratic clock action; in the internal-action family $E(aL+bM)$ with L the proper length and M the step count, refinement invariance on timelike windows holds exactly when $b=0$ for $E\ne 0$, which excludes the per-step additive ledger of the second composite shape at nonzero ledger. The proper-time principle, declared, sets the internal action to E per unit proper length; the rest phase of the declared internal clock advances by the mass parameter per unit proper time and its window increment equals that internal action, so the declared internal clock is the proper-length form, and within the family $E\lambda L$ agreement with the principle holds exactly when $\lambda=1$ for $E\ne 0$. For the length action $(m+E)L$ plus a declared potential pairing, the derivative along every variation exists on timelike windows, and stationarity under endpoint-fixed variations is equivalent to $(m+E)$ times the unit-tangent difference equal to the impulse at every interior node, so the inertial coefficient of the refinement-invariant form is exactly $m+E$; on unit-increment windows the length and quadratic equations of motion coincide (`Lean/Geometry/ProperTimeInternalAction.lean`). The refinement, the proper-time principle, the momentum family, and the covariance requirement are declared, every shape is a declared enrichment by the realized-history Legendre non-identifiability, and nothing derives the identity; its dynamical content is located in these two principles.

The ledger itself composes exactly. Over a product reference the modular energy of any joint state, correlated or not, is the sum of the marginal ledgers; against a general joint reference the binding defect is the expectation of $\log(\tau_1\tau_2/\tau_{12})$, and against the Gibbs reference of a declared total energy $H_1 + H_2 + V$ it is $\beta\langle V\rangle_p + \log(Z_{12}/Z_1Z_2)$, whose constant vanishes at $V = 0$. For a declared attractive $V \leq 0$ and $\beta \geq 0$ the state part is nonpositive while the constant is nonnegative, so the sign of the full defect is not fixed, and a two-point uniform state has the exact positive defect $-1/4+\log((3+e)/4)$; every normalized faithful joint reference is Gibbs for its own modular Hamiltonian, so the defect is always the expectation of a state-independent effective interaction. Under the declared composite shape the inertial coefficient of a composite is $m_1 + m_2 + E_1 + E_2$ plus the defect, a definitional composition whose physical reading is the calibration import (`Lean/Thermodynamics/ModularEnergyAdditivity.lean`). Matter dynamics beyond these declared shapes and observer readout appear in no field. The common-world row therefore stays owed. A physical common action, source-selected carrier identification, matter dynamics, and readout are four named residual categories, not an exhaustive theorem about all required attachments.

2.3 Conditional history weights and the variational bridge boundary

Two standard finite-horizon constructions clarify what the observer axioms must supply before an action principle could be claimed.

Theorem 2.3 (Finite path projection and positive-gap tail)**.** *Let Ω be finite, let $\Gamma = \Omega^{\{0,\dots,N\}}$, and let P, τ be probability laws on Γ, with $\tau > 0$. Let $R > 0$ be a reference weight, let $F_a : \Gamma \to \mathbb{R}$ be finitely many declared observables, and suppose*

$$\log \tau(\gamma) = \log R(\gamma) - \sum_a \lambda_a F_a(\gamma) - \log Z$$

and P and τ have the same F_a-moments. Then

$$D(P\|R) = D(P\|\tau) + D(\tau\|R), \qquad D(\tau\|R) \leq D(P\|R).$$

With

$$A_{\text{eff}}(\gamma) = -\log R(\gamma) + \sum_a \lambda_a F_a(\gamma),$$

one has $\tau(\gamma) = \exp[-A_{\text{eff}}(\gamma) - \log Z]$, so every supplied modal history minimizes A_{eff} over Γ.

Separately, for any nonempty finite set Γ, function $S : \Gamma \to \mathbb{R}$, global minimizer γ_0, and

$$q_\beta(\gamma) = \frac{e^{-\beta S(\gamma)}}{\sum_\delta e^{-\beta S(\delta)}},$$

the mass at least $\Delta > 0$ above that minimum obeys

$$\sum_{S(\gamma)\geq S(\gamma_0)+\Delta} q_\beta(\gamma) \leq |\Gamma| e^{-\beta\Delta}$$

for $\beta \geq 0$, and this displayed mass tends to zero as $\beta \to \infty$. Finiteness also gives the stronger concentration statement

$$\lim_{\beta\to\infty} \sum_{S(\gamma)>S(\gamma_0)} q_\beta(\gamma) = 0,$$

so the total mass outside the full minimizer set vanishes.

Proof. Expand $D(P\|R) - D(P\|\tau)$. The displayed logarithmic form, normalization, and moment matching make the result independent of P on the constraint surface; evaluating it at $P = \tau$ gives the Pythagorean identity. Nonnegativity of $D(P\|\tau)$ gives minimality. Exponentiating the logarithmic form reverses probability and effective-action order. For the tail bound, the partition sum is at least $e^{-\beta S(\gamma_0)}$, while each displayed numerator is at most that quantity times $e^{-\beta\Delta}$; summing and taking the elementary exponential limit proves the fixed-gap statement. Apply that argument to each strictly nonminimal history and sum over the finite set for the final limit. The statements are checked in Lean [25]; they are instances of standard information-projection mathematics [33] and elementary finite exponential estimates. □

The theorem assumes the exponential form and matching moments. It neither constructs R from the OPH source nor proves that an Axiom 3 history optimizer exists, has that form, or is unique. Calling R a probability law additionally requires its normalization.

Theorem 2.4 (Real finite-horizon variational helpers)**.** *Let $L : \mathbb{R}^2 \to \mathbb{R}$ be differentiable and define*

$$\mathcal{S}_L(\gamma) = \sum_{n=0}^{N-1} L(\gamma_n, \gamma_{n+1}) \qquad (\gamma : \{0, \dots, N\} \to \mathbb{R}).$$

At an interior junction $k+1$, suppose γ minimizes $\mathcal{S}_L$ under every replacement of γ_{k+1}. Then

$$\partial_2 L(\gamma_k, \gamma_{k+1}) + \partial_1 L(\gamma_{k+1}, \gamma_{k+2}) = 0.$$

If a differentiable one-parameter family T_s satisfies $T_0 x = x$, has generator $\xi(x) = \partial_s T_s x|_{s=0}$, and obeys $L(T_s x, T_s y) = L(x, y)$, then the segment momentum

$$p_n = \partial_2 L(\gamma_n, \gamma_{n+1})\, \xi(\gamma_{n+1})$$

is equal on the two segments adjacent to that junction. More generally, for a path with $M+2$ records, if the same single-site minimization premise holds at all M interior records, then there is a scalar J such that $p_n = J$ on every one of its $M+1$ segments.

Proof. Changing one interior record changes exactly the two adjacent summands. Differentiating their sum at its declared minimum gives the first identity. Differentiating the invariance relation at $s = 0$ and combining it with that identity gives $p_k = p_{k+1}$. Applying this identity at every interior junction and inducting along the finite chain gives the scalar J. These are standard discrete variational and Noether calculations [34]; their precise premises are checked in Lean [25]. □

Proposition 2.5 (Finite/real variation obstruction)**.** *Fix a real path $\gamma : \{0, \dots, N\} \to \mathbb{R}$, a site i, and any finite family G of real paths. There is an $x \in \mathbb{R}$ for which the path obtained by replacing γ_i with x does not belong to G. Hence no finite real-path family contains every single-site real variation of even one supplied path.*

Proof. The map from x to the updated path is injective because evaluation at i recovers x. Its image is infinite and therefore cannot be contained in finite G. This elementary cardinality obstruction is machine checked in Lean [25]. □

The two theorem domains compose only through a certified receipt. The first uses paths in a finite state set; the second uses arbitrary real single-site variations, and Proposition 2.5 rules out closure under all such variations inside any finite real-path family. The composition that survives this obstruction is conditional: an undercut receipt asserts that every real single-site variation of an embedded path is undercut in action by some embedded finite path, and under that receipt finite minimality transfers to real local minimality and to the real Euler–Lagrange packet. Membership receipts are impossible (that is the obstruction restated); undercut receipts are attainable, and the committed quadratic witness carries one, while a one-path control family provably carries none and the transferred conclusion fails there. All three layers, transfer, witness, and control, are checked in Lean [25].

Which action the history law carries is itself a theorem rather than a choice.

Theorem 2.6 (Derived action of a source chain). *Let Ω be a finite nonempty state set, P a strictly positive row-stochastic kernel, and π a strictly positive normalized initial law. On paths with n transitions define the log-transition action $S_{\log}(\gamma) = -\sum_i \log P(\gamma_i, \gamma_{i+1})$ and the reference $R(\gamma) = \pi(\gamma_0)\,|\Omega|^{-n}$. Then the Markov path law equals the exponential tilt of R by $S_{\log}$ at multiplier one, with partition constant $|\Omega|^{-n}$; and a pair (S', λ') reproduces the path law as its tilt of R if and only if there is a constant c with $\lambda' S' = S_{\log} + c$ pointwise.*

Proof. Exponentiating the negated action recovers the transition product, and row normalization makes the partition sum telescope to $|\Omega|^{-n}$, which cancels against the reference weight. Uniqueness reduces to the statement that two tilts of one strictly positive reference agree exactly when the multiplier-weighted actions differ by an additive constant, proved by taking logarithms at a common path and cancelling the shared normalizer. Both directions of the gauge are realized: per-step additive constants aggregate to a path constant, and the multiplier trades against the action scale, so the bare-action convention chooses $\lambda = 1$ as a representative. It is the unique multiplier for that unrescaled action when the path action is nonconstant; for a uniform kernel the action is constant and normalization erases the multiplier. The statements are checked in Lean [25]. □

The committed source chain instantiates the theorem exactly: the tilt of its reference by its log-transition action reproduces its path law with partition constant $1/4$, through kernel-decided integer identities, and the committed repair-count action reproduces that law at no multiplier, because it takes equal values on two paths that the reference weighs equally and the chain separates. The action of the realized dynamics is therefore forced up to an additive constant and a multiplier rescaling. This multiplier is an action-normalization gauge for the supplied chain; it is not thereby identified with a gauge-sector coupling coefficient. The separate finite Gibbs-kernel construction for one three-dimensional P-family factor shows that a common rescaling of its supplied quadratic cost can occupy the same formal multiplier slot, but it neither derives that kernel from source dynamics nor determines relative couplings between simple factors.

The retained length-three history packet also has an exact finite operator attachment. Let $p_g = n_g/1754$ be its empirical law on the eight binary paths and let $S_g \in \{0, 1, 2\}$ count the two adjacent state changes. The diagonal matrices

$$\rho_{\mathrm{hist}} = \mathrm{diag}(p_g), \qquad H_{\mathrm{hist}} = \mathrm{diag}(S_g)$$

form a strictly positive normalized density and a positive non-scalar self-adjoint Hamiltonian, with

$$[\rho_{\mathrm{hist}}, H_{\mathrm{hist}}] = 0, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{\mathrm{hist}}) = \frac{197}{1754}.$$

On a one-regulator consensus tower, the selected-state construction places this pair on its completed-colimit GNS Hilbert space. The finite-stage representation is injective, the represented Hamiltonian and one off-diagonal commutator response are nonzero, and the mapped exponential flow is unitary and leaves the density stationary. The cyclic unit has nonzero energy, so it is not a ground state or vacuum for this Hamiltonian.

The eight histories also reindex exactly as three binary tensor slots. Their one-slot algebras and the two adjacent interval algebras are isotone, the first and last slot algebras commute, and the Hamiltonian splits without a remainder into positive local bonds,

$$H_{\mathrm{hist}} = H_{01} + H_{12}, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{01}) = \frac{94}{1754}, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{12}) = \frac{103}{1754}.$$

The first commutator of an endpoint observable lies in its adjacent two-slot algebra. The source functional restricts compatibly along all these algebra inclusions, and the locality, bond, and propagation identities survive the injective selected-GNS representation. This supplies a finite history-coordinate local dynamical system on a selected-GNS carrier built from the source-counted diagonal functional. The retained source packet fixes the diagonal probabilities and repair counts. The extension to the full matrix algebra, the tensor-local reading, off-diagonal observables, and the real flow parameter are mathematical adapters. The history slots are not calibrated physical times or Lorentzian regions, and no spectrum condition, continuum/RG limit, physical field, particle, scattering, or detector interpretation is obtained from them [25].

The rational record layer of the same carrier carries a finite-probability conditioning family. Conditioning the window law on any slot region defines a rational-linear, unital, pointwise-positive, localized, idempotent, tower-compatible map that preserves the empirical mean; the walls are fixed points of their own regional maps, and the exact conditional weights of the late slot are 383/415 and 2/89 in two (s_0, s_1) cells of the counts 1149 : 96 and 2 : 87. These counts give a narrowly scoped matrix no-go: no complex-linear map from the full matrix algebra into the fixed first two-slot algebra $\mathcal{A}_{01}$ can both obey the $\mathcal{A}_{01}$-bimodule law and preserve the committed state. They do not rule out ordinary AQFT inclusions with state restriction, a different or enlarged target algebra, ancillary boundary memory, generalized expectations, or non-bimodule positive or completely positive channels [25]. The enlarged-target path is constructed: a unital injective star-embedding reads the slot-two record bit of each history into one qubit ancilla, its range is exactly the ampliated interval algebra joined with the ancilla factor, and it satisfies clause for clause the conjunction the fixed-target no-go kills, with the range clause the single clause whose target changes and the exhibited state a proved non-product extension restricting to the committed source state. Two delimiting theorems establish two necessary features of this escape: a product-form state extension keeps the obstruction for every ancilla matrix, and a map into the ampliated copy alone is obstructed even against the correlated extension. Thus, within this clause family, the ancilla factor and some non-product state extension are necessary. The exhibited record-correlated extension is one witness and is not proved unique. This is a finite-matrix existence statement; regions, time slices, dynamics, and any continuum attachment are not touched by it [25].

The same packet also fixes a bounded overlapping-window action statistic. Among 1754 retained length-three windows, the two domain-wall indicators have count-weighted incidences 94 in the first edge position and 103 in the second, hence mean two-edge action 197/1754. The value 197 counts incidences inside overlapping windows, not distinct underlying repair events. The formal object stores only the eight-path histogram and chosen observables: it contains no window order, successor relation, cadence, or period. Dividing by an externally declared positive real, interpreted as a window duration, is therefore only a conditional unit conversion; that interpretation is not related to the histogram by any theorem. This supplies neither a source-derived clock nor a reduction of the physical calibration input [25].

The time row is located more exactly by two finite results. Accepted repair hosts no periodic clock: every accepted step strictly lowers the mismatch count, so the accepted-repair relation admits no cycle, and the canonical repair iteration admits no periodic orbit other than a fixed point. The committed stationary source chain, by contrast, carries a recurrent process: the first-return law of each state is written in closed form from the kernel, sums to one, and satisfies Kac's identity with the exact mean return times $61511/7155$ and $61511/54356$ steps, whose ratio $54356/7155$ is the same number under every declared tick. The return law assigns positive probability to both one- and two-step returns. Its standard Markov-chain period is therefore one, while no deterministic return interval exists because no return time has probability one. A return count along a trajectory counts distinct visit events, the property the overlapping incidence statistic lacks. One chain step stays a declared tick and no physical duration is identified; the abstract rate non-identifiability result concerns transition systems whose bridge to the repair layer fails, and it forbids nothing here (`Lean/QFT/SourceRecurrenceClock.lean`).

The field sector, by contrast, hosts periodic processes. Every eigenvector v of the local operator $C^{\mathsf{T}}C$ with eigenvalue λ carries, at a declared nonzero step h with $0 \le h^2\lambda \le 4$, the histories $A_n = \cos(n\theta)\, v$ and $A_n = \sin(n\theta)\, v$ with $\theta = \arccos(1 - h^2\lambda/2)$; they solve the zero-current scaled evolution in the temporal gauge, conserve the staggered form, and are bounded. Kernel-checked integer eigenvectors with eigenvalues 2 and 3 join the committed ones with eigenvalues 5 and $3+\sqrt{5}$. At $h^2 = 1/2$ the eigenvalue-two mode returns after six steps, at $h^2 = 2/3$ the eigenvalue-three mode after four, at $h^2 = 1/3$ after six, each step inside the sharp window $h^2 < 2/\varphi^2$, since $3 \pm \sqrt{5} = 2\varphi^{\pm 2}$; the projector traces $1, 5, 4, 4, 6$ read with rank equal to trace give the multiplicities $1, 5, 4, 4$ and $3+3$. Frequency ratios carry no tick: $h\sqrt{\lambda} \le \theta \le h\sqrt{\lambda}/\sqrt{1 - h^2\lambda/4}$, $\theta/h \to \sqrt{\lambda}$ as the step shrinks, and the small-step ratio of the golden mode to the slowest mode is φ^2 exactly. A period p and the mean return time $61511/7155$ of the recurrence clock are counted in one shared step index, with ratio $7155p/61511$ under every declared tick. The step, the gauge, and the kinetic term are declared; the repair-layer no-periodic-orbit theorems and these field-sector periods have disjoint scope (`Lean/Screen/CarrierModeOscillators.lean`).

The discrete step embeds in a continuous flow. On the amplitude-velocity plane of every eigenvector v with eigenvalue λ and $0 < h^2\lambda < 4$, the zero-current temporal-gauge evolution is the symplectic map $T = \left(\begin{smallmatrix} 1 & h \\ -h\lambda & 1-h^2\lambda \end{smallmatrix}\right)$ of determinant one and trace $2 - h^2\lambda$, conjugate by an explicit matrix Q to the rotation by the mode angle; the family $\Phi_t = Q \operatorname{rot}(\theta t/h)\, Q^{-1}$ is a continuous one-parameter group with $\Phi_h = T$ and $\Phi_{nh} = T^n$ that preserves the committed staggered energy, equal to $(\|v\|^2 \sin^2\theta/2)$ times the pulled-back Euclidean form. On the kernel of the local operator, exactly the gradient space, the evolution is the h-step of the shear flow. For any finite family of pairwise orthogonal admissible eigenvectors the componentwise coefficient-state flow is a linear continuous one-parameter group whose h-step is the committed step, instantiated on the eigenvalue-two, three, five, and golden modes and a gradient. The assembled theorem permits zero listed vectors, so a faithful flow on the actual field span additionally requires nonzero linear independence or quotienting redundant coefficient directions. This generic flow theorem alone supplies neither an inner product nor the full-curl selection; the energy construction and explicit nineteen-vector basis below supply those separate inputs. Reading t as physical time remains declared (`Lean/Screen/CarrierEvolutionFlow.lean`).

Inside the strict Courant window $0 < h^2\lambda < 4$ the staggered energy of a carrier mode is a positive definite quadratic form on the state (a, b), and fails to be one exactly at and beyond the boundary: at $h^2\lambda = 4$ the state $(-h/2, 1)$ has zero energy, beyond it negative energy, and on the gradient sector the amplitude direction is null. Its polarization is an inner product on each curl mode, the committed step and the interpolating flow are isometries, and for a nonzero step every step-invariant symmetric bilinear form is a real multiple of it, the committed energy fixing the multiple. The conjugated quarter turn squares to minus one and is orthogonal for this inner product, so each mode carries a complex line on which the flow is the phase $e^{i\theta t/h}$, with orbit derivative $i(\theta/h)$ times the state; the generator is exhibited, not obtained from Stone's theorem.

On a finite orthogonal family of curl modes the direct sum is an inner product whose diagonal is the committed staggered energy at every step, the assembled flow is unitary with diagonal generator θ_i/h. Precisely the static gradient-amplitude states with zero electric component form the radical of the semidefinite extension; gradient velocity/electric directions are non-null. On an equal-eigenvalue block invariance alone does not fix the form and the energy does. The step h is a declared Courant number and t a declared flow parameter (`Lean/Screen/FieldSectorEnergyInnerProduct.lean`).

The curl sector of the committed seam operator has an explicit orthogonal eigenbasis: nineteen seam vectors with entries in $\mathbb{Z}[\varphi]$, five at eigenvalue 2, four at 3, four at 5, three at $3+\sqrt{5}$ and their Galois conjugates at $3-\sqrt{5}$, kernel-checked as eigenvectors with diagonal Gram matrix. The four committed modes are rows of the list. The nineteen span the range of the codifferential, which is the kernel of the boundary; eleven coboundaries of listed port loads span the gradient sector, and the two sectors are complementary. On this basis the coefficient flow extends to the whole seam space: for a declared step with $h^2(3+\sqrt{5}) < 4$, the two-sided admissibility window of the family, every zero-current temporal-gauge solution is the assembled history of its initial state. On all nineteen nonzero curl modes, the weighted energy form is positive definite, its real and Hermitian forms are flow-invariant, its complex coordinates have the explicit phase generator, its diagonal is the field energy, and the coefficient state is recoverable from potential and electric readouts. After conjugating the coefficient orientation, the packet has an exact finite Stone representation for the selected principal-angle interpolation on a separately constructed $M_{19}(\mathbb{C})$ block: the self-adjoint diagonal Hamiltonian has $H_{ii} = \theta_i/h$, its propagator is e^{-iHt}, and conjugation of the coefficient outer product is exactly the outer product of the evolved coefficient state. The vector intertwiner retains absolute phase; the rank-one outer product loses global phase and retains relative phases only. The energy-weighted outer product has trace equal to the assembled field-energy Hermitian norm. A selected mode is driven by the joined scalar clock at discrete steps under the sufficient calibrations $\delta = h$ and $m = \theta_i/h$. The 19-index block is not a relabelling of the committed $M_2(\mathbb{C})$ private carrier, the committed witness fails the duration calibration, and the eigenvalue-two and eigenvalue-three principal-branch frequencies are unequal throughout the strict positive window. Thus no one identical scalar rate equals every selected principal-branch mode frequency. That statement does not exclude a shared time parameter, phase aliases, harmonics, finite-horizon decoding, mode-resolved rates, or operator-valued readout. There are five distinct frequencies among the nineteen coordinates; the logarithm branch is selected rather than uniquely derived from the discrete step, and the two displayed calibrations are sufficient rather than necessary modulo 2π aliases. The coefficient input is arbitrary and is not proved to be the joined record's actual Maxwell history. The rank-one projective image is neither onto $M_{19}(\mathbb{C})$ nor an algebra homomorphism, and it is not an equivalence with the committed private algebra. A physical bridge therefore requires a specified compression/channel, a selected lower-dimensional sector, or an enlarged private carrier. This is not a source-selected public representation or Hamiltonian, physical time calibration, photon Hilbert-space identification, or laboratory Schrödinger law. The step and flow parameter remain declared (`Lean/Screen/CurlSectorEigenbasis.lean`, `Lean/Screen/CurlStoneClockBridge.lean`).

These modes are sorted by the carrier's own symmetry. The sixty listed icosahedral port permutations induce signed permutations of the seams and orientation-preserving permutations of the faces, the signed face–seam incidence is invariant under the simultaneous action on all $60 \times 20 \times 30$ triples, and so the curvature, the codifferential, the local operator, the five spectral projectors, the mode histories, the scaled Ampère evolution for every step, potentials, and current, and the staggered energy commute with the action, all by kernel checks on the committed tables (the intertwining of the operators, fields, evolution, and energy under the same sixty rows was proved with the carrier dynamics compatibility; new here are the explicit signed actions with their composition laws, the projector and eigenspace invariance, the mode transport, and the characters).

The constant face vector spans the fixed eigenvalue-zero line, the element orders are $1, 2, 3, 5$ with class sizes $1, 15, 20, 24$, and the traces of the action on the projector images by element order are 1; $5, 1, -1, 0$; $4, 0, 1, -1$; $4, 0, 1, -1$; $6, -2, 0, 1$, with character norms $60, 60, 60, 120$. Read with the orthogonality relations, an inference outside the checked statements, the face space decomposes as $1 \oplus 5 \oplus 4 \oplus 4 \oplus (3 \oplus 3')$, the two four-dimensional eigenspaces carrying one irreducible representation and the golden sector the two three-dimensional ones; the checked characters do not separate the two order-five classes, so the split of the golden sector by eigenvalue is an observation. The field modes are thus sorted by the same group that sorts the matter sectors, with the identification of a mode with a physical oscillation open (`Lean/Screen/CarrierModeEquivariance.lean`).

The finite six-axis interface also has an abstract group endpoint. The antipodal port bridge is first a pointwise equivalence of indexed action rows. Separately, the canonical center quotient $SL(2, \mathbb{Z}/5) \to PSL(2, \mathbb{Z}/5)$, whose kernel is the center $\{+I, -I\}$ of $SL(2, \mathbb{Z}/5)$, yields a faithful action of $PSL(2, \mathbb{Z}/5)$ on the projective line under the explicit coordinates $[z : 1] \leftrightarrow z$ and $[1 : 0] \leftrightarrow 5$; its image is exactly the committed sixty-element subgroup `A5SixAxes.L60`, and the result is a typed group isomorphism onto that subgroup (`Lean/Screen/A5PortSixAxesBridge.lean`, `Lean/Screen/PSL2F5SixAxesBridge.lean`). The sixty committed twelve-port rotations are then packaged as a typed subgroup of the permutations of the twelve ports, with the multiplication and inverse laws checked on the actual port permutations, and the pointwise port bridge is promoted to a group isomorphism between that port subgroup, the six-axis subgroup, and $PSL(2, \mathbb{Z}/5)$. This is not an abstract $PSL(2, 5) \simeq A_5$ classification, an identification with the binary icosahedral group or McKay E_8, a transport of the golden pieces as typed projective-group representations, a selection of φ or a mass law, or a physical-rotation statement.

The golden split is exact. On the six-dimensional golden sector, where $N^2 - 6N + 4 = 0$, the two spectral projectors with eigenvalues $3 \pm \sqrt{5} = 2\varphi^{\pm 2}$ have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, are symmetric orthogonal idempotents summing to the golden projector, each of trace 3, and the minus projector is the entrywise Galois conjugate ($\varphi \mapsto 1 - \varphi$) of the plus one. On the sixty listed automorphisms the plus character takes the values $3, -1, 0$ on the orders $1, 2, 3$ and φ or $1 - \varphi$ on the twenty-four elements of order five, twelve each; the minus character is its Galois conjugate on every element, the two agree exactly off order five, squaring an order-five element exchanges them, both are class functions commuting with every automorphism, and each has norm 60. Read with the orthogonality relations, these data have the usual two-character interpretation, but this character module does not itself prove irreducibility. The separate span certificate below proves that the two real projector images are three-dimensional and irreducible for the listed incidence action. Scalar extension to complex irreducibility, identification with the abstract icosahedral or A_5 character table, and a physical-oscillation reading remain outside the checked statements (`Lean/Screen/GoldenSectorCharacters.lean`).

Irreducibility of the two golden pieces is a theorem. Each image $W_\pm$ of the golden projectors is invariant under every listed automorphism and has dimension three (an explicit factorization $P = AB$ with $BA = 1$); the real span of the sixty matrices $gP_\pm$ has dimension nine and maps onto all 3×3 real matrices in the basis A (nine listed elements with an explicit two-sided inverse certificate over $\mathbb{Z}[\varphi]$, the minus certificates the Galois conjugates of the plus ones), so every invariant subspace of $W_\pm$ is zero or all of it: both pieces are irreducible real representations of the listed incidence-automorphism action (`Lean/Screen/GoldenSectorIrreducibility.lean`). The pieces stay irreducible after extension of scalars to $\mathbb{C}$: the nine listed automorphisms whose transported tables span the real endomorphism algebra span the complex one with the same inverse-certificate coefficients, so every invariant complex subspace is zero or the whole piece. The two complexified pieces are inequivalent, since on the listed row of order five their traces are $1 - \varphi$ and φ, every complex intertwiner between them is zero, and the real endomorphism algebra of each piece is the real scalars, so both are of real type (`Lean/Screen/GoldenSectorComplexIrreducibility.lean`). Identification with the abstract icosahedral or A_5 table, with physical rotations, or with physical oscillations stays open.

The two attained faces of mechanics are joined by a finite Legendre bridge, also checked in Lean [25]. For discrete two-point Lagrangians the Legendre transform, its involutivity on the strictly convex class, the Fenchel–Young inequality with its equality case, and exact degenerate controls are proved; the discrete Euler–Lagrange condition at a junction holds exactly when one step of the discrete Hamilton flow carries the incoming junction state to the outgoing one, for the quadratic class and for strictly convex Lagrangians with a solver section; single-site minimizers of the log-transition local action coincide with most probable paths of the realized chain, through an exact corner identity; and the constant Noether current of the chain face equals the quadratic Legendre momentum, conserved together with the energy along the free Hamilton orbit that reproduces the committed witness path. One remainder is named: no theorem produces the realized chain from a Hamiltonian flow. What is proved in that direction is narrower. The committed repair-count action reproduces the chain path law at no multiplier and lies outside the additive-constant and rescaling gauge orbit of the derived log-transition action (`sourceRepairAction_no_multiplier`, `sourceRepairAction_not_gauge`), so the naive quadratic-Gibbs identification is refuted on the chain literals, and the derived action above is the only route this package supplies. A direct Hamiltonian-flow derivation of the chain, for example a block Hamiltonian of the Stone converse coarse-grained to the two-state chain, is excluded by no theorem here and stays open.

Theorem 2.7 (Realized histories do not select a real Legendre enrichment)**.** *Let $L_0(x, y)$ be the bilinear real extension of the committed two-state log-transition table. It is affine in y, so its momentum at fixed x has singleton image and admits no global velocity solver. For every $a \in \mathbb{R}$, define*

$$L_a(x, y) = L_0(x, y) + \frac{a}{2}\, y(y - 1).$$

Every L_a agrees with the exact source log-transition action on every realized binary history at every path length. For $a > 0$, however, L_a is strictly convex in y, has an explicit global velocity solver and Legendre transform, and different positive a's give different formulas. In particular the checked $a = 1$ and $a = 2$ members have distinct Lagrangians and Hamiltonians, agree on all source histories but differ by $1/8$ when one middle record is varied to $y = 1/2$.

Proof. The correction $y(y - 1)$ vanishes at both source symbols. Differentiation gives $\partial_y L_a = \partial_y L_0 + ay - a/2$, so $a = 0$ is degenerate and $a \neq 0$ has the displayed affine inverse. Completing the square gives the Legendre transform; the strict-convexity gap is $(a/2)b(1 - b)(y - z)^2$. The path identity, midpoint control, solver, convexity, and the $a = 1, 2$ distinction are checked in Lean. □

The reference measure is characterized within a declared independent-target scrambling normal form: among row-stochastic kernels, invariance under every independent relabeling of transition targets at fixed source, row-constant transition weight, and constant log-transition step action are each exactly equivalent to the uniform kernel, so any independently target-relabeling-invariant Markov reference has the step-uniform path law, and a biased two-state control shows the invariance premise is load-bearing. Ordinary simultaneous source-target relabeling is strictly weaker and admits a positive nonuniform stay-biased control. The stronger condition fixes the reference gauge by an invariance convention, exactly as a coordinate convention is fixed; the invariance principle itself is stated, not source-produced. Feeding the separately declared counting reference and trivial visible datum into the Axiom 3 conditional-resampling construction reproduces the uniform transition kernel exactly. This is a conditional realization, not source selection of those inputs, the initial distribution, or the complete path reference; constant rescalings of the counting mass are another exact ambiguity. The committed reference/action/multiplier receipts also belong to two distinct Gibbs constructions and do not form one source-selected capstone. Within the repair-count construction, the matching polynomial is strictly increasing on the positive exponential-parameter ray, so the supplied empirical target has one unique matching parameter. The target observable and level are declared postprocessing inputs rather than prospectively source-selected constraints.

A three-record concave-action control has a stationary history that is not a minimum and is assigned less positive Gibbs weight than a unit variation. It proves that a positive Gibbs/MAP rule need not make every stationary point a mode or minimizer. It is not a no-go for constrained saddles, complex stationary phase, path-integral interference, or a source-selected real Lagrangian, all of which remain available. The package identifies neither A_{eff} with the local action nor the finite scalar chain constant with a physical current, and supplies no amplitude, clock, or continuum path measure. The theorem above also shows that the realized two-state history law, on the committed chain literals, does not select the off-alphabet curvature needed for a regular Legendre system; finer alphabets of the same run, refinement limits, other source data, and declared principles are untested by that theorem and stay live routes to the curvature. It derives the finite action and conditional bridges at the representation level; the real enrichment and physical attachments are not supplied under the attachment doctrine.

Two composition results sharpen this boundary. First, the two halves of the mechanics package form one theorem (`source_to_hamiltonian_composed`): under the registered path reference, real enrichment, and supplied-dynamics rows, a fixed-endpoint real-extremal embedded history realizes the derived corner action of the same kernel whose exponential tilt over the registered reference is the path law, is an interior most-probable update of that law at every interior site, and satisfies the discrete Hamilton equations. The separate `source_to_hamiltonian_composed_noether` theorem makes the segment momenta constant under supplied symmetry data; the mode/minimizer scoping and the enrichment no-go ride along as conjuncts of the same statement. Second, while the realized law cannot select the enrichment, a declared mode-extremality principle fixes one member inside the registered one-parameter curvature ansatz on the committed chain: interior stationarity of the embedded constant-record history forces the curvature $a_\star = 2\log\bigl(W_{11}^2/(W_{10}W_{01})\bigr) > 0$, positivity being exactly strict interior mode dominance, and at $a_\star$ that history is a genuine fixed-endpoint single-site minimizer at every interior junction and length, so the composed conditional is inhabited on the committed kernel rather than vacuous. This is not uniqueness among regular corner-preserving enrichments, and the residual freedom is exactly characterized inside one declared grammar, in the module `EnrichmentCharacterization`. Within the quadratic two-slot grammar, a sentence vanishes at the four binary corners exactly when it is $\varepsilon\, x(x-1)+\zeta\, y(y-1)$, so the corner-invisible enrichments are exactly the two-parameter family $L_{a,c} = L_0 + a\, y(y-1)/2 + c\, x(x-1)/2$, with injective parameters and no third corner-invisible quadratic direction (`quadPoly_cornerInvisible_iff`). The committed interior-junction stationarity holds on this family exactly on the affine line $a+c = a_\star$ (`chainTwoSlot_stationary_line`); members of that line share the binary corners, the constant-history stationarity, and the single-site variation minimum, while the regular member $a = c = a_\star/2$ has a different velocity curvature. One named velocity-only clause, that the added sentence is a function of the velocity slot alone, restores uniqueness at $(a, c) = (a_\star, 0)$, the committed one-parameter rule with all six grammar coefficients pinned (`quadPoly_selection_unique`). Each clause is load-bearing: the counterfamily point $a = c = a_\star/2$ satisfies every clause except velocity-only and carries a different momentum map (`velocityOnly_clause_necessary`); an explicit velocity-only cubic satisfies corner invisibility and the junction stationarity equation, so quadraticity is load-bearing under stationarity alone (`quadraticity_clause_necessary`); and the committed real fixed-endpoint single-site minimality excludes those cubics and returns the committed rule exactly (`chainCubic_realMin_forces_quadratic`). This cubic exclusion is not global selection. For every $0 < \lambda < a_\star$, the velocity-only quartic continuation

$$L_\lambda(x, y) = L_{a_\star}(x, y) + \lambda y^2(y - 1)^2$$

is corner-invisible, has the same junction stationarity, and keeps the constant-one history globally minimal under every fixed-endpoint single-site variation, with exact gap $(x - 1)^2(a_\star/2 + \lambda x^2)$. Its velocity Hessian is $(a_\star - \lambda) + 12\lambda(y - 1/2)^2 > 0$, so its momentum map is strictly increasing, surjective, and globally invertible, yet it is neither the committed rule nor any member of the quadratic grammar (`quartic_continuation_blocks_global_selection`). Thus the tested clauses are not a proved-minimal or globally selecting axiom system.

The quartic boundary is sharpened by one further named clause, tested in `Lean/Variational/PolynomialEnrichmentSelection.lean` against the full two-slot polynomial grammar at every degree bound. A two-point sentence is a discrete null Lagrangian when no single-site interior replacement changes the sum of the two adjacent terms; by the committed difference identity such sentences contribute only telescoping and boundary terms to every fixed-endpoint local action. The literal boost clause, that every shift residue $q(x + u, y + v) - q(x, y)$ is null, fails at the committed increment itself (`committed_increment_not_boostNull`): a lattice-anchored quadratic has a linear, non-telescoping boost residue. The named candidate clause is second order: every boost difference of a boost difference is null (`BoostNullCovariant`). Every quadratic-grammar sentence satisfies it. Among velocity-only polynomial enrichments of every degree bound the clause holds exactly for the quadratic-affine members (`velocityPoly_boostNull_iff`), and the quartic continuation fails it at an explicit finite witness (`quarticIncrement_not_boostNullCovariant`). In the full polynomial grammar, corner invisibility, the velocity-only clause, the committed stationarity calibration, and the boost clause select exactly the committed point (`polyGrammar_selection_eq_committed`), and the committed point satisfies all four clauses, so the axiom system is inhabited. Every clause is load-bearing: dropping the boost clause readmits the quartic, and in the full two-slot grammar the boost clause does not force quadraticity, since the antisymmetric cubic $(x - y)^3 - (x - y)$ is corner-invisible, satisfies the boost clause, is neutral for the committed stationarity calibration, and equals no sentence of the committed quadratic grammar (`boost_clause_does_not_force_quadraticity`); the velocity-only clause is what removes this direction. The boost clause is a declared candidate axiom rather than a derived symmetry: no boost action on realized histories and no invariance of the source law is constructed, and smooth non-polynomial enrichments are outside the theorem.

The grammar, the mode-extremality principle, and the velocity-only clause are declared rather than source-derived; nothing here states that the source selects the enrichment, and no real enrichment is supplied. The strengthened nonidentifiability boundary remains.

2.4 Thermodynamics from conditional repair in an observer modular parameter

On one finite regulator stage, the four laws form a conditional theorem package once one repaired distinction and two compatible instantiations of Axiom 3 are supplied. Five named source and physical receipts are unsupplied. The Lean development carries three composed named-premise surfaces for these packages. `FourLawAdequacySurface` proves one theorem, `fourLaws_composed`: one typed antecedent bundle whose fields are exactly the registered repair law, the clock-and-energy calibration, and the refinement-uniform gap family, and one conclusion record whose clauses all consume that bundle: the kernel receipts, both zeroth-law forms, the modular and diagonal-heat-channel first laws, the second-law and fluctuation statements, the finite and refinement-uniform third laws threaded to the same calibrated energy through an explicit member identification, and the Landauer bound. The same surface carries an exact dissipation identity: the mean fluctuating entropy production of one repair step equals the relative entropy from the input state to its repaired image, is strictly positive whenever the step changes a strictly positive state, and vanishes exactly on the fibre-conditional-reference fixed points, so one state-changing projection has a strict dissipative orientation rather than a merely nonnegative one. Because this repair kernel is idempotent, repeating the identical step stalls on the fixed manifold after that first projection; a sustained macroscopic arrow requires source dynamics. `HorizonThermalitySurface` carries the cap Clausius inequality and the capacity identities as the horizon-facing composition. `MechanicsAdequacySurface` re-exports the composed source-to-Hamiltonian theorem described below. Three maps stay distinct. The strict-descent normalizer settles public facts; it is terminating and idempotent and carries no entropy inequality, and the two-point normalizer $a \mapsto b$, $b \mapsto b$ sends the uniform law to a deterministic law, dropping Shannon entropy from $\log 2$ to zero. The equilibrium repair operator resamples unresolved degrees inside the fibre of the complete repaired visible datum. The modular flow of a faithful reference supplies the reversible observer ordering and preserves relative entropy. Only the middle map carries the arrow.

A directed coherent family is supplied by `CoherentRefinementFamily`, with per-stage finite spaces and energies, refine maps satisfying identity, composition, energy, and ground-compatibility laws, and one uniform spectral-tail envelope. The envelope yields one inverse-temperature threshold per tolerance for the off-minimum Gibbs mass of every member (`uniform_concentration`). The type is directed, not cofinal: it has no ambient regulator system, cofinal embedding, strict stage growth, or no-maximal-stage field, and a constant or singleton family can inhabit it.

The explicit ladder family has nontrivial truncation maps, satisfies the envelope, and has unbounded stage cardinality (`ladderFamily`, `ladder_uniformGap_isEmpty`). This proves that a spectral-tail envelope does not imply a uniform total-cardinality bound on the same carriers and energies. It does not prove inclusion between the complete structures: `ofUniformGap` requires additional directedness, functor, energy, and ground hypotheses, while the constant conversion installs identity maps rather than preserving arbitrary old refinement maps.

The theorem `fourLaws_composed_coherent` is a companion conjunction, not a rethreading of the supplied family through the old conclusion record. The old `FourLawConclusions` is obtained using a separately constructed degenerate one-member gap family; the supplied directed family separately contributes its envelope and concentration clauses. No source-produced family, ambient cofinal refinement, or physical temperature is attached. Those requirements and the physical energy/clock identification are additional assumptions.

The stronger module `CofinalSpectralTailFamily` supplies the missing formal regulator semantics without relabeling directedness as cofinality. It adds a nonempty ambient regulator preorder, an order-reflecting cofinal stage embedding, no maximal ambient regulator, and the data-level condition

$$\forall r\ \exists s > r, \qquad |X_r| < |X_s|.$$

Consequently the represented stage type is infinite and a constant finite carrier cannot inhabit the interface. A nonconstant $\mathbb{N}$-indexed ladder has strict carrier growth and unbounded cardinality and supplies a nonvacuous inhabitant. Its spectral envelope gives one temperature threshold uniformly over all stages and, for every ambient regulator, over a complete cofinal tail above it.

Most importantly, `fourLaws_composed_cofinal` does not recover the old four-law record through a synthetic gap family. It constructs the refinement-independent finite-law core directly from the declared repair and calibration, and every regulator-family conclusion refers to the same supplied cofinal family: the envelope, uniform and cofinal-tail concentration, strict stage and carrier progress, and infinitely many stages. A separately supplied calibrated-stage energy identification yields exact off-minimum-mass equality and the same envelope at that member; the calibration is not derived. Strict carrier-cardinality growth is one sufficient genuine-refinement route, not a claim of necessity for every physical regulator model. No OPH source produces this family or identifies its regulator, energy, inverse temperature, repair, or continuum meaning, so the uniform low-temperature tail premise is narrowed rather than removed.

Axiom 3 applies separately to states and to transition distributions, and both instantiations are solved exactly. On the state side, with faithful global reference σ_0 and observer-visible conserved quantities Q_a, the information-projection Pythagorean identity $D(\rho\|\sigma_0) = D(\rho\|\tau_\lambda) + D(\tau_\lambda\|\sigma_0)$ holds on the constrained moment surface, so the optimizer, unique by strict convexity of relative entropy, is the exponential family $\tau_\lambda = Z^{-1}\exp(\log\sigma_0 - \sum_a \lambda_a Q_a)$; on the tracial branch this is the Gibbs family. The exact zeroth-law receipt is temperature identifiability: on one nondegenerate finite spectrum, two Gibbs distributions that agree have equal β. It is not a contact-dynamics or equilibration theorem. On the transition side, the feasible set at x is the fibre $F_C(x) = \{y : b_C(y) = b_C(x)\}$ of the complete repaired visible datum b_C, and the minimizer of $D(q\|\pi)$ over the fibre, unique by the same strict convexity, is weighted conditional resampling,

$$P_C(x,y) = \mathbf{1}_{\{b_C(y)=b_C(x)\}}\,\frac{\pi(y)}{\pi(F_C(x))},$$

which is the weighted observation-fiber resampling projector, and any physical identification of a concrete repair law with it passes through the support, equal-fiber-row, and weighted detailed-balance receipt stated above. The kernel is stochastic, idempotent, reversible, and stationary for π, and fixes every fibre-measurable observable, which is dynamical conservation of every protected charge and the content side of the first law. For a finite joint update, the exact bookkeeping identity is $\Delta U = \delta Q + \delta W + \operatorname{Re}\operatorname{tr}(\delta\rho\,\delta H)$. The two-term form is first-order, or exact for an ordered stroke with one of H or ρ held fixed; the composed repair heat stroke holds H fixed, so its work and cross terms vanish. Data processing then gives the transition-side second law $D(pP_C\|\pi) \le D(p\|\pi)$, and the exact split $D(p\|\tau) = \langle K\rangle_p - S(p)$ with $K = -\log\tau$ turns it into the modular Clausius inequality $\Delta S \ge \Delta\langle K\rangle$ for every τ-preserving repair channel, which specializes to $\Delta S \ge \beta Q$ on the identified energy branch. The Landauer bound is a one-line corollary: a repair channel that lowers the record entropy by c expels at least c/β of energy, so erasing one bit costs at least $k_B T\log 2$. At every finite regulator the excited Gibbs mass obeys $q_\beta \le \frac{d-g_0}{g_0}e^{-\beta\Delta}$, with the quantitative threshold that any target mass ε is reached once $\beta\Delta$ exceeds $\log\frac{d-g_0}{g_0\varepsilon}$, and the entropy limit $k_B\log g_0$ is the standard finite consequence of the displayed mass bound and finite continuity of entropy; the current composed Lean record contains the mass bound, not a separate entropy-limit field. The repair kernel preserves full support, since every diagonal entry is positive, while the zero-temperature state is rank deficient for $g_0 < d$; finitely many repair or pinching steps therefore cannot reach it, which is the fixed-regulator third law. Public quantum-record pinching and averaging have explicit normalized Kraus families and preserve trace. A separate module proves these channels completely positive and trace preserving through the Kraus criterion; this result's own packet reports Kraus and trace identities. The record tower $S(\rho) \le S(\mathcal{E}_\pi\rho) \le S(\mathcal{A}_\pi\rho)$ separates conditioned learning from the nonselective commit, so observers convert uncertainty into correlations and durable records without violating the second law.

The second law refines to an exact identity. For one repair step the fluctuating entropy production $\sigma(x,y) = \log\frac{p(x)}{\pi(x)} - \log\frac{(pP_C)(y)}{\pi(y)}$ obeys three exact relations. The integral identity $\langle e^{-\sigma}\rangle = 1$ holds for every kernel that preserves the reference; under detailed balance the pointwise relation $p(x)\,P_C(x,y) = e^{\sigma(x,y)}\,(pP_C)(y)\,P_C(y,x)$ holds at every pair of states; and summing it over a level set gives the forward weight at entropy production s as e^s times the reversed weight. The mean of σ is exactly the certified descent $D(p\|\pi) - D(pP_C\|\pi)$, so the second law is the Jensen shadow of an identity rather than a primitive inequality. Detailed balance also gives finite Onsager reciprocity.

Because $e^{-\sigma}$ is a ratio of rationals, the fluctuation certificate closes exactly over the rationals, with no numerical logarithm.

The reversible symmetry and correlation representation below are finite relatives of standard Onsager and Green–Kubo theory [31, 32]. The OPH-specific statement is the resulting model-selection obstruction for the full-fibre publicization projector.

Theorem 2.8 (Finite transport kernel and repair-projector obstruction)**.** *Let K be a finite stochastic kernel reversible under a strictly positive weight π, let $L = I - K$, and let R be a linear solver of $LRj = j$ on the π-centered currents. For any finite current family $\{j_a\}$,*

$$\Gamma_{ab} = \langle j_a, Rj_b \rangle_\pi$$

is symmetric and positive semidefinite. At every finite cutoff N,

$$\langle j, Rk \rangle_\pi = \sum_{n=0}^{N} \langle j, K^n k \rangle_\pi + \langle j, K^{N+1} Rk \rangle_\pi.$$

Thus a Green–Kubo coefficient is a correlation sum plus an exact displayed remainder, rather than a silently assumed infinite series.

For the full-fibre conditional-resampling kernel P_C, idempotence gives $P_C^n = P_C$ for every $n \geq 1$. Positive-lag correlations are therefore constant. A fibre-centered current with $P_C j = 0$ has no positive-lag memory, while a nonzero one-step correlation makes consecutive partial sums differ by the same nonzero amount. One full-fibre repair projector cannot be a decaying, nonzero-memory transport dynamics.

On a finite closed oriented graph, typed positive cell volumes and heat capacities convert concentration to particle amount and temperature to energy. Fick and Fourier fluxes then define canonical one-step updates whose total changes equal the clock increment times the total source and hence vanish when the source sums vanish. Nonnegative conductance makes the flux–gradient pairing nonpositive; explicit negative-conductance controls reverse that sign.

Proof. Detailed balance makes L self-adjoint, and the exact Dirichlet identity expresses $\langle f, Lf \rangle_\pi$ as a nonnegative weighted sum of squared differences. Applying this to linear combinations of the Poisson solutions proves symmetry and the full finite quadratic-form inequality. The cutoff formula telescopes the Poisson equation. Projector idempotence gives the positive-lag dichotomy. Graph summation by parts and cancellation of internal oriented edges give the constitutive and global-balance statements. Every step, including two-state and two-vertex witnesses and negative controls, is checked in Lean [25]. □

The theorem does not select a physical transition generator, distance, clock, boundary condition, conductance, or readout. It proves no stability, hydrodynamic limit, or measured transport coefficient. In particular, the projector obstruction requires a source-derived nonidempotent local or random-scan effective evolution before OPH can make a long-memory transport claim.

The same package carries the entropy-balance shape used by the gravitational branch. For any state and faithful reference the exact identity $S(p) - S(\tau) = \langle K \rangle_p - \langle K \rangle_\tau - D(p \| \tau)$ holds, so the entanglement first law $\delta S = \delta \langle K \rangle$ is this identity with the quadratic-order deficit dropped; through the central split $K = 2\pi B + Z$ it reads

$$\Delta S = 2\pi \Delta \langle B \rangle + \Delta \langle Z \rangle - \Delta D(\,\cdot\, \| \tau),$$

where $\Delta D = D(p' \| \tau) - D(p \| \tau)$. For the reference-preserving repair step, data processing gives $\Delta D \le 0$, and one repair step with a central charge measurable through the repaired visible datum obeys the cap Clausius inequality $2\pi\Delta\langle B\rangle \le \Delta S$, since the kernel also conserves every fibre-measurable charge and hence $\Delta\langle Z\rangle = 0$. The finite first-law premise package of the Einstein branch is discharged on this model: on the tangent space of the probability simplex the entropy differential at the reference equals the modular pairing as a theorem, the modular pairing distributes through the central split, and the central pairing is the edge differential, so the branch's finite bulk-edge-central first law applies with no assumed premise beyond the split of the reference. Repair displacements are mass preserving with vanishing central pairing, so the first-order entropy change of one repair step is purely bulk-modular. The physical attachment of the split stays with the energy-clock receipt.

The Lean development carries the exact finite statements: relative entropy nonnegativity, the log-sum inequality, data processing, the conditional-resampling optimizer and its kernel package, the Gibbs Pythagorean identity and minimizer, the first-law split, the low-temperature bound, the fluctuation identities with reciprocity, the finite Green–Kubo matrix and exact cutoff remainder, the repair-projector memory obstruction, the typed Fick and Fourier updates, the exact-remainder cap first law, the discharged Einstein-branch first-law premises, and the Kraus completeness of partition pinching; the modules are `FiniteConditionalRepair`, `FirstLawIdentity`, `FluctuationTheorems`, `GreenKubo`, `GraphDiffusion`, `CapFirstLaw`, `EinsteinPremiseLink`, `StationaryRealization`, and `PartitionPinchingCP` in the Lean thermodynamics and event-algebra libraries, and the exact-rational certificate replays the kernel algebra and fluctuation identities and records the entropy checks [25]. The source-realization interface needs less than reversibility: every stochastic kernel that preserves a faithful reference contracts relative entropy to that reference. Detailed balance is not a premise. The exact lazy directed three-cycle is stationary and violates detailed balance, yet satisfies this second law; the Lean module `StationaryRealization` proves the separation. Detailed fluctuation relations and Onsager reciprocity therefore remain additional receipts rather than consequences of entropy descent.

The pinned source-counted collar table has also been exhausted within its declared coordinate-projection grammar. All $2^4 - 1 = 15$ syntactic maps obtained by retaining a nonempty subset of its four committed packet fields were audited on 1028 observers and 31744 counted transitions, without an additional observer-patch run. Because the record-family and S_3-sector fields are constant on this run, those maps induce only four distinct partitions. The repair-load count aggregation is an eight-state stochastic kernel that is irreducible and aperiodic and has a full-support stationary law. It is not a certified Markov quotient of the fine chain: its strong-lumpability defect—the largest outgoing target-block mass difference between two fine states in one repair-load block—is 0.911587983 on the pinned floating-point table. On that table its detailed-balance defect is $2.009924089 \times 10^{-4}$; from the uniform initial law the sampled relative entropy falls from 5.00947 to 1.84009 over sixteen steps. This is a nonreversible finite H-theorem probe, not a microscopic-reversibility receipt. It does not close the realization: the only observed record-family value makes protected-charge preservation vacuous, and the stationary law is not identified with the state optimizer's source reference. On the fine twenty-state quotient the only closed communicating class is the singleton freezeout state, so the alternative recurrent-class restriction is a trivial equilibrium rather than a nontrivial relaxation model. These results exhaust only the declared coordinate maps and the fine-chain recurrent restriction. They do not enumerate arbitrary partitions or nonlinear, statistical, stochastic, weakly lumpable, or history-dependent quotient maps of the retained artifact. The audit excludes none of those routes, nor changed source dynamics or a bounded run with a nonconstant protected record.

The state table/reference was locally pre-specified and hash-pinned; the recurrent transition chain was extracted post hoc from retained run data. This does not repair the failure. The state-side conditional-resampling action is idempotent, whereas the exact two-state recurrent transition action has the nonconstant eigenpair

$$Pv = \frac{665437}{726948}v, \qquad v = (54356, -7155), \qquad 0 < \frac{665437}{726948} < 1.$$

If an idempotent action H and this transition action P were related by $HT = TP$, applying idempotence twice gives $\lambda(\lambda - 1)Tv = 0$, hence $Tv = 0$. Thus every such intertwiner erases the exact extracted mixing mode. The transition stationary mass 7155/61511 also lies strictly between the adjacent empirical masses 1905/16384 and 1906/16384; no deterministic coarse graining of the 16384-sample state reference produces it. A freely chosen stochastic coupling could evade the denominator test, but would be a new premise rather than source evidence. These exact obstructions exclude the two audited direct bindings (a mixing-mode-preserving linear intertwiner into the idempotent heat bath and a deterministic empirical pushforward) on the current artifact. They do not exclude stochastic, nonlinear, reverse-direction, or enriched-source constructions. The committed random-scan preflight decides the random-scan instance of the stochastic route within its certified grammar: on every computed subset (the eleven subsets of at least two of the four committed packet fields on the 26-state alphabet, under the visit-count reference, and the record/companion pair on the realized 256-state structure, under the occupation reference), the random-scan mixture of conditional-resampling projectors under both declared schedulers is row-stochastic and exactly stationary for the one shared reference of its arena. The committed uniform-scheduler mixtures are additionally certified non-idempotent, so they escape the idempotent spectral obstruction; the exact fixed space of every computed mixture under either scheduler is one-dimensional, so the only observables it protects are constants. The remaining step-field subsets contain a certified-constant field; their mixtures carry the strictly positive full-space resampler as a component, which forces the same one-dimensional fixed space without any claim of non-idempotence. The only retained step field beyond the packet fields is certified constant on the run, so the retained step record admits no enrichment; the retained non-step cell fields are outside the certified grammar. The recorded continuations are a fresh export of a dynamically conserved label whose join with a committed field disconnects, or a dilated construction outside the random-scan grammar.

The conditional package retains five typed requirements: the global representation of the weighted local objective for both optimizer instantiations; the common source-derived reference shared by state and transition optimizers, which forbids manufacturing the transition matrix from a desired equilibrium output; the identification of the actual source-derived collar transition matrix with the conditional-resampling kernel through the existing matrix receipt; stochasticity, stationarity, and protected-charge preservation alone certify only the weaker stationary H-theorem branch and do not discharge that full Axiom-3 receipt, while detailed balance is additionally required wherever microscopic reversibility is claimed; the identification of one modular charge with physical energy and a calibrated clock, of which the finite central-interface split $K_C = 2\pi B_C + Z_C$ is the attained cap-branch part; and uniform low-temperature spectral-tail control on a genuinely cofinal family for a continuum third law. The committed cofinal interface provides the ambient order, cofinal embedding, no-terminal-stage theorem, strict carrier growth, and same-family concentration theorem, while supplying no source or physical continuum interpretation. Thus a source-produced family inhabiting that interface and its physical regulator attachment are unsupplied. The finite gap-plus-cardinality theorem remains one sufficient route rather than a necessary condition. The derivation carries a structural point worth stating plainly: every proof in the package is short and elementary once the framework supplies its two inputs, the repair fibre as the transition feasible set and the reference shared with the state optimizer. The four-law identities then require no additional thermodynamic axiom or deep theorem; the work lies in the decomposition, which the certified strict-descent counterexample shows is necessary. On any future source object discharging the named receipts, ordinary finite thermodynamic identities follow from the instantiated observer model rather than being postulated separately.

3 Physical Implementation and A_5/E_8 Significance

The implementation question is what must exist below the observer-level axioms so they have a physical carrier. OPH's answer is a fixed-cutoff federation of finite patch carriers. Each carrier exposes overlap ports, record registers, repair maps, and checkpoint interfaces. Three geometric objects must stay distinct. A local carrier boundary may have twelve-port icosahedral incidence. The federation screen is the finite federation and its overlap nerve. The support screen is the observer-facing S^2 chart produced from the repaired quotient only on the spherical receipt branch. The implementation is the finite algebraic machinery that makes compare, write, repair, and re-read operations available.

The term "sphere folding" has a restricted technical meaning here. It is the observer-facing spherical presentation of the repaired quotient normal form:

$$\mathrm{Fold}_{S,r}(s) = \chi_{S,r}(n_r(\pi_r(s)))\,.$$

The quotient map removes hidden carrier presentation, the normal-form map performs accepted repair, and the spherical chart displays caps, collars, cuts, edge sectors, and boundary records. Folding is therefore the geometric face of overlap repair on the screen chart. On the consensus surface this repair is defined first as the quotient operator $\mathrm{Rep}_\lambda : Q \to Q$, with representative-level maps only serving as lifts, so hidden carrier labels do not become additional physical data.

Spherical geometry carries several pieces of the construction at once on the certified branch. An observer-accessible cut has a closed two-dimensional angular chart after the global support incidence, mesh, cross-ratio, and normalization receipts hold. Caps and collars on that chart supply the cut data used by modular flow, entropy variation, and overlap comparison. The orientation-preserving conformal group of the same S^2 chart is the celestial-sphere realization of $\mathrm{SO}^+(3,1)$, so the sphere is the kinematic bridge between finite screen cuts and the emergent $3+1$-dimensional Lorentz branch once the controlled cap-pair theorem is satisfied. Finite cellulations of the chart provide the regulator surface on which ports, edge sectors, and local comparison data can be made explicit; they are not by themselves a Lorentz invariant continuum.

The echosahedral carrier is a concrete local realization of the observer-patch interface. Its multi-port boundary supplies discrete directions, face and edge incidence, exposed readout channels, record slots, and repair channels. Recurrent toroidal subchannels can expose winding and phase-sensitive observables. A physical phase-locking law would have to produce the accepted overlap repair relation, its confluence receipts, and its noise bounds. No theorem supplies that producer. Phase locking alone leaves BW modular ordering and operational clock calibration unproduced.

A federation of local icosahedral carriers does not automatically produce the global S^2 support screen. That step needs a quotient-visible carrier-to-screen map whose federation nerve has the spherical incidence receipt and whose cap data survive refinement. Local A_5 symmetry can therefore coexist with a nonspherical global nerve. The spacetime and Einstein paper's topology countermodels mark this boundary.

The A_5 and E_8 labels mark different roles in that machinery. A_5-icosahedral symmetry is the local finite-screen language used when a patch carrier is made concrete: it organizes ports, faces, and overlap data with enough discrete symmetry to stabilize observer-facing cuts. E_8-type language is the exceptional closure language used by the high-symmetry branch. It names the root-lattice and affine exceptional structure in which the icosahedral data can sit. The term is representation-closure language for the branches that call for it.

The compact side of that closure language is classified at the bracket level. On the canonical oriented carrier, the compact real locus of the fourteen-parameter A_5-equivariant Jacobi variety is exactly three families, all inside the derivation-free slice: one closed two-parameter plane, generically $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ plus an abelian complement, and two mirror three-parameter cells of $\mathfrak{su}(3) \oplus \mathfrak{so}(3)$ type, each cut out by one strict sign condition. Every numerical step is exact over $\mathbb{Q}(\sqrt{5})$, fifteen certified sample points inhabit the strata, and an independent mutation-tested verifier replays the certificate. The classification is conditional on three named textbook lemmas, the compact-type Killing criterion, the compact semisimple dimension list, and A_5 simplicity; it selects no source, preferred bracket, or holonomy [25]. A compactly closing equivariant bracket therefore lands on the $\mathfrak{su}(3) \oplus \mathfrak{so}(3)$ shape up to the mirror choice and the closed plane.

This distinction matters for the synthesis paper. The physical claim lives at the observer-visible quotient: finite patches must expose the same records and shared observables after allowed implementation-hiding changes. The microphysics paper gives the concrete reference architecture for that quotient surface, including the federated patch-carrier model, fixed-cutoff edge heat-kernel/Casimir law, central records, the declared algebra-state measurement interface, the declared two-wing Bell/CHSH surface, and the checkpoint/restoration package. See Ref. [18].

The A1–A2 finite theorem and the transportable-sector/Tannaka route are mathematically distinct. The first forces the abstract local Standard Model Lie type. The second is a conditional compact-group classification. Under the declared matrix-current and matter packet, the common $\mathbb{Z}_6$ kernel and maximal faithful image are exact. The physical global quotient is not source-selected. Physical source binding of the first route and equality as one physical gauge-current object enter as premises. Separate band premises select rank three. Physical family interpretation and extra-sector exclusion require separate identifications. Tensoring the response band with the generation table gives a conditional rank-45 candidate. A distinct local-domain receipt checks a declared tensor-identity operator; it does not source-select the matter action or transport the twelve-port Spin packet. Physical matter-pole, continuum, seam-selection, persistence, and symmetry-descent receipts are required.

Famous equations and structural outputs recovered

The table below maps headline formulas by logical role. Some rows are core theorem results, some are short corollaries of recovered actions, and some are standard limits once the parent OPH branch is established.

Recovered result	OPH role	Famous display	Support boundary
Three-dimensional observer-frame chart, Lorentz kinematics, and invariant light cone	core theorem	$q(\Omega) = (1, \Omega)$, $n_C = (\cot\alpha, \csc\alpha\,\mathbf{c})$, $n_{gC} = \Lambda_g n_C$, $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$	scaling-limit geometric subnet; chart population and neutral bulk are separate premises

Recovered result	OPH role	Famous display	Support boundary
Record-conditioned H^3 frame estimate	conditional theorem	$R_i(C,t,O) \to F(X_i)+e$, $\alpha\bar{d} \le \lvert F(X)-F(Y)\rvert_W$, $S_i(t) \subseteq B_H(\widehat{X}_i, r_i)$	frame-local calibrated cap responses, compact frame domain, bounded error, conservative frame enclosure, and positive gap for unique finite output; no event position is inferred
Einstein equation	conditional composition theorem	$G_{ab}+\Lambda g_{ab} = 8\pi G\,\langle T_{ab}\rangle$	one source-derived common-domain tower with uniform asymptotics, universal coupling, a vacuum reference, and independent scale readouts; the tower and its certification are consumed as premises
Newton-Poisson gravity	inherited weak-field limit	$\nabla^2\Phi = 4\pi G\rho$, $\ddot{\mathbf{x}} = -\nabla\Phi$	absolute Einstein-branch premises, weak field, and slow motion
Maxwell equations	electromagnetic corollary	$dF_Q = 0, \quad d{*}F_Q = g_Q^2{*}J_Q$	ordinary $\mathrm{U}(1)_Q$ branch plus explicit Maxwell action/current hypothesis
Discrete Maxwell evolution on the screen carrier	exact conditional finite theorem	$B_{n+1}-B_n = -hCE_n$, $E_{n+1}-E_n = h(C^{\mathsf{T}}B_{n+1}-J_n)$, $\rho_{n+1}-\rho_n+h\,\partial J_n = 0$, $h^2(3+\sqrt{5}) < 4$	the step-scaled update and the Gauss constraint are the Euler–Lagrange equations of a declared discrete action, the committed local face action plus a declared kinetic term; for nonzero h, every zero-current datum has uniformly bounded electric seam energy exactly in the strict window, where the magnetic face energy is also bounded; equality and larger steps admit explicit unbounded electric-energy modes, and the unit step is unstable. No gauge-potential bound follows. Physical time and sources, covariance, continuum control, and readout are not supplied

Recovered result	OPH role	Famous display	Support boundary
Yang–Mills equations	compact-gauge corollary	$DF = 0, \quad D{*}F = g^2{*}J$	compact-gauge connection branch plus explicit Yang–Mills action/current hypothesis
Born rule and selected Lüders update	conditional fixed-cutoff theorem	$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E)$, $\rho\|_E = P_E \rho P_E / \mathrm{Tr}(\rho P_E)$	declared finite algebra-state embedding and declared Lüders instrument; the effect table does not select the instrument
Tsirelson bound	fixed-cutoff theorem	$\|S_{\mathrm{CHSH}}\| \leq 2\sqrt{2}$	declared commuting wing algebras, binary settings and readouts, and joint source state; this is an upper bound and supplies no source-derived violation. Reachable states on the committed pair carriers are separable across the declared slot split and obey the classical bound 2 against every slot-local unit-interval readout, jointly diagonal or not, a diagonality-preserving map cannot reach the declared Bell-state witness, and a product-diagonal state paired with cross-commuting settings outside both slots reaches $2\sqrt{2}$, so slot membership relative to the split carries the classical bound and readouts outside the split lie outside it
Charge quantization	core theorem	$Q = T_3 + Y$, color singlets have $Q \in \mathbb{Z}$	quarks are confined fractional-charge fields

Recovered result	OPH role	Famous display	Support boundary
Gyromagnetic ratio two	exact algebra theorem	$(\sigma\cdot X)^2 = X\cdot X - q\,(\sigma\cdot B)$, $X\cdot X := \sum_i X_i X_i$, for $[X_i, X_j] = i\,q\,\epsilon_{ijk}B_k$	the sum is ordered algebraic multiplication, not an adjoint norm. The commutation data, coupling, and carrier are declared; the physical Hamiltonian and $g = 2$ reading additionally require a Hermitian kinetic-momentum realization and gauge attachment
Classical carrier-mode poles	conditional action theorem	$K_X^{\mathrm{phys}} \propto (\omega^2 - c_\star^2\lvert\mathbf{k}\rvert^2)\Pi_X$	Maxwell, perturbative pure-Yang–Mills, or pure-Einstein action/phase receipt; the quantum particle claim is a separate premise
Primitive-port propagation fingerprint	frozen prospective branch prediction	$\omega^2 = k^2 - (a^2/20)k^4 + (a^4/840)k^6 + (2a^4/7875)k^6 I_6 + O(a^6k^8)$	the complete equal-weight twelve-port orbit is the physical directional support; the tested sector realizes the scalar symbol, with equal action on both transverse polarizations for light; intrinsic coefficients must be isolated from source, medium, gravitational, and instrumental effects
Source-seam propagation fingerprint	exact internal action and frozen conditional branch	$\widehat{\Lambda} = q^2 - q^4/20 + (1/840 - I_6/12600)q^6 + O(q^8)$	the source fixes the D_6 seam carrier and response metric; named naturality and unique-minimizer premises select the homogeneous action; no physical position, field, clock, frequency, frame, readout, or nuisance attachment is supplied

Recovered result	OPH role	Famous display	Support boundary
Coherent-matter scalar susceptibility	conditional scalar theorem	$\chi_\nu^{\mathrm{can}} = 1 - P_\chi/24 = 0.9320429912748350\ldots$	nondegenerate OPH-coherent material source, Scalar Edge-Center Exhaustion, and protected-reserve collar closure; force additionally requires repair charge, an external gradient, and field-momentum closure

On the separate global capacity branch, the Bekenstein–Hawking / de Sitter area law keeps the display $N_{\mathrm{scr}} = A/(4\ell_P^2) = 3\pi/(G\Lambda)$ with screen-capacity closure as its boundary.

Particle result scope

The particle paper treats the closure matrix explicitly. The non-hadron bundle has the following support levels:

The target-free but declared and incomplete running/map packet evaluates to the electroweak chart coordinates (80.330, 91.119) GeV; the source does not select this map. A separate strict one-loop consumer and interval stack act on an external Standard Model fixture. They exclude scalar zeros on declared principal-sheet boxes and isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet and prove no unique continuation, sign bridge, full-matrix Laurent residue, current amplitude, or independent numerical replay. The external fixture is not composed with the OPH chart. A physical W/Z statement requires an OPH-native source-to-pole packet, a strict source root, an independently physical $E_\star$, finite quotient transport, a renormalization-group matching and scheme prescription fixed independently of the comparison, branch rigidity, and propagated uncertainty.

Exact flavor algebra and its boundary. The twenty-face orbit has a regular C_3 corner fiber. Every equivariant Hermitian response on that fiber is a circulant $C = aI + bR + \bar{b}R^2$. When all three eigenvalues are nonnegative and read as square-root masses, the root-of-unity identities give

$$Q = \frac{1}{3} + \frac{2}{3}\left(\frac{|b|}{a}\right)^2, \qquad Q = \frac{2}{3} \Longleftrightarrow \frac{|b|}{a} = \frac{1}{\sqrt{2}}.$$

Equal rank-two blocks in the finite tracial-GNS packet give the balanced modulus exactly under the packet premises. The phase, and therefore the two mass ratios it jointly controls, is not fixed. The face fiber also lacks a constructed physical chiral-family attachment. The target-informed response coordinate is a diagnostic.

The five-dimensional traceless-symmetric family space sharpens the dynamical gap. Threefold and fivefold residual invariance forces a double eigenvalue. The twofold fixed locus has two parameters after overall scaling and admits simple spectrum, leaving exactly enough freedom to carry two mass ratios. Symmetry alone therefore supplies no numerical ratio prediction; a specific screen-derived invariant potential must select the orbit. Ref. [19] gives the proofs and scope.

Conditional comparison coordinates. The declared, not source-selected zero-selector map outputs the running/chart coordinates $(80.330, 91.119)$ GeV. The separate strict one-loop fixture has principal-sheet scalar zero-exclusion receipts and one simple scalar zero for each of W and Z in channel-specific lower-half pole boxes and algebraic charts. No certified continuation identity or composition joins that fixture to the OPH chart, so it supplies no physical W/Z pole or mass comparison. The Higgs/top pair is read on the double-criticality branch, a zero-continuous-parameter family whose comparison-exposed boundary-scale candidate gives $(m_H, m_t) = (125.77, 172.63)$ GeV at two loops; its declared calibration surface is a data-anchored fit that validates the formula stack and never predicts. The quark reciprocal-ray candidate fails its common-scale audit with a 21.556% held-out minimax error. The register-Clebsch candidate supplies six assignments and the distinct light-family coefficient-ratio menu $\{1/9, 1/3, 3, 9\}$ under the declared alphabet and the two invariant channel assignments. The adopted ordering is target-informed and uniquely least discrepant. Both FLAG 2024 rows reject every assignment under the retrospective conservative experimental-only decision rule. The unavailable covariance and absent OPH theory uncertainty preclude a covariance-aware significance, so the result does not constitute a preregistered theory-wide falsification. It excludes only the declared common-transport assignment family. Different coefficient relations, alphabets, charged-family attachments, or generation-dependent threshold transport lie outside this class. The retained results are the conditional channel-pairing theorem, the target-free unordered multiset under the declared rules, and the exact positive-chamber Koide identity. The pairing theorem supplies no physical coefficient equality or source-derived generation order. The corresponding $\sqrt{m_d/m_s} = 0.2086$ display is an algebraic Gatto–Sartori–Tonin estimate from the same failed ratio, without the up/down matrices and relative eigenbasis needed for a mixing prediction. All of these comparison coordinates, including the selected-carrier, value-law, adapter, two-loop, pole variants, and exact Particle Data Group convention maps, are excluded from the physical particle outputs. Ref. [19] gives the detailed coordinate tables and verification.

Observable-identification boundary

The arguments use one no-relabeling rule throughout. A finite diagnostic or calibration field does not become a physical observable merely because it is renamed. Capacity bookkeeping needs an independent energy readout before it can count as mass; an archive needs a physical source, propagation, and detector channel before it can count as radiation; a finite repair spectrum needs a continuum operator/readout bridge before it can count as a physical spectrum; and a finite reconstruction threshold needs physical entropy and clock data before it can count as a Page-time claim. The spacetime and Einstein paper records this as the source-separation rule behind the support-level table.

Sector-dimension shock boundary. The declared repair generator

$$L^{\text{rep}} = \sum_v (I - P_v)$$

uses single-site heat-bath conditional expectations on $K_r = L^2(X_r, \pi_r)$ at fixed sector structure. Since the P_v are orthogonal projections,

$$\langle f, L^{\text{rep}} f\rangle = \sum_v \|(I - P_v)f\|^2,$$

so $\ker L^{\text{rep}} = \bigcap_v \operatorname{Ran} P_v$. Identifying this intersection with the constants requires a separate irreducibility result. A horizon shock changes the sector dimensions d_α in $L_C = \sum_\alpha (\log d_\alpha) P_\alpha$, thereby changing X_r. It is outside the domain of this fixed-sector repair generator. The repair gap therefore neither constrains nor produces a sector-dimension shock mode.

A separate finite-capacity calculation gives a one-sided sign statement. Uniform transfer of a fraction f from a fixed horizon–observer budget changes the extremal sector entropy and logarithmic area observable by $\log(1-f) < 0$ on every admissible integer depletion, and the positive-real interpolation is strictly decreasing. Its analytic Hessian has negative homogeneous curvature and positive curvature on the fixed-horizon-capacity tangent space, so the relevant maximum is the transfer boundary rather than an interior stationary point. Identifying this capacity transfer with horizon-area transfer gives a conditional mechanism for the de Sitter time-advance sign reported by Chen, Stanford, Tang, and Yang [58, 23]. Identifying the finite capacity-ledger transfer with observer mass is a separate physical dictionary. The normalized icosahedral spectrum additionally requires exact gauge transport of the rotation triplet and identification of the kinetic term with the scaled nearest-neighbour port or edge-sector Laplacian. Its coefficient and physical attachment are separate premises. This mechanism supplies no static-patch trace and leaves the cited trace obstruction in force. That obstruction concerns the positive cyclic trace proposal tested in the cited work. Nontrace horizon-screen descriptions, separately constructed dimension-changing generators, and kinetic operators beyond the scaled nearest-neighbour graph lie outside the tested classes. None of these scope boundaries disproves OPH; the pure-de-Sitter normalization, finite entropy and transfer identities, and regular line-graph theorem survive.

The χ_ν scalar-channel identity is theorem-grade for nondegenerate OPH-coherent material sources once Scalar Edge-Center Exhaustion is imported. Device-scale force, cosmological abundance, and finite covariant dark stress require their own receipt packages.

Frozen primitive-port propagation prediction. On the real reciprocal finite-range cosine branch whose complete hop support is the primitive twelve-port orbit, with no independent kinetic term through the displayed order, proper carrier covariance forces equal weights and continuum normalization fixes the displayed propagation symbol. Writing its corrections as $C_4k^4 + B_0k^6 + B_6k^6I_6$, the exact scale-free relations are

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}, \qquad C_4 < 0.$$

The intrinsic anisotropic ranks one through five vanish. Rank six has one icosahedral harmonic shape up to an orientation in $\mathrm{SO}(3)/A_5$. Here "spin six" names spherical-harmonic rank, unrelated to particle spin. Once a negative C_4 is resolved on this branch, the scale and both sixth-order amplitudes are fixed; the residual freedom is one three-parameter orientation class in $\mathrm{SO}(3)/A_5$. For the complete equal-weight cosine kernel and $0 < |ak| \le 1$, an exact three-chart interval cover and local singular-value bounds prove that its stationary set consists of precisely the twelve vertex maxima, twenty face-center minima, and thirty edge-center saddles. This finite-range theorem uses the declared kernel. Source selection and physical readout remain in the branch premises [25].

The certified repair operator acts on thirty internal carrier seams, not on translated field sites. The finite source supplies no map from that operator to a spatial hop stencil or a physical readout. Exact equal-weight stencils on the vertex, face, and edge direction orbits have different rank-six rays. Carrier transitivity fixes equal weights within a selected orbit and does not select the vertex orbit. The displayed prediction therefore remains a named physical-branch statement.

The first refinement layer does admit an exact active internal readout. The twelve inherited values Qx, together with one labeled pair-averaging response for each of the thirty midpoint values, form an invertible 42×42 rational map. Its decoder is $x_m = 2(QE_{u,m}x)_u - (Qx)_u$. The minimal thirty-event selector is not A_5-invariant. Retaining both endpoint choices gives a sixty-event family stable under the proper A_5 action and rank 42 once Q is included. This is a different protocol from the passive history $(Q, QL_f, \ldots, QL_f^{41})$, whose rank remains 29. The active responses must share one pre-event state. A classical record-assisted implementation satisfies this condition by restoring each probed pair before the next probe. No source-selected physical instrument, spatial translation, laboratory observable, or sky map follows (`Lean/Screen/LabeledEventReadout.lean`; [24]). This symmetry statement concerns the scalar probe support. It asserts no equivariance of mixed scalar/seam-potential measurements under signed cochain transformations.

Theorem 3.1 (Serial feedback readout and finite Maxwell action)**.** *Assume exact classical scalar records and writable coordinates on $\{1, \ldots, 12\} \sqcup \{1, \ldots, 30\}$. Retain the inherited baseline $b_u = x_u$. For a labeled endpoint–midpoint pair (u, m), apply pair averaging, record its exposed endpoint response r, and replace the pair by $(b_u, 2r - b_u)$, fixing every other coordinate. Every finite sequence of these cycles preserves x, including repeated and overlapping pairs. If each midpoint is probed, the records reconstruct x exactly.*

Explicitly type a packet as $x_n = (\phi_n, A_n)$, with twelve scalar potentials and thirty oriented seam potentials. Serial decoding at each slice commutes with

$$E_n = -\frac{A_{n+1} - A_n}{h} - D\phi_n, \qquad B_n = CA_n,$$

where $h \neq 0$, D is the committed port coboundary and C its oriented face incidence. It preserves the entire declared neutral-pair action, including its values on varied fields and paths.

Proof. The response is $r = (b_u + x_m)/2$, so $2r - b_u = x_m$. The feedback restores both written coordinates and leaves the rest fixed. Induction on the probe word gives preservation at every prefix; each retained response therefore refers to the identical initial state. The explicit inverse then reconstructs every coordinate. Applied separately at each potential slice, this identity gives the displayed fields and equality of any action evaluated on the decoded potentials. It holds for every input, hence for every allowed variation. These identities and their application to the rational stationary neutral-pair history are kernel-checked in `Lean/Screen/SerialMaxwellReadout.lean`. □

The executable instrument uses one evolving 42-slot state, retained baseline and response records, port-local feedback writes, and a public evidence bundle. It reads all sixty endpoint–midpoint pairs at each of three slices. Only A_0, A_1 are supplied as initial seam data. At $h = 1/2$, the decoded records and the current of the opposite charged paths produce

$$A_2 = 2A_1 - A_0 - h^2 C^{\mathsf{T}} CA_1 + h^2 J_0 - hD(\phi_1 - \phi_0).$$

An independent implementation advances E by Ampère's equation and then recovers A_2; it obtains the same rational stationary history. It checks all 54 free field derivatives of the finite action, all eight admissible closed two-step path replacements, Gauss, Faraday and source continuity. Each of two gauge-related executions has 585 semantic events and 180 completed feedback cycles. Replay reconstructs each operation's read set and checks its pre-write values and writer versions; declared parent lists cannot manufacture causal reads. The gauge control changes raw potentials and records while preserving fields and, with zero endpoint gauge, the coupled action. The exact execution receipt and its independent verifier are reproducible from the accompanying code [25].

This instrument assumes classical read/write access and exact memory. A baseline error ϵ_b and response error ϵ_r give midpoint error $2\epsilon_r - \epsilon_b$ in one cycle; this is not a stability bound for a long noisy execution. Pair averaging alone does not implement the feedback, and quantum state copying is not asserted. The feedback-cycle counter counts executed operations. Its relation to the separately declared Lorentz clock unit $\tau = 3$, laboratory time, source-selected geometry or a spatial continuum refinement is not supplied. A scalar first-refinement slot becomes a potential only under the displayed typing assumption.

From decoded cochains to fields on a solid. The twelve vertices, thirty edges and twenty faces form a closed surface. A concrete volume reconstruction is obtained by coning its declared Euclidean realization to the centre. Use the registered coordinates $(0, \pm 1, \pm\varphi)$, $(\pm\varphi, 0, \pm 1)$, $(\pm 1, \pm\varphi, 0)$, with their fixed port labels and $\varphi = (1 + \sqrt{5})/2$. Each boundary edge has length two; each of the twenty positively oriented tetrahedra has volume $(3 + \sqrt{5})/6$. The cone has $13, 42, 50, 20$ simplices in degrees zero through three. This geometric realization and its numerical length unit are supplied.

Theorem 3.2 (Gauge-covariant cone reconstruction)**.** *Let D and C denote the signed boundary gradient and curl. For $\Pi_m = \mathbf{1}_m \mathbf{1}_m^{\mathsf{T}}/m$, set*

$$G = (D^{\mathsf{T}} D + \Pi_{12})^{-1} - \Pi_{12}, \qquad H = (CC^{\mathsf{T}} + \Pi_{20})^{-1} - \Pi_{20}.$$

Order cone edges radially first and cone faces by boundary faces first. Then its incidence maps and trace-preserving extensions are

$$\widetilde{D} = \begin{pmatrix} -\mathbf{1}_{12} & I_{12} \\ 0 & D \end{pmatrix}, \qquad \widetilde{C} = \begin{pmatrix} 0 & C \\ -D & I_{30} \end{pmatrix}, \qquad \widetilde{B} = \begin{pmatrix} I_{20} & -C \end{pmatrix},$$

$$P_0 = \begin{pmatrix} \mathbf{1}_{12}^{\mathsf{T}}/12 \\ I_{12} \end{pmatrix}, \qquad P_1 = \begin{pmatrix} GD^{\mathsf{T}} \\ I_{30} \end{pmatrix}, \qquad P_2 = \begin{pmatrix} I_{20} \\ C^{\mathsf{T}} H \end{pmatrix}.$$

They satisfy

$$\widetilde{D} P_0 = P_1 D, \qquad \widetilde{C} P_1 = P_2 C, \qquad \widetilde{B} P_2 = \Pi_{20}.$$

A boundary two-cochain has a closed cone extension if and only if its total flux vanishes. In particular, every $B = CA$ has such an extension.

Let W_k be Whitney interpolation on this geometric cone. On a uniform time grid $t_n = nh$, $h > 0$, interpolate $P_1 A_n$ linearly and $P_0 \phi_n$ constantly on each open slab. For $s = (t - t_n)/h$, the fields are

$$\begin{aligned} \mathcal{E}(t) &= W_1 P_1 \left[-(A_{n+1} - A_n)/h - D\phi_n \right], \\ \mathcal{B}(t) &= W_2 P_2 \left[(1 - s) C A_n + s C A_{n+1} \right]. \end{aligned}$$

The discrete gauge rule lifts to the ordinary potential gauge rule. On the interior of the supplied solid and time window, $\partial_t \mathcal{B} + d\mathcal{E} = 0$ and $d\mathcal{B} = 0$ hold distributionally.

Proof. The exact Green identity gives $GD^{\mathsf{T}} D = I - \Pi_{12}$. The verified boundary complex has $\ker C = \operatorname{im} D$ and $\operatorname{im} C = \mathbf{1}_{20}^{\perp}$, hence $C^{\mathsf{T}} H C = I - DGD^{\mathsf{T}}$ and $CC^{\mathsf{T}} H = I - \Pi_{20}$. Block multiplication proves the three displayed identities. Closedness of a cone two-cochain with boundary value B requires $B = Cr$ for its radial component r, equivalent to zero total flux. The radial potential $GD^{\mathsf{T}} A$ has zero mean and changes by $\chi - \overline{\chi}$ under $A \mapsto A + D\chi$, exactly matching the apex scalar value $\overline{\chi}$.

In barycentric coordinates, the local forms are

$$\begin{aligned} w_i = \lambda_i, \qquad w_{ij} &= \lambda_i \, d\lambda_j - \lambda_j \, d\lambda_i, \\ w_{ijk} &= 2 \sum_{\mathrm{cyc}(i,j,k)} \lambda_i \, d\lambda_j \wedge d\lambda_k. \end{aligned}$$

Their exterior derivatives commute with signed incidence. Their matching traces give the assembled identities in the weak sense [41]. Vector potentials continuous and piecewise affine in time, with constant slab scalar potentials, give the displayed fields. For affine $\chi(t)$, the scalar gauge change is $-\partial_t\chi$; differentiation and $d^2 = 0$ prove the claims. The magnetic field is continuous in time, so its derivative acquires no additional time-interface impulse. □

The radial reconstruction is global on the boundary graph: its 12×30 matrix has 240 nonzero entries. It is a reconstruction rule, not a derivation of local radial transport. Its cone coefficients are determined by the boundary data and introduce no independent radial degrees of freedom. The finite cochain and zero-flux statements are formalized in `Lean/Screen/ConeCochainBridge.lean`. That implementation uses a dual-tree section followed by the cycle projector for the radial two-form; uniqueness among co-closed radial solutions proves agreement with $C^{\mathsf{T}}H$ on the zero-flux domain. The Whitney field and distributional statements are the paper argument above.

The metric and time defects of the actual history. Let M_1, M_2 be the Gram matrices of W_1, W_2 in the supplied Euclidean metric, with unit constitutive coefficients. Their exact integrals use $\int_T \lambda_i\lambda_j = |T|(1+\delta_{ij})/20$. These positive matrices specify geometric field energy. They are distinct from counting inner products. Extend source covectors as $\widetilde{\rho}_n = (0, \rho_n)$ and $\widetilde{J}_n = (0_{12}, J_n)$, preserving the source pairing. Slabwise constant charge has jumps at the knots. Its compatible temporal current functional is $\sum_n h\widetilde{J}_n\delta_{t_{n+1}}$, since $\rho_{n+1} - \rho_n = -hD^{\mathsf{T}}J_n$. This identity uses interior test functions; an extension beyond the window uses the initial charge before zero and the terminal charge after the last current impulse. Ordinary constant slab current has a different pairing.

Write $\widetilde{A}_n = P_1A_n$, $\widetilde{\phi}_n = P_0\phi_n$, $\widetilde{E}_n = P_1E_n$, $\widetilde{B}_n = P_2CA_n$, and $Q(b) = b^{\mathsf{T}}M_2b$. The exactly integrated field-plus-source action is

$$S_{\mathrm{pr}} = \sum_{n=0}^{N-1}\Big\{\frac{h}{2}\widetilde{E}_n^{\mathsf{T}}M_1\widetilde{E}_n - \frac{h}{6}\Big[Q(\widetilde{B}_n) + \widetilde{B}_n^{\mathsf{T}}M_2\widetilde{B}_{n+1} + Q(\widetilde{B}_{n+1})\Big] + h\widetilde{J}_n^{\mathsf{T}}\widetilde{A}_{n+1} + h\widetilde{\rho}_n^{\mathsf{T}}\widetilde{\phi}_n\Big\}.$$

Let S_{right} use the same electric and source terms but right-endpoint magnetic quadrature. Direct integration and telescoping give

$$S_{\mathrm{pr}} - S_{\mathrm{right}} = \frac{h}{4}\Big[Q(\widetilde{B}_N) - Q(\widetilde{B}_0)\Big] + \frac{h}{12}\sum_{n=0}^{N-1} Q(\widetilde{B}_{n+1} - \widetilde{B}_n). \tag{1}$$

The second term equals $h^2\int Q(\partial_t\widetilde{B})\,dt/12$. It is second order on a fixed time window only under uniform temporal regularity; the total action defect includes the displayed endpoint term. Fixed-endpoint variations remove that term. Nonuniform time grids need additional terms. The separate spatial defect relative to the counting action is

$$S_{\mathrm{right}} - S_{\mathrm{count}} = \frac{h}{2}\sum_n\Big[E_n^{\mathsf{T}}(P_1^{\mathsf{T}}M_1P_1 - I)E_n - B_{n+1}^{\mathsf{T}}(P_2^{\mathsf{T}}M_2P_2 - I)B_{n+1}\Big].$$

The clock action is excluded from all three field-plus-source actions.

For the derivatives, allow every cone potential coefficient to vary independently and evaluate afterward at the reconstructed history; variations need not stay in the images of P_1 and P_0. With fixed vector-potential endpoints and free spatial traces, the full finite-element derivatives of S_{pr} are

$$\begin{aligned}
\frac{\partial S_{\mathrm{pr}}}{\partial \widetilde{\phi}_n} &= h\left(\widetilde{\rho}_n - \widetilde{D}^{\mathsf{T}} M_1 \widetilde{E}_n\right), \\
\frac{\partial S_{\mathrm{pr}}}{\partial \widetilde{A}_m} &= M_1(\widetilde{E}_m - \widetilde{E}_{m-1}) + h\widetilde{J}_{m-1} \\
&\quad - \frac{h}{6}\widetilde{C}^{\mathsf{T}} M_2(\widetilde{B}_{m-1} + 4\widetilde{B}_m + \widetilde{B}_{m+1}), \qquad 0 < m < N.
\end{aligned}$$

The second line differs from the metric right-endpoint derivative by $h\widetilde{C}^{\mathsf{T}} M_2(2\widetilde{B}_m - \widetilde{B}_{m-1} - \widetilde{B}_{m+1})/6$. This temporal correction does not remove the spatial metric/source defect. The coefficient identities and telescoping are formalized in `Lean/Screen/WhitneyTimeBridge.lean`; analytic integration and distributional source interpretation are stated here separately.

The authenticated serial history gives the following dimensionless values for both endpoint-fixed gauge representatives:

Counting field-plus-source action	−14.42982115034
Volume action, right-endpoint magnetic term	−14.25989506948
Exactly time-integrated volume action	−10.21278655268
Spatial action defect	0.16992608086
Temporal action defect	4.04710851680
Maximum absolute scalar residual, divided by h	0.64544633271
Maximum absolute interior vector residual	1.57903507514

The rational cochain identities are exact. The volume Gram matrices and displayed values use double-precision analytic moments and independent tetrahedral quadrature, compared with absolute and relative tolerances 10^{-10}; they are not interval certificates. The independent verifier differentiates all $42 + 2 \cdot 13 = 68$ free coefficients of the action, including the twelve radial vector variations omitted by a boundary-only test. A separate exact counting-metric control gives temporal action defect 663629599/63369648 and first vector residual 44749/55152. These are mathematical controls on recorded inputs, not measurements.

Whitney one-forms have matching tangential traces; their normal jumps can contribute to electric divergence. Magnetic tangential jumps and temporal electric jumps likewise contribute to the sourced equations. The assembled weak derivatives retain these effects; element-interior tests are insufficient. The source covectors specify finite-element test functionals, not arbitrary-test physical densities. Under the conventional smooth potential signs, D^{T} represents weak negative divergence, so the coupling $+J \cdot A + \rho_{\mathrm{load}}\phi$ corresponds to $\rho_{\mathrm{physical}} = -\rho_{\mathrm{load}}$ if a density identification is made. The supplied cone yields continuous-coordinate fields and explicit defects. It supplies neither sourced stationary volume dynamics nor a source-selected geometry, physical clock, spatial refinement family, intertwiner with the translation-symbol field group, or laboratory law.

Stationary evolution in the full volume. The preceding interpolation need not be stationary. A different, explicitly declared initial-value problem uses every volume edge as an independent coordinate. In this paragraph write $D = \widetilde{D}$, $C = \widetilde{C}$, $M = M_1$, $K = C^{\mathsf{T}} M_2 C$ and $F = M + h^2 K/6$. These are the geometric matrices above, not counting pairings. Since $M > 0$, $K \geq 0$ and $CD = 0$, one has $F > 0$ and $KD = 0$. Sources have the same covector and temporal impulse interpretation as before.

Theorem 3.3 (Full-volume prism evolution and constraint preservation)**.** *For $h > 0$, the interior vector-potential equations of the exactly integrated prism action are equivalent to*

$$FA_{n+1} = \left(2M - \frac{2h^2}{3}K\right) A_n - FA_{n-1} + h^2 J_{n-1} - hMD(\phi_n - \phi_{n-1}). \tag{2}$$

They determine a unique next vector potential for every supplied scalar potential and pair of previous vector potentials. With $E_n = -(A_{n+1} - A_n)/h - D\phi_n$, the Gauss residual $g_n = D^\mathsf{T} M E_n - \rho_n$ satisfies

$$g_n - g_{n-1} = -hD^\mathsf{T} J_{n-1} - (\rho_n - \rho_{n-1}).$$

Thus conserved sources propagate the initial constraint, including any initial error. For neutral ρ_0, a given initial electric field E_{old} can instead be replaced by

$$E_0 = E_{\text{old}} + Dz, \qquad z = (D^\mathsf{T} MD + \Pi_{13})^{-1}(\rho_0 - D^\mathsf{T} M E_{\text{old}}). \tag{3}$$

This is the minimum M-norm correction satisfying Gauss's law. It preserves the curl of the initial electric field and defines a new initial condition.

Proof. Substitute the definitions of E_n and $B_n = CA_n$ in the full action derivatives above and collect the next-potential coefficient. Positive definiteness of F proves existence and uniqueness in the finite-dimensional edge space. Applying D^T to the vector equation removes every magnetic term and gives the displayed residual identity. The connected cone has $\ker D = \operatorname{span}\{\mathbf{1}\}$, so $D^\mathsf{T} MD$ is invertible on mean-zero vertex vectors. Neutrality places the right-hand side of (3) in that subspace. Any competing correction differs from Dz by w with $D^\mathsf{T} Mw = 0$; hence $(Dz)^\mathsf{T} Mw = 0$. Pythagoras proves minimality. Finally, $CDz = 0$. □

Theorem 3.4 (Work identity and sharp positive-mode stability)**.** *Let $\overline{B}_n = (B_n + B_{n+1})/2$. Along the full recurrence,*

$$H_n = \frac{1}{2} E_n^\mathsf{T} \left(M - \frac{h^2}{12}K\right) E_n + \frac{1}{2}\overline{B}_n^\mathsf{T} M_2 \overline{B}_n,$$

$$H_n - H_{n-1} = -\frac{h}{2} J_{n-1}^\mathsf{T}(E_n + E_{n-1}).$$

For a positive generalized mode $Kv = \lambda Mv$, all source-free modal solutions are bounded exactly when $0 < h^2\lambda < 12$. At $h^2\lambda = 12$, the double characteristic root -1 admits linearly growing alternating solutions. Above twelve an exponentially growing solution exists. Source-free zero modes have constant electric field and can have linearly drifting gauge potentials.

Proof. The recurrence also reads $F(E_n - E_{n-1}) = hKA_n - hJ_{n-1}$; here $KD = 0$ removes the scalar-potential terms. Faraday's identity is $B_{n+1} - B_n = -hCE_n$. Polarize the two quadratic terms, use these identities, and cancel the magnetic cross terms. Equivalently, $H_n = E_n^\mathsf{T} FE_n/2 + B_n^\mathsf{T} M_2 B_{n+1}/2$, which makes the telescoping cancellation immediate. This proves the work law in every gauge.

For $z = h^2\lambda$, the scalar characteristic equation is

$$r^2 - 2a_z r + 1 = 0, \qquad a_z = \frac{1 - z/3}{1 + z/6}.$$

For positive z, the strict inequality $|a_z| < 1$ holds exactly on $z < 12$. Its two distinct unit-circle roots give bounded solutions. At twelve, $(-1)^{n+1}n$ is a solution and its electric difference is unbounded. Above twelve, the negative real reciprocal roots include one with modulus greater than one. In temporal gauge the zero-mode recurrence is a second difference, giving affine potentials. These facts do not bound gauge potentials by field energy. □

The finite action expansions, weak Noether identity, work law, uniqueness and sharp scalar alternatives are formalized in `Lean/Screen/WhitneyMaxwellDynamics.lean`. Stable initial-value evolution does not imply a nonsingular fixed-endpoint boundary problem: for two slabs and one mode with $h^2\lambda = 3$, zero endpoint potentials leave the interior potential arbitrary when its source is zero, and are incompatible with a nonzero interior source. The initial-value coefficient $1 + h^2\lambda/6$ remains positive there.

An exact bound for the supplied cone. Every tetrahedron has three apex rays with Gram matrix $2I_3 + \varphi\mathbf{1}\mathbf{1}^\mathsf{T}$. Divide each local Whitney mass and curl-energy matrix by its positive tetrahedron volume. Exact arithmetic in $\mathbb{Q}(\sqrt{5})$ gives an LDL^T factorization of $24M_T - K_T$ with strictly positive diagonal. An independent verifier reconstructs the barycentric gradients from the supplied vertex coordinates, checks the matrix identity and encloses $\sqrt{5}$ between rational bounds to certify each sign. The twenty signed local-to-global edge maps cover every global edge, so assembly proves $K < 24M$, without a floating-point eigenvalue bound. At $h = 1/2$, therefore,

$$M - h^2K/12 > M/2, \qquad E_n^\mathsf{T} M E_n \leq 4H_n, \qquad B_n^\mathsf{T} M_2 B_n \leq 16H_n$$

on a source-free interval. The magnetic estimate also holds at the other endpoint of each slab: use $B_n = \overline{B}_n + hCE_n/2$ and $h^2E_n^\mathsf{T} K E_n \leq 6E_n^\mathsf{T} M E_n$. The factorization is an exact algebraic certificate; the trajectory below is a separate numerical calculation.

A recorded stationary volume history. Retain the inherited initial vector potential and prescribed neutral source covectors, apply (3), and use (2). The correction changes the initial electric field by approximately 0.71279026 in M-norm. Consequently this history is distinct from the earlier counting-action history and its cone interpolation. Each of three slices occupies thirteen scalar and forty-two edge registers. For each edge, read both endpoint pair averages, retain scalar baselines, and restore the registers by $(b, 2r - b)$. The resulting bounded observer-like software patch has writable ports, readback, records, feedback and a public evidence bundle. Its 805 recorded events include 252 feedback cycles. Every read names its last writer and exact value; the volume advance consumes the decoded registers and source covectors. The dense metric projection and evolution rule are supplied software operations, not consequences of pair averaging or local causal ancestry.

The two endpoint-fixed gauge representatives have field-plus-source action approximately -1.14785515542. Independent spacetime quadrature and complex-step differentiation verify all $42 + 2 \cdot 13 = 68$ free action derivatives, including radial edges and the apex. Their absolute residuals are below 10^{-12}. The metric evolution uses double precision and declared 10^{-9} comparison tolerances; register values are exact rational encodings of the numerical outputs, and feedback restoration is exact for those encodings. No interval or exact-real trajectory certificate is asserted.

A separate 64-slab numerical continuation starts from the decoded temporal-gauge data. Its maximum Gauss, vector-equation and work residuals are below 10^{-12}, as is the source-free energy drift. Its departure from the fixed radial extension reaches approximately 0.45218, showing that the volume recurrence exercises independent radial degrees of freedom. Only the first three slices have the serial execution trace. The remaining continuation is a numerical field calculation. Prescribed currents are not dynamical charged matter. Stationarity against all finite-element coordinates also does not imply stationarity against arbitrary smooth continuum tests. Spatial refinement, physical clock and metric selection, and laboratory identification require further input.

The Hilbert space of the same radiative modes. Consider the source-free neutral sector of this same geometric action, in continuous time and temporal gauge, with the spatial mesh fixed.

The constraint is $D^{\mathsf{T}}M\dot{A} = 0$. A time-independent gauge transformation removes the constant longitudinal potential, giving the transverse space $V_\perp = \ker(D^{\mathsf{T}}M)$. The cone is contractible: its explicit curl blocks imply $\ker C = \operatorname{im} D$, of dimension twelve. Thus $V_\perp$ has dimension thirty and K is positive definite there. The finite-dimensional spectral theorem gives a complete real frame $v_1, \ldots, v_{30}$ with

$$v_i^{\mathsf{T}} M v_j = \delta_{ij}, \qquad K v_i = \omega_i^2 M v_i, \qquad \omega_i > 0.$$

For $A = \sum_i q_i v_i$, the exact spatially semidiscrete action is

$$S_0 = \frac{1}{2}\int \left(\dot{A}^{\mathsf{T}} M \dot{A} - A^{\mathsf{T}} K A\right) dt = \frac{1}{2}\int \sum_i (\dot{q}_i^2 - \omega_i^2 q_i^2)\, dt.$$

The spatial edge-element action is standard [42], as are the finite normal-mode theorem and canonical field quantization. The attachment here is to these geometric matrices.

Theorem 3.5 (Canonical quantum realization of the volume radiative sector)**.** *Supply canonical bosonic quantization and $\hbar > 0$. The preceding sector has Hilbert space $\mathcal{H} = \ell^2(\mathbb{N}^{30})$. On its finite-occupation domain, let*

$$a_i|n\rangle = \sqrt{n_i}\,|n - e_i\rangle, \qquad\qquad a_i^\dagger|n\rangle = \sqrt{n_i + 1}\,|n + e_i\rangle,$$

$$\widehat{q}_i = \sqrt{\frac{\hbar}{2\omega_i}}(a_i + a_i^\dagger), \qquad\qquad \widehat{p}_i = i\sqrt{\frac{\hbar\omega_i}{2}}(a_i^\dagger - a_i).$$

Then $[\widehat{q}_i, \widehat{p}_j] = i\hbar\delta_{ij} I$, and

$$\widehat{H} = \frac{1}{2}\sum_i (\widehat{p}_i^2 + \omega_i^2 \widehat{q}_i^2), \qquad E_n = \hbar \sum_i \omega_i (n_i + 1/2).$$

The diagonal Hamiltonian on $\mathcal{D}(\widehat{H}) = \{\psi : \sum_n E_n^2 |\psi_n|^2 < \infty\}$ is self-adjoint and generates the strongly continuous unitary group $(U_t\psi)_n = e^{-itE_n/\hbar}\psi_n$. The Heisenberg equations are $\dot{\widehat{q}}_i = \widehat{p}_i$ and $\dot{\widehat{p}}_i = -\omega_i^2 \widehat{q}_i$. Consequently $\widehat{A} = \sum_i v_i \widehat{q}_i$, $\widehat{E} = -\sum_i v_i \widehat{p}_i$ and $\widehat{B} = C\widehat{A}$ use the same classical mode shapes and satisfy the source-free semidiscrete Maxwell equations on the common invariant domain.

Proof. Start with polynomials in thirty complex variables. Give monomials the inner product $\langle X^n, X^m\rangle = n!\delta_{nm}$, where $n! = \prod_i n_i!$. Multiplication by X_i and differentiation by X_i are adjoint on this domain and satisfy $[a_i, a_j^\dagger] = \delta_{ij} I$. Normalizing monomials gives the displayed orthonormal occupation basis and its ℓ^2 completion. The quadratic Hamiltonian and commutators follow by expansion. Finite occupation vectors are analytic for each position and momentum operator. If N bounds a vector's occupation support, its k-th iterate has norm at most $C_\psi C^k \sqrt{(N + k)!/N!}$, with fixed constants C_ψ, C. This makes the analytic-vector series converge for sufficiently small argument. Thus these symmetric operators have unique self-adjoint closures by the analytic-vector theorem [43].

The adjoint of the real diagonal operator has exactly the stated weighted square-summability domain, proving self-adjointness of $\widehat{H}$. Truncation in the occupation basis proves that finite-support vectors are a core. The same finite-tail argument proves strong continuity of U_t. Finally $[\widehat{H}, \widehat{q}_i] = -i\hbar\widehat{p}_i$ and $[\widehat{H}, \widehat{p}_i] = i\hbar\omega_i^2 \widehat{q}_i$ give the equations, and the complete spectral frame pulls them back to the actual mass and stiffness forms. $\square$

`Lean/Screen/WhitneyQuantumBridge.lean` formalizes the complete normal-frame action pullback, polynomial commutation relations, occupation energies and Heisenberg identities. Its frame is supplied; existence here uses the finite spectral argument above. The Hilbert completion and self-adjoint-domain argument are the analytic proof just given, not Lean results. This is canonical quantization of the free radiative sector. The recorded history with nonzero prescribed charge additionally needs an affine longitudinal sector and source coupling. Classical readback does not select a quantum state, Born statistics, $\hbar$ or a physical clock, and the finite mesh supplies no relativistic quantum field net.

A controlled temporal limit. The finite-step prism dynamics and the continuous-time oscillator have different frequencies. For one mode, put $z = h^2\lambda$, $f = 1 + z/6$ and choose $0 < z < 12$. The discrete Legendre momenta give the exact symplectic map

$$\begin{pmatrix} q' \\ p' \end{pmatrix} = \frac{1}{f} \begin{pmatrix} 1 - z/3 & h \\ -h\lambda(1 - z/12) & 1 - z/3 \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix}.$$

Define $\theta \in (0, \pi)$ by $\cos\theta = (1 - z/3)/f$. Since $\sin^2\theta = z(1 - z/12)/f^2$, this is the time-$h$ flow of the positive modified Hamiltonian

$$H_h = \frac{\theta}{2f\sin\theta}\left[p^2 + \lambda(1 - z/12)q^2\right].$$

On any fixed bounded positive spectral window, $\theta/h = \sqrt{\lambda} + O(h^2)$ and the quadratic coefficients of $H_h - H_0$ are $O(h^2)$, uniformly as $h \to 0$. On bounded time intervals the corresponding classical propagators therefore converge at order two for bounded canonical initial data. The initial pair of potentials must be obtained from the discrete Legendre map for those data; an arbitrary first-order initial displacement does not provide the same order claim. This is a finite-dimensional temporal convergence argument with the spatial mesh fixed. Canonical quantization of H_h supplies its own unitary oscillator flow, with frequency θ/h; its continuous Hamiltonian path fixes the metaplectic lift. Neither its generator nor its frequency equals $\widehat{H}$ at fixed h. No uniform operator-norm error bound for the unbounded quantum generators, spatial convergence theorem or physical time calibration follows from this calculation.

A joint action with dynamical charged matter. Prescribed currents can be replaced by a dynamical complex scalar on the same solid. This requires an additional matter action and coupling; neither is selected by the classical readback protocol. Gauge-invariant simplicial actions have established precedents [44]. The construction below specifies a potential-dependent scalar trial space on this particular volume and retains the complete dependence of the scalar interpolation on the gauge potential.

Let $a \in \mathbb{R}^{42}$, $\phi \in \mathbb{R}^{13}$ and $\psi \in \mathbb{C}^{13}$ be variable coefficients, and supply a real charge e. On a tetrahedron, write $a_{ij} = -a_{ji}$ for the oriented edge coefficient, set $a_{ii} = 0$, and define

$$A_h = W_1 a, \qquad \phi_h = W_0\phi, \qquad \chi_h = W_0\chi,$$
$$\theta_i(x) = \int_{v_i}^{x} A_h = \sum_j a_{ij}\lambda_j(x), \qquad \Psi_a(x) = \sum_i \lambda_i(x)e^{ie\theta_i(x)}\psi_i.$$

The paths in θ_i are straight segments within the tetrahedron. The coefficients a_{ij} are real unwrapped integrals; compact link phases alone do not determine these interpolants. Let $D_t\Psi = \partial_t\Psi + ie\phi_h\Psi$, $D_x\Psi = \nabla\Psi - ieA_h\Psi$, and $E = -\dot{a} - D\phi$. With supplied $m^2, g \geq 0$, use the autonomous action

$$S = \int dt \left\{\frac{1}{2}E^{\mathsf{T}}ME - \frac{1}{2}a^{\mathsf{T}}Ka + \int_\Omega \left[|D_t\Psi_a|^2 - |D_x\Psi_a|^2 - m^2|\Psi_a|^2 - \frac{g}{2}|\Psi_a|^4\right] dx\right\}. \tag{4}$$

Every occurrence of Ψ_a, including its time derivative, is differentiated when varying a. The field-dependent trial space makes this requirement substantive.

Theorem 3.6 (Gauge-covariant charged field and joint evolution)**.** *On the fixed nondegenerate cone, Ψ_a is a continuous $H^1(\Omega)$ field with nodal values ψ_i. The action (4) is invariant under every smooth nodal gauge transformation*

$$a \longmapsto a + D\chi, \qquad \phi \longmapsto \phi - \dot{\chi}, \qquad \psi_i \longmapsto e^{ie\chi_i}\psi_i.$$

In temporal gauge, its joint real Euler–Lagrange equations for $a, \operatorname{Re}\psi, \operatorname{Im}\psi$ have a unique global classical solution for every finite initial position and velocity. If all thirteen Gauss equations hold initially, they hold for all time. Nonzero locally charged, globally neutral compatible initial data exist whenever $e \neq 0$. These statements concern the full nonlinear finite-dimensional action and do not assume a prescribed external current.

Proof. Pullback of a Whitney edge form to the straight segment from v_i to x gives the displayed formula for θ_i. Since $\sum_j \lambda_j = 1$, a gauge transformation changes it by $\chi_h(x) - \chi_i$, and consequently $\Psi_{a+D\chi}[e^{ie\chi}\psi] = e^{ie\chi_h}\Psi_a[\psi]$. Both covariant derivatives acquire this same unit phase. The electric and magnetic fields are unchanged, proving action invariance. At a vertex only its own basis function survives. On a common face only its vertices contribute, and the barycentric functions and tangential Whitney field have identical traces from either tetrahedron. The scalar traces therefore agree. Piecewise smoothness on a finite mesh proves H^1 conformity.

Write $\psi = u + iv$ and define the Euler expressions with the sign convention $\mathcal{E}_y = \partial L/\partial y - d(\partial L/\partial \dot{y})/dt$, while $\mathcal{E}_\phi = \partial L/\partial \phi$. Gauge invariance, integrated against an arbitrary compactly supported $\chi(t)$, gives the off-shell identity

$$\dot{\mathcal{E}}_\phi + D^\mathsf{T}\mathcal{E}_a + e(u \odot \mathcal{E}_v - v \odot \mathcal{E}_u) = 0. \tag{5}$$

Here $\odot$ means coordinatewise multiplication. In particular,

$$(\mathcal{E}_\phi)_i = -(D^\mathsf{T}ME)_i + \rho_i, \qquad \rho_i = 2e\int_\Omega \lambda_i \operatorname{Im}(\overline{\Psi_a} D_t \Psi_a)\, dx.$$

Thus the joint matter and vector equations propagate Gauss's law. Free variation of every scalar coefficient requires $\sum_i \rho_i = 0$ on this closed-load convention; a nonzero total charge would require a different boundary-flux or source prescription. As for the prescribed source action, conventional physical charge has the opposite load sign.

To prove existence, fix temporal gauge. Write the scalar interpolation as $W(a)\psi$. Its time derivative is $W(a)\dot{\psi} + S(a,\psi)\dot{a}$, with S the derivative of W in the gauge-potential direction. The quadratic velocity Hessian is

$$\delta\dot{a}^\mathsf{T} M \delta\dot{a} + 2\|W(a)\delta\dot{\psi} + S(a,\psi)\delta\dot{a}\|_{L^2}^2.$$

It is positive definite: vanishing first forces $\delta\dot{a} = 0$, and then nodal interpolation forces $\delta\dot{\psi} = 0$. All coefficients are smooth real functions, so the Euler equations form a regular smooth first-order system in positions and velocities and have unique local solutions.

The conserved energy is the sum of the positive electric and magnetic energies and $\|\partial_t\Psi_a\|_{L^2}^2 + \|D_x\Psi_a\|_{L^2}^2 + m^2\|\Psi_a\|_{L^2}^2 + g\|\Psi_a\|_{L^4}^4/2$. It bounds $|\dot{a}|$, hence keeps a in a compact set on each finite time interval. On that set the positive scalar Gram matrix $W(a)^*W(a)$ has a uniform positive lower bound. Smoothness and linearity in ψ give $\|S(a,\psi)\dot{a}\|_{L^2} \leq C|\psi||\dot{a}|$. Consequently $|\dot{\psi}| \leq C_T(\sqrt{H} + |\psi|)$, where H is the initial energy. Gronwall's inequality bounds ψ and $\dot{\psi}$ on every finite interval. Smooth Hessian inversion then excludes finite-time escape, proving global existence. This argument does not assert a time-uniform bound on gauge potentials or require a globally smooth quotient by the gauge group.

For an explicit charged initial condition, take $a_0 = 0$, $\psi_i(0) = 1$, and $\dot\psi_i(0) = if_i$, where the real nonzero nodal function $f_h = \sum_i f_i\lambda_i$ has zero integral. Antisymmetry gives $\sum_{ij}\lambda_i\lambda_j\dot a_{ij} = 0$, so the dressing contribution to $\partial_t\Psi_a$ cancels at these data for every $\dot a$. Hence $\rho_i = 2e\int\lambda_i f_h$ has zero total and is nonzero by positivity of the scalar mass matrix. Solve $D^{\mathsf T}MDz = \rho$ with mean-zero z, and set $E_0 = Dz$, $\dot a_0 = -Dz$. This satisfies all Gauss equations without a circular source definition. On the regular cone an explicit choice is $f_0 = 3$ at the centre and $f_i = -1$ at all twelve boundary vertices. If $V_T = (3+\sqrt5)/6$ is the common tetrahedron volume, then $\rho_0 = 6eV_T$ and $\rho_i = -eV_T/2$. Writing $\beta = e(2+3\varphi)/10$, take $z_0 = 12\beta/13$ and $z_i = -\beta/13$. Each radial electric coefficient is $-\beta$ and every boundary electric coefficient is zero. The apex gradient norm $|\nabla\lambda_0|^2 = 3/(2+3\varphi)$, simplex incidence and the exact barycentric moments verify all thirteen Gauss equations. □

The finite holonomy and interpolation gauge algebra is formalized in `Lean/Screen/WhitneyChargedMatter.lean`. Conformity, the variational identity and global existence are the analytic argument above. In particular, the action-derived edge force includes the change of Ψ_a when a varies. The Whitney projection of the continuum scalar current generally omits these terms even when the nodal matter equations hold. This finite approximation has dynamical matter and a joint stationary action. Its geometry, scalar species, charge, mass, interaction coefficient and continuum matter Lagrangian are supplied. The analytic classical theorem does not itself certify a numerical history or select physical matter content. Its finite equations, spatial action consistency and interacting Hilbert realization have distinct numerical and analytic statements below. The free radiative Hilbert space describes a sector; quantization of the coupled action requires its full kinetic metric. No Standard Model identification follows from the supplied scalar law.

An executed charged solution of the same finite action. The preceding initial data lie in a useful invariant sector of (4). Let every outward radial edge have coefficient α, every boundary edge have coefficient zero, and let the scalar coefficient be $c \in \mathbb{C}$ at the centre and $b \in \mathbb{C}$ at all boundary vertices. This sector has five real configuration coordinates. On each tetrahedron write $s = \lambda_0$, $C = e^{ie\alpha}c$, and $U = C - b$. Then

$$
\begin{aligned}
A_h &= -\alpha\nabla s, & \Psi_a &= e^{-ie\alpha s}\big(sC + (1-s)b\big),\\
D_x\Psi_a &= e^{-ie\alpha s}U\nabla s, & \partial_t\Psi_a &= e^{-ie\alpha s}\left[se^{ie\alpha}\dot c + (1-s)\dot b + ie\dot\alpha s(1-s)U\right].
\end{aligned}
\tag{6}
$$

The last term retains the time derivative of the scalar basis. Although A_h is a spatial gradient and the magnetic field vanishes, the electric field $\dot\alpha\nabla s$, local charge and matter motion are nonzero. Removing this electric field by a time-dependent spatial gauge introduces a scalar potential; it does not remove the interaction.

Proposition 3.7 (Full variational lift of the symmetric sector)**.** *The five-coordinate Euler–Lagrange equations obtained by restricting* (4) *in temporal gauge imply all sixty-eight real temporal-gauge Euler–Lagrange equations. Initial satisfaction of Gauss's law implies all thirteen Gauss equations along this solution. On this sector the action integrands are polynomials of degree at most four in barycentric coordinates after cancellation of their common unit phase. Their unrestricted first variations, and the full Euler expressions evaluated on a twice differentiable symmetric path, have polynomial degree at most five.*

Proof. The sixty proper rotations of the regular icosahedron act on nodal coefficients by permutation and on oriented edge coefficients by signed permutation. They preserve tetrahedral volume, the Euclidean metric and straight paths, hence preserve the complete dressed action. Vertex transitivity fixes one boundary scalar coefficient. The radial edges form one orbit. Each boundary edge is reversed by a half-turn, forcing its invariant coefficient to vanish. Thus the fixed configuration space is precisely the stated five-dimensional real space.

At an invariant position, velocity and acceleration the full Euler covector is invariant. Its value on any variation equals its value on the average of that variation over the finite rotation group. This average lies in the five-dimensional fixed space. Consequently vanishing of the restricted Euler covector implies vanishing on every real variation. Equivalently, regularity and uniqueness of the full Euler system make its evolution preserve this fixed space. Gauss propagation follows from (5) with the compatible initial data, without discarding the scalar variations.

Antisymmetry gives the first line of (6); direct differentiation gives the second line. After removal of the common phase, the scalar field has degree one and its time derivative degree at most two. The spatial covariant derivative has degree zero. An unrestricted edge variation of the scalar field has degree at most two, its time derivative at most three, and its spatial covariant derivative at most two. These degree counts give the claims for the action and its first variation. The same counts apply after the time integration by parts: the scalar acceleration has degree at most three. Four-point tensor Gauss–Legendre quadrature after a Duffy map integrates these degree-five tetrahedral polynomials exactly in real arithmetic. The Jacobian adds at most two powers in any quadrature coordinate. For the restricted action one can instead integrate over s with the exact marginal weight $3(1-s)^2$. □

A bounded numerical evolution supplies $e = 1/4$, $m^2 = 1/2$ and $g = 1/4$, starts from the charged initial condition above, and evolves the supplied dimensionless action parameter over $[0, 2]$. It stores 81 samples of all position, velocity and acceleration coefficients, electric and magnetic cochains, nodal charge loads, and scalar and covariant-derivative fields at all twenty tetrahedral centroids. The independent verifier reconstructs the oriented mesh and unrestricted scalar Jacobians, checks all 68 Euler and 13 Gauss equations at every sample, and re-integrates the trajectory using full local Jacobians pulled back to the fixed space. This reconstruction does not use the producer's one-dimensional phase-cancelled formulas.

The largest full Euler residual is below 6×10^{-14}, the largest Gauss residual below 4×10^{-15}, and energy drift below 7×10^{-14}. Independent trajectory reconstruction agrees within 3×10^{-14}. Fixed-step fourth-order Runge–Kutta endpoint errors at 80, 160 and 320 steps are approximately 6.61×10^{-6}, 4.32×10^{-7} and 2.75×10^{-8}, respectively, against a tighter reference. Four- and five-point tetrahedral quadratures agree within 5×10^{-14}; underintegration with three points instead gives a full Euler defect of about 3×10^{-4}. Freezing the scalar basis during variation gives a defect exceeding 0.34. The maximum nodal charge-load change exceeds 2.25, so this is a coupled charged trajectory, not repetition of the initial constraint check. These are observed floating-point residuals and refinement comparisons, not a rigorous enclosure of trajectory error.

The producer and independent verifier are `code/electromagnetism/whitney_charged_dynamics.py` and `code/electromagnetism/verify_whitney_charged_dynamics.py`; the versioned data are in `code/electromagnetism/runtime/whitney_charged_dynamics_receipt.json`. The data identify the mesh, oriented basis, gauge, complex encoding, couplings, clock, source hashes and field sampling locations. They contain no authenticated observer readback or quantum state. The construction executes the supplied classical action on one fixed geometry; it does not identify a physical clock, select the matter theory from source histories, or establish a spatial continuum limit.

A rigorous time enclosure of the charged trajectory. The finite trajectory admits a rigorous error bound on the whole supplied action-time interval $[0,2]$. The enclosure uses the same charged initial data and full dressed action as the numerical execution. It controls time integration on the fixed cone, separately from spatial convergence, observer provenance and physical clock calibration.

Put $r = \kappa/|\Omega| = 6/(7+3\sqrt{5})$, and divide the action by $|\Omega|$, which does not change its Euler equations. In the symmetric sector set $C = e^{ie\alpha}c$, $x = (\operatorname{Re} C, \operatorname{Im} C, \operatorname{Re} b, \operatorname{Im} b)$, and $Z(s) = sC + (1-s)b$. The time kinetic term is

$$\tfrac{1}{2} r\dot{\alpha}^2 + \int_0^1 3(1-s)^2 \left|\dot{Z} - ie\dot{\alpha}sZ\right|^2 ds.$$

Thus α is cyclic. The constant scalar velocity block and its inverse are

$$H = \begin{pmatrix} 1/5 & 3/10 \\ 3/10 & 6/5 \end{pmatrix} \otimes I_2, \qquad H^{-1} = \begin{pmatrix} 8 & -2 \\ -2 & 4/3 \end{pmatrix} \otimes I_2.$$

Writing the full kinetic metric as $\left(\begin{smallmatrix} d & \ell^{\mathsf{T}} \\ \ell & H \end{smallmatrix}\right)$, one obtains

$$\begin{aligned}
\ell &= \frac{e}{10}(C_i + b_i, -C_r - b_r, C_i + 2b_i, -C_r - 2b_r), \\
w = H^{-1}\ell &= (e(3C_i + 2b_i)/5, -e(3C_r + 2b_r)/5, e(b_i - C_i)/15, e(C_r - b_r)/15), \\
D = d - \ell^{\mathsf{T}} H^{-1}\ell &= r + \frac{2e^2}{525}|C - b|^2 \ \geq r > \frac{2}{5}.
\end{aligned}$$

Before volume normalization, this is the positive denominator $\kappa + 2e^2|\Omega||C-b|^2/525 \geq \kappa$. All coefficients follow from the exact moments $\int_0^1 3s^n(1-s)^2 ds = 6/[(n+1)(n+2)(n+3)]$.

Let $p \in \mathbb{R}^4$ be the scalar canonical momenta of the normalized action. The initial cyclic momentum is exactly zero and remains zero. The normalized potential is

$$W(x) = r|C-b|^2 + \tfrac{1}{2}\langle |Z|^2 \rangle + \tfrac{1}{8}\langle |Z|^4 \rangle, \qquad \langle f \rangle = \int_0^1 3(1-s)^2 f(s)\, ds.$$

For $X = |C|^2$, $Y = \operatorname{Re}(\overline{C}b)$, $B = |b|^2$, the moments are

$$\langle |Z|^2 \rangle = X/10 + 3Y/10 + 3B/5, \qquad \langle |Z|^4 \rangle = \frac{X^2 + 3XY + 2XB + 4Y^2 + 10YB + 15B^2}{35}.$$

The reduced Hamiltonian is $\mathcal{H} = \frac{1}{2}p^{\mathsf{T}} H^{-1} p + (w^{\mathsf{T}} p)^2/(2D) + W$. Together with cyclic-coordinate reconstruction, its equations are the rational system

$$\begin{aligned}
a &= -w^{\mathsf{T}} p/D, \qquad \dot{\alpha} = a, \qquad \dot{x} = H^{-1}p - wa, \\
\dot{p}_j &= a\,\partial_j(w^{\mathsf{T}} p) + \tfrac{1}{2}a^2 \partial_j D - \partial_j W.
\end{aligned} \tag{7}$$

Its nine initial coordinates are

$$y(0) = (\alpha, x, p) = (0, 1, 0, 1, 0, 0, 3/10, 0, -3/10).$$

They give $\dot{\alpha}(0) = 3/(40r)$, $\dot{C}(0) = i(3 + e\dot{\alpha}(0))$ and $\dot{b}(0) = -i$, exactly the preceding charged temporal-gauge initial condition. The normalized conserved energy is $49/40 + 9/(3200r)$.

Proposition 3.8 (Uniform time enclosure for the finite charged action)**.** *For $e = 1/4$, $m^2 = 1/2$, $g = 1/4$, the exact initial-value solution of* (7) *exists uniquely on* $[0, 2]$. *The supplied piecewise degree-32 polynomial P, on eighty intervals of exact length* $1/40$, *satisfies*

$$\sup_{0\le t\le 2} \|y(t) - P(t)\|_\infty \le 10^{-20}. \tag{8}$$

At shared interval endpoints either adjacent polynomial may be used; the approximants need not join exactly. Under the inverse phase map, the enclosed solution lifts to all 68 real temporal-gauge Euler equations and all 13 Gauss constraints. At the 81 nominal times $j/40$, its five original positions and five original velocities differ from the stored binary64 samples by at most 10^{-10} *in each coordinate.*

Proof. The Hamiltonian reduction above follows by completing the scalar kinetic square. Its positive Schur denominator makes the rational vector field smooth on all real canonical coordinates. The exact initial cyclic momentum is zero. Canonical equivalence and the full symmetry lift therefore identify the enclosed solution with the same finite-action solution; the compatible initial Gauss law propagates by the action's Noether identity.

Here is the numerical certificate argument, including continuous times. On a step starting from interval enclosure Y_j, the verifier checks a closed box B_j and the strict Picard inclusion

$$Y_j + [0, h]F(B_j) \subset \operatorname{int} B_j, \qquad h = 1/40.$$

Local existence, smoothness and the first-exit argument keep every relevant solution in this box for the whole step. A contraction estimate is not required. Formal power-series substitution in $\dot{y} = F(y)$ produces intervals $A_k(Y_j)$ containing the normalized derivatives $y^{(k)}(t_j)/k!$. The same recurrence initialized on the entire box B_j bounds $y^{(33)}(t)/33!$ at every time in the step. Taylor's theorem consequently gives, for every $u \in [0, h]$,

$$y(t_j + u) \in \sum_{k=0}^{32} A_k(Y_j)u^k + A_{33}(B_j)u^{33}.$$

At $u = h$ this interval is checked to lie in the next recorded enclosure. For each supplied polynomial coefficient $P_{j,k}$, the verifier also bounds $|A_k(Y_j) - P_{j,k}|$ coordinatewise and sums those radii times h^k, together with the remainder radius times h^{33}. Every resulting bound is at most 10^{-20}. Induction over the eighty steps proves existence and the displayed uniform error estimate, including both choices of approximant at a join.

Interval endpoints are integers divided by 2^{256}; all elementary operations round outwards by integer division. The exact rational step is enclosed rather than rounded to a point. An integer square-root inequality encloses $\sqrt{5}$, and each division checks separation from zero. Polynomial coefficients are exact multiples of 2^{-96}. The verifier independently reconstructs the scalar mass block, Schur complement and potential from rational simplex moments, and does not import the producer or a floating-point ODE solver.

For the historical comparison, each stored binary64 value is interpreted as its exact dyadic rational. Interval sine and cosine evaluations use degree-40 Taylor polynomials and the real Lagrange remainder $|\theta|^{41}/41!$. Applying these to $c = e^{-ie\alpha}C$ and its time derivative encloses the original position and velocity coordinates at each nominal sample time. Their checked differences give the stated 10^{-10} bound. The historical samples play no role in constructing or proving the trajectory enclosure. □

The producer, independent verifier and exact interval data are `code/electromagnetism/whitney_charged_enclosure.py`, `code/electromagnetism/verify_whitney_charged_enclosure.p` y and `code/electromagnetism/runtime/whitney_charged_enclosure_receipt.json`. Source identities, coefficient encodings, initial data and interpretation are part of the checked packet. The enclosure certifies this finite classical initial-value problem, not the truth of its supplied physical inputs. It neither calibrates the action parameter in seconds nor certifies the separate polygonal Jacobi-clock quadrature, charged spatial continuum limit, observer execution or quantum evolution. The interval argument is implemented and independently replayed; it is not a Lean formalization.

A self-reading execution of the charged action. The five-coordinate invariant sector also admits a bounded software execution with its own local states, ports, records and feedback. Each patch holds one coordinate and its velocity. Five ring ports connect these computational patches; their locations are not identified with physical observers or with points of the supplied cone. The numerical integrator reads the preceding decoded records before each advance. Historical trajectory samples enter only a subsequent comparison.

Proposition 3.9 (Exact record-assisted restoration)**.** *Let each patch hold a pair in $\mathbb{Q}^2$. Record its initial pair, then probe a ring edge by replacing both endpoint pairs x, y with $r = (x + y)/2$. Retain the response r, and restore the endpoints using the first endpoint's retained baseline b as $(b, 2r - b)$. Processing the five edges serially restores every patch exactly. Each patch's pair can be recovered from its predecessor's response and baseline and agrees with its own retained baseline.*

Proof. At the first probe, $b = x$, hence $2r - b = y$. Inductively every probe starts from the original endpoint pairs and restores them, so the same identity holds on every edge. Each patch has one predecessor in the ring, which proves the stated decoding rule. □

The implementation in `code/electromagnetism/whitney_charged_instrument.py` encodes numerical coordinates as exact rational representations of binary64 values. Its destructive probes, responses and feedback therefore obey the proposition exactly, while its dynamical advances remain numerical. The recorded execution has 81 decoded states, 405 completed repair cycles and 1782 events. The action time step 1/40, couplings and initial data are supplied. Every advance consumes the previous decoded state; a distinct verifier reconstructs event read versions, write values and hash links, then independently integrates the action from those records. It checks all 68 configuration equations, all 13 Gauss equations, energy and the full coupled field readouts on the decoded samples. These checks retain the scalar basis derivatives.

The public frames expand the decoded coordinates by the proved symmetry map and provide the Whitney connection, electric cochains, complex matter coefficients, local charge and centroid covariant derivatives. They support synchronized visualization of this software computation and its declared field interpretation. The repair-cycle counter measures completed protocol operations. The conversion from a sweep to an action-time step is supplied; it is neither a laboratory calibration nor a derivation of physical time. Hash-pinned replay establishes internal software provenance, without an external witness or an identification of the patches with physical observers. The exact register restoration is separate from numerical trajectory error, spatial convergence and quantum-state evolution.

A clock read from the changing field configuration. A repair count and a time coordinate are different objects. The same autonomous action does, however, supply a dynamical duration functional. This is an application of the classical Jacobi–Maupertuis construction [36], not a new quantization rule. Write a natural mechanical Lagrangian as $L = \frac{1}{2}G_q(\dot{q}, \dot{q}) - V(q)$, using the complete coupled kinetic metric, or its Gauss-reduced metric on an admissible horizontal path. Fix an energy E and an oriented regular path in the open region $V < E$. Define

$$d\tau = \sqrt{\frac{G_q(dq, dq)}{2(E - V(q))}}, \qquad J_E[q] = \int \sqrt{2(E - V(q))G_q(q', q')}\, ds. \tag{9}$$

Proposition 3.10 (Dynamical duration on a nonturning path)**.** *For smooth positive-definite G and smooth V, the duration above is invariant under orientation-preserving regular reparameterization. A regular stationary path of J_E, parameterized by τ, solves the Euler–Lagrange equations of L with energy E. Conversely, every solution of that energy with positive kinetic energy is such a path, and its τ increments equal its original action-time increments. For fixed G, V, E, positive energy-compatible timing of that oriented path is unique up to its origin.*

Proof. Set $T_s = G_q(q', q')/2$ and $N = \sqrt{T_s/(E - V)}$. Positive reparameterization multiplies both q' and N by the same positive derivative, proving duration invariance. The lapse action

$$I[q, N] = \int (T_s/N + N(E - V))\, ds$$

has lapse equation $T_s/N^2 = E - V$. Its unique positive solution gives $I[q, N(q, q')] = J_E[q]$. Substitution preserves the configuration Euler equation because the lapse variation vanishes. With $d\tau = N\, ds$, that equation is

$$\frac{d}{d\tau}(G_{ij}\dot{q}^j) - \frac{1}{2}\partial_i G_{jk}\dot{q}^j\dot{q}^k + \partial_i V = 0.$$

The lapse equation is the required energy identity. Conversely this identity and the Euler equation give the two equations for I. Uniqueness follows directly from its positive square root. The assertion does not extend through a turning point where numerator and denominator both vanish, or to a constant equilibrium path. □

This construction is covariant under changes of configuration coordinates. For gauge independence one must use the reduced metric or keep the scalar potential in the full covariant tangent. Arbitrary time-dependent gauge changes followed by discarding the scalar potential do not preserve the kinetic energy. The numerical construction below uses the specified temporal-gauge representative throughout. Nor does a Jacobi metric on configuration space itself define a Lorentzian spacetime metric.

There is also a useful approximation statement. Let a fixed C^3 regular path remain in a compact nonturning region, with G, V smooth there. Its polygonal interpolants at parameter spacing at most δ have duration error $O(\delta^2)$. To see this, write the duration integrand as $F(q, v) = \sqrt{G_q(v, v)/(2(E - V(q)))}$. On each short segment, the configuration error is $O(h^2)$, the velocity error is $O(h)$, and the latter has zero integral. Expand F to first order: freezing its velocity derivative at the midpoint cancels that integral, and the derivative's variation and quadratic remainder both integrate to $O(h^3)$. Summing gives the assertion. Smoothness is used away from zero velocity and $E - V = 0$. This bound compares exact curve integrals; floating-point quadrature and errors in the supplied samples require separate control.

The executable consumer reads only the ordered decoded configuration coordinates from a bounded self-reading software episode. Local patch states, port responses, retained records and feedback restoration belong to that episode's independent provenance check. The clock consumer uses neither the recorded velocities, action timestamps nor repair counts. It retains the same complete scalar-basis derivative in its kinetic form and supplies the initial energy $E = 1969/160 +$

$(493/120)\sqrt{5}$. Piecewise linear interpolation gives a duration on each configuration segment. A separate implementation integrates unrestricted tetrahedral fields instead of the producer's one-dimensional symmetric formulas; a nonlinear reparameterization of each segment checks duration invariance. The producer, verifier and data are `code/electromagnetism/whitney_ephemeris_clock.py`, `code/electromagnetism/verify_whitney_ephemeris_clock.py` and `code/electromagnetism/runtime/whitney_ephemeris_clock_receipt.json`.

The clock is internal to the supplied autonomous model: its rate is reconstructed from change and the action-energy convention. Selecting that action, energy and physical preparation from observer histories, assigning physical units, and comparing this duration with laboratory clocks remain additional tasks. It neither constructs a quantum clock observable nor identifies an arbitrary event ordering with proper time.

Spatial consistency of the same coupled action. The dressed scalar construction admits an action and first-variation estimate on the original solid under spatial refinement. The estimate uses the same edge integrals, Whitney potentials and potential-dependent scalar basis, rather than replacing them by a different finite-element family. Whitney interpolation and its commuting exterior derivative are standard [41]; the additional point here is to control the nonlinear dressing and its full derivative in this particular action.

Let $\mathcal{T}_0$ be the twenty-tetrahedron cone, $I = [0, T]$, and let $\mathcal{T}_\delta$ be conforming refinements respecting every macro-face, with maximum diameter $\delta \le 1$ and a common bound on diameter divided by inradius. Time remains continuous. Give $U = (A, \phi, \psi)$ the norm

$$\|U\|_X = \max_{K\in\mathcal{T}_0}\big(\|U\|_{L^\infty(I;W^{2,\infty}(K))} + \|\partial_t U\|_{L^\infty(I;W^{2,\infty}(K))}\big).$$

Use continuous representatives on each closed macro-cell and require $\phi, \psi \in H^1(\Omega)$ and $A \in H(\mathrm{curl};\Omega)$ at each time, with the same trace conditions on their time derivatives. The componentwise interpretation applies to complex ψ. Let $a_{ij} = \int_{v_i}^{v_j} A \cdot dx$, take nodal samples of ϕ, ψ, and write $A_\delta = I_1 A$, $\phi_\delta = I_0\phi$, $\Psi_\delta = \sum_i \lambda_i e^{ie\theta_i}\psi(v_i)$, where $\theta_i = \int_{v_i}^{x} I_1 A \cdot dx = \sum_j a_{ij}\lambda_j$. These paths integrate the interpolated potential, not the original smooth potential. Continuous tangential traces agree on shared faces, hence on their edges; connectivity of the cells incident to an edge makes its oriented integral single valued. Scalar face traces agree as well. No global continuity of the normal component of A is required. Define $S_\delta(U)$ by exact space and time integration of (4) with these fields. Define $S(U)$ by the same continuum scalar-electrodynamics density with A, ϕ, ψ, $E = -\partial_t A - \nabla\phi$ and $B = \mathrm{curl}\, A$.

Theorem 3.11 (Smooth-window action consistency)**.** *Fix finite T, R, e, m^2, g, the macro-mesh and a shape-regularity bound. There is a constant C_R, independent of the refinement, such that for every $\|U\|_X \le R$,*

$$|S_\delta(U) - S(U)| + \sup_{\substack{V\in X\\ \|V\|_X\le 1}} |DS_\delta(U)[V] - DS(U)[V]| \le C_R\delta. \tag{10}$$

Variations obey the same trace requirements. The derivative is the full real derivative with respect to all three sampled fields, including the dependence of the scalar basis on the edge potential. This estimate is a consistency statement on bounded smooth test fields; it does not assert convergence of nonlinear trajectories or of arbitrary finite-energy data.

Proof. All following bounds are uniform on each small tetrahedron of diameter h, with constants depending only on the displayed data. Shape regularity gives $|\nabla\lambda_i| \le C/h$. The usual elementary Taylor estimates for nodal interpolation give $\|I_0 f - f\|_\infty \le Ch^2\|f\|_{W^{2,\infty}}$ and $\|\nabla(I_0 f - f)\|_\infty \le$

$Ch\|f\|_{W^{2,\infty}}$. Whitney edge interpolation reproduces constant vector fields, is locally bounded in the sup norm, and commutes with the exterior derivative by Stokes's theorem. Consequently

$$\|I_1 A - A\|_\infty + \|\operatorname{curl}(I_1 A - A)\|_\infty \le Ch\|A\|_{W^{2,\infty}}.$$

For clarity, the curl estimate also follows without a general projection theorem: subtract the affine Taylor polynomial of A. Both its curl and the interpolated curl are constant; their fluxes through every face agree by Stokes's theorem and exact sampled edge circulations. The face normals span $\mathbb{R}^3$, so the curls agree. The remaining edge integrals are $O(h^3)$, while curls of edge basis functions are $O(h^{-2})$. The same estimates apply to $\partial_t A$, and to a variation $V = (B, \eta, \zeta)$, by linearity. Thus the electric and magnetic fields and their variations have $O(h)$ errors.

For the scalar field, antisymmetry of a_{ij} gives the exact cancellation

$$\sum_i \lambda_i \theta_i = \sum_{ij} \lambda_i a_{ij} \lambda_j = 0.$$

The same identity holds after differentiating in space, time or a potential direction. Put $r_i = \psi(v_i) - \psi(x)$ and $q(s) = e^{ies} - 1 - ies$. The exact error decomposition is

$$\Psi_\delta - I_0\psi = \psi(x) \sum_i \lambda_i q(\theta_i) + \sum_i \lambda_i (e^{ie\theta_i} - 1) r_i.$$

Here $\theta_i, \partial_t\theta_i, r_i, \partial_t r_i = O(h)$, their spatial gradients are bounded, and $q(\theta_i) = O(h^2)$, $q'(\theta_i) = O(h)$. Differentiating the displayed identity gives

$$\|\Psi_\delta - \psi\|_\infty + \|\partial_t(\Psi_\delta - \psi)\|_\infty \le C_R h^2, \qquad \|\nabla(\Psi_\delta - \psi)\|_\infty \le C_R h.$$

In particular the time estimate retains the phase velocity, and does not hold by simply deleting it from the definition.

To check that differentiation in the fields preserves these estimates, write $b_{ij} = \int_{v_i}^{v_j} B \cdot dx$ and $\beta_i = \sum_j b_{ij}\lambda_j$. The full derivative is

$$Z_\delta := D\Psi_\delta(U)[V] = \sum_i \lambda_i e^{ie\theta_i} (\zeta(v_i) + ie\beta_i \psi(v_i)).$$

The first term has the scalar interpolation estimates just proved. For the second, use $\sum_i \lambda_i \beta_i = 0$ to replace $e^{ie\theta_i}\psi(v_i)$ by $e^{ie\theta_i}\psi(v_i) - \psi(x) = O(h)$. Since $\beta_i, \partial_t \beta_i = O(h)$ and their spatial gradients are bounded, the resulting sum is $O(h^2)$, its time derivative is $O(h^2)$, and its spatial gradient is $O(h)$. These estimates are proportional to $\|V\|_X$. In particular $Z_\delta - \zeta$ has those same three bounds. Its time derivative includes all four terms

$$\partial_t Z_\delta = \sum_i \lambda_i e^{ie\theta_i} \left[\dot\zeta_i + ie\dot\beta_i \psi_i + ie\beta_i \dot\psi_i + ie\dot\theta_i(\zeta_i + ie\beta_i\psi_i) \right].$$

No second time derivative is used.

The covariant quantities and their variations are

$$\begin{aligned} Q_\delta &= \partial_t \Psi_\delta + ie\phi_\delta \Psi_\delta, & P_\delta &= \nabla \Psi_\delta - ieA_\delta \Psi_\delta, \\ \delta Q_\delta &= \partial_t Z_\delta + ie(I_0\eta)\Psi_\delta + ie\phi_\delta Z_\delta, & \delta P_\delta &= \nabla Z_\delta - ie(I_1 B)\Psi_\delta - ieA_\delta Z_\delta. \end{aligned}$$

They are uniformly bounded and differ from their continuum counterparts by at most $C_R h$, with the variation bounds proportional to $\|V\|_X$. Finally the action density is $\frac{1}{2}|E|^2 - \frac{1}{2}|B|^2 + |Q|^2 -$

$|P|^2 - m^2|\psi|^2 - g|\psi|^4/2$. Its first variation is the sum of the corresponding bilinear pairings, with matter potential contribution $-2(m^2 + g|\psi|^2)\,\mathrm{Re}(\overline{\psi}\,\zeta)$. On the bounded range just obtained both expressions are Lipschitz in their arguments. Integrating their $C_R h$ differences over $I \times \Omega$, and using $h \le \delta$, proves the result. The estimates hold almost everywhere for the stated Sobolev time regularity; all spatial operations use the Lipschitz representatives on each macro-cell. $\square$

Two controls delimit the argument. On the tetrahedron with vertices

$$0, \qquad (h,0,0), \qquad (0,h,0), \qquad (0,0,h),$$

let $A = \nabla(x_1^2)$. The weighted original-potential path sum is $x_1^2 - hx_1$, whereas the weighted Whitney-potential path sum is exactly zero. Interchanging these two path definitions therefore changes the interpolant. For a smooth bump b supported near an interior vertex with $b(0) = 1$, the fields $b_\epsilon(x) = b(x/\epsilon)$ satisfy $\|b_\epsilon\|_{H^1}^2 = \epsilon^3\|b\|_{L^2}^2 + \epsilon\|\nabla b\|_{L^2}^2 \to 0$, while the nodal value stays one. Thus this nodal construction cannot inherit an estimate for arbitrary three-dimensional H^1 data by point sampling.

`Lean/Screen/WhitneySpatialConsistency.lean` checks the finite antisymmetric cancellation and error-decomposition algebra; the spatial estimate itself is the analytic proof above. The executable refinement probe in `code/electromagnetism/whitney_spatial_consistency.py` uses manufactured nonzero-curvature fields on refinements of the same cone, with independent quadrature and directional finite-difference checks. These numerical checks test the implementation, not the uniform theorem or a physical observer history. The controlled comparison retains a supplied geometry, time coordinate and scalar-electrodynamics action.

A controlled continuum trajectory in the same matter action. The charged action contains a nonlinear real-matter sector for which a spatial solution estimate can be proved. The electric and magnetic fields vanish in this sector; the scalar quartic interaction remains active. The construction first specifies a uniformly regular refinement of the actual solid, then compares its trajectories with a supplied smooth continuum solution. It does not assume that consistency on smooth test functions controls arbitrary finite-element errors.

Proposition 3.12 (A uniform conforming cone refinement)**.** *There is a nested conforming refinement $\mathcal{T}_n$, $n = 2^k$, of the twenty-tetrahedron cone with $20n^3$ cells. It respects every macro-face and has*

$$\frac{1}{2n} < \operatorname{diam} K < \frac{6}{n}, \qquad r_K > \frac{1}{12n}, \qquad \frac{\operatorname{diam} K}{r_K} < 72 \quad (K \in \mathcal{T}_n).$$

Here r_K is the inradius. The constants are independent of n.

Proof. Give the macro-vertices one global order. On an ordered tetrahedron (v_0, v_1, v_2, v_3), use cumulative barycentric coordinates

$$x = v_0 + By, \qquad B = (v_1 - v_0,\ v_2 - v_1,\ v_3 - v_2), \qquad 0 \le y_3 \le y_2 \le y_1 \le 1.$$

Intersect this ordered simplex with the Kuhn triangulation of the $1/n$ cubical lattice. In a cube based at b/n, its tetrahedra have vertices

$$\frac{b}{n}, \quad \frac{b + e_{\pi(1)}}{n}, \quad \frac{b + e_{\pi(1)} + e_{\pi(2)}}{n}, \quad \frac{b + (1,1,1)}{n}, \qquad \pi \in S_3.$$

Retain those contained in the ordered simplex. The hyperplanes $y_i = j/n$ and $y_i - y_l = j/n$ describe this triangulation. They include the ordered-simplex boundary and subdivide its interior into disjoint open tetrahedra. Their volume is $1/(6n^3)$, giving n^3 cells per macro-tetrahedron. The hyperplanes at scale n occur at scale $2n$, so these refinements are nested. On a macro-face a cumulative coordinate becomes an endpoint or two adjacent coordinates agree. The induced arrangement is precisely the lower-dimensional construction in the inherited vertex order. Thus both incident macro-cells give the same face subdivision. This is the standard edgewise construction [37]; the coordinate description makes its compatibility explicit here.

Every cell is a translate of $BP\widehat{K}/n$, where P is a coordinate permutation and $\widehat{K}$ is one fixed Kuhn tetrahedron. Its diameter is $\sqrt{3}$ and inradius is $1/(2+2\sqrt{2})$. The centre is the first vertex of every ordered macro-cell. With $\varphi = (1+\sqrt{5})/2$, its matrix satisfies $\|B\|_F^2 = \varphi + 10 < 12$ and $|\det B| = 3+\sqrt{5} > 5$. Hence $\sigma_{\max}(B) < \sqrt{12}$ and $\sigma_{\min}(B) > 5/12$: use $|\det B| \le \sigma_{\min}(B)\sigma_{\max}(B)^2$. Linear maps bound diameters above and below by these singular values, and map an inscribed ball to an ellipsoid containing a ball of radius $\sigma_{\min}(B)r_{\widehat{K}}$. The displayed estimates follow. ☐

Lemma 3.13 (The real sector is invariant under the full action)**.** *On any conforming tetrahedral refinement, set $a = \phi = 0$ and take real scalar coefficients and real velocities. Solutions of*

$$(\ddot{u}_n, v_n) + (\nabla u_n, \nabla v_n) + m^2(u_n, v_n) + g(u_n^3, v_n) = 0 \quad \text{for all } v_n \in V_n \tag{11}$$

solve all equations of the original charged action, for any real e. Here V_n is the continuous piecewise-affine real space on $\mathcal{T}_n$, including its boundary degrees of freedom. All Gauss constraints and electromagnetic fields vanish exactly.

Proof. At $a = 0$ the dressed interpolation is exactly $u_n = W_0 u$. Its variation in any real edge direction is $ie\sum_i \lambda_i \beta_i u_i$, which is purely imaginary. Its time and spatial derivatives are also purely imaginary, including the terms containing the edge-direction velocity and the real scalar velocity. The additional term $-ieB_n u_n$ in the covariant spatial variation is imaginary. Every scalar kinetic, gradient and potential first variation pairs these terms with real fields and takes the real part, giving zero. Variations in imaginary scalar coefficients and in the scalar potential vanish for the same reason. Maxwell terms vanish because both field strengths are zero. Real scalar variations give the displayed wave equation after cancelling the common factor two. This argument tests all variations of the full action; it does not drop the dressing derivative. Uniqueness of the finite classical equations then makes this sector invariant. Its local charge and current vanish, even when $g > 0$ gives nonlinear scalar motion. ☐

Let Ω be the fixed solid and let $a_1(v, w) = (\nabla v, \nabla w) + (v, w)$. Define the H^1 Ritz projection by

$$a_1(R_n v, v_n) = a_1(v, v_n) \qquad (v_n \in V_n). \tag{12}$$

The added mass term controls the constant mode; no boundary value is fixed by this definition. The uniform mesh gives

$$\|R_n v - v\|_{H^1} \le Cn^{-1}\|v\|_{H^2}, \qquad \|R_n v\|_{H^1} \le \|v\|_{H^1}. \tag{13}$$

Indeed, orthogonal projection is a contraction in a_1, and its error is no larger than the piecewise-affine interpolation error. The latter follows by scaling the fixed reference tetrahedra and the usual H^2-to-H^1 affine interpolation estimate. In three dimensions H^2 has continuous representatives, so its nodal values are defined. No L^2 duality estimate or elliptic regularity assertion is needed.

Theorem 3.14 (Conditional nonlinear matter trajectory convergence)**.** *Fix $T < \infty$, $m^2 > 0$ and $g \ge 0$. Suppose a real continuum solution satisfies*

$$u_{tt} - \Delta u + m^2 u + gu^3 = 0, \qquad \partial_\nu u = 0 \text{ on } \partial\Omega, \qquad u \in C^2([0,T]; H^2(\Omega)). \tag{14}$$

The equation and boundary condition may equivalently be stated weakly against every $H^1(\Omega)$ test function. Let u_n solve the finite equation with $u_n(0) = R_n u(0)$ and $\dot{u}_n(0) = R_n u_t(0)$. Then

$$\sup_{0 \le t \le T} (\|u_n(t) - u(t)\|_{H^1} + \|\dot{u}_n(t) - u_t(t)\|_{L^2}) \le C_T n^{-1}. \tag{15}$$

The constant depends on the fixed domain, couplings, time window and displayed reference norms, but not on the refinement. These are trajectories of the full charged action in its invariant real sector. The theorem assumes the indicated smooth continuum reference exists.

Proof. The finite equation conserves

$$E_n = \tfrac{1}{2}\|\dot{u}_n\|_2^2 + \tfrac{1}{2}\|\nabla u_n\|_2^2 + \tfrac{m^2}{2}\|u_n\|_2^2 + \tfrac{g}{4}\|u_n\|_4^4.$$

Its initial value is uniformly bounded: use Ritz contraction and the fixed-domain embeddings $H^1 \hookrightarrow L^4, L^6$. Positive m^2 therefore bounds $\|u_n\|_{H^1}$ and $\|\dot{u}_n\|_2$ uniformly on the whole interval. For each finite mesh the positive mass matrix and this bound also give global classical existence of the polynomial ordinary differential equation.

Put $\eta = R_n u - u$, $\theta = u_n - R_n u$. The time-independent bounded projection commutes with both time derivatives. Equation (13) bounds η, η_t, η_{tt} in H^1 by C/n. Subtracting the full continuum weak equation gives, for every $v_n \in V_n$,

$$(\ddot{\theta}, v_n) + a_1(\theta, v_n) = -(\eta_{tt}, v_n) - (m^2 - 1)(\theta + \eta, v_n) - g(u_n^3 - u^3, v_n).$$

The Ritz identity cancelled the spatial derivative of η; there is no inverse estimate on $\dot{\theta}$. The cubic factorization and Hölder's inequality imply

$$\|v^3 - w^3\|_2 \le C(\|v\|_{H^1}^2 + \|w\|_{H^1}^2)\|v - w\|_{H^1}.$$

Apply this with $v = u_n, w = u$, using the uniform bounds above. Testing with $\dot{\theta}$ and writing $\mathcal{E} = \frac{1}{2}\|\dot{\theta}\|_2^2 + \frac{1}{2}\|\theta\|_{H^1}^2$ yields $\dot{\mathcal{E}} \le C\mathcal{E} + Cn^{-2}$. The chosen initialization gives $\mathcal{E}(0) = 0$. Gronwall's inequality and the projection estimates prove the claim. For other uniformly energy-bounded initializations, the same proof adds $C_T(\|u_n(0) - R_n u(0)\|_{H^1} + \|\dot{u}_n(0) - R_n u_t(0)\|_2)$ to the right-hand side. □

The smooth-reference class has nonzero localized examples. Choose nonzero C_c^∞ real data in a ball strictly inside the cone and first solve on $\mathbb{R}^3$. The standard local semilinear Klein–Gordon theorem and finite propagation speed [38] give a smooth solution on a positive interval. At the regularity needed here, local existence also follows directly from the Klein–Gordon Duhamel map in $CH^4 \cap C^1 H^3$, since the cubic is locally Lipschitz from H^4 to H^3; the equation gives $u_{tt} \in CH^2$. Take the time window shorter than both this existence interval and the distance of the initial support to the cone boundary. Finite propagation then leaves an open zero-field collar at that boundary, so restriction to Ω satisfies the stated homogeneous Neumann condition. This uses standard local continuum PDE input, not a source-derived physical state.

This result supplies an actual nonlinear spatial trajectory comparison, with zero electromagnetic current. Electrically active complex matter requires a further stability argument for its field-dependent kinetic metric and constraints. The present theorem supplies neither that argument nor physical geometry, a calibrated clock or a physical matter species.

`code/electromagnetism/whitney_real_continuum.py` constructs the ordered refinements, checks a quadratic Ritz comparator with all boundary variations free, and integrates a compact real pulse under the autonomous finite quartic-wave equation. Its exact polynomial cell integrals are independently reconstructed by the verifier. Finite mesh and numerical time-step comparisons test the implementation; the uniform geometric and continuum error statements are the analytic proofs above, not numerical error certificates or Lean formalizations.

Complex matter in a fixed magnetic background. The dressed scalar action also admits a controlled complex continuum limit with a nonzero magnetic field. Here the electromagnetic potential is prescribed and held fixed in the variation. This is a matter equation in an external field, rather than an invariant sector of the complete matter–Maxwell equations.

On the fixed solid Ω, take

$$A(x) = c + \tfrac{1}{2} B \times x, \qquad \phi = 0, \qquad D_A = \nabla - ieA,$$

where $c, B \in \mathbb{R}^3$ and $e \in \mathbb{R}$ are fixed. Use the conforming uniform meshes $\mathcal{T}_n$, $n = 2^k$, of Proposition 3.12. For every oriented edge set $a_{ij} = \int_{v_i}^{v_j} A \cdot dx$. On a tetrahedron define

$$b_i^A(x) = \lambda_i(x) e^{ie\theta_i(x)}, \qquad \theta_i(x) = \sum_j a_{ij}\lambda_j(x), \qquad J_n^A u = \sum_i b_i^A u(v_i).$$

Let V_n^A be the complex space obtained from these functions with one coefficient at each global vertex, including boundary vertices.

Lemma 3.15 (Conforming magnetic approximation)**.** *The Whitney interpolant of A equals A on every cell, and $\theta_i = \int_{v_i}^{x} A \cdot dx$. The dressed space V_n^A is a subspace of $H^1(\Omega;\mathbb{C})$. There is a constant independent of n such that*

$$\|J_n^A u - u\|_{H^1} \le C n^{-1} \|u\|_{H^2} \qquad (u \in H^2(\Omega;\mathbb{C})). \tag{16}$$

Proof. Write $A(x) = c + Lx$, where $L^{\mathsf{T}} = -L$. The local Whitney vector space is exactly $\{d + Kx : K^{\mathsf{T}} = -K\}$. To check this directly, $\lambda_i \nabla\lambda_j - \lambda_j \nabla\lambda_i$ has skew linear part, and its six oriented edge integrals give the identity matrix. Thus the six-dimensional space is unisolvent for those integrals, and reproduces A. This is the lowest-order edge space in the usual Whitney construction [41]. Skewness gives $\int_{v_i}^{x} A \cdot dx = A(v_i) \cdot (x - v_i)$, an affine function of x, whose vertex values are a_{ij}. On a shared face, the barycentric functions for vertices off that face vanish. The remaining phases use the same straight paths within the face, and their traces agree. The resulting piecewise smooth functions are continuous and therefore belong to H^1. Their vertex values are the coefficients, so the global basis is linearly independent.

For the approximation bound, fix a cell K of diameter h and a vertex x_0. Put $A_0 = A(x_0)$, $\chi(x) = A_0 \cdot x$, and $w = e^{-ie\chi} u$. Then on that cell

$$J_n^A u = e^{ie\chi(x)} \sum_i \lambda_i(x) e^{ie\delta_i(x)} w(v_i), \qquad \delta_i = \int_{v_i}^{x} (A - A_0) \cdot dx.$$

Uniform shape regularity gives $\|\delta_i\|_\infty \le Ch^2$, $\|\nabla\delta_i\|_\infty \le Ch$, and $\|\nabla\lambda_i\| \le C/h$. Consequently, with $r = \sum_i \lambda_i (e^{ie\delta_i} - 1) w(v_i)$,

$$\|r\|_{L^2(K)} \le Ch^2 |K|^{1/2} \Big(\sum_i |w(v_i)|^2\Big)^{1/2}, \quad \|\nabla r\|_{L^2(K)} \le Ch |K|^{1/2} \Big(\sum_i |w(v_i)|^2\Big)^{1/2}.$$

The constants may absorb the fixed upper bound on cell diameters. Point evaluation on the reference tetrahedron is bounded on H^2 in three dimensions. Scaling therefore bounds the last weighted nodal norm by $C(\|w\|_{L^2(K)} + h\|\nabla w\|_{L^2(K)} + h^2 \|D^2 w\|_{L^2(K)})$. The ordinary affine interpolation bound for $I_0 w - w$, together with the estimate for r, gives $\|J_n^A u - u\|_{H^1(K)} \le Ch\|u\|_{H^2(K)}$. Multiplication by $e^{\pm ie\chi}$ has uniformly bounded H^2 operator norm, since A_0 remains bounded on the fixed solid. Sum the squared local estimates and use $h < 6/n$. No point evaluation on arbitrary H^1 data is used. $\square$

For $m^2 > 0$, use the Hermitian form (linear in its first entry)

$$a_A(v,w) = \int_\Omega D_A v \cdot \overline{D_A w} + m^2 v\overline{w}\,dx.$$

Its norm is equivalent to H^1, with constants depending only on $m^2, e, \|A\|_\infty$. Indeed, $\|\nabla v\|_2 \leq \|D_A v\|_2 + |e|\|A\|_\infty\|v\|_2$, and the reverse inequality follows by the same triangle inequality. Define the complex magnetic Ritz projection by

$$a_A(R_n^A u, v_n) = a_A(u, v_n) \quad (v_n \in V_n^A).$$

Orthogonal projection, norm equivalence and (16) imply

$$\|R_n^A u\|_{H^1} \leq C\|u\|_{H^1}, \qquad \|R_n^A u - u\|_{H^1} \leq Cn^{-1}\|u\|_{H^2}. \tag{17}$$

Theorem 3.16 (Complex magnetic matter trajectory convergence)**.** *Fix the preceding background, $m^2 > 0$, $g \geq 0$, and a finite time interval $[0,T]$. Suppose a complex reference $u \in C^2([0,T]; H^2(\Omega))$ satisfies*

$$(u_{tt}, v) + a_A(u,v) + g(|u|^2 u, v) = 0 \quad (v \in H^1(\Omega;\mathbb{C})). \tag{18}$$

Here $(v,w) = \int v\overline{w}$. In strong notation this is $u_{tt} + D_A^* D_A u + m^2 u + g|u|^2 u = 0$, with magnetic Neumann condition $\nu \cdot D_A u = 0$. Let $u_n \in V_n^A$ satisfy the same equation against all $v_n \in V_n^A$, with $u_n(0) = R_n^A u(0)$, $\dot{u}_n(0) = R_n^A u_t(0)$. Then the finite solution exists for every time and

$$\sup_{0\leq t\leq T}(\|u_n - u\|_{H^1} + \|\dot{u}_n - u_t\|_{L^2}) \leq C_T n^{-1}. \tag{19}$$

The constant is independent of refinement. These are precisely the scalar Euler equations of the original dressed action with A held fixed, $\phi = 0$, and exact spatial integration.

Proof. The fixed basis has a positive definite mass matrix and no phase velocity. Taking the real part of the equation tested with $\dot{u}_n$ shows conservation of

$$E_n = \tfrac{1}{2}\|\dot{u}_n\|_2^2 + \tfrac{1}{2}a_A(u_n,u_n) + \tfrac{g}{4}\|u_n\|_4^4.$$

The initial energies are uniformly bounded by Ritz stability and the fixed-domain embedding $H^1 \hookrightarrow L^4$. Coercivity yields uniform H^1 field and L^2 velocity bounds. At each fixed mesh these also prevent finite-time escape of the finite polynomial ordinary differential equation, proving global existence.

Put $\eta = R_n^A u - u$, $\theta = u_n - R_n^A u$. The projection is time independent and bounded; it commutes with both time derivatives. Thus η, η_t, η_{tt} are $O(n^{-1})$ in H^1, uniformly in time. Ritz orthogonality gives the exact error equation

$$(\theta_{tt}, v_n) + a_A(\theta, v_n) = -(\eta_{tt}, v_n) - g(|u_n|^2 u_n - |u|^2 u, v_n).$$

For complex numbers the inequality $\big||z|^2 z - |w|^2 w\big| \leq 2(|z|^2 + |w|^2)|z-w|$, followed by Hölder and $H^1 \hookrightarrow L^6$, yields

$$\||v|^2 v - |w|^2 w\|_2 \leq C(\|v\|_{H^1}^2 + \|w\|_{H^1}^2)\|v - w\|_{H^1}.$$

All norms in its coefficient are uniformly bounded by the energy and the reference hypothesis. Take $v_n = \dot{\theta}$, take real parts, and set $\mathcal{E} = \frac{1}{2}\|\dot{\theta}\|_2^2 + \frac{1}{2}a_A(\theta,\theta)$. Coercivity and Young's inequality give $\dot{\mathcal{E}} \leq C\mathcal{E} + Cn^{-2}$. The initial error energy is zero. Gronwall and the projection estimates prove the bound. Uniformly energy-bounded alternative initializations add their initial H^1 field and L^2 velocity distances from the Ritz data, multiplied by C_T. This proof uses no inverse inequality or mesh-dependent stability constant. $\square$

The reference hypothesis admits localized complex examples. Extend the background smoothly with bounded derivatives outside a neighborhood of the solid, without changing it there. In the magnetic wave equation the additional connection and potential terms have at most one spatial derivative. The usual local wave Duhamel contraction in $CH^4 \cap C^1H^3$, with forcing in CH^3, therefore applies to smooth compactly supported complex data. The cubic is locally Lipschitz in this space. The equation gives $u_{tt} \in CH^2$. The standard local wave energy argument gives finite propagation speed one, since the principal part is unchanged [38]. For a short interval before the support reaches the boundary, restriction to the solid has a zero-field boundary collar and obeys the magnetic Neumann condition. For example, $u(0) = f$, $u_t(0) = i\omega f$ with a nonzero real compact bump and $e\omega \neq 0$ has $\mathrm{Im}(\overline{u}u_t) = \omega f^2 \neq 0$. The matter charge need not vanish.

Time independence makes the magnetic mass matrix fixed, and exact Whitney reproduction removes a potential-consistency defect. Those two facts are the essential restrictions of this result. In particular, such charged data generally violate the Gauss equation for $E = 0$; the prescribed field is not varied, and its sources and backreaction are not derived. The theorem does not establish the self-consistent charged continuum or select a physical background, preparation or laboratory clock.

`code/electromagnetism/whitney_magnetic_continuum.py` evaluates nonzero-curvature complex Ritz errors on the same refinements and evolves a finite nonlinear complex pulse. The independent verifier reconstructs the mesh, phases and covariant derivatives using a different quadrature and checks an independent time integration. Gauge changes, shared-face traces and omitted phase derivatives supply falsifying controls. These are finite numerical comparisons with declared tolerances; (19) is an analytic paper proof, not an interval simulation certificate or a Lean formalization. The three pointwise norm estimates in `Lean/Screen/MagneticMatterStability.lean` are Lean-checked; they include the stronger cubic constant 3/2. The Sobolev, Ritz and continuum trajectory arguments above remain analytic proofs.

A Hilbert realization of the coupled finite action. The charged action also admits an interacting quantum Hamiltonian once a quantization prescription is supplied. The kinetic energy depends on the matter field and its gauge-dependent interpolation. Its reduction therefore uses that kinetic metric, rather than the Maxwell mass matrix alone. The construction retains the fixed cone, its unwrapped real edge coefficients, the autonomous scalar action with $m^2, g \geq 0$, and $e \neq 0$.

Write $y = (a, \psi) \in \mathcal{Y} = \mathbb{R}^{42} \times \mathbb{C}^{13}$, regarded as a real manifold. Let $W(a)\psi = \Psi_a$, $S(a, \psi)b = \partial_a(W(a)\psi)[b]$, and define

$$G_y(v, v) = v_a^{\mathsf{T}} M v_a + 2\|W(a)v_\psi + S(a, \psi)v_a\|^2_{L^2(\Omega)}, \tag{20}$$

$$V(y) = \frac{1}{2}a^{\mathsf{T}} K a + \|D_x \Psi_a\|^2_{L^2} + m^2\|\Psi_a\|^2_{L^2} + \frac{g}{2}\|\Psi_a\|^4_{L^4}. \tag{21}$$

The factor two in G follows from the normalization of the scalar kinetic term. Nodal interpolation and $M > 0$ imply that G is a smooth positive metric, including at $\psi = 0$. The potential is smooth and nonnegative.

Theorem 3.17 (Gauge reduction and a neutral interacting Hilbert space)**.** *The mean-zero real gauge group has a global smooth quotient*

$$\mathcal{Q} = \ker(D^{\mathsf{T}} M) \times \mathbb{C}^{13} \cong \mathbb{R}^{30} \times \mathbb{C}^{13}.$$

Eliminating the twelve mean-zero scalar-potential coefficients from (4) *defines a smooth positive metric γ on $\mathcal{Q}$. In fixed Euclidean-orthonormal slice coordinates there are constants $c_*, C_* > 0$ such that*

$$\frac{c_*|w|^2}{1 + |\psi|^2} \leq \gamma_y(w, w) \leq C_*(1 + |\psi|^2)|w|^2. \tag{22}$$

This metric is complete. With $J_y = (0, i\psi)$ and c the remaining constant scalar potential, the reduced action is

$$L_{\mathrm{red}}(y, \dot{y}, c) = \frac{1}{2}\gamma_y(\dot{y} + ecJ_y, \dot{y} + ecJ_y) - V(y). \tag{23}$$

Its Hamiltonian at $c = 0$ is $H_{\mathrm{cl}}(y,p) = \frac{1}{2}\gamma_y^{-1}(p,p) + V(y)$, with the remaining Gauss constraint $p(J_y) = 0$.

Supply $\hbar > 0$, the Riemannian volume $d\mu = \mathrm{dvol}_\gamma$, and Laplace–Beltrami quantization. Then $\mathcal{H} = L^2(\mathcal{Q}, d\mu)$ has a nonnegative self-adjoint Hamiltonian $\widehat{H}$: the operator $-\hbar^2\Delta_\gamma/2 + V$ on $C_c^\infty(\mathcal{Q})$ is essentially self-adjoint, and its unique self-adjoint extension is its Friedrichs realization. The closed, nonzero subspace

$$\mathcal{H}_0 = \{f \in \mathcal{H} : f(a, e^{i\alpha}\psi) = f(a,\psi) \text{ in } L^2 \text{ for every } \alpha \in \mathbb{R}\} \tag{24}$$

reduces $\widehat{H}$. Its restriction $\widehat{H}_0$ is self-adjoint on $\mathcal{D}(\widehat{H}) \cap \mathcal{H}_0$ and generates the strongly continuous unitary evolution $\exp(-it\widehat{H}_0/\hbar)$. Thus a single interacting finite action supplies both a classical constrained Hamiltonian and, under the declared prescription, a compatible neutral quantum state space.

Proof. Let $\mathfrak{g}_0 = \{\xi \in \mathbb{R}^{13} : \sum_i \xi_i = 0\}$. Its action is $(a,\psi) \mapsto (a + D\xi, e^{ie\xi}\psi)$. Because the incidence matrix has kernel the constants, $D^\mathsf{T} M D$ is positive on $\mathfrak{g}_0$. Every orbit therefore meets $\mathcal{Q}$ exactly once: solve $D^\mathsf{T} M D\xi = -D^\mathsf{T} M a$ in $\mathfrak{g}_0$. This solution depends smoothly on a. Together with the inverse gauge action it gives a global product description $\mathcal{Y} \cong \mathcal{Q} \times \mathfrak{g}_0$, so no singular quotient is needed for these twelve gauge directions.

For a gauge velocity define $R_y\xi = (D\xi, ie\xi \odot \psi)$. Differentiating the exact interpolation gauge identity gives

$$S(a,\psi)D\xi + W(a)(ie\xi \odot \psi) = ie(W_0\xi)\Psi_a. \tag{25}$$

Consequently the full Lagrangian is exactly $L = G_y(\dot{y} + R_y\phi, \dot{y} + R_y\phi)/2 - V(y)$. This identity retains the derivative of the dressing; discarding it would change both the scalar-potential equation and the reduced metric. Time-independent gauge transformations are isometries of G and preserve V, by the same pointwise unit-phase covariance.

Choose any basis of $\mathfrak{g}_0$. At $y \in \mathcal{Q}$ let R denote its vertical matrix and put $I_y = R^\mathsf{T} G_y R$. The matrix I_y is positive: the Maxwell contribution gives $\xi^\mathsf{T} D^\mathsf{T} M D\xi > 0$ for nonzero $\xi \in \mathfrak{g}_0$. For $w \in T_y\mathcal{Q}$, set

$$\begin{aligned}\eta_*(w) &= -I_y^{-1}R^\mathsf{T} G_y w,\\ \gamma_y(w,w) &= G_y(w,w) - (R^\mathsf{T} G_y w)^\mathsf{T} I_y^{-1}(R^\mathsf{T} G_y w).\end{aligned}$$

Completing the square yields the exact identity

$$G_y(w + R\eta, w + R\eta) = \gamma_y(w,w) + (\eta - \eta_*)^\mathsf{T} I_y(\eta - \eta_*).$$

Thus γ is independent of the chosen vertical basis and smooth. If $\gamma_y(w,w) = 0$, positivity of G gives $w = -R\eta_*$. Since w is tangent to the Coulomb slice, $D^\mathsf{T} M D\eta_* = 0$, so $\eta_* = 0$ and $w = 0$.

The stronger bound uses the actual dressed basis. On a tetrahedron T, choose a node maximizing $|z_i|$. Where $\lambda_i \geq 3/4$, unit-modulus phases and the triangle inequality give $|W(a)z| \geq (2\lambda_i - 1)|z_i| \geq |z_i|/2$. This region has volume $|T|/64$, so

$$\|W(a)z\|^2_{L^2(T)} \geq \frac{|T|}{1024}|z_T|^2.$$

Summing over tetrahedra gives $\|W(a)z\|^2 \geq c_W|z|^2$, uniformly in all unwrapped a, with $c_W = \min_T |T|/1024 > 0$. The fixed path coefficients also give $\|W(a)\| \leq C$ and $\|S(a,\psi)\| \leq C|\psi|$. Let T_c and B be orthonormal bases of the Coulomb and mean-zero spaces, and write $w = (T_c b, z)$. For every η, the vertical identity and $T_c^{\mathsf{T}} MD = 0$ give

$$\begin{aligned} E_\eta &:= G(w + R_y(B\eta), w + R_y(B\eta)) \\ &= b^{\mathsf{T}} T_c^{\mathsf{T}} M T_c b + \eta^{\mathsf{T}} B^{\mathsf{T}} D^{\mathsf{T}} M D B\eta + 2\|F_\eta\|^2, \\ F_\eta &= Wz + ST_c b + ie(W_0 B\eta)\Psi_a. \end{aligned}$$

Both displayed Maxwell matrices are positive. The remaining two terms in $F_\eta - Wz$ have norm at most $C|\psi|(|b| + |\eta|)$. Consequently $|b|^2 + |z|^2 \leq C(1 + |\psi|^2)E_\eta$ for every η. Taking the minimum proves the lower bound in (22); taking $\eta = 0$ proves the upper bound. Constants concern this fixed mesh. In global coordinates q, the lower bound implies $\gamma \geq c_*(1 + |q|^2)^{-1} I$. An escaping curve therefore has length at least $\sqrt{c_*}$ times its radial arsinh $|q|$ variation and cannot have finite length. Smooth positivity on compact sets then proves completeness.

Write $\phi = \eta + c\mathbf{1}$, with $\eta \in \mathfrak{g}_0$. The twelve η equations are precisely the preceding minimization, with $w = \dot{y} + ecJ_y$. Substitution proves (23). Elimination is variationally valid because the eliminated derivative $\partial L/\partial\eta$ vanishes identically at its unique solution. The Legendre transform gives

$$p = \gamma_y(\dot{y} + ecJ_y, \cdot\,), \qquad H(y,p,c) = \frac{1}{2}\gamma_y^{-1}(p,p) + V(y) - ec\, p(J_y).$$

Variation of c, with $e \neq 0$, gives the remaining constraint. It is the constant component of the original thirteen Gauss equations, not an extra neutrality assumption added after quantization.

For the analytic construction use the Hermitian form, initially on $C_c^\infty(\mathcal{Q})$,

$$q(f,k) = \frac{\hbar^2}{2}\int_{\mathcal{Q}} \gamma^{-1}(d\overline{f}, dk)\, d\mu + \int_{\mathcal{Q}} V\overline{f}k\, d\mu. \tag{26}$$

Its domain is dense because μ has a smooth positive density in the global Euclidean coordinates. The form is nonnegative and closable. Indeed, if $f_n \to 0$ in L^2 and f_n is form-Cauchy, its differentials and $\sqrt{V} f_n$ have respective L^2 limits. Integration against compactly supported smooth vector fields, using the metric divergence, shows that the first limit is zero. Local boundedness of V shows that the second limit is zero on every compact set and hence globally. Therefore $q(f_n, f_n) \to 0$, proving closability.

Let $\overline{q}$ be its closure. The closed-form representation theorem gives a unique nonnegative self-adjoint operator for this particular closed form; this is the Friedrichs realization of the displayed differential operator. Explicitly, with the inner product linear in its second argument,

$$\mathcal{D}(\widehat{H}) = \{f \in \mathcal{D}(\overline{q}) : \exists b \in \mathcal{H}\ \forall k \in \mathcal{D}(\overline{q}),\ \overline{q}(k,f) = \langle k, b\rangle\}, \qquad \widehat{H}f = b.$$

The representation theorem and its symmetry version are standard functional analysis [40]; the reduction and coefficient hypotheses needed here have been verified above. This argument identifies the Friedrichs realization. Completeness strengthens this identification: apply the essential-self-adjointness theorem of Shubin [39, Theorem 1.1], with zero magnetic one-form and scalar potential $2V/\hbar^2$. The manifold and measure are smooth, the metric is complete, and this potential is smooth and nonnegative. Thus the minimal operator has exactly one self-adjoint extension.

The residual circle $T_\alpha(a,\psi) = (a, e^{i\alpha}\psi)$ preserves the slice, its vertical spaces, G, and V. It therefore preserves γ, μ, and q. The pullbacks $\mathcal{U}_\alpha f = f \circ T_{-\alpha}$ are unitary and strongly continuous, first on compactly supported smooth functions by dominated convergence and then on $\mathcal{H}$ by density. They preserve the closed form and its domain. Uniqueness in the displayed operator-domain characterization gives $\mathcal{U}_\alpha \widehat{H} = \widehat{H}\mathcal{U}_\alpha$ with domain preservation. Their bounded group average $P_0 = (2\pi)^{-1}\int_0^{2\pi} \mathcal{U}_\alpha \, d\alpha$ is the orthogonal projection onto $\mathcal{H}_0$ and commutes with the resolvent of $\widehat{H}$. Hence this subspace reduces the operator, and its restriction is self-adjoint. Smooth compactly supported functions of a and $\sum_i |\psi_i|^2$ give nonzero invariant vectors; the subspace is infinite-dimensional. The space $P_0 C_c^\infty(\mathcal{Q})$ is an operator core for the restriction: averaging preserves smoothness, its support lies in the compact circle-saturation of the original support, and averaging is contractive in the graph norm because it commutes with $\widehat{H}$. Apply P_0 to the compactly supported smooth graph approximants supplied by essential self-adjointness. It is also a form core. Spectral calculus supplies the stated strongly continuous unitary group. For initial states in $\mathcal{D}(\widehat{H}_0)$ it solves $i\hbar\partial_t f = \widehat{H}_0 f$ in the strong sense; for every Hilbert state it supplies norm-preserving evolution. □

With the convention $\mathcal{U}_{es} = \exp(-is\widehat{Q}/\hbar)$, the self-adjoint charge generator acts on compactly supported smooth functions as

$$\widehat{Q} = -i\hbar e \sum_i \left(u_i \frac{\partial}{\partial v_i} - v_i \frac{\partial}{\partial u_i} \right), \qquad \psi_i = u_i + iv_i.$$

Thus $\mathcal{H}_0 = \ker \widehat{Q}$ expresses the quantum constant Gauss constraint. It allows matter amplitudes and neutral correlations; it does not freeze the scalar field at zero. The point $\psi = 0$, where the circle has a stabilizer, remains an ordinary point of $\mathcal{Q}$. Taking circle-invariant Hilbert vectors avoids imposing a smooth structure on a singular full orbit space.

Proposition 3.18 (A normalized neutral initial state)**.** *In the orthonormal* 56*-dimensional slice coordinates write $d\mu = \rho(q)\, dq$, with $\rho = \sqrt{\det \gamma}$. For any supplied $\sigma > 0$, the function*

$$f_\sigma(q) = \rho(q)^{-1/2}(2\pi\sigma^2)^{-14} \exp\left(-\frac{|q|^2}{4\sigma^2} \right) \tag{27}$$

is a normalized vector in $\mathcal{D}(\widehat{H}_0)$. Its initial multiplication observables satisfy

$$\mathbb{E}\|\Psi_a\|_{L^2}^2 = \frac{4}{5}\sigma^2|\Omega|, \qquad \mathbb{E}\|\Psi_a\|_{L^4}^4 = \frac{48}{35}\sigma^4|\Omega|, \tag{28}$$

$$\mathbb{E}\frac{a^\mathsf{T} K a}{2} = \frac{\sigma^2}{2}\operatorname{tr}(T_c^\mathsf{T} K T_c). \tag{29}$$

Proof. The measure $|f_\sigma|^2 d\mu$ is exactly $N(0, \sigma^2 I_{56})$. The residual circle acts orthogonally and preserves ρ, so f_σ is normalized and neutral.

For the operator-domain condition, every derivative of W in a is uniformly bounded, and the corresponding derivatives of S are linear in ψ. The vertical Gram matrix has the fixed positive lower bound $B^\mathsf{T} D^\mathsf{T} M D B$. Differentiating its inverse and the Schur formula shows polynomial growth of all metric derivatives. The global metric bound gives polynomial growth of γ^{-1} and its derivatives; $\partial_j \log \rho = \frac{1}{2}\operatorname{tr}(\gamma^{-1}\partial_j\gamma)$ gives the same control for derivatives of $\log \rho$. The potential and its derivatives grow polynomially, with $V(q) \le C(1+|q|)^4$. Thus f_σ and the differential expression $(-\hbar^2\Delta_\gamma/2 + V)f_\sigma$ lie in $L^2(d\mu)$: after multiplication by $\sqrt{\rho}$, each differentiated term is bounded by a polynomial times a Gaussian. Integration by parts against C_c^∞ puts the state in the adjoint domain; essential self-adjointness identifies that domain with $\mathcal{D}(\widehat{H})$.

Conditional on a, independent circular Gaussian nodal coefficients give pointwise second and fourth moments $2\sigma^2 \sum_i \lambda_i^2$ and $8\sigma^4(\sum_i \lambda_i^2)^2$. The simplex averages of $\sum_i \lambda_i^2$ and $(\sum_i \lambda_i^2)^2$ are $2/5$ and $6/35$, respectively. This proves the matter identities; the magnetic identity follows from the transverse Gaussian covariance. □

This is a selected initial state, without a ground-state, physical-preparation, or computed-time-history assertion.

The metric coefficients and their Schur reduction are approximated by positive element quadrature in `code/electromagnetism/whitney_interacting_quantum.py`; independent tests check the actual cone, gauge covariance, scalar normalization, Gauss elimination, and the nondegenerate zero-matter limit. These finite checks compare exact simplex moments at $a = 0$ and quadrature orders four, five and six at a fixed nonzero configuration. The latter comparison measures quadrature sensitivity, not a certified integration error. The tests do not prove closability or self-adjointness; those are the analytic arguments above. The quantum prescription chooses a measure, operator ordering and $\hbar$; the extension is unique for that operator. The code evaluates Gaussian half-densities and exact moment formulae; its curved-measure amplitudes use the approximate metric density. No equivalence with quantization before reduction, unique quantization, quantum trajectory computation, continuum interacting field theory, physical-state selection, or Born-rule selection follows. The established attachment is to the same finite charged action and its full kinetic metric.

A trial quantum history with a global norm bound. The complete reduced metric also permits an explicit, error-controlled trial history in the full interacting Hilbert space. The bound below is deliberately conservative: its time interval is extremely short in the supplied dimensionless units. It does not furnish an ordinary-physics benchmark, a physical clock, or an exact computed Hamiltonian trajectory.

Retain the same fixed cone and set $e = 1/4$, $m^2 = 1/2$, $g = 1/4$, $\hbar = 1$, and $\sigma = 1/2$. Let H be the neutral Hamiltonian of Theorem 3.17 and $f = f_\sigma$ the normalized state of Proposition 3.18. All configuration coordinates remain present: $q = (a, \psi) \in \mathcal{Q} \cong \mathbb{R}^{30} \times \mathbb{C}^{13}$, with Euclidean-orthonormal coordinates on the Coulomb subspace. Write $X = |a|^2$, $Y = |\psi|^2$, and $\rho = \sqrt{\det \gamma}$.

Theorem 3.19 (Controlled potential-phase trial)**.** *The normalized neutral trial path*

$$v(t, q) = e^{-itV(q)/\hbar} f(q)$$

belongs to $\mathcal{D}(H)$ and is differentiable in the Hilbert norm. If $u(t) = e^{-itH/\hbar} f$, then

$$\|u(t) - v(t)\|^2_{L^2(\mathrm{dvol}_\gamma)} \leq \frac{(\overline{E} + \overline{K})|t| + \overline{B}|t|^3/6}{\hbar}, \tag{30}$$

where the following rational constants are admissible:

$$\overline{K} = 135829605095437, \qquad \overline{E} = \frac{2173273682146345}{16}, \qquad \overline{B} = \frac{24441816695556087}{4}.$$

In particular, $\|u(t) - v(t)\| \leq 1/10$ for every $|t| \leq 2^{-55}$. The trial configuration probability measure is exactly $N(0, \sigma^2 I_{56})$ at every time; its phase changes, while its configuration density stays fixed.

Proof. The argument separates a form comparison from explicit coefficient bounds. Define

$$K_0 = \frac{\hbar^2}{2} \int |df|^2_{\gamma^{-1}} \, d\mu, \qquad E_0 = K_0 + \int V|f|^2 \, d\mu, \qquad B_0 = \int |dV|^2_{\gamma^{-1}} |f|^2 \, d\mu.$$

The polynomial derivative bounds used for the initial state also apply to $e^{-itV/\hbar} f$ on bounded time intervals. Thus $v(t)$, its formal Hamiltonian image, and $Vv(t)$ lie in $L^2(d\mu)$. The adjoint-domain characterization and essential self-adjointness put $v(t)$ in $\mathcal{D}(H)$; differentiation under its Gaussian envelope gives $i\hbar\dot{v} = Vv$. Gauge invariance of V gives neutrality, and the phase has unit modulus.

Let $\mathfrak{t}$ denote the kinetic quadratic form. Since f is real and positive, $\mathfrak{t}(v(s), v(s)) = K_0 + s^2 B_0/2$. Positivity of V and conservation of the exact Hamiltonian energy give $\mathfrak{t}(u(s), u(s)) \le E_0$. Differentiating the overlap and using the form Cauchy inequality yields

$$\left|\frac{d}{ds}\|u(s) - v(s)\|^2\right| \le \frac{2}{\hbar}\sqrt{E_0}\sqrt{K_0 + s^2 B_0/2} \le \frac{E_0 + K_0 + s^2 B_0/2}{\hbar}.$$

Integration proves (30) for any valid upper bounds on these three constants. This comparison uses the full kinetic form and its Riemannian measure.

The canonical geometry has volume at most 18, tetrahedron volumes at least 5/6, squared edge lengths at most 4, and $|\nabla\lambda_i| \le 2$. Every node belongs to at least five tetrahedra and every edge to at least two. These inequalities are checked in $\mathbb{Q}(\sqrt{5})$ from the canonical vertices and incidence lists. They imply the useful uniform bounds

$$M \ge \frac{1}{81} I, \qquad W(a)^* W(a) \ge \frac{1}{128} I. \tag{31}$$

For the first, an affine vector field F on a tetrahedron satisfies $\int_T |F|^2 \ge |T| \sum_i |F(v_i)|^2/20$, while its six edge integrals obey $\sum_{ij} |a_{ij}|^2 \le 6 \sum_i |F(v_i)|^2$. Summing gives $M \ge 1/72$, which implies the displayed bound. For the second, choose a largest local nodal coefficient $|z_i|$. On $\lambda_i \ge 1/2$, $|Wz| \ge (2\lambda_i - 1)|z_i|$, independently of the phases, and

$$\frac{1}{|T|}\int_{\lambda_i \ge 1/2} (2\lambda_i - 1)^2\, dx = 3\int_{1/2}^{1} (2s-1)^2(1-s)^2\, ds = \frac{1}{80}.$$

Thus the summed lower bound is 5/384, which exceeds 1/128.

The radial edges give $|D\xi|^2 \ge |\xi|^2$ for mean-zero ξ: their contribution is $\sum_{i=1}^{12} (\xi_i - \xi_0)^2 = |\xi|^2 + 13\xi_0^2$. The mean-zero vertical Gram matrix I consequently satisfies $I \ge 1/81$. For a slice velocity (b, z) and mean-zero gauge velocity η, let $E_\eta = G(w + R\eta, w + R\eta)$. Maxwell orthogonality gives $|b|^2 + |\eta|^2 \le 81 E_\eta$. The vertical interpolation identity and $\|W\| \le \sqrt{18}$, $\|S\| \le \sqrt{18}|\psi|/4$ give $\|Wz\|^2 \le (1 + 729Y/2)E_\eta$. Applying (31) and minimizing over η proves

$$\gamma^{-1} \le A(Y) I, \qquad A(Y) = 209 + 46656Y. \tag{32}$$

The density derivative can be bounded without differentiating a matrix inverse twice. The combined slice and vertical tangent matrix has constant determinant: after permuting columns its edge block is the fixed matrix $[T_c, DB]$ and its scalar diagonal block is the identity. Schur factorization therefore gives $\log\rho = \text{constant} + \frac{1}{2}\log\det G - \frac{1}{2}\log\det I$. Put $J = (S, W)$. For any full velocity $w = (w_a, w_\psi)$, (31) implies

$$|w_a| \le 9\sqrt{G(w,w)}, \qquad |w_\psi| \le (8 + 108|\psi|)\sqrt{G(w,w)}.$$

Each real phase-path row has Euclidean norm at most one. Differentiating J and using $2\|Jw\|^2 \le G(w, w)$ yields, for a configuration direction h,

$$|dG[h](w,w)| \le 3(17 + 441|\psi|/4)|h| G(w,w).$$

For $\chi = W_0 B$, the identity $I = I_0 + 2e^2 \int \chi^{\mathsf{T}}\chi |\Psi_a|^2$ similarly gives $|dI[h](\eta,\eta)| \le 27(1 + |\psi|/4)|h| I(\eta,\eta)$. Taking traces in dimensions 68 and 12 proves

$$|\nabla \log\rho| \le 1896 + 11286\sqrt{Y}. \tag{33}$$

All gradients here are in the orthonormal slice coordinates; bounds for unit full-configuration directions restrict to those coordinates.

For completeness, the remaining potential bounds follow directly from the same geometry. Whitney curl has squared operator bound 384 on each cell. The pointwise estimates

$$|\Psi_a| \le |\psi|, \qquad |D_x\Psi_a| \le (4+9|a|/4)|\psi|,$$
$$|d(D_x\Psi_a)[b,z]| \le (4+9|a|/4)|z| + (13/4+9|a|/16)|\psi||b|$$

retain the derivative of the gauge-dependent dressing. The phase-gradient bound uses $|\nabla p_i\cdot a| \le 4|a|$ and $|\mathbf{A}(a)| \le 5|a|$. Squaring and integrating gives

$$\begin{aligned} V \le V_*(X,Y) &:= 3456X + 585Y + \frac{729}{4}XY + \frac{9}{4}Y^2, \\ |\nabla V|^2 \le P(X,Y) &:= 4\cdot 6912^2 X + 2313^2 Y^2 + 4\cdot 1152^2 X^2Y^2 + \frac{81}{4}Y^4 \\ &\qquad + 3\cdot 1170^2 Y + 3\cdot 1152^2 X^2 Y + 243Y^3. \end{aligned}$$

For example, the vector and scalar gradient norms are bounded by $6912\sqrt{X}+(2313/2)Y+1152XY+(9/4)Y^2$ and $1170\sqrt{Y}+1152X\sqrt{Y}+9Y^{3/2}$; Cauchy's inequality for four and three summands gives the displayed P.

Under $|f|^2 d\mu$, the independent variables X, Y have laws $\sigma^2\chi^2_{30}$ and $\sigma^2\chi^2_{26}$. The Gaussian integration-by-parts recurrence gives exactly

$$\mathbb{E}[(\sigma^2\chi^2_d)^k] = \sigma^{2k}\prod_{j=0}^{k-1}(d+2j).$$

Since $df = -f(d\log\rho + q\cdot dq/\sigma^2)/2$, admissible constants are the finite polynomial expectations

$$\overline{K} = \frac{1}{4}\mathbb{E}\Big[A(Y)\{2\cdot 1896^2 + 2\cdot 11286^2 Y + 16(X+Y)\}\Big],$$
$$\overline{E} = \overline{K} + \mathbb{E}[V_*] = \overline{K} + \frac{619353}{16}, \qquad \overline{B} = \mathbb{E}[A(Y)P(X,Y)].$$

Exact rational evaluation gives the stated values. The right side of (30) is increasing in $|t|$; at 2^{-55} it is less than $0.007541 < 1/100$, whereas the preceding dyadic time 2^{-54} fails that target. This proves coverage of the entire stated interval, including negative times. □

The packet `code/electromagnetism/runtime/whitney_quantum_history_receipt.json` records five trial times and four configuration probes. Potential values and the relative phase angles at those probes are exact elements of $\mathbb{Q}(\sqrt{5})$; an independent verifier reconstructs their simplex integrals and all envelope constants. The sampled phases illustrate the trial function; the norm bound concerns the full Hilbert space. Global polynomial integration includes the entire Gaussian tail, with no Monte Carlo, quadrature, or floating-point allowance used to certify the error. A separate numerical evaluator retains the full metric density and marks its approximate coefficients explicitly. The certificate inherits the supplied action, geometry, quantum prescription, initial state and time units. Its tiny conservative horizon is not physical state preparation, authenticated observer activity, spatial convergence, or an interacting continuum quantum field theory.

Charged phase-space data and neutral Gaussian packets. The charged trajectory can supply packet parameters only after its positions and velocities are expressed in the Coulomb quotient. This is a preparation construction in the full interacting Hilbert space of Theorem 3.17. A family of prepared packets at classical sample times does not constitute quantum propagation.

Let B have orthonormal columns spanning the mean-zero gauge space, let T_c span $\ker(D^\top M)$ orthonormally, and put $L_0 = B^\top D^\top MDB$. For a differentiable full configuration $y(t) = (a(t), \psi(t))$, define

$$\begin{aligned} \xi &= -BL_0^{-1}B^\top D^\top Ma, & \dot\xi &= -BL_0^{-1}B^\top D^\top M\dot a, \\ a_C &= a + D\xi, & \psi_C &= e^{ie\xi}\psi, \\ \dot a_C &= \dot a + D\dot\xi, & \dot\psi_C &= e^{ie\xi}(\dot\psi + ie\dot\xi \odot \psi). \end{aligned} \tag{34}$$

Products and exponentials on nodal coefficients are componentwise. The quotient coordinates are $q = (T_c^\top a_C, \operatorname{Re}\psi_C, \operatorname{Im}\psi_C) \in \mathbb{R}^{56}$, with velocity $w = \dot q$. The unit phase generator is $J(a,u,v) = (0,-v,u)$; its parameter is the phase angle, without the charge factor e.

Proposition 3.20 (Coulomb phase-space preparation)**.** *For a temporal-gauge trajectory satisfying all thirteen Gauss equations, the transformed scalar potential is $\phi_C = -\dot\xi$, its constant component vanishes, and the Schur minimizer is exactly $-\dot\xi$. The reduced cotangent is $p = \gamma_q w$ and satisfies $p(Jq) = 0$. For the recorded symmetric sector, with radial coefficient α,*

$$\xi = \frac{\alpha}{13}(12, -1, \ldots, -1), \qquad a_C = 0. \tag{35}$$

At its exact initial datum, with $\mathcal{V} = |\Omega| = 10 + 10\sqrt{5}/3$,

$$q_a = p_a = 0, \quad \psi_C = \mathbf{1}, \quad p_{\operatorname{Re}\psi} = 0, \quad p_{\operatorname{Im}\psi} = \mathcal{V}(3/10, -1/40, \ldots, -1/40). \tag{36}$$

Proof. The gauge law for the temporal covariant derivative is $\phi \mapsto \phi - \dot\xi$; differentiating the nodal gauge law gives (34). The transformed velocity plus vertical scalar-potential term is the image of the original velocity under the time-independent gauge differential. Gauge covariance therefore preserves every scalar-potential equation. Positivity of its mean-zero Hessian makes $-\dot\xi$ the unique minimizer. Since $\sum \xi_i = 0$, the constant potential remains zero. The reduced Legendre formula and constant Gauss equation give the cotangent and its zero moment map. Without Gauss, the displayed rechart remains valid but the equality of its transformed potential and Schur minimizer need not hold.

The radial incidence equation is $\xi_j - \xi_0 = -\alpha$, and its mean-zero solution gives (35). Initially the edge dressing derivative vanishes because all nodal coefficients equal one. Write A_0 for the scalar nodal mass matrix. The transformed covariant velocity is the original $i(3,-1,\ldots,-1)$, so the scalar cotangent is $2A_0 i(3,-1,\ldots,-1)$. Simplex integration gives $(A_0)_{00} = \mathcal{V}/10$, $(A_0)_{0j} = \mathcal{V}/80$, and the boundary row sum $\mathcal{V}/16$. These identities prove (36). Maxwell orthogonality $T_c^\top MD = 0$ gives the zero edge cotangent. Its constant charge is $\mathcal{V}(3/10 - 12/40) = 0$. □

Supply a width $\sigma > 0$, momentum p, center q, and $\hbar > 0$. In Lebesgue measure on all 56 quotient coordinates, let the normalized seed be

$$g_{q,p,\sigma}(x) = (2\pi\sigma^2)^{-14} \exp\left(-\frac{|x-q|^2}{4\sigma^2} + \frac{i}{\hbar} p \cdot (x-q)\right). \tag{37}$$

Its position covariance is $\sigma^2 I_{56}$; no transverse direction has been removed. The unitary half-density map is $f \mapsto \sqrt{\rho} f$, where $\rho = \sqrt{\det\gamma}$. The circle acts orthogonally on the scalar coordinates and fixes the thirty edge coordinates, so its average P_0 has the same pullback expression in both measures.

Proposition 3.21 (A nonvanishing neutral projection)**.** *Write $q_s, p_s \in \mathbb{R}^{26}$ for the scalar parts and set*

$$A = \frac{|q_s|^2}{4\sigma^2} + \frac{\sigma^2|p_s|^2}{\hbar^2}, \qquad B_* = \frac{p(Jq)}{\hbar}. \tag{38}$$

Then $A \geq |B_*|$ and

$$n^2 := \|P_0 g_{q,p,\sigma}\|^2 = e^{-A} I_0\Big(\sqrt{A^2 - B_*^2}\Big) > 0. \tag{39}$$

Consequently $F = \rho^{-1/2}P_0 g/n$ is a normalized vector in $\mathcal{D}(\widehat{H}_0)$. At the exact initial datum and $\hbar = 1$, each width $\sigma \in \{1/4, 1/2, 1\}$ satisfies

$$B_* = 0, \quad A = \frac{13}{4\sigma^2} + \frac{39\sigma^2\mathcal{V}^2}{400} < 64, \qquad n^2 > \frac{1}{64}. \tag{40}$$

Proof. The Cauchy and arithmetic–geometric mean inequalities give $|B_*| \leq |p_s||q_s|/\hbar \leq A$. Completing the Gaussian square in $\langle g, \mathcal{U}_\theta g\rangle$ gives

$$\exp\big[-A(1-\cos\theta) - iB_*\sin\theta\big].$$

Its circle average is n^2. Expanding $\exp[(A-B_*)e^{i\theta}/2 + (A+B_*)e^{-i\theta}/2]$ and retaining equal powers gives $e^{-A}\sum_{k\geq 0}((A^2 - B_*^2)/4)^k/(k!)^2$, proving (39). This also defines I_0 and proves positivity, including the equality case $A = |B_*|$. For $B_* = 0$ it is the usual real circle-integral representation.

Polynomial coefficient and density-derivative bounds from Proposition 3.18 apply to every translated, modulated Gaussian. Thus $\rho^{-1/2}g$ belongs to $\mathcal{D}(\widehat{H})$. The bounded circle projection preserves that domain and commutes with the Hamiltonian. Division by $n > 0$ proves the domain claim. The same argument gives finite moments of every polynomial multiplication observable.

Initially $|q_s|^2 = 13$ and $|p_s|^2 = 39\mathcal{V}^2/400$. Since $\mathcal{V} < 18$, direct rational substitution at the three stated widths gives $A < 64$. On $|\theta| \leq 1/8$, $A(1-\cos\theta) \leq A\theta^2/2 < 1/2$. Hence the integral is larger than $e^{-1/2}/(8\pi) > 1/64$, using $e^{-1/2} > 1/2$ and $\pi < 4$. □

Projection changes the state: a nonzero center is a parameter for an orbit, not the scalar one-point expectation. In the Lebesgue half-density representation, circle invariance gives $\langle x_s\rangle = \langle -i\hbar\nabla_s\rangle = 0$. For $B_* = 0$, direct Gaussian integration gives the useful invariant nodal-radius observable

$$\langle |x_s|^2\rangle_F = 26\sigma^2 + \frac{|q_s|^2}{2}(1+r) - \frac{2\sigma^4|p_s|^2}{\hbar^2}(1-r), \qquad r = \frac{I_1(A)}{I_0(A)}. \tag{41}$$

Indeed, inserting $|x_s|^2$ in the overlap adds $26\sigma^2 + |q_s|^2(1+\cos\theta)/2 - 2\sigma^4|p_s|^2(1-\cos\theta)/\hbar^2$; the imaginary term is proportional to B_*. The cosine average is I_1/I_0. For arbitrary B_*, the same calculation gives $26\sigma^2 + |q_s|^2/2 - 2\sigma^4|p_s|^2/\hbar^2 + 2\sigma^2 z I_1(z)/I_0(z)$, where $z = \sqrt{A^2 - B_*^2}$; its last term is zero when $z = 0$. The thirty edge coordinates retain their seed covariance, independently of this scalar projection. This nodal observable is not a supplied laboratory measurement operator.

The residual required for propagation. The half-density change of measure does not flatten the Hamiltonian. Writing $a^{ij} = (\gamma^{-1})^{ij}$ and $\ell = \log\rho$, its Lebesgue expression is

$$\widetilde{H} = -\frac{\hbar^2}{2}\partial_i(a^{ij}\partial_j) + V + U_\rho, \qquad U_\rho = \frac{\hbar^2}{4}\partial_i(a^{ij}\partial_j\ell) + \frac{\hbar^2}{8}(\partial_i\ell)a^{ij}(\partial_j\ell). \tag{42}$$

This follows by differentiating $\rho^{-1/2}g$; the two first-order cross terms cancel by symmetry of a^{ij}. For any smooth seed path in the strong domain, let $r_g = i\hbar\dot{g} - \widetilde{H}g$. Where $n(t) = \|P_0 g(t)\| > 0$, the normalized projection has residual norm at most $2\|r_g(t)\|/n(t)$: differentiating the projected norm and using self-adjointness gives $|\dot{n}| \le \|P_0 r_g\|/\hbar$. Duhamel's formula therefore gives

$$\left\| e^{-it\widehat{H}_0/\hbar} F(0) - F(t) \right\| \le \frac{2}{\hbar} \int_0^{|t|} \frac{\|r_g(s)\|}{n(s)} \, ds \tag{43}$$

for forward time, and with the reversed path for negative time. The initial bound on n alone does not bound this integral. Continuous center and covariance defects, the full variable coefficients and U_ρ, their numerical errors, and all Gaussian tails must enter an actual residual certificate. The prepared samples supply none of those propagation estimates. In particular, no error bound on a $1/40$ interval at $\hbar = 1$ follows from the preparation or from a semiclassical limit.

The executable packet at `code/electromagnetism/runtime/whitney_quantum_packet_receipt.json` recharts two adjacent classical samples and checks all mean-zero multipliers and reduced cotangents. Its independent checker uses exact-degree simplex moments at $a_C = 0$, including the complete 68-coordinate kinetic metric and all 56 reduced directions. Matrix operations and the second classical sample are numerical, while the displayed initial cotangent and projection lower bound are analytic identities. The packet specifies three full-dimensional seeds and their neutral projections. It does not select a physical preparation, evolve their covariances, identify them with a computed quantum history, or establish interacting continuum field theory.

Primitive-port comparison. The coefficient manifold and its decision rule are frozen before a qualifying comparison [26]. The dated cosmic-microwave-background template search and every data product examined in it are an excluded exposure class; the linked coefficient manifold has no eligible physical comparison. If C_4 is negative at five or more standard deviations with enough sensitivity, exclusion of the linked B_0/B_6 terms or the rotated I_6 vector at five or more standard deviations, after the fixed $\mathrm{SO}(3)/A_5$ profile and with calibrated joint coverage, rejects the primitive-port physical branch. An isolated positive C_4 or nonzero intrinsic anisotropy at ranks one through five also rejects the branch at the same threshold and under the same calibrated coverage. A calibrated joint likelihood that excludes the complete branch manifold at five or more standard deviations also rejects it. Null, underpowered, incomplete-covariance, frame-indeterminate, polarization-split, and non-isolated results are inconclusive. Support requires exclusion of the zero-coefficient baseline at five standard deviations or more, agreement with the complete linked branch within two, rejection of the named systematic alternatives, and an independent eligible replication. Minimal locally Lorentz-invariant Standard Model physics with General Relativity gives the zero-coefficient baseline. Nonminimal effective operators can imitate the pattern. A physical test therefore requires the sector bridge, equal action on both transverse polarizations for a photon test, coherent carrier-frame transport, and a nuisance model that isolates the intrinsic coefficient vector. Derivation of these bridges from the repair law is a separate premise. Failure reaches OPH as a whole only if such a derivation proves the branch forced and exclusive.

Frozen source-seam propagation branch. The signed source-incidence map sends the complete thirty-seam system onto D_6, and its pullback response metric completes to the same three-dimensional carrier as the source loads. Under the named naturality, objective, and unique-minimizer premises, the sixty directed seam weights are $1/60$. The resulting internal operator is the exact homogeneous Dirichlet generator. With $q = ak$ and unit direction n, its normalized spatial character is

$$\widehat{\Lambda}(q, n) = \frac{1}{5} \sum_{j=1}^{30} \left[1 - \cos(q \, w_j \cdot n)\right].$$

Its coefficient ray is

$$C_4 = -\frac{a^2}{20}, \qquad B_0 = \frac{a^4}{840}, \qquad B_6 = -\frac{a^4}{12600},$$

so $B_0/C_4^2 = 10/21$, $B_6/C_4^2 = -2/63$, and $B_6/B_0 = -1/15$. Intrinsic anisotropies of ranks one through five vanish, while rank six has one icosahedral harmonic up to orientation.

Define

$$P_6(q,n) = q^2 - \frac{q^4}{20} + \left(\frac{1}{840} - \frac{I_6(n)}{12600}\right) q^6.$$

The exact eighth moment and alternating cosine bounds give, uniformly for $-5/9 \le I_6(n) \le 1$ and $0 \le q \le 1$,

$$\left|\widehat{\Lambda} - P_6\right| \le \frac{7}{388800} q^8, \qquad \frac{19}{20} q^2 \le \widehat{\Lambda} \le q^2.$$

This is a target-free bound for the finite spatial symbol. A separately declared basis-free oscillator completion has a two-dimensional momentum-orthogonal fiber at every nonzero momentum. The scalar symbol acts equally on both directions, and the declared generator gives

$$A_T'' + \Lambda_a A_T = 0, \qquad \omega_a^2 = \Lambda_a, \qquad \omega_a(0) = 0.$$

The corresponding quadratic energy has zero algebraic first variation on that generator. These are conditional oscillator statements. They do not construct a flow, gauge redundancy, Gauss law, Maxwell dynamics, a physical clock, a photon Hilbert space, or a massless physical photon.

An exact synthetic calibration enumerates all $2^{12} = 4096$ sign-error vectors in a fixed twelve-row design at noise scale $1/200$. It rejects a zero leading coefficient in every replica, and its nominal 95 percent interval covers the injected value in 3904 replicas. The linked higher-order pair is detected in 209 replicas; this stress law does not resolve it. No experimental response or sensitivity follows from this synthetic model.

The nonnegative auxiliary root of the complete positive cosine symbol obeys $|\Omega(k) - \Omega(p)| \le |k - p|$ at all momenta in the selected Euclidean carrier chart. The exact theorem uses the full symbol and no Taylor truncation. It supplies no physical position, scale, or detector map. If the same seam symbol is identified with physical photon frequency squared, then for $\mathbf{k} = kn$, $k \ge 0$, and $|n| = 1$, its complete cosine form obeys $0 \le \Omega_\gamma(k,n)^2 \le k^2$ in the Euclidean carrier metric. Ordinary additive energy-momentum conservation and positive-mass electrons and positrons with Lorentz-invariant positive-energy dispersion then exclude photon decay into an electron-positron pair and place the seam-current incoming-energy budget inside the Lorentz-invariant photon budget at fixed incoming momenta. With the soft background photon Lorentz invariant at leading order, use $E_i^2 = p_i^2 + m_i^2 + \delta_{i,2} E_i^4$, with $i \in \{\gamma, +, -\}$, $m_\gamma = 0$, and $m_+ = m_- = m_e$. Independent leading hard-photon, positron, and electron coefficients give a head-on, collinear threshold with fixed positron energy share $0 < x < 1$ that depends on

$$\delta_{\gamma,2} - x^3 \delta_{+,2} - (1-x)^3 \delta_{-,2}.$$

At equal sharing this is $\delta_{\gamma,2} - (\delta_{+,2} + \delta_{-,2})/8$, a rank-one readout with a two-dimensional coefficient fiber. On the Lorentz-invariant-lepton branch, equal sharing uniquely maximizes the leading head-on, collinear threshold residual. No general independent-lepton or full anisotropic optimization is supplied.

One universal subcase is nevertheless exact at the same retained order. If the hard photon, electron, and positron share the negative coefficient $\delta_{\gamma,2} = \delta_{+,2} = \delta_{-,2} = -d$, $d > 0$, and $u = x(1-x) \in (0, 1/4]$, the leading residual vanishes precisely when

$$\epsilon = \frac{m_e^2}{4Eu} + \frac{3dE^3}{4} u.$$

Lean proves the reciprocal-linear lower envelope, the equal-share endpoint regime $3dE^4 \le 16m_e^2$, the absence of a solution below the envelope, and attainment by an open physical share at and above the transition. The exact constrained minimum at this retained order is

$$\epsilon_{\min}(E) = \begin{cases} m_e^2/E + 3dE^3/16, & 3dE^4 \le 16m_e^2, \\ (\sqrt{3}/2)m_e E\sqrt{d}, & 3dE^4 \ge 16m_e^2. \end{cases}$$

The branches agree at equality; in the second branch the minimizer is $u_* = m_e/(\sqrt{3d}\,E^2)$. For the conditional scale $a = \ell_P$, $d = a^2/20$, representative CMB energies 6.34×10^{-4} and 3.0×10^{-3} eV give leading kinematic windows $[4.12 \times 10^{14}, 7.82 \times 10^{19}]$ eV and $[8.70 \times 10^{13}, 3.70 \times 10^{20}]$ eV, respectively [25].

These conditional kinematic statements supply no charged-lepton action, pair-production vertex, cross section, opacity, source model, shower response, or detector response.

The source-seam coefficient ray and decision rule are frozen before an eligible physical comparison [25]. Physical position, sector, frequency, clock, frame and boost law, wave-packet propagation, cosmological transport, detector readout, and nuisance isolation are premises. A branch-falsifying null additionally requires a positive lower bound on the physical scale for the same action and preregistered power to exclude the complete admitted manifold. A failed qualifying comparison rejects this physical propagation branch. It reaches OPH as a whole only if a separate theorem makes the branch forced and exclusive.

Evidence boundary. Two source-bound public-data replays and one deterministic solver replay sharpen the empirical boundary without earning confirmation credit. Hash-checked processing of the four official DESI DR2 chains in each default-CMB w_0w_aCDM combination gives posterior weights in the conditional nondecreasing-capacity subset on $0 \le z \le 2$ of 0.00705%, 0.1125%, 0.01206%, and 0.02054% for BAO+CMB, Pantheon+, Union3, and DESY5, with only 6, 126, 15, and 11 raw tail rows [27, 25]. These are seen-data posterior fractions under the collaboration likelihoods and priors, not branch probabilities or exclusions. The local electroweak-closure replay uses an outward-rounded implicit-function certificate in place of finite-difference stability. On $|\Delta \ln \alpha| \le 10^{-5}$, it proves the selected declared branch continuously differentiable and encloses $\mathrm{d}\ln N/\mathrm{d}\ln\alpha$ in $[-0.214173865, -0.206176031]$ and its reciprocal in $[-4.850224325, -4.669103775]$, after excluding zero. A cosmological reading requires (B1) identity of the two capacities, (B2) an epochwise physical closure law, and (B3) physical selection of that root branch and co-variation convention together with the homogeneous-α and clock time map. With $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, only B1–B3 together translate the optical-clock result into $|1 + w_0| \le 6.94 \times 10^{-10}$ [28, 25]; the replay therefore does not eliminate the nonconstant monotone branch.

A separate replay of the empirical full SPARC radial-acceleration interpolation authenticates the CDS/VizieR inputs and reproduces the published 153-galaxy parent selection. The strict point cuts retain 2696 points in 147 contributing galaxies, three more than the published census. At fixed disk and bulge mass-to-light values 0.5 and 0.7, the unweighted log-residual fit is $a_0 = (1.1613 \pm 0.0802) \times 10^{-10}\,\mathrm{m\,s^{-2}}$ with 0.1327 dex scatter, where the width is a galaxy-bootstrap standard deviation [29, 30, 25]. Equal-galaxy and velocity-error-only weighting move the fitted scale by −9.58% and +15.49%, exposing estimator sensitivity. The interpolation, scale, and dataset predate this replay; a_0 is fitted and the nuisance likelihood is incomplete. This is reproducible calibration, not OPH-specific evidence.

The Ward-projected hadronic transport gives $S_{\text{hadronic}} \in [0.5578, 1.0543]$ for the two electromagnetic endpoint residuals. This interval contains the zero-electroweak diagnostic 0.8954 and spans about 1.16×10^8 declared tolerance widths. Its evaluation pixel differs from the comparison pixel, so it is a comparison diagnostic rather than a physical prediction.

The fixed-cutoff direct correctable-public-record map has exact record restrictions, public reachability, publicness, global checkpoint coupling, correctable-code, carrier, extension, refinement, and sewing receipts. A bounded all-rung counterfamily has incompatible zero sets under shared base, positivity, carrier, and executable finite controls. The finite-rung complete-packet lift retains those incompatible zero sets on the six audited rungs. It does not prove that the executable candidates inhabit the complete source class at every rung or bind them to the all-rung Lean completion. A positive direct map requires completion of those bridges or an additional named source law, together with a universe-level carrier attachment. The de Sitter and electroweak interpretations also require horizon–record and common screen/electroweak load identifications. The interval at $3.5321315434 \times 10^{122}$ belongs to the conditional electroweak bridge. It does not construct the direct map F, and it differs from the Planck base-ΛCDM comparison coordinate by about 6.6 percent. The finite-presence and exponential reserve candidates give $3.2920978773\ldots \times 10^{122}$ and $3.3000722254\ldots \times 10^{122}$. They are separately typed and retrospective. Exact neutral and multiplicative completions of the same local survival datum obey the declared positive composition and regrouping laws but have different global effects. The finite source selects no action and no blocked-event semantics, so the named-law and horizon readings are not evaluable without a stronger source-derived global action.

The fixed 96-entry one-loop electroweak menu excludes its own entries. It says nothing about higher-order, matching, threshold, tadpole, scale, or field-content prescriptions outside that menu. Twelve fixed flavor candidates are likewise excluded without exhausting the broader selector family. These comparisons use known targets and therefore do not count as target-blind predictions.

Companion Papers

The detailed results used throughout this synthesis are:

1. *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems* [14], which gives the presentation-invariant same-source/cross-source quantifier split, endpoint and refinement bounds, collar-repair obstructions, and the finite conditional-resampling receipt used by the repair papers.

2. *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency* [15], which gives the exact port-record metric completion together with the support-incidence, modular, source-causal continuum, stress, generalized-entropy, and Einstein derivation chain. Every finite, continuum, scaling, and physical premise is stated at the arrow where it enters.

3. *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [16], which gives the categorical and finite-carrier routes. It proves the complete-response Lie-type theorem and gives the conditional matrix-current, matter, and tensor-kernel theorems. Identification of the two routes requires a physical current intertwiner.

4. *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [17], which carries the finite patch-net fixed-point, defect, quotient, and operator-record consensus construction. Its semantic-complete transaction theorem derives the local diamond from coherent aggregate gluing and full acceptance dependencies; repair completeness is a separate premise. Quantum error-correcting code/min-cut, spectral, Byzantine fault-tolerance, and hardware-speedup statements require their respective certificates.

5. *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [18], which carries the regulated federated patch-carrier architecture, the fixed-cutoff edge heat-kernel / Casimir theorem, exact finite modular and rate statements, the $A_5 \times \mathbb{Z}_2$ register classification, and the measurement and observer checkpoint/restoration packages.

6. *Deriving the Particle Zoo from Observer Consistency* [19], which carries the particle derivations, masses, couplings, and sector calculations.

7. *Observer-Patch Holography and the Dark Sector* [20], which studies modular charge on overlap collars under a physical stress attachment, with conditional recovery bounds, compact-source saturation and deep-galaxy scaling laws.

8. *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics* [21], which isolates the controlled compact-gauge repair mechanism and states the continuum certificate needed before the finite repair gap becomes a Clay-facing four-dimensional Yang–Mills gap.

9. *Observer-Patch Holography as a String-Vacuum Selector* [22], which treats string theory as an effective edge language and applies the OPH acceptance criteria to critical-string candidates; its rank certificate classifies the Bouchard-Donagi row as a structural benchmark rather than a selected witness.

The complete conditional spacetime and Einstein implication chain, with its physical and continuum premises stated at the arrows, is proved in Ref. [15]; the two gauge-reconstruction routes and their identification boundary are proved in Ref. [16]. The synthesis therefore concentrates on their shared observer interpretation, quantitative closures, and cross-branch claim boundary.

Physical scope of the icosahedral Standard Model and W/Z results

The finite icosahedral package contains an axiom-forced Lie-type theorem and an exact conditional recognition result. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence determines the antipode J. Under the explicit contract that an admissible response is a signed central involutive graph automorphism implementing inverse-port readback, the responses are exactly $\pm J$; their common sign is conventional. Conditional also on the matrix current and rank-15 matter contract with its unique charge-conjugate projector pair, the anomaly and tensor-descent certificates fix the $3+2$ block structure, hypercharge lattice, common $\mathbb{Z}_6$ kernel, and maximal faithful matter image. The kernel is computed on every declared tensor and is insensitive to the projector representative. The source does not select the matrix current, matter action, or physical global quotient. This finite calculation supplies no laboratory identification of its current or flux sectors and no physical seam action. Equality with the independently reconstructed Tannaka current, attachment of the band to three physical chiral families, exclusion of extra light sectors, four-dimensional topological attachment, scalar attachment and dynamics, and construction of a chiral quantum field theory are not supplied.

The conditional field-theory implications are explicit. A finite local action gives an exact finite gauge-invariance and locality theorem, and the familiar electroweak tree kernel is conditional on a separate canonical continuum and action-normalization bridge. An exact finite measure criterion and an exact finite Hamiltonian criterion are two parallel branches over that action. A separate formal perturbative branch carries the strict finite-order W/Z pole theorem. A nonperturbative continuum completion gives an observable-sector reconstruction implication and a distinct continued-sheet resonance-stability implication. The measure and perturbative branches are parallel descendants of the finite local action. Neither implies the other, and a perturbative pole does not imply the continuum completion.

These quantum field theory (QFT) theorems state what follows from typed action and quantization packets. They do not show that the target-free source emits those packets. The source-selected action and normalization, complete measure, target-clean perturbative matching, physical current amplitudes, source law and covariance, uncertainty enclosure, operational clock, and continuum tower with its continued-sheet packet are not supplied. The declared external Standard Model action and effective-field-theory interval provide bounded validation inputs for the perturbative protocol. Interval receipts exclude scalar zeros in the declared principal-sheet boxes and isolate, for each of W and Z, one simple scalar zero with derivative and scalar-residue balls in its declared lower-half pole box on a channel-specific algebraic chart. They identify neither chart with the physical resonance sheet and prove no unique continuation or self-energy sign bridge. Proof-bearing physical validation also requires two genuinely independent raw loop engines, counterterm and Becchi–Rouet–Stora–Tyutin generation from the complete action, Nielsen identities, an artifact-resolving third verifier, full-matrix contours and Laurent data, dressed-current amplitudes, and complete independent replay. The external packet is a validation input rather than an OPH source producer. No source-native dimensionless or physical-unit W/Z pole follows from it.

4 Consensus, Defects, and Implementation Hiding

The fixed-cutoff consensus package admits an equivalent finite patch-net formulation. Local patch descriptions agree on overlaps, local recovery-derived repair rules remove mismatches, and the physical output is the schedule-independent quotient normal form together with the induced terminal expectation functionals on the declared physical observable algebras. This section records the fixed-point, defect, and gauge-quotient statements in that language.

The consensus branch also carries a simple finite-candidate law-selection model: once one equips candidate repair rules with a fitness functional, replicator dynamics gives a clean toy picture in which more successful reconciliation laws dominate a competing pool. This is a finite-candidate monotonicity result. It supplies no gravity or gauge theorem, universality result, or literal cosmological dynamics.

On its declared fixed-cutoff branch, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [17] supplies the following results:

1. **Constraint-code firewall.** A bare finite overlap net is a finite constraint code: its codewords are exactly the globally consistent states $C = \Phi^{-1}(0)$. It is not automatically a QECC/topological code, and its graph min-cut does not determine code distance. Distance/min-cut, Knill–Laflamme resilience, spectral convergence, BFT liveness, and hardware speedup enter only through separate certificates.

2. **Asynchronous confluence.** For the declared accepted repair law, the local-fit contract makes Φ a Lyapunov functional, hence gives termination on the finite patch net. The fixed-cutoff union-collar gluing package supplies the local diamond on the physical quotient, and only that confluence condition together with repair completeness yields a unique schedule-independent normal form from a fixed initial quotient state. Same-boundary uniqueness requires the additional unique consistent extension condition in the preserved boundary/sector fiber.

3. **Cycle obstruction and higher-gauge defects.** On the abelian branch, global consistency holds exactly when cycle holonomy vanishes; on the genuinely noncentral branch, the crossed-

module gluing orbit

$$q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$$

labels the full fixed-cutoff gluing orbit. It can admit multiple strict representatives related by H_Σ-valued edge changes and need not determine one ordinary G_Σ-valued 1-cocycle class. Associator strictifiability is the separate condition that this orbit admit a representative $(g^{\mathrm{str}}, 1)$; strict endpoint-only transport additionally requires at least one allowed strict representative with trivial represented loop holonomy.

4. **Gauge quotient and observable-level confluence.** The repair law descends to the overlap-invariant quotient, and the induced terminal state on the declared physical observable algebras is unique there even when microscopic representatives differ by gauge or sector relabelings inside one quotient-local glued state.

5. **Record algebra and stability.** On the declared fixed-cutoff observer-accessible operator surface, a separately declared algebra-state representation and Lüders instrument make the central record projectors carry the corresponding Born/Lüders rules. The projectors alone do not select that instrument, while approximate record projectors inherit explicit $(\varepsilon, \delta_{\mathrm{rec}})$ stability bounds on that same event surface. This is a theorem about the stated algebraic record interface, not a requirement that OPH first rebuild every mathematical ingredient from operational records alone.

The consensus paper is equally explicit about the imported repair-law data and what sits outside the fixed-cutoff theorems. The declared repair step includes the touched-overlap local-fit contract and the union-collar gluing package on the physical quotient; the branch conditions above that step are repair completeness and, on the Petz branch, the stated support/CPTP clause. On that same fixed-cutoff surface, exact normal-form computation is finite-state and decidable, automatic approximate stability is only collar-local through the splice and record estimates, and long-run noisy approximate consensus is theorem-grade only after a fair-block contraction certificate is supplied for the chosen exported patch-net family. That certificate boundary is not a dependency for the support-visible BW theorem, local Einstein branch, receipt-conditional compact-gauge reconstruction, or realized Standard Model branch.

5 Particle-Spectrum Branch

The particle branch follows the same logical order as *Deriving the Particle Zoo from Observer Consistency* [19]. The complete twelve-port response in A1 and endogenous holonomy in A2 first force the abstract local algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$, without an ambient continuous gauge group. The explicit inverse-port contract separately gives the signed response. Applying the declared matrix-current and rank-15 matter contracts, followed by anomaly balance and tensor descent, gives the exact conditional Standard Model charge lattice and maximal faithful matter image. The one-Higgs clauses leave the window $3 \le N_g \le 5$. Under the named complete-band and operational-cost premises, an exact screen theorem selects the rank-three response band, and the declared unitary response places its residue at the lowest positive generator frequency. Identifying that finite band with three physical matter families and excluding extra light sectors remain separate physical attachments. The independent transportable-sector and Tannaka reconstruction remains a conditional classification route.

Structural carrier and physical premises. Under the declared response and matter contracts, the structural carrier roles and maximal faithful matter image follow from those contracts alone:

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6},\qquad N_c=3,\qquad 3\le N_g\le 5.$$

On a declared trace-balanced block carrier $V=C\oplus W$, with $\dim C=3$, $\dim W=2$, and block hypercharges $(-1/3,1/2)$, the selected exterior package

$$\Lambda^2V\oplus\Lambda^4V=Q\oplus u^c\oplus e^c\oplus d^c\oplus L$$

is an exact one-generation Standard Model representation witness. It has the three one-Higgs invariant lines, cancels the gauge and mixed anomalies, and contains four weak doublets per generation. Hence every additive isomorphism-invariant load normalized by $L_P(\mathbb{C})=P$ gives $L_P(\mathrm{Hom}_{SU(2)}(W,M_1))=4P$. A positive unital map between one-dimensional order-unit load lines is unique once the physical order units are identified. Complete reversible response and endogenous overlap transport force the abstract local Standard Model gauge Lie algebra. Under the conditional matrix-current and rank-15 matter contracts, determinant balance, the non-vacuum exterior package, and the common central kernel are exact finite implications. The scalar scan fixes compatible charges and Yukawa channels, not scalar multiplicity. The finite source does not select the matrix current, matter action, or physical $\mathbb{Z}_6$ quotient. Laboratory current and flux identification, removal of the omitted Λ^0V singlet and other light sectors, attachment of the A_5 face module to physical families, scalar attachment and dynamics, four-dimensional instanton normalization, and continuum quantum field theory are not supplied.

Theorem 5.1 (A1–A2 Lie-type theorem with conditional physical landing)**.** *Assume the declared quotient-visible echosahedral carrier lineage together with its integer atom-counting grammar and normalized central-readback Hilbert–Schmidt cost. Then the counting theorem supplies the twelve unit ports and exact gap, while the oriented-incidence theorem supplies inverse pairing, proper A_5 action, the rank-three frame, and refinement/relabeling naturality. On the named twelve-register realization, atomic signed record events generate integer loads. Conservative whole-unit repairs terminate by strict descent of $V(N)=\sum_i N_i^2$, and their minimum move count is natural under rotations and the declared refinements. A half-unit display rescales the same event graph and threshold. The quadratic readback cost, dynamic move cost, load-square Lyapunov function, seam quadratic, and A3 Hessian are distinct objects. Their selection on an arbitrary A1–A3 carrier is not implied by this named realization. The complete-response clause in A1 and endogenous proper-carrier transport in A2 force the abstract local Standard Model gauge Lie algebra $\mathfrak{u}(1)\oplus\mathfrak{su}(2)\oplus\mathfrak{su}(3)$. Assume further the explicit contract that inverse-port readback is a signed central involutive graph automorphism, together with the declared matrix current, rank-15 matter contract, and tensor-descent receipt. Incidence then determines J, the admissible inverse-port responses are $\pm J$, and the conditional finite certificates verify the matrix realization, refinement maps, anomaly-forced determinant balance, Spin lift, rank-15 exterior matter package, and axis/center deck descent. These implications produce the trace-balanced $3+2$ block carrier with hypercharges $(-1/3,1/2)$ and fix the common $\mathbb{Z}_6$ tensor kernel on the declared matter table. The scalar scan identifies compatible charges and Yukawa channels, not $H=W$ or scalar multiplicity. For a physical realization, assume that a source packet reconstructs the matrix current and matter action; that a complete source character lattice and same-source carrier-loop-to-kernel map select the global quotient; that a source-bound refinement-natural commuting square identifies the A_5 current group and action with the independently reconstructed Tannaka group and its realized matter action; that laboratory current and flux identification passes; that the matter contract is source-bound; that a scalar-multiplicity/one-Higgs* *receipt lands; that a source-complete no-extra-light-sector receipt lands; and that the A_5 multiplicity is physically attached to three families with the selecting symmetry hidden, broken, or forgotten on the Yukawa surface. On a receipt-certified compact-gauge continuum/QFT landing, the resulting low-energy chiral package has the Standard Model physical global form*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

with the exact Standard Model hypercharge lattice, one Higgs doublet, three colors, three attached generations, perturbative anomaly cancellation, even Witten parity, and no simple-GUT X/Y gauge channel.

Proof. The declared counting and incidence theorems produce the finite port and frame data on the declared carrier lineage. The A1–A2 response clauses and compact classification force the abstract local Standard Model gauge Lie algebra. The declared matrix-current and matter contracts make the exact representation, refinement, rank-15 matter, determinant, Spin, and descent calculations applicable. Exterior branching gives the five chiral multiplets and the scalar scan gives the compatible cubic invariant lines without a scalar-multiplicity claim. The anomaly and center calculations fix the hypercharge lattice, color triplet, and common $\mathbb{Z}_6$ tensor kernel from those calculations alone. The source character and loop-to-kernel premises select the quotient. The laboratory flux premise attaches it to the physical spectrum. The source-bound commuting square identifies that group with the group acting through the Tannaka current. The stated completion, sector, and family receipts remove the otherwise available singlet, extra-sector, and family-attachment ambiguities. The continuum quantum-field premise completes the physical implication. □

This theorem is a logical implication under its premise list. Each named screen-to-current, cross-route identity, carrier, sector, family, and continuum condition is independently removable. Removing one limits the conclusion at the corresponding boundary.

This does not determine quantum-particle masses. The specialist derivation papers separately prove that explicit Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic actions have transverse or TT classical massless modes on their stated backgrounds and phases. Photon, gluon, or graviton particle language additionally requires a positive-energy physical quantization, positive-residue two-point pole, and the appropriate asymptotic/deconfinement receipt. Separate complete-band and cost-order premises select the rank-three response band. Tensoring it with the declared generation table gives a conditional rank-45 candidate; it is not a physical attachment theorem. Chirality and the diagonal $\mathbb{Z}_6$ action come from that table. A distinct local-domain receipt checks the declared tensor-identity operator and conditional gap inheritance without source-selecting the action or transporting the twelve-port Spin packet. Promotion to three physical families requires matter-pole, continuum Spin/locality, physical seam, persistence, excluded-band, and complement-complete refinement receipts.

Proof route. The proof route is the one developed in *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency* and *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [15, 16] and *Deriving the Particle Zoo from Observer Consistency* [19]. Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence determines J, while the explicit signed-central-involutive inverse-port response contract restricts admissible responses to $\pm J$. Together with the conditional matrix current and rank-15 matter contract, anomaly-forced determinant balance and tensor descent give the exact hypercharge lattice, common $\mathbb{Z}_6$ kernel, maximal faithful matter image, and $N_c = 3$. The target-blind producer derives the inverse-port response, without selecting the matrix current. Physical laboratory-current and flux attachment are not supplied. Under the explicit compact-gauge refinement receipt, coherent bosonic pullback ladder, symmetry, and forgetful-fiber conditions of the Standard Model gauge paper, the surviving transportable edge-sector category separately reconstructs a compact internal symmetry group. The one-Higgs clauses give $3 \le N_g \le 5$. The exact screen-band theorem selects rank three under its complete-band and operational-cost premises, and the declared unitary response places the selected residue at the lowest positive generator frequency. No physical family identification or extra-sector exclusion is supplied. The particle branch does not bypass the unsupplied laboratory current, cross-route identity, or physical family attachment.

Sector	Result class	What is fixed here	Requirements for physical interpretation
Carrier roles and classical modes	conditional group/content theorem plus conditional quadratic-action theorem	declared quotient, $N_c = 3$, the conditional window $3 \le N_g \le 5$, and the exact rank-three screen-band selection under its named premises; two Maxwell transverse modes, $2 \dim G$ perturbative Yang–Mills modes, and two Einstein TT modes on stated branches	the rank-three band is a candidate for three physical families; physical identification requires matter-pole and laboratory attachment, while the mode counts are classical kernels and quantum particles require the Hilbert-space, pole-residue, and phase premises
Electroweak bosons	exact selected-carrier chart + conditional quotient-transport implication + measured-reference inverse adapter	no nonzero source-only physical mass emitted	the selected chart, conditional value law, and inverse adapter have different provenance. The finite-carrier certificate is not emitted, the adapter is a separate reference-fitted comparison map, and no physical complex-pole pair is identified
Fine-structure endpoint	source witness + mixed diagnostic + empirical closure + measurement	four distinct coordinates, detailed in the endpoint comparison	the external-data empirical interval misses the measured endpoint. A physical source-only theorem requires a no-target-leak source-derived hadronic spectral payload, target-independent map selection, and the typed same-quantity bridge
Higgs/top stage	conditional downstream split theorem on the declared running, matching, and threshold surface + compare-only exact calculation	no nonzero source-only physical mass emitted	a physical mass requires the source root, physical scale, transport, running, matching, rigidity, provenance, uncertainty, and complex-pole premises. The auxiliary direct-top codomain is comparison-only
Quark family	common-scale reciprocal-ray falsification + exact generic interface	no nonzero source-only physical mass emitted	reciprocal-ray closure fails across the tested common scales. Its sub-percent residuals use a target-anchored mixed-convention chart, and a lower-order ablation fits that chart better. The simulator has no Yukawa coupling or Yukawa information. Generation-blind flavor-singlet inputs produce no physical flavor-orbit selector

Sector	Result class	What is fixed here	Requirements for physical interpretation
Charged leptons	conditional construction + exact witness + face-carrier theorem	exact A_5/C_3 face orbit; conditional contraction theorem; engineered finite model of the declared charged face-quotient schema	no nonzero source-only physical mass is emitted. The bounded model supplies local register state, event readback, and a central accepted/rejected record, establishing fixed-cutoff schema existence. Its dimensions, automaton, grading, clock, and response are authored inputs; no frozen receipt excludes target dependence. A separate nature/pole bridge assumes both the physical Yukawa response and singularity readout. No physical source selection, family/Yukawa attachment, interacting kernel, cofinal refinement, or pole-scheme input is supplied
Neutrinos	rejected target-informed weighted-cycle candidate + diagnostic calculations	no nonzero source-only physical mass emitted; no source-level PMNS matrix, physical ordering, Majorana phases, or absolute masses are supplied	the candidate fails the NuFIT 6.1 correlated profile; its family kernel is a template, no charged-basis construction is supplied, the template basis is nearly degenerate, and its shared-basis recovery is tautological
Hadrons	backend-gated nonperturbative continuation	stable-channel and readout architecture are defined	source-only masses require one production backend export bundle, then executed unquenching, runtime receipt, and production systematics

Theorem boundary. Structural carrier roles and the realized Standard Model branch are theorem-bearing outputs; classical propagation is conditional on the displayed action, background, and phase premises, and quantum-particle interpretation requires a physical Hilbert space and pole residue. The electroweak quantitative branch separates an exact selected-carrier chart, a conditional value law, and a reference-fitted inverse adapter. The quotient-transport assumptions imply the value law exactly, but the finite carrier does not emit their certificate. The color-balanced $(\sqrt{N_c}/2, N_c)$ amplitude/loop construction defines a distinct alternative model rather than the complete transport law. The hierarchy theorem fixes the dimensionless ratio $v/E_\star$; an independently physical $E_\star$ and a pole receipt govern any mass in GeV. A physical W/Z mass conclusion also requires a strict source root, a frozen RG/matching/scheme packet, branch rigidity, target-independent precommitment, and uncertainty control. The Higgs/top stage is conditional on the declared downstream surface and inherits the same source-root, transport, scale, scheme, rigidity, provenance, uncertainty, and complex-pole premises. Measured electroweak quantities are comparison data. The executed empirical hadron closure integrates external spectral input and misses the measured Thomson endpoint by a certified same-scheme interval. That payload tests the map; it does not source-close the Ward-projected hadronic spectral transport needed for a source-only theorem.

The $\alpha_U(P)$ source proof record. The electroweak hierarchy used by the clock-scale branch depends first on a source readout of the unified diffusion coupling

$$\alpha_U(P_\star).$$

For a trial pixel P, define

$$M_U(P) = E_\star e^{-2\pi} P^{1/6}, \qquad E_{\mathrm{cell}}(P) = \frac{E_\star}{\sqrt{P}}.$$

On the realized color branch,

$$N_c = 3, \qquad \beta_{\mathrm{EW}} := N_c + 1 = 4,$$

and the transmutation scale for a candidate coupling a is

$$v(P, a) = E_{\mathrm{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\mathrm{EW}} a}\right].$$

The exterior representation theorem fixes the weak-doublet multiplicity $3 + 1 = 4$, which also passes Witten parity on the realized branch. Its use as the electroweak transmutation coefficient $\beta_{\mathrm{EW}} = 4$ requires the common screen/electroweak load-carrier identification and remains declared. The SU(2) one-loop coefficient of the realized one-Higgs branch is 19/6 and the declared unification packet's coefficient is 1, and neither equals 4. The integer is exact structural mathematics; no physical load-attachment receipt is supplied. It is not derived by dividing the 24-slot register by six. The source running family is

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \qquad i = 1, 2, 3.$$

The beta-coefficient packet (b_1, b_2, b_3), matching convention, threshold packet, and renormalization convention are part of the source proof record. They are declared before any comparison with measured electroweak or gravitational data.

The hypercharge coupling is

$$\alpha_Y(\mu; P, a) = \frac{3}{5}\alpha_1(\mu; P, a).$$

The source Z-scale is a fixed point of the source equations, not an inserted measured mass:

$$\mu_Z(P, a) = \frac{v(P, a)}{2}\sqrt{4\pi\alpha_2(\mu_Z; P, a) + 4\pi\alpha_Y(\mu_Z; P, a)}.$$

For $G = \mathrm{SU}(2), \mathrm{SU}(3)$, define the finite representation heat-kernel sums

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \qquad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

The source heat-kernel parameters are

$$t_2(P, a) = 4\pi^2\alpha_2(\mu_Z(P, a); P, a), \qquad t_3(P, a) = 4\pi^2\alpha_3(\mu_Z(P, a); P, a).$$

The unified-coupling residual is

$$\Phi_U(P, a) := \bar{\ell}_{\mathrm{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\mathrm{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The source value is the zero

$$\Phi_U(P, \alpha_U(P)) = 0.$$

Definition 5.2 (No-hidden-free-variable admissibility)**.** *The unified-coupling proof record is admissible only when the representation cutoffs, beta packet, matching prescription, threshold packet, and renormalization convention are frozen before comparison with measured electroweak or gravitational data:*

$$N_2, \quad N_3, \quad (b_1, b_2, b_3), \quad \textit{matching prescription}, \quad \textit{threshold packet}, \quad \textit{renormalization convention.}$$

None of these may be selected by minimizing the error against

$$M_W, \quad M_Z, \quad \alpha_s(M_Z), \quad \sin^2\theta_W, \quad v_{\text{exp}}, \quad G_{\text{exp}}.$$

Tuning any of these objects after comparison with observed electroweak or gravitational data makes the hierarchy row a calibration.

Theorem 5.3 (No-measured-input unified-coupling proof record)**.** *Let*

$$\mathcal{R}_U = (P_\star, N_2, N_3, I_U, \Phi_U, K_U, a_U^{(0)}, r_U, m_U, \mathrm{DAG}_U)$$

be an admissible unified-coupling proof record. Here N_2, N_3 are the finite representation cutoffs for $\mathrm{SU}(2)$ *and* $\mathrm{SU}(3)$*, $I_U \subset \mathbb{R}_+$ is the declared physical interval for α_U, Φ_U is the heat-kernel closure residual, K_U is an interval-Newton or Krawczyk operator, $a_U^{(0)}$ is the displayed candidate, r_U is the residual bound, m_U is a lower derivative bound, and* DAG_U *is the dependency graph.*

Assume

$$K_U(I_U) \subset \mathrm{int}(I_U),$$
$$|\Phi_U(P_\star, a_U^{(0)})| \le r_U,$$
$$|\partial_a \Phi_U(P_\star, a)| \ge m_U > 0 \qquad (a \in I_U),$$

and assume that DAG_U *contains no directed path from measured electroweak masses, measured low-energy gauge couplings, measured G, Planck area, Planck mass, measured Λ, or any equivalent gravity-calibrated or electroweak-calibrated scale. Then there is a unique source value*

$$\alpha_U(P_\star) \in I_U$$

such that

$$\Phi_U(P_\star, \alpha_U(P_\star)) = 0,$$

and

$$|\alpha_U(P_\star) - a_U^{(0)}| \le \frac{r_U}{m_U}.$$

Consequently, the electroweak hierarchy

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{(N_c+1)\alpha_U(P_\star)}\right]$$

is no-measured-input on this proof record.

Proof. The residual $\Phi_U(P_\star, a)$ is built from the source-side pixel $P_\star$, declared representation cutoffs, source transmutation law, source running law, and source Z-scale fixed-point equation. The dependency condition excludes measured electroweak data and all gravity-calibrated scales. The interval-Newton/Krawczyk inclusion proves existence and uniqueness of a zero of $\Phi_U(P_\star, a)$ inside I_U. The residual estimate and derivative lower bound give

$$|\alpha_U(P_\star) - a_U^{(0)}| \le \frac{|\Phi_U(P_\star, a_U^{(0)})|}{m_U} \le \frac{r_U}{m_U}.$$

Substitution into the declared transmutation law gives the displayed hierarchy. The dependency graph contains no measured G, Planck unit, measured M_Z, measured M_W, measured $\alpha_s(M_Z)$, or measured low-energy electroweak coupling ancestor, so the emitted hierarchy is source-side on $\mathcal{R}_U$. □

The theorem fixes a dimensionless ratio. It supplies neither an independently physical $E_\star$ nor a mass in GeV. On the populated branch $P_\star$ is the unique root of the declared local self-read map, so the downstream source expressions inherit that root. Their physical endpoint interpretation requires target-independent map selection, the typed same-quantity bridge, and the corresponding transport and pole constructions.

Lemma 5.4 (Exponential sensitivity of the hierarchy row)**.** *On the transmutation branch*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U}\right],$$

and on a clock row whose dominant scale dependence enters through $m_e^2 \propto v^2$ in the leading hyperfine scaling $\varepsilon_{\mathrm{Cs}} \sim \mathcal{C}_{\mathrm{Cs}}\, \alpha_\star^4\, m_e^2/(m_p E_\star)$, the induced gravity readout satisfies approximately

$$\frac{\partial \ln G}{\partial \alpha_U} = \frac{2\pi}{\alpha_U^2}.$$

At $\alpha_U \simeq 0.0411$,

$$\frac{\partial \ln G}{\partial \alpha_U} \simeq 3.7 \times 10^3.$$

Therefore a certified relative gravity precision η_G requires

$$|\delta\alpha_U| \lesssim \frac{\eta_G}{3.7 \times 10^3}.$$

Proof. For $\beta_{\mathrm{EW}} = 4$,

$$\ln(v/E_\star) = -\frac{1}{2}\ln P_\star - \frac{2\pi}{4\alpha_U}.$$

Thus

$$\frac{\partial \ln(v/E_\star)}{\partial \alpha_U} = \frac{2\pi}{4\alpha_U^2} = \frac{\pi}{2\alpha_U^2}.$$

The clock gap carries the square of the electroweak scale: in the leading hyperfine scaling $\varepsilon_{\mathrm{Cs}} \propto m_e^2 \propto v^2$ at fixed Yukawa, fine-structure, and strong-sector ratios, so

$$\frac{\partial \ln \varepsilon_{\mathrm{Cs}}}{\partial \alpha_U} = 2 \cdot \frac{\pi}{2\alpha_U^2} = \frac{\pi}{\alpha_U^2}.$$

Since the gravity row is quadratic in the clock scale,

$$G \propto \varepsilon_{\mathrm{Cs}}^2,$$

one obtains

$$\frac{\partial \ln G}{\partial \alpha_U} = 2 \cdot \frac{\pi}{\alpha_U^2} = \frac{2\pi}{\alpha_U^2}.$$

□

Remark 5.5 (Digits policy)**.** *The exponential branch allows precision only to the extent that the source proof records carry interval bounds. The number of printed digits in G_{SI} may not exceed the precision certified by $\mathcal{R}_U$, $\mathcal{R}_{\mathrm{QCD}}$, $\mathcal{R}_{\mathrm{flav}}$, and $\mathcal{R}_{\mathrm{Cs}}$. A display such as*

$$\varepsilon_{\mathrm{Cs}} = 2\pi\nu_{\mathrm{Cs}}\sqrt{\hbar G_{\mathrm{ref}}/c^5}$$

is a calibration checksum unless the full source construction emits the same interval with a no-G dependency graph.

No-G clock hierarchy for the gravity scale. The numerical gravity row uses the dimensionless cesium clock gap

$$\varepsilon_{\rm Cs} := \frac{\hbar\omega_{\rm Cs}}{E_\star}, \qquad E_\star := \frac{\hbar c}{\ell_\star}, \qquad \omega_{\rm Cs} = 2\pi\nu_{\rm Cs}.$$

The size of this gap is a hierarchy problem for the OPH scale branch. The gap is represented as a factorized source-side readout. The source-only scale branch is accepted as no-G only when the gap factorizes through source-side dimensionless physics:

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\rm abs} + \mathcal{R}_{\rm QCD/nuc}^{\rm 133Cs} + \mathcal{R}_{\rm atom}^{\rm 133Cs}.$$

Here $\mathcal{R}_U$ emits $\alpha_U(P_\star)$ and the electroweak scale, $\mathcal{R}_\alpha$ emits the electromagnetic coupling used by the atomic Hamiltonian, $\mathcal{R}_e^{\rm abs}$ emits the electron absolute mass ratio, $\mathcal{R}_{\rm QCD/nuc}^{\rm 133Cs}$ emits the cesium nuclear source packet, and $\mathcal{R}_{\rm atom}^{\rm 133Cs}$ emits the hyperfine spectral gap from those dimensionless data.

At leading OPH source level, the electroweak hierarchy is generated by the declared transmutation law

$$\frac{v}{E_\star} = P_\star^{-1/2}\exp\!\left[-\frac{2\pi}{(N_c+1)\alpha_U(P_\star)}\right], \qquad N_c = 3.$$

Thus the weak-to-UV ratio is generated from the dimensionless source data $P_\star$, N_c, and $\alpha_U(P_\star)$. The cesium gap then has the schematic source form

$$\varepsilon_{\rm Cs} = \mathcal{H}_{\rm Cs}\left(\alpha_\star, \frac{v}{E_\star}, \frac{\Lambda_{\rm QCD}}{E_\star}, y_e, y_q, \text{nuclear data}, \text{atomic corrections}\right),$$

where all arguments are dimensionless source outputs on the declared branch. In leading hyperfine scaling this contains the familiar structure

$$\varepsilon_{\rm Cs} \sim \mathcal{C}_{\rm Cs}\,\alpha_\star^4 \frac{m_e^2}{m_p E_\star},$$

with relativistic, many-body, recoil, QED, finite-nuclear-size, and nuclear-moment corrections absorbed into $\mathcal{C}_{\rm Cs}$.

Definition 5.6 (Cesium source packet)**.** *A source-only cesium packet is*

$$\mathcal{D}_{\rm Cs}^{\rm src} = \left(\alpha_\star,\, \frac{m_e}{E_\star},\, \frac{\Lambda_{\rm QCD}}{E_\star},\, \{y_q\},\, \mathcal{N}_{133},\, \mathcal{B}_{\rm atom}^{\rm Cs}\right).$$

$\mathcal{N}_{133}$ contains the source-side nuclear data for ^{133}Cs*, including nuclear spin, magnetic moment, magnetization distribution, charge radius, recoil data, and finite-size data. $\mathcal{B}_{\rm atom}^{\rm Cs}$ contains the atomic many-body, relativistic, QED, recoil, and finite-nucleus correction package. Every object in $\mathcal{D}_{\rm Cs}^{\rm src}$ is dimensionless and has no dependency path from measured G, Planck area, Planck mass, Planck time, measured Λ, or an algebraically equivalent gravity-calibrated scale.*

Definition 5.7 (OPH cesium Hamiltonian)**.** *Given $\mathcal{D}_{\rm Cs}^{\rm src}$, define the dimensionless cesium Hamiltonian*

$$\widehat{H}_{\rm Cs}^{\rm OPH} = \widehat{H}_{\rm DC} + \widehat{H}_{\rm Breit} + \widehat{H}_{\rm QED} + \widehat{H}_{\rm recoil} + \widehat{H}_{\rm nuc} + \widehat{H}_{\rm manybody}.$$

The hats indicate that every term is measured in units of $E_\star$, not in SI units and not in Planck units inferred from measured G.

Theorem 5.8 (No-G OPH clock-hierarchy theorem)**.** *Assume the local pixel proof selects $P_\star$. Assume the unified-coupling proof record $\mathcal{R}_U$ of Theorem 5.3 emits $\alpha_U(P_\star)$ and*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{(N_c+1)\alpha_U(P_\star)}\right].$$

Assume the electromagnetic endpoint branch $\mathcal{R}_\alpha$ emits $\alpha_\star$, the charged-lepton absolute-scale branch $\mathcal{R}_e^{\mathrm{abs}}$ emits

$$\frac{m_e}{E_\star} = \frac{y_e}{\sqrt{2}}\frac{v}{E_\star},$$

the QCD/nuclear branch $\mathcal{R}^{133\mathrm{Cs}}_{\mathrm{QCD/nuc}}$ emits $\mathcal{N}_{133}$, and the atomic branch $\mathcal{R}^{133\mathrm{Cs}}_{\mathrm{atom}}$ supplies isolated spectral enclosures

$$J_{F=3}, \qquad J_{F=4}$$

for the source-only Hamiltonian

$$\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}} = \widehat{H}_{\mathrm{DC}} + \widehat{H}_{\mathrm{Breit}} + \widehat{H}_{\mathrm{QED}} + \widehat{H}_{\mathrm{recoil}} + \widehat{H}_{\mathrm{nuc}} + \widehat{H}_{\mathrm{manybody}}$$

with disjoint intervals and certified hyperfine gap

$$\varepsilon_{\mathrm{Cs}} = \lambda_{F=4}(\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}}) - \lambda_{F=3}(\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}}).$$

Assume further that the combined dependency graph

$$\mathrm{DAG}_{\mathrm{Cs}} = \mathrm{DAG}\left(\mathcal{R}_U, \mathcal{R}_\alpha, \mathcal{R}_e^{\mathrm{abs}}, \mathcal{R}^{133\mathrm{Cs}}_{\mathrm{QCD/nuc}}, \mathcal{R}^{133\mathrm{Cs}}_{\mathrm{atom}}\right)$$

contains no directed path from

$$G_{\mathrm{exp}}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \qquad m_P = \sqrt{\frac{\hbar c}{G}}, \qquad t_P = \sqrt{\frac{\hbar G}{c^5}}, \qquad \Lambda_{\mathrm{exp}}, \qquad \frac{3\pi c^3}{\hbar G},$$

or from any algebraically equivalent gravity-calibrated scale. Then $\varepsilon_{\mathrm{Cs}}$ is a no-G hierarchy readout. Consequently

$$\gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c} = \frac{\varepsilon_{\mathrm{Cs}}}{2\pi}$$

is non-circular, and

$$G_{\mathrm{SI}} = \frac{c^5}{4\pi^2\hbar\nu_{\mathrm{Cs}}^2}\varepsilon_{\mathrm{Cs}}^2$$

has no measured-G ancestor.

Proof. $\mathcal{R}_U$ emits the small weak ratio from $P_\star$, N_c, and $\alpha_U(P_\star)$, all of which are dimensionless branch data. $\mathcal{R}_\alpha$ emits the electromagnetic coupling used by the atomic Hamiltonian. $\mathcal{R}_e^{\mathrm{abs}}$ emits the electron mass ratio. $\mathcal{R}^{133\mathrm{Cs}}_{\mathrm{QCD/nuc}}$ emits the cesium nuclear source packet. Therefore every coefficient of $\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}}$ has no measured-G ancestor. The atomic branch gives isolated spectral enclosures for the two hyperfine levels, so the gap

$$\varepsilon_{\mathrm{Cs}} = \lambda_{F=4}(\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}}) - \lambda_{F=3}(\widehat{H}^{\mathrm{OPH}}_{\mathrm{Cs}})$$

is well-defined and inherits the no-G dependency condition of the Hamiltonian.

Finally,

$$\varepsilon_{\mathrm{Cs}} = \frac{\hbar(2\pi\nu_{\mathrm{Cs}})}{\hbar c/\ell_\star} = \frac{2\pi\nu_{\mathrm{Cs}}\ell_\star}{c},$$

so

$$\gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c} = \frac{\varepsilon_{\mathrm{Cs}}}{2\pi}.$$

Substitution into $G_{\mathrm{SI}} = c^3 \ell_\star^2/\hbar$ gives the displayed expression for G_{SI}. The only non-display parent of the expression is the source-side $\varepsilon_{\mathrm{Cs}}$ readout, so the gravity row is non-circular under the stated hypotheses. □

Remark 5.9 (Clock-certificate boundary)**.** *The theorem above is not discharged by printing*

$$\varepsilon_{\mathrm{Cs}} = 3.1139305134\ldots \times 10^{-33}.$$

That decimal is source-predictive only if it is emitted by

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}} + \mathcal{R}_{\mathrm{atom}}^{133\mathrm{Cs}}.$$

If instead it is computed as

$$\varepsilon_{\mathrm{Cs}}^{\mathrm{cal}} = 2\pi \nu_{\mathrm{Cs}} \sqrt{\frac{\hbar G_{\mathrm{exp}}}{c^5}},$$

then it is a gravity-side calibration checksum, not a source-side clock prediction. The paper surface supplies the algebraic certificate shape and the $\mathcal{R}_U$ numerical witness. The full source-only $\mathcal{R}_{\mathrm{Cs}}$ branch requires the source-only electromagnetic endpoint, charged-lepton absolute-scale premise, source-side cesium nuclear packet, and atomic spectral enclosure.

Remark 5.10 (Hadronic and nuclear certificate boundary)**.** *The no-G SI gravity row requires an unsupplied hadronic/same-scheme endpoint payload in $\mathcal{R}_\alpha$ and the source-side cesium QCD/nuclear packet in $\mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}}$. The displayed endpoint and $\varepsilon_{\mathrm{Cs}}$ rows therefore fix those payloads for checksum and comparison purposes. They are not counted as source-only OPH emissions.*

The source-only completion route is a Ward-projected QCD/nuclear spectral certificate from a dedicated OPH optical-compute backend, or an equivalent nonperturbative backend with manifest provenance and systematics. Ordinary CPU/GPU runs in this paper are not the declared source certificate for that hadronic payload.

Direct public-record closure for total capacity. The global counterpart of the local P-closure begins with a supplied positive integer capacity-carrier dimension

$$D = \dim \mathcal{H}_{\mathrm{cap},r,D}, \qquad N = \log D.$$

The universe-level readback equation is

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

Here the argument fixes the type. In every occurrence, M_0 is a multiplicative code size. Once the complete terminal fiber scalarizes, the universe-level notation means

$$M_0(\mathfrak{U}_N) := \widehat{F}_{r,0}(e^N).$$

Thus N is the logarithmic capacity read back from the trial universe, while $M_0(\mathfrak{U}_N)$ is the number of correctable public records. The equation is the logarithmic form of stable direct capacity closure, not a new producer or an appeal to a measured target. The carrier type is frozen in the branch contract. Boundary, total, edge-center, and other Hilbert spaces are not interchangeable. Let

$\widetilde{\Omega}_{r,D}$ be the terminal physical quotient fiber reached from the source-derived trial universe $\mathfrak{U}_{r,D}$. Its members are repair-normal and pass the declared local, obstruction, and provenance gates. Membership contains no capacity-equals-D predicate.

For $q \in \widetilde{\Omega}_{r,D}$, observer O has a finite commutative record algebra $\mathcal{R}_O(q)$ with atom set $X_O(q) = \operatorname{At}\mathcal{R}_O(q)$. Each shared interface e has an atom set $X_e(q)$ and a source-derived record-atom readout

$$r_{Oe} : X_O(q) \longrightarrow X_e(q).$$

This atom-level construction is used because the generic visible maps on full observer algebras need not be unital *-homomorphisms in the direction required by an algebra equalizer. The compatible public global sections are

$$X_{\text{pub}}(q) = \{(x_O)_O : r_{Oe}(x_O) = r_{O'e}(x_{O'}) \text{ on every shared interface}\}, \qquad \mathcal{R}_{\text{pub}}(q) = C(X_{\text{pub}}(q)).$$

Theorem 5.11 (Public global sections)**.** *For every finite observer record diagram, $X_{\text{pub}}(q)$ is finite and $C(X_{\text{pub}}(q))$ is a finite commutative unital C^*-algebra. Every isomorphism of observer record diagrams induces a bijection of public global sections and a *-isomorphism of their function algebras.*

Proof. The public sections form the subset of the finite product $\prod_O X_O(q)$ satisfying finitely many equalities. A diagram isomorphism transports the atom sets and interface maps, hence compatible tuples bijectively; pullback transports their function algebras. $\square$

Only endogenously writable records count. Define

$$X_{\text{reach}}(q) = \{x \in X_{\text{pub}}(q) : x \text{ terminates an admissible endogenous semantic history}\}.$$

The history may use declared source data, accepted repair, semantic continuation, and quotient-visible writes. It may not use a target record, supplied-capacity metadata, measured Λ, or the electroweak bridge. The publicness policy is a frozen nonempty family

$$\mathfrak{P}(q) \subseteq 2^{\mathcal{O}(q)} \setminus \{\varnothing\}$$

of authorized observer subsets. Collective, universal-local, and quorum publicness are distinct theorem branches.

For every $x \in X_{\text{reach}}(q)$, authorized set $A \in \mathfrak{P}(q)$, and allowed same-interface semantic continuation κ, the source construction must emit a joint stochastic kernel

$$K_{A,\kappa}(y \mid x), \qquad y \in Y_{A,\kappa},$$

whose observer marginals agree with the local checkpoint packets. The joint kernel is an independent receipt. It cannot be manufactured from local marginals.

Theorem 5.12 (Local marginals do not determine public capacity)**.** *There are two binary-input joint checkpoint channels with identical observer-by-observer marginals and different public capacities.*

Proof. Let U be uniform. The channel $(Y_1, Y_2) = (U, U \oplus x)$ has uniform input-independent marginals, while the joint parity recovers x. Independent uniform outputs have the same marginals and carry no information about x. Thus the joint capacities are two and one, respectively. $\square$

For a channel K, write $S_K(x) = \{y : K(y \mid x) > 0\}$. A code $C \subseteq X_{\text{reach}}(q)$ is zero-error correctable for K when a decoder d_K satisfies

$$K(d_K^{-1}(x) \mid x) = 1 \qquad (x \in C).$$

Theorem 5.13 (Support criterion and compound zero-error capacity)**.** *A code is zero-error correctable for K exactly when the supports $S_K(x)$, $x \in C$, are pairwise disjoint. Let $\mathfrak{K}(q)$ contain every authorized checkpoint channel and every declared finite composition. Define the compound confusability graph G_q on $X_{\text{reach}}(q)$ by joining $x \neq x'$ whenever their output supports overlap for at least one $K \in \mathfrak{K}(q)$. Then the exact public-record capacity is*

$$\boxed{M_0(q) = \alpha(G_q)},$$

where α is the graph independence number.

Proof. A successful decoder cannot assign one output to two inputs, which proves the support criterion in one direction. Pairwise disjoint supports admit the decoder that returns the unique supporting input. A code valid for every declared channel is therefore exactly an independent set in the union of the channel confusability graphs. □

For exact indefinite continuation, replace each kernel by its support relation $R_K = \{(x, y) : K(y \mid x) > 0\}$. Boolean relation composition generates at most $2^{|X_{\text{reach}}|^2}$ relations, so the support semigroup and its compound confusability graph close after finitely many steps.

Theorem 5.14 (Reversible fast branch)**.** *If every authorized continuation is a known injective deterministic map on $X_{\text{reach}}(q)$, then*

$$M_0(q) = |X_{\text{reach}}(q)|.$$

Proof. Injectivity keeps every pair of inputs distinguishable for every continuation, so G_q has no edges. □

Checkpoint invariance is not the capacity definition. A cyclic permutation of m labels has a one-dimensional fixed function algebra, while its inverse decodes all m labels. Thus a fixed-projector count can undercount the correctable capacity arbitrarily.

For a frozen finite channel family or continuation horizon and tolerance ε, let $M_\varepsilon(q)$ be the largest code for which every declared channel has a decoder with worst-input error at most ε.

Theorem 5.15 (Approximate-capacity stability and zero-error discontinuity)**.** *If corresponding channel rows obey*

$$\sup_{K,x} d_{\text{TV}}(K(\cdot \mid x), \widehat{K}(\cdot \mid x)) \le \delta,$$

then

$$M_{\varepsilon+\delta}(\widehat{\mathfrak{K}}) \ge M_\varepsilon(\mathfrak{K}).$$

Zero-error capacity is discontinuous at every identity channel on more than one symbol: adding arbitrarily small full-support noise collapses M_0 to one.

Proof. Total-variation control changes the probability of each correct-decoding event by at most δ. For the second statement, $(1-\delta)I + \delta U$, with U full support, makes every pair of inputs confusable for every $\delta > 0$. □

The capacity carrier is an exact finite Hilbert space $\mathcal{H}_{\text{cap},r,D}$ of dimension D. Each reachable public record class has a nonzero projection P_x on this carrier, and distinct records have orthogonal projections.

Theorem 5.16 (Carrier bound and saturation rigidity). *Every correctable public code C satisfies*

$$|C| \le D, \qquad \log M_\varepsilon(q) \le \log D = N.$$

If $M_0(q) = D$, the code projections are rank one and sum to the identity.

Proof. The ranks of nonzero orthogonal projections add and their sum has rank at most D. Equality with D codewords forces every rank to be one and the sum to have full rank. □

The primary finite readback remains set-valued:

$$\boxed{\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.}$$

A scalar $\widehat{F}_{r,\varepsilon}(D)$ exists exactly when the nonempty terminal fiber has one common defined value. The evaluator must distinguish an empty readback, an ambiguous readback, and a singleton scalar readback. Existential closure means $D \in \mathfrak{F}_{r,0}(D)$; selector-free stable closure means

$$\boxed{\mathfrak{F}_{r,0}(D) = \{D\}.}$$

Equivalently, the unclosed kernel

$$K_r(D, m) = |\{q \in \widetilde{\Omega}_{r,D} : M_0(q) = m\}|$$

has support only at $m = D$. One saturated branch is insufficient when the terminal fiber is ambiguous.

The full capacity packet consists of observer lineage, record atoms, interface maps, reachable public sections, publicness policy, global kernels, and carrier projections. An isomorphism preserving that packet transports every exact confusability edge, approximate decoder error, and projection rank. Hence M_ε is implementation invariant under packet isomorphism.

Once scalarized, the active map $f(D) = \widehat{F}_{r,0}(D)$ is deflationary. It cannot attract a positive fixed point from below. If f is total on a declared finite admissible chain, monotone, and deflationary, iteration from its top element stabilizes at the greatest fixed point. This is a conditional order theorem, not a uniqueness or cutoff-independence theorem.

Theorem 5.17 (Bounded capacity-continuation counterfamily). *At rung k, each of the twenty-four visible base records has k copies, so $D_{\mathrm{pub}}(k) = 24k$. In the spectator control the raw carrier is $D_{\mathrm{raw}}(k, s) = 24ks$. A target-clean counterfamily shares base agreement, positivity, the carrier bound, and executable finite incidence, publicness, reversible-action, semantic-projection, composition, extension, and oriented-record fiber-product controls. Its reversible identity completion has*

$$M_{\mathrm{id}}(k) = 24k.$$

Copy collapse and a two-class completion have

$$M_{\mathrm{erase}}(k) = 24, \qquad M_2(k) = 24 \min(k, 2).$$

A hidden spectator of multiplicity s leaves the public capacity at $24k$. The corresponding slack-zero sets are every positive k, $\{1\}$, $\{1, 2\}$, and every positive k only for $s = 1$. The shared bounded structure does not entail a unique zero.

Proof. Each collapse channel yields a disjoint union of complete confusability components. One representative per component is an independent set, and completeness of each component gives the matching upper bound. The collapse maps are idempotent, commute with the forty reversible slot actions, and preserve the old induced graph under rung extension. Lean proves the all-rung arithmetic consequence of typed base agreement, positivity, and the carrier bound, including the disagreement between the displayed zero sets. The executable certificate checks the finite incidence, action, projection, composition, extension, and sewing controls. A separate simulator implementation reconstructs held-out finite sample graphs without importing the producer. □

This bounded counterfamily is not a complete source-class nonidentifiability theorem. The complete A1–A3 terminal fibers, observer and interface atom maps, joint checkpoint kernels, meaning maps, feasible sets, and regulator controls are absent across capacity. The result does not reject strange-loop self-identification. Equality follows after the simulating and simulated readings are proved to be readings of the same invariant quantity.

Capacity extension and regulator refinement are distinct. For $D \leq D'$, an injection of reachable records that reflects confusability,

$$e_{DD'}(x) \sim_{G_{D'}} e_{DD'}(x') \Longrightarrow x \sim_{G_D} x',$$

maps every coarse independent set to a fine independent set and gives $M_0(D) \leq M_0(D')$. At fixed D, refinement injections with the same property make $M_{0,r}(D)$ a nondecreasing integer bounded by D, hence eventually constant along each cofinal sequence.

For regions A, B sewn along public seam S, exact public records form the fiber product

$$X_{A\cup B} = X_A \times_{X_S} X_B, \qquad |X_{A\cup B}| = \sum_{z\in X_S} |r_A^{-1}(z)|\,|r_B^{-1}(z)|.$$

This gives an exact CSP, tensor-network, or model-counting route on the reversible branch. A cosmological conclusion additionally requires an exact transfer recurrence, exact fiber-product recursion, or a seam theorem with a certified subleading term.

Define the finite-size slack

$$\boxed{s_r(D) = \log D - \log M_{0,r}(D).}$$

If $\limsup_{D\to\infty} \log M_0(D)/\log D < 1$, closure is excluded at all sufficiently large dimensions. Unit asymptotic density is insufficient: $M(D) = D$ and $M(D) = D - 1$ have the same limiting density but opposite fixed-point behavior. Full physical N-closure requires a source-derived finite-size law proving

$$s(D_\star) = 0, \qquad s(D) > 0 \quad (D \neq D_\star)$$

on the declared physical domain, or an equally explicit physical selector. Such a law is a separately named physical premise unless it is derived from a stronger source architecture. The common $D = 24$ screen packet is not a universe-level selection.

Two independent physical identifications follow only after stable direct closure. The first hypothesis identifies the same capacity carrier with the de Sitter horizon record:

$$N_\star = \log D_\star = \frac{A_{\mathrm{dS}}}{4\ell_\star^2}, \qquad A_{\mathrm{dS}} = \frac{12\pi}{\Lambda} \quad \Longrightarrow \quad \boxed{\Lambda\ell_\star^2 = \frac{3\pi}{N_\star}}.$$

The second hypothesis identifies the screen load with the electroweak load by a positive, unital, refinement-natural map

$$\Xi_r : \mathcal{L}_{\mathrm{screen},r} \longrightarrow \mathcal{L}_{\mathrm{EW},r}.$$

Given the independent screen-sieve and electroweak source laws,

$$\Gamma_{\rm scr} = \frac{P}{12}\log\frac{N_\star}{\pi}, \qquad \log\frac{E_{\rm cell}}{v} = \frac{\pi}{2\alpha_U(P)}, \qquad \Xi_r(\Gamma_{\rm scr}) = \log\frac{E_{\rm cell}}{v},$$

it gives the conditional Higgs/electroweak residual and coordinate

$$\boxed{R_{\rm EW} = \alpha_U(P)\log\frac{N_\star}{\pi} - \frac{6\pi}{P} = 0, \qquad N_{\rm bridge} = \pi\exp\left[\frac{6\pi}{P\alpha_U(P)}\right].}$$

Neither physical identification constructs the public-record map. The independently frozen operational-resolution residual

$$R_\rho = \log M_0 - \frac{\pi}{\rho_{\rm op}^2}$$

is likewise a downstream test and never a producer definition.

The finite global-section, correctable-code, compound-capacity, approximate-stability, carrier-bound, scalarization, order, extension, refinement, and sewing implications are theorems. The fixed-$D = 24$ packet realizes its complete terminal fiber and the associated atom, publicness, joint-kernel, carrier, scalarization, extension, and refinement controls. Physical closure requires a complete capacity-indexed lift of those objects, including the A2 meaning maps, A3 feasible sets, and regulator controls; an exact finite-size slack law with one physical zero; the universe-level carrier attachment; the horizon–record identification; and the common screen/electroweak load-carrier identification. No QCD or hadronic backend is a dependency of this program.

Outer/inner self-consistency for the pixel ratio. This synthesis paper is the canonical place where the local pixel ratio P is defined, so the compact and particle papers can cite one derivation rather than duplicate it. The construction has six steps.

1. **Outer pixel ratio.** The outer-side local screen datum is the dimensionless area ratio

$$P := \frac{a_{\rm cell}}{\ell_\star^2}.$$

Here $\ell_\star^2 = 3\pi/B_\star$ is supplied by the selected scale certificate. After that scale is emitted, it is displayed in SI language as the Planck area.

2. **Exact equilibrium benchmark.** The total/bulk/edge hierarchy has one exact self-similar balance point. Writing

$$x(C) := \frac{S_{\rm gen}(C)}{S_{\rm bulk}(C)} = 1 + \frac{\langle L_C\rangle}{S_{\rm bulk}(C)},$$

the self-similar balance condition

$$\frac{S_{\rm gen}(C)}{S_{\rm bulk}(C)} = \frac{S_{\rm bulk}(C)}{\langle L_C\rangle}$$

gives $x^2 - x - 1 = 0$ and hence $x = \varphi = (1+\sqrt{5})/2$. The realized branch must sit near, but not exactly at, this point because exact equilibrium is too symmetric to support durable records, structure, and dynamics.

3. **Outer detuning variable.** On the declared closure surface we parametrize the outer-side detuning by

$$\alpha_{\rm ext}(P) := \frac{P-\varphi}{\sqrt{\pi}}, \qquad \text{equivalently} \qquad P = \varphi + \alpha_{\rm ext}(P)\sqrt{\pi}.$$

4. **Declared quantitative anchor on the trial domain.** Let $I \subset \mathbb{R}_+$ be the declared branch interval and let
$$F_{\mathrm{EW}} : I \longrightarrow \mathcal{D}_{\mathrm{EW}}$$
denote the declared electroweak source map. For any trial $P \in I$, the forward electroweak branch emits
$$P \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\mathrm{tr}}(P)) \longmapsto (t_2(P), t_3(P), v(P)) \longmapsto A_Z(P) = \alpha_{\mathrm{em}}^{-1}(m_Z^2; P).$$

5. **Ward-projected Thomson transport.** Let T_Q be the required Ward-projected $\mathrm{U}(1)_Q$ transport operator from the m_Z-anchor to the Thomson limit. The complete source-only operator is not emitted. A comparison branch may insert the measured endpoint $\alpha^{-1}(0) = 137.035999177(21)$, while the separate empirical branch carries the published hadronic compilation and returns 136.3827548175 on $[136.3670480603, 136.3984651934]$. For either declared branch, define
$$A_T(P) := T_Q(A_Z(P), F_{\mathrm{EW}}(P)) = \alpha_{\mathrm{Th}}^{-1}(P).$$
The inner coupling used in the outer/inner closure is the coupling, not the inverse coupling:
$$\alpha_{\mathrm{in}}(P) := \frac{1}{A_T(P)}.$$

6. **Conditional closure.** If the typed same-quantity bridge, target-independent map-selection premise, and same-scheme endpoint transport hold, the physical pixel ratio is the fixed point for which the outer detuning equals the inner observation coupling:
$$\alpha_{\mathrm{ext}}(P) = \alpha_{\mathrm{in}}(P).$$
Equivalently,
$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)} = 0, \qquad \text{or} \qquad P = \varphi + \frac{\sqrt{\pi}}{A_T(P)} = \varphi + \alpha_{\mathrm{in}}(P)\sqrt{\pi}.$$

Writing
$$G(P) := \varphi + \frac{\sqrt{\pi}}{A_T(P)},$$
the closure problem is the fixed-point condition $G(P) = P$. On any declared branch interval where this induced map is a self-map and a contraction, its mathematical fixed point is locally unique. Equivalently, the solver can use
$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}$$
and accept the unique zero of H on the declared branch interval. Observer-facing data may localize that interval. The measured endpoint cannot replace the closure solve.

Conditional on the typed bridge, the closure interpretation assigns one screen cell two coupled roles. On the outer side it is one cell of the screen whose displacement above the exact self-similar equilibrium φ sets the geometric detuning. Conditional on a typed bridge identifying the returned electromagnetic coupling as another reading of that cell, closure requires the outer detuning and inner reading to agree. The pixel ratio P is the size of that cell in units of the scale-certificate area $\ell_\star^2$. The fixed-point equation is the mathematical form of that conditional consistency statement.

Neither the exact fixed-point mathematics nor the source certificates construct the bridge or select either certified P map. On the measured-endpoint comparison branch, the root is

$$P_C \simeq 1.6309682094, \qquad \alpha^{-1}(0) = 137.035999177(21).$$

This is a comparison coordinate inferred from the measured endpoint. It does not define the pixel used by the theory. The declared source map has one mathematical fixed point. The same-quantity bridge and target-independent map selection would promote it to a physical closure point. The difference between the source-map root and the comparison coordinate is the loop residual: 3.0×10^{-4} relative on the source chain and 2.5×10^{-6} relative (about 1.6×10^4 measurement sigma) for the certified self-consistent gauge-width fixed point $\alpha^{-1} = 137.035660136946577\ldots$, a certified fixed point of the declared map with no supplied Ward-projected hadronic transport, map selection, or same-quantity bridge. Distinct coordinates describe the comparison. The certified source/root witness is $136.994835177413\ldots$. The mixed-provenance no-hadron diagnostic is $137.0359595136\ldots$; it mixes the inner value at P_{fwd} with α_U at the comparison pixel, is no fixed point, and is excluded from physical output. The external-data empirical closure is 136.3827548175 on $[136.3670480603, 136.3984651934]$, and the measured endpoint is 137.035999177(21). The empirical interval requires a same-scheme correction in $[0.6198609041, 0.6505569679]$ inverse-alpha units, and the standard on-shell reference deficit 0.631 lies inside that certified interval. A zero-momentum laboratory measurement sees the dressed $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme endpoint matching. A physical source-only conclusion requires the missing source-derived hadronic spectral transport and its dependency certificate, target-independent map selection, and the typed same-quantity bridge.

The source-side trunk, mixed diagnostic, empirical endpoint, measurement, and interval have distinct provenance. Downstream quantities inherit the provenance of whichever declared pixel branch they use. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. The same closure equation also admits an inverse observational use case in which observers infer P and total screen capacity from inside the world.

One source-only route studies the closure directly at zero momentum. It requires a self-contained hadronic transport law rather than the external empirical compilation used by the comparison.

Non-hadron results. The non-hadron results consist of conditional classical carrier modes without quantum mass claims, the exact electroweak selected-carrier chart, the conditional value-law implication, the distinct reference-fitted inverse adapter, the conditional downstream Higgs/top coordinate, a same-family charged witness, and the common-scale rejection of the reciprocal-ray quark texture. The generic quark interface has six scalar coordinates, and no source-derived flavor-orbit selector is emitted. The rejected weighted-cycle neutrino candidate is a comparison-only falsification result. Ref. [19] gives the exact values, premises, diagnostics, and sector boundaries.

Extended derivation. The structural-to-family route and the complete per-sector theorem boundaries are developed in *Deriving the Particle Zoo from Observer Consistency* [19].

6 Screen Microphysics, Records, and Observer Continuation

The screen-microphysics branch gives a fixed-cutoff engineering realization. It asks whether some microscopic model might exist in principle and gives one explicit regulated carrier architecture: a federation of finite observer patches with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap data, records, repair instruments, and observer-facing interfaces all made concrete at fixed cutoff. A spherical screen is used there only as an observer-facing regulator chart for support-visible cuts, not as a claim that the universe is a literal spherical quantum computer.

The branch has five theorem-bearing components. First, the reference architecture itself is explicit. Second, the regulated patch-net embedding theorem applies on its fixed-cutoff object/local-interface domain, and the edge-sector thermal/Casimir branch carries an explicit fixed-cutoff law with a separate compact lift. Third, the fixed-cutoff measurement/Born-rule theorem applies on its declared operator and record-event surface. Fourth, the fixed-cutoff Bell/CHSH theorem applies on the declared two-wing surface. Fifth, the checkpoint/restoration observer theorem applies. The additional premises are packet-level quotient closure where one wants an autonomous exported dynamics, compact-group/Peter–Weyl bookkeeping on the electroweak and compact surfaces, fair-block contraction certificates for long-run noisy approximate consensus on exported packet nets, and model-family universality. The support-visible BW / geometric-modular / local-Einstein closure lives on the companion recovered-core theorems rather than on this fixed-cutoff engineering branch. A reconstruction of Hilbert, C^*- or von Neumann algebra structure, Born probabilities, trace structure, and entropy from operational records alone is a separate question outside this branch.

Reference-architecture takeaway. The synthesis-level point is that the regulated screen-side architecture supplies a concrete habitat in which local observables, overlap observables, record registers, and synchronization maps can all be written without pretending to identify a unique final UV completion. Hardware evidence enters only through a public, hash-stable evidence bundle. The OPH-FPE simulator provides finite-run receipts and replayable experiment bundles [24]. Its public large-run archive contains a resolved level-six icosahedral configuration with 81,920 patch rows, 122,880 seams, twelve local slots per row, and full patch-state custody. Three slots per row route the cell adjacencies and nine remain exposed or reserved. A precommitted array gives each row one distinct immutable authority value. The larger-authority endpoint is preserved on every inconsistent seam. The 102,415 enabled repairs each lower the mismatch count by one, and sixteen shuffled replays agree on one authority-bound terminal hash. A standalone verifier reconstructs S_3, the exact terminal state, and every claim-bearing hash from the archived primitive arrays without simulator imports. The result is conditional on the authority-decorated source. Its axiomatic selection and refinement naturality remain open, and its 2,048 observer neighborhoods provide custody without record-driven feedback.

A separate archive supplies literal feedback on eight twelve-port carriers. It contains eight full integer records and 96 probe/read/write events. The record-conditioned write restores each probed coordinate exactly; removing feedback leaves the probe displacement, and changing one read record coordinate changes the later write. A producer-free verifier recomputes the events, 720 A_5 covariance squares, 96 idempotence checks, and 528 disjoint commutations. Integration with the large run, axiomatic selection of the signed source law, and physical realization remain open.

Imported conditional theorem from Ref. [18] (Fixed-cutoff record algebra and declared Born–Lüders package). For each completed compare/write/verify slice of the regulated microphysics, the declared pointer and overlap-sector projectors generate a finite commutative central record algebra

$$\mathcal{Z}_{\mathrm{rec}}.$$

Given the supplied density-state valuation and declared Lüders instrument, every observer-accessible event E in that algebra has probability

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E),$$

and the selected conditioning on E gives the operational post-measurement state

$$\rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

The effect projectors do not determine this instrument; same-effect non-Lüders instruments remain possible. If a practical readout instead uses projectors Q_a with commuting central reference projectors $\widehat{Q}_a$ on the same declared slots and

$$\delta_{\mathrm{rec}} = \max_a \|Q_a - \widehat{Q}_a\|,$$

then

$$\|[Q_a, Q_b]\| \leq 4\,\delta_{\mathrm{rec}},$$

and any accessible-state perturbation $\|\widetilde{\rho} - \rho\|_1 \leq \varepsilon$ changes each declared elementary record-event probability by at most $\varepsilon + \delta_{\mathrm{rec}}$.

The charged-family construction contains one engineered witness for this distinction. Inside a bounded finite patch, eight local register graphs and eight declared paths contain 6,467 matrix units. A noncentral event projection is read into a central two-valued accepted/rejected record, and sixty A_5 charts verify the declared equivariance. The evidence bundle proves that the stipulated fixed-cutoff schema is nonempty. Its register sizes, path automaton, signs, clock, and response are authored, while inert ancillary stabilization does not supply a cofinal physical refinement. The witness therefore connects event readback to a public record at fixed cutoff; it does not select the charged source, recovery law, family attachment, or physical pole.

Imported theorem from Ref. [18] (Fixed-cutoff Bell / CHSH package). On the declared fixed-cutoff physical observable algebra, fix commuting left/right wing subalgebras, central binary setting registers, binary projective readouts on each wing, and a physical source state on the joint wing algebra. Then the compare slice carries the joint law

$$p(a, b \mid x, y) = \mathrm{Tr}\big(\rho_{LR}\, P^{(L)}_{a|x} P^{(R)}_{b|y}\big),$$

the local marginals are independent of the remote setting, the Bell correlators satisfy

$$E(x, y) = \mathrm{Tr}\big(\rho_{LR}\, A_x B_y\big),$$

and the CHSH combination obeys

$$\big|E(0,0) + E(0,1) + E(1,0) - E(1,1)\big| \leq 2\sqrt{2}.$$

If the source family contains an explicit two-qubit branch with the stated Pauli readouts, the same fixed-cutoff surface saturates $2\sqrt{2}$ exactly on that branch. The source-state input is explicit: the theorem derives the Bell law and the Tsirelson bound from the stated fixed-cutoff operator surface, while Bell-pair preparation and two-qubit factor structure are stated branch conditions.

Imported theorem from Ref. [18] (Checkpoint/restoration and observer backup). For an observer patch P_O, let the checkpoint data consist of the observer-facing record algebra, the accessible local state, the future update schedule, and the externally visible overlap interface data. Exact restoration reproduces the full future law of observer-accessible events, while an ε-accurate restoration changes that future law by at most ε in total variation.

Observer-facing conclusion. The combined theorems supply a fixed-cutoff measurement interface, a fixed-cutoff checkpoint/restoration package, and an operational observer-identity criterion on observer-accessible event algebras. Stronger substrate-selection or strange-loop closure claims are outside the recovered theorem package.

Extended derivation. The fixed-cutoff edge-law, measurement, Born-rule, Bell/CHSH, and checkpoint/restoration packages are developed in *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [18].

7 Worldsheet/String Branch

The string/worldsheet result is a theorem-level bridge from the fixed-cutoff edge-sector law to the two-dimensional Yang–Mills partition function, followed by a controlled large-N_{edge} effective-description theorem on the stated branch. The conditional support-visible four-dimensional Yang–Mills form and repair-gap theorems are separate results whose continuum identification and gap transport require their stated receipts.

Imported theorem (Heat-kernel edge-sector bridge). Import the fixed-cutoff same-overlap edge-law package from *Federated Echosahedral Screen Microphysics* [18]. Under the same overlap-gauge realization and local-Gibbs/MaxEnt edge branch, together with the edge-Hamiltonian and one-sided-algebra hypotheses (EH-1)/(EH-2) of Theorem 6.20 (the sector probabilities are the one-sided d_R convention; the closed partition below is the two-sided d_R^2 convention), the edge-sector weights satisfy

$$p_R(t) \propto d_R e^{-tC_2(R)}.$$

Consequently the edge partition function matches the standard two-dimensional Yang-Mills heat-kernel form. Osterwalder–Schrader and Wightman-strength continuum reconstruction for the four-dimensional Euclidean Yang–Mills form and the mass-gap statement require the separate support-visible compact-gauge premises.

Imported proof route. The microphysics source surface carries the fixed-cutoff thermal/Casimir law on the declared overlap interface. Peter–Weyl decomposition [81] then identifies the resulting sum with the standard heat-kernel expression for two-dimensional Yang-Mills on the compact-group refinement lift.

Compact-carrier theorem. Writing the closed edge partition function as

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)},$$

the compact carrier states the exact bridge itself as a named theorem: Peter–Weyl identifies it with the compact-group heat kernel at the identity,

$$Z_{\mathrm{edge}}(t) = K_t(1).$$

The Chapman–Kolmogorov gluing law for K_t is the precise sense in which collar sewing matches the two-dimensional Yang–Mills heat-kernel surface before any large-$N_{\rm edge}$ continuation is invoked. The compact carrier is a partition-identity theorem on its stated branch. The four-dimensional OPH claim identifies the compact-gauge branch with the Euclidean Yang–Mills form and applies the repair-dynamics theorem on that same support-visible branch.

Controlled large-$N_{\rm edge}$ branch. Fix a distinct large-$N_{\rm edge}$ realization, with $N_{\rm edge} \neq N_c = 3$, and the fixed-$\tau$ variable

$$\tau = tN_{\rm edge}$$

on a compact window I. If the resulting edge free energy satisfies the Standard Model gauge paper's large-$N_{\rm edge}$ criterion, with remainder control on I, then the Gross-Taylor rewriting [80] is a controlled theorem-level worldsheet effective description of the edge dynamics on that branch. Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching are outside the recovered core theorem package.

Higher-gauge continuation slot. The genuinely noncentral crossed-module gluing data on cuts provide the natural continuation-level slot for B-field / gerbe data in any string-style reorganization of the OPH edge sector. This is a structural analogy only; it is not an open/closed-string boundary formalism or a critical-string theorem.

8 Theorem Boundaries and Conditional Continuations

The synthesis has the following theorem boundary:

1. The fixed-cutoff collar, higher-gauge, consensus, record, Bell/CHSH, checkpoint, and restoration packages are theorem-bearing on their stated finite-regulator surfaces.

2. One supplied charged-scalar/Maxwell action with positive mass squared and nonnegative quartic coupling has a nonlinear real sector whose trajectories converge at first order to smooth Neumann solutions under uniform refinement and Ritz initialization (Theorem 3.14). On its fixed volume mesh, the full interacting kinetic metric is complete; the declared Laplace–Beltrami operator has a unique self-adjoint closure and neutral strong-domain Gaussian states (Theorem 3.17 and Proposition 3.18). Exact software restoration supports numerical charged histories, and a Jacobi duration recovers model time on regular nonturning solutions at supplied energy (Propositions 3.9 and 3.10). Physical state preparation, calibration and an interacting continuum quantum field theory are separate from these constructions.

3. Physical UV uniqueness is quotient-level and stable under inert ancillas. OPH fixes terminal values on declared physical observables, not a unique microscopic representative.

4. The Lorentz and null-modular chain is closed on the support-visible refinement/scaling branch by the finite cap-normal BW scaling theorem on the geometric subnet. Its two inputs are the finite cap-normal support/flow certificate and an independently complete multiresolution algebra-state comparison package on the same tower. Fixed-collar Markov replacement with carried $r_{\rm FR}$, $\delta^{\rm M}$, and regularized modular remainders, support-readable modular covariance, support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and

independently normalized geometric 2π-KMS convergence make the cap modular automorphism geometric. The compact cap-normal theorem then identifies S^2 with projective future null rays $q(\Omega) = (1, \Omega)$, represents oriented round caps by $n_C = (\cot\alpha, \csc\alpha\,\mathbf{c})$, proves the signed incidence formula, and obtains $n_{gC} = \Lambda_g n_C$ and $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$. This is not a finite-cell Lorentz-invariance claim, a one-shot upgrade from small CMI to exact Markov geometry, a proof that bare finite consensus produces the cap-normal certificate, or a populated/neutral-bulk claim. The absolute Einstein implication additionally requires one source-derived common-domain tower carrying the event, stress, entropy, uniform-asymptotic, universal-coupling, vacuum-reference, and independent-scale premises. No such tower is constructed or certified.

5. Record-conditioned observer-frame estimation in H^3 is a separate conditional theorem: calibrated modular cap responses select a frame value only when they factor through frame-local H^3 data on a compact domain and the finite cap frame has $\alpha > 0$. Exact data identify one frame value; finite noisy data produce a frame ball, and a unique finite value requires a positive residual gap Δ_{loc}. This does not locate an event, populate the event base, derive particle species or stress-energy, or construct chart-blind neutral bulk.

6. The cosmological constant branch has a closed conditional implication layer and a bounded capacity counterfamily. The descended semantic checkpoint packet defines exact public capacity as the independence number of the compound confusability graph at finite integer dimension. Under a faithful carrier representation and whole-fiber scalarization it is deflationary; under a confusability-reflecting capacity embedding, top-down iteration selects its greatest fixed point, while fixed-capacity refinement injections imply eventual exact stabilization. A source-derived packet realizes the finite receipts at $D = 24$ inside its declared finite source category. Reversible, copy-collapse, two-class, and hidden-spectator continuations share base agreement, positivity, the carrier bound, and executable finite controls while giving different exact fixed sets. The exact verdict applies to this bounded completion class. No all-rung membership in the complete A1–A3 capacity-source contract or executable-to-Lean bridge is supplied. Direct N is not evaluable on the incomplete source antecedent. A positive cosmic value requires a complete source antecedent, one physical zero, and a universe-level carrier attachment. The horizon–record identification then gives $\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$, and the common screen/electroweak load-carrier map separately gives the electroweak bridge. A diagonal normal-form count constructs neither the off-diagonal map nor deterministic closure. The operational resolution experiment is an independent test rather than the producer.

7. Complete reversible response and endogenous overlap transport force the abstract local Standard Model gauge Lie algebra. Under the explicit inverse-port contract, incidence and target-blind port readback derive the signed response. Under the conditional matrix current and matter representation, anomaly-forced determinant balance and tensor descent give exact hypercharge, $N_c = 3$, the common $\mathbb{Z}_6$ kernel, and the maximal faithful matter image. The source does not select the matrix current, matter action, or physical $\mathbb{Z}_6$ quotient. Laboratory identification of the current and flux sectors is not supplied. Compact gauge reconstruction uses the theorem-produced combined transportability criterion and fixed-cutoff tensor-generated bosonic sector category. On a cofinal tail carrying the explicit compact-gauge refinement receipt, the bosonic refinement ladder and compatible forgetful fibers reconstruct a compact group from sectors with strictifiable central or higher-associator defects and at least one allowed trivial-holonomy strict representative. That is a distinct conditional classification route. Physical family identification and extra-sector exclusion require separate identifications. No

laboratory current identification, equality with the Tannaka current, or physical matter-pole and continuum interpretation of the conditional rank-45 family candidate is supplied.

8. The particle branch is sector-split. Electroweak W/Z has an exact selected-carrier chart, a conditional value law whose finite-carrier certificate is not emitted, and a separate reference-fitted inverse adapter. An external strict one-loop fixture has principal-sheet scalar zero-exclusion receipts and one simple scalar zero for each of W and Z in channel-specific lower-half pole boxes and algebraic charts. No certified continuation identity or composition joins it to the OPH chart, so it supplies no physical pole or mass comparison. Higgs/top is conditional on its declared downstream surface. The reciprocal-ray quark candidate fails on dimensionless common-scale Yukawas, while the exact generic interface has six scalar coordinates and no source-derived flavor-orbit selector. Its sub-percent residuals use a target-anchored mixed-convention chart. The particle simulator contains no Yukawa coupling and returns a null calibration result. The weighted-cycle neutrino candidate is rejected, no charged-lepton source attachment is supplied despite the engineered fixed-cutoff record witness, and source-only hadron masses require a production OPH backend. The empirical hadron endpoint is an external-data validation surface rather than that backend.

9. The string/worldsheet branch is a controlled heat-kernel effective-description branch. A critical-string conclusion requires a separate selector premise.

Local unification boundary. One local unification identity is explicit. There is no common theorem for c, G, W, Z, and H. The declared local input P governs the electroweak/Higgs trunk and fixes the matching between cell area and edge entropy. The gravity normalization is emitted by the selected scale certificate, while the invariant causal speed c is structural from the Lorentz branch and gains its SI label only on the local readout package.

Quantity	Value	Derivation chain	Caveat
$c_\star$	299792458 m/s by SI definition	structural Lorentz/BW branch $\to$ common invariant null cone/speed $\to$ SI unit convention	the common cone and dimensionless speed equality are structural; the decimal magnitude is not predicted
G	$6.674299995910528 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ on the stated local extension surface	$\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, $B_\star = 3\pi/\ell_\star^2$, $a_{\mathrm{cell}} = P\ell_\star^2$, $\bar{\ell}_{\mathrm{shared}} = P/4$, $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$	the pixel fixes the shared cell/edge identity and cancels in the Newton area law; the strict classical-regime clause is explicit
M_W	no nonzero source-only physical mass emitted	selected carrier $\to$ exact chart; quotient-transport assumptions $\Rightarrow$ conditional value law; inverse target adapter separate	the finite-carrier certificate is not emitted, and no pole mass is identified
M_Z	no nonzero source-only physical mass emitted	same chart, conditional law, and separate adapter	same source-certificate and pole-mass boundary
M_H	no nonzero source-only physical mass emitted	double-criticality family from the gauge sector $\to$ frozen boundary-scale candidate $(m_H, m_t) = (125.77, 172.63)$ GeV	inherits the boundary-scale selection theorem plus the source-root, physical-scale, running, matching, rigidity, provenance, uncertainty, and complex-pole premises

Conditional electroweak comparison coordinates. The evaluated strictest target-free W/Z branch is a declared, not source-selected zero-selector map, $(80.330, 91.119)$ GeV, with no target input and a single discrete two-law choice. This is a running/chart coordinate, not a physical pole-mass prediction. The source does not close the renormalized-vev, tadpole, field-content, threshold, matching, running, complex-pole, or theory-covariance map needed to compare it with measured resonance parameters, so no physical comparison is defined. For the PDG-2026 reference pair, the exact convention map gives energy-pole masses $(M_W, M_Z) = (80.3411410, 91.1623040)$ GeV, while the separate legacy coordinates $\sqrt{\mathrm{Re}\, s}$ are $(80.3340218, 91.1537725)$ GeV. The running/chart coordinates have no OPH theory covariance, and no physical readout contract is supplied. A separate strict one-loop fixture has principal-sheet scalar zero-exclusion receipts and one simple scalar zero for each of W and Z in channel-specific lower-half pole boxes and algebraic charts. No certified continuation identity or composition joins that fixture to the OPH chart. The selected carrier, conditional value law, and reference-fitted inverse adapter give the downstream comparison coordinates

$$\begin{aligned} \text{selected carrier} \qquad & (80.3862916924, 91.1829044467)\ \text{GeV}, \\ \text{conditional value law} \qquad & (80.3770000154, 91.1879780779)\ \text{GeV}, \\ \text{reference-fitted inverse adapter} \qquad & (80.3625, 91.1879)\ \text{GeV}. \end{aligned}$$

The Higgs/top pair is read on the double-criticality branch, whose frozen boundary-scale candidate $E_\star e^{-\pi} P^{-1/6}$ gives $(m_H, m_t) = (125.77,\ 172.63)$ GeV at two loops, with $m_H = 125.72$ GeV on the fit-free curve at the measured top; the declared calibration surface (125.1995304097, 172.3523553288) GeV is a target-anchored fit. These coordinates are comparison-only and are not physical particle outputs.

The local finite package supplies the strict classical-regime clause and the exact selected-carrier chart. The electroweak value law is conditional on the finite quotient-transport certificate. The familiar-unit display split on that same surface is

$$L_{\mathrm{loc}} = \sqrt{a_{\mathrm{cell}}}\, \widehat{L}(P), \qquad t_{\mathrm{loc}} = \frac{\sqrt{a_{\mathrm{cell}}}}{c}\, \widehat{T}(P),$$

$$E_{\mathrm{loc}} = \frac{\hbar c}{\sqrt{a_{\mathrm{cell}}}}\, \widehat{E}(P), \qquad \Theta_{\mathrm{loc}} = \frac{\hbar c}{k_B \sqrt{a_{\mathrm{cell}}}}\, \widehat{\Theta}(P),$$

with dimensionless $\widehat{L}, \widehat{T}, \widehat{E}, \widehat{\Theta}$. The gravity-side shared-edge identity fixes the cancellation of P, while the local SI readout comes from $\ell_\star^2$. Meters, seconds, GeV, and Kelvin are downstream display conventions built from the structural c row and the familiar constants $\hbar$, k_B.

Companion papers. Detailed derivations appear in Refs. [15, 16, 17, 18, 19, 21, 22].

9 Comparison with other unification approaches

The case for OPH is the explanatory reach of its observer architecture. A useful comparison records the starting objects, the physical structures forced by them, and the experiments that could distinguish the resulting models. Mathematical consistency, recovery of known physics and a successful prospective prediction answer different questions. The constructions below illustrate these distinctions without assigning a numerical probability of physical truth to a research program.

String theory and holographic duality. String theory has concrete results that a unification proposal must take seriously. Strominger and Vafa count microscopic states reproducing the area entropy of a class of five-dimensional extremal black holes [96]. Maldacena's construction relates specified superconformal theories to anti-de Sitter gravity and proposes their full duality [97]. Thus a fixed bulk spacetime is not a universal primitive of string-based descriptions. OPH asks a different reconstruction question: which public records, symmetries and effective physical laws follow from bounded patches and their agreement constraints? Its finite gauge result fixes a Lie type under complete response and endogenous transport. That result must be compared with a specified string construction, including its compactification and matter inputs, rather than with an unrestricted menu of vacua.

Loop quantum gravity. The canonical loop construction quantizes general relativity using connection variables on a differentiable manifold, without choosing a fixed background metric [98]. Discrete area and volume operators are substantive quantum-geometric results [99]. OPH's finite antecedent instead starts with patch records and authenticated causal dependence. Its rank-three response quotient and a real axis give an ambient Lorentz carrier; an observer-facing spacetime requires a physically identified refinement family. A meaningful comparison concerns the controlled recovery of geometry and matter on each construction's stated domain. A finite Lorentz carrier alone cannot settle that comparison.

Wolfram's rewriting approach. Hypergraph and multiway rewriting supply executable discrete structures, and observers are explicitly treated as systems within those structures [100]. Gorard, Namuduri and Arsiwalla embed quantum ZX-diagram rewrites into multiway systems and prove compatibility of the associated monoidal structures [101]. OPH's specific construction uses bounded self-reading patches, protected records and typed overlap repair. Its confluence theorem makes accepted public readouts independent of authorized repair scheduling, while authenticated read-from relations define record order. The comparison therefore concerns these exact observer and update semantics and their physical realization, rather than whether either approach mentions observers or performs computation.

Geometric Unity. Eric Weinstein's 2021 draft starts its Einsteinian construction with a smooth four-manifold and the bundle of metrics over it, using differential geometry to organize gravitational and gauge data [102]. Its observer maps are part of that geometric construction. OPH's primitive observer carries bounded local state, a boundary and writable records, with geometric readouts attached through additional maps. These are different starting objects. Their physical content must be compared after specifying dynamics, quantization and the measured observables represented by each construction.

Hopf fibrations and geometric state representations. A Hopf fibration is a mathematical construction used in several physical settings. For example, Mosseri and Dandoloff use it to represent qubit phases and two-qubit entanglement [103]. It does not name a unique competing theory of everything. OPH also uses familiar bundle and representation geometry. The substantive question is which dynamics, probabilities and physical identifications a particular use of that geometry derives. A symmetry or fibration identity alone supplies no laboratory law.

Causal and operational reconstruction. Causal-set theory starts from locally finite order [1, 2]; OPH's authenticated record relation constructs one finite informational order from specified transactions. Both physical readings require a justified relation between order, number and continuum geometry. Operational quantum reconstructions and entanglement-based gravity also share parts of OPH's mathematical vocabulary. Their results are credited in the relevant derivations. Finite discreteness, relational observables and emergent geometry are therefore shared research ideas; the OPH contribution is the particular construction joining them and the implications it proves.

Concrete reasons to investigate OPH. Protected overlap agreement gives an explicit criterion for a public fact and, under termination, local-diamond and completeness premises, a unique normal form from each initial quotient state. The same architecture constrains internal symmetry: complete reversible response and endogenous transport force the local Standard Model gauge Lie algebra. The fermionic matter representation and its charge normalization are additional inputs to the conditional hypercharge and common $\mathbb{Z}_6$ kernel theorem. Lie type, matter content and physical current identification are separately checkable statements.

One supplied charged-scalar/Maxwell action also connects finite dynamics to ordinary calculational objects. Its invariant real zero-current sector has first-order spatial trajectory convergence for smooth Neumann references and Ritz initial data with positive mass squared and nonnegative quartic coupling. At fixed mesh and nonzero charge, the full reduced kinetic metric is complete; the declared Laplace–Beltrami quantization has a unique self-adjoint closure and a neutral Hilbert sector. These results concern the same supplied action. Their physical couplings, preparation, geometry and quantization prescription retain their stated input status.

The gravity construction gives an explicit implication from realized null balance, stress conservation and continuum control to the Einstein equation up to a cosmological constant. Finite null tomography isolates the metric-proportional ambiguity. Subsystem boundaries are treated with algebras and edge sectors, a problem also addressed directly in gauge theory and gravity [104]. Small conditional mutual information controls specified recovery estimates. Local modular stress, alignment and refinement require the additional hypotheses printed in the corresponding theorems; small conditional mutual information alone does not supply them.

Proofs, exact certificates and independent replay make these implications inspectable. The primitive-port dispersion branch adds a prospective test: its linked coefficients and first anisotropic harmonic are fixed before an eligible comparison. A resolved violation rejects that physical branch; extension to the whole framework requires the branch to be forced and exclusive. Target-informed closure diagnostics retain a different evidential role. Theorem counts and numerical proximity do not substitute for this test.

The resulting argument for OPH is constructive: a specified observer architecture supports linked reconstruction results and explicit ways to check them. Demonstrating one physical realization carrying the records, geometry, matter and quantum predictions is the decisive test of the proposed unification. The standard applies equally to every candidate.

10 Conclusion

The three-axiom basis, gravity and gauge branches, consensus formulation, particle-spectrum calculation, regulated screen microphysics, fixed-cutoff measurement and observer results, and controlled large-N_{edge} worldsheet effective description fit into one typed construction.

The finite analytic core establishes schedule-independent consensus on the declared quotient, the conditional relativity and Einstein branches, and the local Standard Model gauge Lie algebra. The finite matrix current, matter image, and common $\mathbb{Z}_6$ kernel are exact conditional realizations. The fixed-cutoff microphysics gives explicit record, measurement, Bell, checkpoint, and restoration interfaces. The worldsheet result is a controlled heat-kernel effective description under its stated large-N_{edge} premises.

The numerical maps have no continuously tunable fit dials. Interval arithmetic proves that each declared fixed-point map has exactly one fixed point on its stated domain. The declared finite candidate family comprises the seventeen-row menu for the executed capacity families, the two declared P maps, one direct fixed-cutoff control, the common-load baseline, reserve maps, RC-LOAD, the existing construction aggregate, and five hierarchy packets. Fifteen of the thirty entries have certified unique fixed points. Across these thirty entries, none supplies target-independent selection, same-quantity construction, and the complete source return map together. This finite family does not exhaust other source laws. Physical conclusions require target-independent map selection, the relevant same-quantity bridges, the Ward-projected hadronic transport, the capacity readback map F, the forward electroweak source chain, a source-derived flavor-orbit selector, and the corresponding pole, scale, family, and nonperturbative attachments. Strong CP, full flavor and CKM closure, charged-lepton source landing, and source-only hadron masses are not supplied.

Refs. [15, 16, 17, 18, 19] give the sector-by-sector proofs and quantitative details. The shared numerical surface is concise: the fixed point $P_\star$ organizes the bosonic mass trunk and the gravity-side shared edge-entropy identity beneath the emitted local G row, while the same familiar-unit package reads meters and seconds from the single ruler $\sqrt{a_{\text{cell}}}$ together with the structural invariant speed c, and reads GeV and Kelvin from the inverse local ruler through $\hbar$ and k_B.

A Candidate Microphysics Finite Reference Checks

The synthesis paper records the finite echosahedral reference-model checks in an appendix. The reference object is a typed carrier/federation data structure, not a physical federation source and not a realization of observer patches, overlap observables, or a support S^2. Its verified content is finite and structural: one shared twelve-port 12/30/20 carrier template with antipode and A_5 action, typed collar bijections, orientation reversal, canonical finite matrix-algebra maps on individual seams, external-boundary coverage, presentation relabeling invariance, and a finite reference-tower commutation check.

Proposition (Finite carrier and seam conformance). For the declared finite reference objects, the checker recomputes local carrier conformance, seam bijectivity and orientation, endpoint algebra-schema agreement, the permutation-induced matrix-algebra isomorphism on each seam, boundary coverage, and the presentation firewall. The two-carrier reference fixture has no composable seam triangle. Its higher-overlap Čech condition is therefore true only vacuously, and its nonvacuous higher-overlap witness is false. A separate coherent-triangle control exercises the cocycle checker, but deliberately fails the positive federation-sewing conditions and cannot be combined with the reference fixture as a physical witness.

Consequently, the physical echosahedral-federation realization and source-instrument premises are false. The level-zero through level-two naturality calculation is a finite diagram for the shared regular-icosahedron template; it selects no physical support chart and supplies no carrier-to-support S^2, H^3, event, BW, or KMS receipt.

Proof sketch. Each positive finite statement follows by exact recomputation on the declared template, seam maps, canonical matrix schemas, boundary declarations, and reference refinement maps. The absent composable seam triangle prevents a nonvacuous higher-overlap witness. No measured physical carrier-to-patch identification or quotient-visible spherical-support map is present. The physical federation, source, and support-screen premises are therefore not satisfied.

Proposition (Fixed-cutoff edge heat-kernel calculation). For a separately declared finite sector kernel satisfying weighted proposal symmetry and the Casimir acceptance rule, the stationary sector law is

$$\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)},$$

and along a separately assumed refinement ladder converging to a compact-group Peter–Weyl decomposition [81] the same weights converge cylinderwise to the heat-kernel/Casimir law used in the OPH edge-sector branch.

Proof sketch. The finite proposal kernel is chosen so that the weighted proposal symmetry $d_\alpha q_{\alpha\beta} = d_\beta q_{\beta\alpha}$ holds. Detailed balance with the Casimir acceptance rule then fixes the stationary distribution in finite dimension. The compact-group heat-kernel law is the conditional refinement lift. Neither calculation supplies the missing physical federation, source-instrument, or support-S^2 receipt.

Extended derivation. The finite candidate model, validation package, and conditional theorems are developed in *Federated Echosahedral Screen Microphysics* [18].

B Interpretive Epilogue: State-and-Law Habitat

This appendix records the OPH state-and-law habitat available inside the framework. A strange-loop closure theorem is outside this appendix.

The OPH inputs used here are:

1. the patch net $P \mapsto A(P)$ of von Neumann algebras together with isotony and overlap restriction maps;

2. a compatible local state family, plus the separately declared global-extension interface used by this appendix to represent it on the inductive-limit algebra;

3. the finite Axiom-3 information-projection branch and the separate optimizer-pushforward interface, which together supply a common finite family of gauge-invariant local constraint observables across cutoffs;

4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained A1-generated constraint family used by Axiom 3. That choice furnishes a compact convex state-and-law habitat for the framework.

Theorem (Internal state-and-law habitat theorem). Fix finitely many screen patches $P_1, \dots, P_N$ in the standard OPH setup. For each i, let

$$M_i := A(P_i), \qquad M_{ij} := A(P_i \cap P_j),$$

where M_i is the patch von Neumann algebra and $M_{ij} \subset M_i, M_j$ is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of M_i, endowed with the weak* topology $\sigma(M_i^*, M_i)$.

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the A1-generated Axiom-3 constraint grammar, with each O_a supported in one of the chosen patches; write $i(a)$ for an index such that $O_a \in M_{i(a)}$.

Define the overlap-consistent state sector

$$X_{\mathrm{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^{N} S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\mathrm{ov}} \to \mathbb{R}^m, \qquad c(\omega_1, \ldots, \omega_N) := (\omega_{i(1)}(O_1), \ldots, \omega_{i(m)}(O_m)).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \ldots, \omega_N, \ell) \in \left(\prod_{i=1}^{N} S_i \right) \times \mathbb{R}^m : (\omega_1, \ldots, \omega_N) \in X_{\mathrm{ov}},\ \ell = c(\omega_1, \ldots, \omega_N) \right\}.$$

Then:

1. the ambient product
$$E := \left(\prod_{i=1}^{N} M_i^* \right) \times \mathbb{R}^m$$
is a Banach space for any product norm;
2. $X_{\mathcal{O}}$ is nonempty and convex;
3. $X_{\mathcal{O}}$ is norm-closed in E;
4. $X_{\mathcal{O}}$ is compact in the product topology
$$\tau := \left(\prod_{i=1}^{N} \sigma(M_i^*, M_i) \right) \times \text{Euclidean topology on } \mathbb{R}^m;$$

Proof. Each M_i is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual M_i^* is Banach. A finite product of Banach spaces is Banach, hence so is E.

For each i, the state space

$$S_i = \{\omega \in M_i^* : \omega \geq 0,\ \omega(1) = 1\}$$

is convex and weak* compact by Banach–Alaoglu, because it is a weak* closed subset of the dual unit ball. For every pair (i, j), restriction along $M_{ij} \subset M_i$ and $M_{ij} \subset M_j$ defines affine weak* continuous maps

$$r_{ij} : S_i \to S(M_{ij}), \qquad r_{ji} : S_j \to S(M_{ij}).$$

Hence X_{ov} is an intersection of affine equalizer sets, so it is convex and τ-closed inside $\prod_i S_i$. Since $\prod_i S_i$ is compact and convex in the product weak* topology, X_{ov} is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions

$$(\omega|_{M_1}, \ldots, \omega|_{M_N})$$

belong to X_{ov} because the restrictions agree on every overlap by construction.

Each coordinate $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$ is affine and weak* continuous, since evaluation at a fixed algebra element is weak* continuous. Thus c is affine and τ-continuous. The graph map

$$G : X_{\mathrm{ov}} \to E, \qquad G(x) := (x, c(x)),$$

is affine and continuous, so its image $X_{\mathcal{O}} = G(X_{\mathrm{ov}})$ is convex and τ-compact.

To see norm-closedness in E, let $(x_n, c(x_n)) \to (x, \ell)$ in norm. Norm convergence implies weak* convergence on each coordinate, so $x \in X_{\mathrm{ov}}$ because X_{ov} is weak* closed. Since c is weak* continuous, $c(x_n) \to c(x)$, while norm convergence in $\mathbb{R}^m$ also gives $c(x_n) \to \ell$. Hence $\ell = c(x)$, so $(x, \ell) \in X_{\mathcal{O}}$.

Finally, $X_{\mathcal{O}}$ is a compact convex subset of the locally convex topological vector space E equipped with τ. $\square$

The additional inputs needed for a strange-loop closure theorem are:

1. a closure map T built from internal operations such as overlap repair, Axiom-3 information reprojection, an explicit collar-recovery interface, and record-sector coarse-graining;
2. a proof that the stronger "observer-supporting" or "selected-world" criteria carve out a nonempty T-invariant observer-supporting subset of $X_{\mathcal{O}}$;
3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem supplies only the ambient Banach/compact-convex setting. It does not define a canonical internal self-map, it does not identify any nonempty invariant observer-supporting subset, and it does not prove uniqueness or stability. In particular, nonemptiness and compact-convexity of the ambient habitat do not by themselves imply the existence of a nonempty invariant observer-supporting sector. This does not erase the exact finite packet branch proved on the consensus surface: there the quotient normal-form map pushes forward to a continuous affine idempotent self-map of the finite packet simplex, with fixed points exactly the packets supported on quotient normal forms. Any further strange-loop reading of the ambient habitat as a closure story for existence is interpretive only and is not part of the recovered theorem package.

B.1 Additional Conditional Results

Using the theorem classification of Section 3, the broader claims read as follows. "Declared-branch corollary" means proved under the named branch conditions and cited companion results.

Claim	Classification	Boundary
Quantum gravity consistency	Conditional theorem under explicit receipts	The Einstein relation follows from one source-derived common-domain tower carrying the geometric modular, null, event, entropy, certified-tail, coupling, vacuum, and scale premises. No such tower is constructed or certified.
UV completion / microscopic uniqueness	Separate scope boundary	Fixed-cutoff theorem packages exist, but a unique microscopic representative is not identified. The support-visible BW scaling theorem and the receipt-certified realized combined-zero-obstruction bosonic compact-gauge branch (central or higher-associator strictification plus an allowed trivial-holonomy strict representative) hold under their declared premises; the separate boundary is microscopic representative uniqueness rather than a missing recovered-core lift.
Measurement problem	Separate scope boundary	Ref. [18] closes the fixed-cutoff central-record / Born-Lüders interface, but a full philosophical or continuum-level measurement closure is not derived here.
Cosmological principle and horizon homogeneity	Separate scope boundary	MaxEnt symmetry inheritance is suggestive only; an independent derivation of cosmological isotropy and homogeneity sits outside the recovered-core theorems.
Observer-relative modular ordering	Declared-branch corollary	On the BW_{S^2} geometric branch, the cap modular automorphism parameter supplies a dimensionless ordering for the observer's accessible algebra/state pair. Physical time additionally requires an observer-readable transition, event correspondence, and calibrated clock instrument.
Global / operational problem of time	Separate scope boundary	OPH does not prove a separate operational-clock theorem or a global solution of the problem of time.
Black-hole information paradox	Separate scope boundary	Edge-center sectorization and recoverability motivate an internal resolution template. A physical black-hole-information theorem requires the radiation algebra, exterior time calibration, evaporation dynamics, and entropy readout.

Claim	Classification	Boundary
Cosmological-capacity closure	Conditional corollary	Conditional on a complete capacity-source antecedent selecting one physical zero of the finite-size slack, a universe-level carrier attachment, stable whole-fiber closure $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, and horizon–record identification $N_{\mathrm{CRC}} = S_{\mathrm{dS}}$, one gets $\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$. The fixed-$D = 24$ packet is exact inside its declared finite source category. The bounded completion class has incompatible fixed sets. Universal all-rung membership in a complete A1–A3 capacity-source contract and an executable-to-Lean bridge are absent, so the incomplete source antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. Exact positive compositional completions separately show that the declared finite reserve datum selects no global action or blocked-event semantics. The static-patch SI display additionally uses the selected OPH scale certificate. The uncorrected electroweak/Higgs bridge differs from the weighted Planck base-ΛCDM coordinate by 6.6 percent. Separately premised finite-presence and Poisson formulas give retrospective residuals of -0.63 and -0.39 percent. The finite source selects neither formula, and neither comparison establishes the common-carrier or horizon receipt.
Bare cosmological-constant problem from local null data	Separate scope boundary	The local null-data route determines the Einstein branch only modulo Λg_{ab} and does not derive the screen-capacity identification from bare OPH axioms.
Magnetic cocharacter and simple-GUT monopole boundary	Conditional matter-image and global-form arithmetic	Complete reversible response and endogenous overlap transport force the local Standard Model gauge Lie algebra. Incidence and target-blind port readback derive the signed response. Under the conditional matrix current and rank-15 matter representation, the common central kernel is $\mathbb{Z}_6$ and the quotient by it is the maximal faithful image. The cover and its $\mathbb{Z}_2$, $\mathbb{Z}_3$, and $\mathbb{Z}_6$ quotients carry the same local tensors. The six-axis menu matches the $\mathbb{Z}_6$ quotient only after its coefficient relations are declared, so it does not select the physical global form. The cocharacter $(1, 1, 1/6)$ and its electromagnetic multiple apply to the declared image. The Dirac-pairing commutant theorem selects the electric polarization uniquely inside the declared line lattice. Theta periodicity requires four-dimensional instanton-sector data. The product adjoint removes the simple-GUT X/Y production route, not all possible 't Hooft lines.
Gauge-mediated proton-decay boundary	Product-adjoint corollary	The connected product adjoint has no X/Y generator, so the standard simple-GUT channel is absent; general proton stability is not claimed.
Proton spin fraction	Separate scope boundary	The proton-spin claim requires nonperturbative quantum chromodynamics.
Dark matter phenomenology	Separate scope boundary	Modular-anomaly response laws are conjectural beyond the recovered gravity chain.
Baryon asymmetry scale	Separate scope boundary	Suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism.
Three generations	Conditional window and rank-three response candidate	CKM capability and the weak-sector clause give $3 \leq N_g \leq 5$. Under the named band premises, rank three is selected. Tensoring it with the declared generation table gives a conditional rank-45 candidate whose chirality and diagonal $\mathbb{Z}_6$ action come from the table. A separate local-domain receipt checks a declared tensor-identity operator and conditional gap inheritance without selecting the matter action or transporting the Spin packet. Physical matter-pole and continuum identification, extra-sector completeness, and the charged-lepton and electroweak quantitative branches carry their own declared premises.
Downstream flavor structure, Koide, and charged-lepton fits	Separate scope boundary	Companion flavor constructions require extra ansätze and sit outside the recovered core.

Claim	Classification	Boundary
Edge-to-2D-YM / controlled large-N_{edge} worldsheet branch	Declared-branch corollary	On the stated overlap-gauge, local-Gibbs/MaxEnt, compact-group, and large-N_{edge} conditions, the edge-sector heat-kernel bridge and controlled Gross–Taylor worldsheet effective description are theorem-level. This is a two-dimensional partition and worldsheet-effective branch, separate from the four-dimensional compact-gauge theorem.
Support-visible compact-gauge Yang–Mills form and mass gap	Conditional theorem under explicit receipts	The finite compact-gauge cylinder system has a proof-bearing projective weak-* / GNS extraction. Identification with the four-dimensional Euclidean Yang–Mills state and the equality $\Delta_{\text{YM}} = \Delta_{\text{rep}}$ additionally require the renormalized Yang–Mills, finite ground-state-transform/cross-fiber, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts; extraction alone does not transport the gap. The legacy commuting Lean theorem takes positive collar rates directly, with no formal bridge from its separate Lemma 7.2 uniform-fiber scalar result, and its stripped fixed-space witness is not a compact-gauge or continuum construction.
Critical superstring / worldsheet CFT lift	Separate scope boundary	Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching require extra ingredients outside the recovered core.
Why anything exists / strange-loop closure	Separate scope boundary	Appendix B provides only the habitat theorem and fixed-point setting. The OPH closure map, invariant observer-supporting sector, and uniqueness/stability proofs are not part of this theorem package.
Fixed-cutoff checkpoint/restoration/backup	Fixed-cutoff theorem	Ref. [18] proves same-interface checkpoint/restoration and backup for observer-accessible event algebras with explicit future-law error control.
Substrate-transfer / stronger observer continuation	Separate scope boundary	Stronger substrate-selection, redesigned-environment continuation, strange-loop closure, uniqueness, and stability claims are outside this theorem package.

C Observer Continuation and Backup

This appendix uses the fixed-cutoff checkpoint/restoration/error/identity theorem package proved in Ref. [18] and summarizes its algebraic interface in observer language. It does not define a strange-loop closure map on an invariant observer-supporting sector or establish uniqueness and stability for such a map. Nor does it extend the fixed-cutoff theorems to continuation across redesigned environments: the proved package is same-interface restoration on observer-accessible event algebras with explicit error control.

C.1 Observer as Algebraic Pattern

For this appendix, use the support-local algebra-state-record reduct

$$O_{\text{red}} = (P, \mathcal{A}(P), \rho, R),$$

where P is a support-screen patch, $\mathcal{A}(P)$ is its local algebra, ρ is the local state, and R is the record algebra. The full operational observer also requires overlap interface algebras and restriction maps, allowed update or repair instruments, durable checkpointed records, and feedback/readback that can affect subsequent admissible behavior. Records are carried exactly by central record projectors and, on practical readout surfaces, by approximately commuting projectors in overlap centers, so they are shareable without violating no-cloning constraints for generic quantum states.

Ref. [18] proves a conditional fixed-cutoff observer-facing measurement interface: a finite central record algebra, a supplied Born valuation for its event projectors, and a declared Lüders instrument on that same commuting record algebra. The projectors alone do not select the instrument. This appendix uses that fixed-cutoff measurement package in observer language. If a practical readout uses projectors that are δ_{rec}-close in operator norm to that central reference algebra and the accessible state is perturbed by at most ε in trace norm, each declared elementary record-event probability shifts by at most $\varepsilon + \delta_{\mathrm{rec}}$.

C.2 Markov Collar Factorization

For a collar tripartition A-B-D, the exact Markov normal form is used only when $I(A : D|B) = 0$ holds literally, or along a controlled fixed-collar family for which the exact-Markov replacement modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$. In that exact or controlled limit one obtains

$$\rho_{ABD} = \bigoplus_{\alpha} p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D}.$$

The sector label α is classical center data. This decomposition is the mathematical basis for extracting an interior observer state with a controlled boundary interface. Small CMI by itself supplies a recovered comparison state, while an exact Markov state requires the exact or controlled limiting premise stated above.

C.3 Checkpoint and Restoration Map

A checkpoint is the tuple

$$\mathcal{C} = (R, \alpha, \rho^{(\alpha)}_{\mathrm{int}}),$$

with $\rho^{(\alpha)}_{\mathrm{int}}$ the interior state after fixing α. Given a compatible target environment state $\sigma^{(\alpha)}_{\mathrm{env}}$, a restored state is

$$\rho^{(\alpha)}_{\mathrm{new}} = \rho^{(\alpha)}_{\mathrm{int}} \otimes \sigma^{(\alpha)}_{\mathrm{env}}.$$

For approximate Markov collars, recovery is controlled by the standard bound

$$\|\rho_{ABD} - (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})\|_1 \le 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Here ε is measured in nats. This gives quantitative trace-distance control on the recovered comparison state under finite CMI (finite collar error). The exact HJPW splice form instead requires literal exact Markovity or the separate fixed-collar replacement modulus tending to zero, in each case together with the Markov-split alignment hypothesis of Section 2.3 identifying the HJPW factors with the preselected edge factors. Interpreting either control as observer continuation requires additional modeling assumptions beyond the bound itself.

The fixed-cutoff microphysics claim is stronger than the bare recoverability estimate written above. In Ref. [18], exact checkpoint restoration preserves the full future law of observer-accessible events, while an ε-accurate restoration changes that future law by at most ε in total variation. The stronger strange-loop / substrate-transfer closure story is outside this theorem.

C.4 Physical Meaning

At fixed cutoff, Ref. [18] proves a checkpoint/restoration theorem and backup corollary for observer-accessible event algebras. The proved package stops at same-interface backup/restoration with explicit future-law error control; stronger continuation and substrate-selection claims are outside it. This appendix states the observer-facing meaning of that theorem and its algebraic prerequisites. The stronger substrate-selection, strange-loop closure, uniqueness, and stability package is separate from the fixed-cutoff theorem.

D Cosmology, Horizons, and Modular-Anomaly Continuations

This appendix carries the cosmology- and continuation-facing technical statements for the synthesis paper. They do not enlarge the recovered core; they make the structural continuation boundary explicit on the synthesis surface.

D.1 Observer Registry and Clock-Naturality Certificate Contract

For finite exported observer systems, a theorem-faithful evidence bundle separates executor metadata from observer semantics. It contains:

(i) a semantic history DAG or labeled poset $\mathsf{H}_O = (E_O, \preceq_O, \ell_O)$, with event keys computed from semantic payload, observer token, visible footprint, and semantic parents;

(ii) a provenance envelope containing worker ID, repair iteration, retry count, queue position, timestamp, and packet latency, none of which may enter the semantic event key unless declared as physical input;

(iii) local observer registry groupoids with disjoint patch-observer, cap-observer, and future-observer namespaces, explicit birth-event classes, continuation arrows, split/merge lineage arrows, and overlap functors preserving those data;

(iv) a global registry descent certificate: cocycle agreement, trivial monodromy, no duplicate global observer identifiers, no mixed patch/cap/future namespace collisions, and no reuse of local anchor-patch identifiers as local observer indices;

(v) observer-algebra extraction data

$$\mathsf{ObsAlg}(\widehat{q}_{\mathrm{nf}}) = (\mathcal{A}_O, \omega_O, \mathcal{R}_O, \ldots)$$

whose outputs are state-preserving isomorphic for history-equivalent or implementation-equivalent terminal augmented normal forms;

(vi) a support-cap chart map J_O natural with those observer-algebra isomorphisms; and

(vii) a clock instrument Θ_O with affine reparameterization and residual bound for state, calibration, and record-readout errors.

Theorem D.1 (Finite observer-clock certificate soundness)**.** *On a finite exported model, suppose the augmented repair relation on $\widehat{Q} = \{(q, [\mathsf{H}])\}$ terminates, terminal states are exactly the consistent physical states with completed semantic records, every augmented critical pair has a history-coherent join, implementation steps are linearizable to semantic commits or stutters, registry descent passes the groupoid cocycle and monodromy checks, and the observer-algebra/chart/clock maps satisfy the state-preserving and affine-residual certificates above. Then worker count, queue order, retry delivery, scheduler counters, and packet latency that is not declared physical input do not change the observer-readable history class, registry class, or clock law beyond the declared affine clock residual.*

Proof. Termination plus the augmented local diamond gives confluence by Newman's lemma for terminal history classes rather than only physical quotient states. Linearizability says every implementation execution projects to the same semantic relation up to stutters, so executor metadata cannot create new semantic events. Registry descent identifies the same global observer groupoid across local presentations. State-preserving observer-algebra extraction and natural support-cap charts transport the modular flow along the same observer identity, while the clock instrument certificate bounds the operational readout after the declared affine reparameterization. □

Remark D.2 (Falsifiers). *The certificate fails if semantic event identifiers depend on worker IDs, timestamps, queue positions, repair iteration counters, or retries; if duplicate retry delivery creates two visible record commits; if two namespaces collide in the global registry; if registry cocycles or lineage arrows fail on overlaps; if histories from two executors are not labeled-poset isomorphic; or if clock residuals exceed the declared bound. These are public evidence-bundle failures that exceed field-name mismatches.*

D.2 Cosmological Principle

Lemma 4.1 (MaxEnt symmetry inheritance). Let G act on the screen by automorphisms α_g. If the constraint set is G-invariant and von Neumann entropy is G-invariant, then the MaxEnt optimizer ω can be taken G-invariant. If the optimizer is unique, it is automatically G-invariant.

Theorem 4.2 (Isotropy from SO(3)-invariant constraints). Under SO(3)-invariant constraints and MaxEnt uniqueness, the reference state satisfies

$$\omega \circ \alpha_g = \omega \qquad \forall g \in \mathrm{SO}(3).$$

The corresponding stress tensor has perfect-fluid form:

$$\langle T_{ab} \rangle = \rho\, u_a u_b + p(g_{ab} + u_a u_b).$$

Theorem 4.3 (Schur-type homogeneity). Let (Σ, h_{ij}) be a three-dimensional Riemannian manifold. If at every point $p \in \Sigma$, the curvature tensor is SO(3)-invariant, then

$$R_{ijkl}(p) = K(p)(h_{ik}h_{jl} - h_{il}h_{jk}),$$

and the second Bianchi identity forces $\nabla_m K = 0$, so K is constant.

Corollary 4.4. If OPH supplies isotropy for all observers, spatial geometry is a constant-curvature space form: S^3, $\mathbb{R}^3$, or H^3.

Theorem 4.5 (FLRW emergence). With the semiclassical Einstein equation from the declared finite generalized-entropy, stationarity, stress, and continuum interfaces, a perfect-fluid stress tensor from the rotationally invariant information-projection branch, and positive Λ from finite screen capacity, the emergent geometry is FLRW:

$$ds^2 = -dt^2 + a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right),$$

with Friedmann equations

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \qquad \dot{H} - \frac{k}{a^2} = -4\pi G(\rho + p).$$

D.3 Horizon-Problem Control

Theorem 6.1 (Conditional collar-CMI bound). For a finite-range Gibbs tripartition A-B-D with collar width δ, assume the uniform strong conditional matrix-mixing premise of Theorem 2.5. Then

$$I(A : D \mid B) \le c \cdot |\partial C|_{\mathrm{UV}} \cdot e^{-\delta/\xi},$$

with the explicit constants stated there. Ordinary two-point clustering does not supply the premise. On the exact central-interface branch, $I(A : D \mid B) = 0$.

Corollary 6.2. For bounded observable O with $\|O\| \sim 1$,

$$|\Delta\langle O\rangle| \leq 2\sqrt{1-e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}$$

when ε is CMI in nats. On the CMB anisotropy target $\delta T/T \lesssim 10^{-5}$, this gives the familiar ε-scale bound used in the horizon-problem bookkeeping:

$$\varepsilon \lesssim 2.5 \times 10^{-11} \text{ nats}, \qquad \frac{\delta}{\xi} \gtrsim \ln\!\left(\frac{c \cdot |\partial C|_{\mathrm{UV}}}{\varepsilon_{\max}}\right) \approx 24.4 + \ln\!\big(c|\partial C|_{\mathrm{UV}}\big).$$

The number 24.4 is only the unit-prefactor value. The boundary count cannot be dropped. With $\xi \approx \sqrt{P}\,\ell_\star \approx 1.28\,\ell_\star$, where $\ell_\star$ is displayed as the usual Planck length after the selected scale certificate, the corresponding collar condition is

$$\delta_{\mathrm{CMB}} \gtrsim 1.28\,\ell_\star \left[24.4 + \ln\!\big(c|\partial C|_{\mathrm{UV}}\big)\right].$$

This is a state-recovery benchmark for a bounded observable, not a derivation of CMB anisotropy. On a continuum family the separate rate condition $\delta/\xi - \log|\partial C|_{\mathrm{UV}} \to +\infty$ is required.

Theorem 6.3 (Homogeneity as the MaxEnt default). If MaxEnt constraints are uniform across UV cells with no marked points, the MaxEnt state is invariant under triangulation automorphisms. In the continuum limit this gives SO(3) invariance.

Reading rule. The cosmological-principle and horizon-homogeneity package is a branch-local theorem package on the realized symmetric MaxEnt branch. The exact theorem-level content is:

1. SO(3)-invariant constraint data plus MaxEnt uniqueness force an isotropic reference state and a perfect-fluid stress tensor.
2. If the same isotropy condition holds for all observers, the Schur/Bianchi argument forces constant-curvature spatial slices.
3. Combined with the semiclassical Einstein branch and positive Λ from finite screen capacity, the metric takes FLRW form.
4. Markov control supplies an explicit patch-overlap anisotropy bound and the collar benchmark used in the CMB homogeneity bookkeeping.

The package does not prove from the bare OPH axioms alone that the realized cosmological branch must satisfy SO(3)-invariant constraint data, that MaxEnt uniqueness holds in every cosmological sector, or that the observed universe saturates the displayed CMB benchmark.

D.4 Conditional Screen-Spectrum Continuation

Theorem 6.4 (Conditional OPH screen spectrum). Let $(\mathcal{S}_r)$ be a cofinal finite spherical OPH screen system with a schedule-independent quotient-normal-form scalar q_r, removal of the background and dipole sector, and a target-free positive quadratic repair operator K_r. If $K_r \to K$, the source-selected finite scalar release energy satisfies $2E^{\mathrm{src}}_{q,r}/d_r \to A_q$, and local MaxEnt is imposed at fixed expected quadratic release energy, then q_r converges in finite harmonic distributions to the centered Gaussian screen field with covariance $A_q K^{-1}$. Exact per-sample release energy would instead give a microcanonical ellipsoid. If the certified repair-scale measure is $t^{\theta/2}\,dt/\Gamma(1+\theta/2)$, then

$$K_{\mathrm{asy}} = (-\Delta_{S^2})^{1+\theta/2}, \qquad C^q_{\ell,\mathrm{asy}} = A_q[\ell(\ell+1)]^{-1-\theta/2},$$

as the large-ℓ asymptotic model. The theorem-grade finite-ℓ conformal-shell family uses the normalized gamma-ratio precision

$$\kappa_\ell(\theta) = \frac{\Gamma(\ell+2+\theta/2)}{\Gamma(\ell-\theta/2)},$$

with

$$C^q_\ell = A_q \frac{\Gamma(\ell-\theta/2)}{\Gamma(\ell+2+\theta/2)}.$$

If q is additionally certified as the thin-shell pullback of a homogeneous curvature field with $\Delta^2_\zeta(k) = A_\zeta(k/k_\star)^{-\theta}$, this same operator supplies the exact finite-ℓ lift; the pure fractional Laplacian is only the asymptotic scaffold. Suppose the same source construction emits a strongly continuous full-collar survival cocycle with generator density $P_\star/24$, together with the orientation-reversal half-collar identity. Then

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}, \qquad \kappa^{\mathrm{edge}}_{\mathrm{rep}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

A finite one-step survival value determines θ through $-\log u_q(\log b)/\log b$; it does not supply the infinitesimal generator receipt. If a scale-natural source embedding transports this cocycle to the physical covariance through $D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta$, the source family is $\Delta^2_\zeta(k) = A_\zeta(k/k_\star)^{-\theta}$. A single shell has an infinite-dimensional radial kernel, including positive ambiguities. Complete radial cross-covariances provide the independent tomography route.

Receipt boundary. The conditional theorem fixes the exact angular family. Its radial theorem gives physical source-dilation and cross-covariance tomography as separate uniqueness routes. A finite OPH source construction that emits the primitive collar ensemble, full-collar generator density, half-collar identity, conformal precision, and physical dilation-intertwiner or tomography receipt is not supplied. Fawzi–Renner/Markov-collar observable control supplies upper bounds on release observables; it is not an amplitude equality. Finite-width windows require an exact Bessel-kernel error certificate. Physical temperature and polarization transfer and a frozen likelihood contract are not supplied.

D.5 Cosmological-Constant / Screen-Capacity Closure

Theorem 6.5 (Local/global cosmological-capacity closure). On the gravity branch, assume the local Einstein equation has been recovered only modulo Λg_{ab}, stable direct public-record closure at a carrier dimension

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star,$$

and the independent horizon–record identification readout

$$N_{\mathrm{CRC}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c}.$$

Then:

1. local null data determine the Einstein branch only modulo Λg_{ab};
2. with the selected scale certificate, the same branch has the global display

$$G_{ab} + \frac{3\pi}{GN_{\mathrm{CRC}}}\, g_{ab} = 8\pi G\, \langle T_{ab}\rangle;$$

3. the same capacity closure fixes the entropy and dimensionless capacity relations

$$S_{\mathrm{dS}} = N_{\mathrm{CRC}}, \qquad A_{\mathrm{dS}} = 4GN_{\mathrm{CRC}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c};$$

4. the observed cosmic age is a downstream FLRW benchmark rather than an additional theorem output.

So the cosmological-constant package is one local/global theorem chain rather than a split local argument plus an unrelated global readout.

Scope boundary. The hypotheses are the local Einstein branch, stable whole-fiber public-record closure, horizon–record identification, the de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that value is fixed only on a stable direct-capacity branch with a physical finite-size selector and horizon–record identification. The conditional capacity closure fixes $\Lambda_{\mathrm{CRC}}\ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$; the SI static-patch scale additionally requires the selected scale certificate.

Capacity self-closure target. At finite regulator use the frozen capacity-carrier dimension D, with $N = \log D$. For each terminal state in the unclosed fiber, form compatible global sections of the local record-atom diagram, retain the endogenously reachable sections, freeze the authorized publicness policy, and use the source-derived joint checkpoint kernels. Their compound confusability graph G_q gives

$$M_0(q) = \alpha(G_q), \qquad \mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

The scalar active map exists only when the whole nonempty terminal fiber has one common defined value. Fixed checkpoint projectors are not the capacity: a cyclic permutation fixes only constant functions while preserving every record label. Local checkpoint marginals are also insufficient to determine the required joint channel.

A faithful capacity-carrier representation gives $M_\varepsilon(q) \le D$, and stable closure is

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}.$$

Equality at zero error forces a rank-one complete public record basis. With a confusability-reflecting embedding, capacity extension is monotone, so a total scalar deflationary map on a declared finite chain reaches its greatest fixed point from the top. Fixed-D refinement then stabilizes exactly. None of these implications supplies cutoff independence or a unique physical zero of

$$s(D) = \log D - \log M_0(D).$$

The fixed-$D = 24$ packet realizes the complete terminal fiber inside its declared finite source category. A bounded all-rung counterfamily shares base agreement, positivity, the carrier bound, and executable finite incidence, action, projection, composition, extension, and sewing controls while giving different exact zero sets. Lean proves the all-rung arithmetic consequence of the base, positivity, and carrier-bound structure. The executable certificate checks the remaining finite controls, and an independent simulator reconstructs held-out finite rungs. The complete capacity-indexed A1–A3 packet lift is absent, so this result does not settle the full source class. A positive direct result also requires a source law selecting the continuation when necessary and a universe-level carrier attachment.

The finite kernel

$$K_r(D, m) = |\{q \in \widetilde{\Omega}_{r,D} : M_0(q) = m\}|$$

distinguishes one closed state from deterministic whole-fiber closure. A diagonal argmax is an extra selector. After direct active closure, the horizon–record identification yields the area-law display, while the common screen/electroweak load-carrier identification yields the electroweak bridge. The operational scale ρ_{op} is an independent estimator tested through $\log M_0 - \pi/\rho_{\mathrm{op}}^2$; it does not define the map. The Λ-located value remains a measured-side comparison coordinate.

Capacity normalization. The branch uses N_{scr} as the de Sitter entropy capacity. The bare horizon ratio is

$$N_{\mathrm{patch}} = \left(\frac{r_{\mathrm{dS}}}{\ell_P}\right)^2,$$

so

$$N_{\mathrm{scr}} = \pi N_{\mathrm{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

For the observed late-time scale, $N_{\mathrm{patch}} \simeq 1.05 \times 10^{122}$ and $N_{\mathrm{scr}} \simeq 3.31 \times 10^{122}$. These are benchmark displays, not high-precision inputs for the Newton row.

D.6 Black-Hole Structural Package and Continuation Boundary

The AMPS-style information trilemma assumes:

1. an outgoing mode B entangled with an interior partner A (smooth horizon),
2. late radiation B entangled with early radiation R (unitarity), and
3. monogamy of entanglement.

The false assumption in this setting is the naive tensor factorization

$$\mathcal{H} \overset{?}{=} \mathcal{H}_{\mathrm{inside}} \otimes \mathcal{H}_{\mathrm{outside}} \otimes \mathcal{H}_{\mathrm{radiation}}.$$

Theorem 7.1 (Edge-center black-hole decomposition). On the fixed-cutoff edge-center collar branch, assume the black-hole collar state lies in the exact Markov class or in the idealized exact-recoverability limit used elsewhere in the paper. Then the collar decomposition gives

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \left(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}\right).$$

The glue between inside and outside is the edge-sector label α in the center. Given α, inside and outside factorize.

Theorem 7.2 (Hawking/KMS normalization). On the geometric modular branch where the horizon generator carries the standard 2π KMS normalization, the outside algebra is in a KMS state at inverse temperature

$$\beta = \frac{2\pi}{\kappa}$$

with respect to the horizon Killing generator.

Corollary 7.3 (Recoverability-style interior encoding). On the same fixed-cutoff collar carrier, when $I(A_\delta : D_\delta \mid B_\delta) \leq \varepsilon$, there exists a recovery channel $\mathcal{R}_{B_\delta \to A_\delta B_\delta}$ such that

$$\|\rho_{A_\delta B_\delta D_\delta} - (\mathcal{R}_{B_\delta \to A_\delta B_\delta} \otimes \mathrm{id}_{D_\delta})(\rho_{B_\delta D_\delta})\|_1 \leq 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Here ε is in nats. So the interior collar data are encoded in the exterior collar together with the shared cut data, rather than supplied by an independent tensor factor.

Corollary 7.4 (Discrete area spectrum and Schwarzschild transition identity). The same edge-center package carries a discrete area operator

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha, \qquad A_\alpha = 4G_{\mathrm{geom}} \log d_\alpha = 4\ell_\star^2 \ln d_\alpha,$$

and for a small Schwarzschild sector transition $d_{\mathrm{before}} \to d_{\mathrm{after}}$ the differential first law gives, to leading semiclassical order,

$$\Delta_{\mathrm{BH}}(Mc^2) = k_B T_H \ln(d_{\mathrm{after}}/d_{\mathrm{before}}).$$

On the imported emission branch $d_{\mathrm{before}} = k d_{\mathrm{after}}$, with integer $k > 1$ dividing the initial dimension, the signed entropy change is $-\ln k$ and the emitted line-energy magnitude is

$$E_{\mathrm{emit}} = k_B T_H \ln k \, [1 + O(\ln k / S_{\mathrm{BH}})].$$

Scope boundary. The retained structural theorem package is exactly Theorem 7.1, Theorem 7.2, Corollary 7.3, and Corollary 7.4. Integer-transition combs, QNM selectors, Page-type linewidth estimates, PBH burst templates, Kerr/LIGO horizon spectroscopy templates, and any Page-curve or island closure are outside that structural core. Those items require continuation inputs such as an integer-transition selection rule, a semiclassical evaporation-power model, a QNM/transition identification, greybody matching, or an explicit island prescription.

The same interpretation rule applies here. A finite capacity register is not an exterior mass, a stored record is not Hawking radiation, a finite repair spectrum is not a GR quasinormal-mode spectrum, and an exact finite recovery threshold is not a Page time unless the corresponding independent physical bridge supplies source-separated readout, calibration, residuals, controls, and fixed validation references.

D.7 Modular-Anomaly Continuation

Theorem 9.1 (Modular additivity defect). Define the collar modular additivity defect by

$$\Delta K_\delta := K_{ABD} - K_{AB} - K_{BD} + K_B.$$

Then

$$\langle \Delta K_\delta \rangle_\omega = -I(A : D \mid B)_\omega.$$

This is a scalar recovery identity. It can enter the modular-anomaly dark-sector continuation only after a separate matching theorem or hypothesis identifies it with an anomalous local modular-energy source. By itself it constructs neither a rank-two tensor nor a dark-matter theorem.

The continuation therefore reserves an anomalous tensor slot only for a finite covariant source packet that supplies source localization, CMI-to-modular-source matching when that route is used, rank-two reconstruction, conservation, normalization, and universal coupling. Once those receipts pass, the modified Einstein equation is

$$G_{ab} + \Lambda g_{ab} = 8\pi G \left(\langle T_{ab} \rangle + \langle T_{ab}^{\mathrm{anom}} \rangle \right),$$

with rest-frame normalization

$$\langle T_{00}^{\mathrm{anom}} \rangle := \frac{15}{8\pi^2} \frac{\delta \langle K_C^{\mathrm{anom}} \rangle}{\ell^4}.$$

On that receipt-certified continuation surface the tensor gravitates and is covariantly conserved. Calling it dark additionally requires the no-electromagnetic-coupling receipt. Central or recoverability data alone do not establish any of these properties.

The dark-sector continuation maps scalar repair occupation to a canonical pair (n, θ). Conditional on the proposed action, its dilute homogeneous phase scales as pressureless matter and its cubic condensed phase gives $a_R = \sqrt{a_b a_0}$ in the spherical deep regime. The action supplies a repair current and metric stress rather than a virtual source.

Reading rule. The scalar modular-additivity identity is a recovery theorem. The canonical repair pair, baryonic source map, dimensional couplings, complete constitutive law, relativistic refinement limit, abundance, and physical likelihoods are not supplied. Every cosmological quantity on this surface is diagnostic.

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Data Availability declaration: Source manuscripts, public PDFs, simulation code, and implementation notes are available through the OPH repository and its linked public artifacts. The curated large-run evidence archive publishes its protected authority, resolved icosahedral configuration, seeds, mismatch trace, primitive and terminal arrays, full patch state, receipts, standalone exact verifier, and all observer rows in losslessly compressed form [24]. A separate archive publishes the literal record-feedback receipt and its producer-free verification. Some hardware and prototype evidence requires a sanitized ancillary audit bundle before public disclosure because raw logs may include device configuration, operational metadata, or private infrastructure details. The intended audit bundle contains source hashes, simulator configs, finite-receipt JSON, verifier code, hardware acceptance records, calibration logs, and selected run traces sufficient to replay the stated implementation claims.

Code Availability declaration: The OPH simulator, finite-consensus harnesses, visualization payload schemas, and public paper build sources are maintained in the OPH project repositories. The archived large run pins the simulator revision that produced it and supplies byte-hash and simulator-independent normal-form verification for the curated evidence subset. Submission artifacts include a standalone source file and a source bundle so that the manuscript can be rebuilt without relying on private local paths.

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Domain	Output	Mathematical role	Physical boundary
Carrier modes	two transverse Maxwell modes; $2\dim G$ perturbative Yang–Mills modes; two Einstein TT modes	symmetry-protected massless classical kernels on the stated action branches	no quantum zero-mass particle row without the physical Hilbert-space, pole-residue, and phase receipts
W/Z	target-free declared-map chart coordinates (80.330, 91.119) GeV; no source-only physical mass emitted	declared zero-selector map; exact selected-carrier chart; separate strict scalar consumer with principal-sheet zero exclusion and one simple channel-specific lower-half algebraic-chart zero for each of W and Z	the source does not select the map and the external fixture is not composed with the OPH chart. Neither declared chart is identified with the physical resonance sheet; unique continuation, sign bridge, full-matrix Laurent residue, current amplitude, independent replay, and W/Z mass comparison are absent
Fine structure	four distinct coordinates	source witness, mixed diagnostic, empirical closure, and measurement	the empirical interval misses the measured endpoint. A physical source-only statement requires the source-derived hadronic spectral backend, target-independent map selection, and the typed same-quantity bridge
Higgs/top	no nonzero source-only physical mass emitted	double-criticality family from the gauge sector; boundary-scale candidate fixed before comparison; declared-surface calibration fit kept separate	a physical statement requires the boundary-scale selection theorem together with the source root, physical scale, running, matching, rigidity, provenance, uncertainty, and complex-pole construction
Charged leptons	no nonzero source-only physical mass emitted	exact positive-chamber C_3 circulant identity; conditional finite tracial-GNS balance; target-informed response diagnostic; conditional nature/pole transport	the exact identity gives $Q = 1/3 + (2/3)(\lvert b\rvert/a)^2$, and the finite packet gives $\lvert b\rvert/a = 1/\sqrt{2}$ under its declared event-block hypotheses. Physical family attachment, phase, mass ratios, source selection, interacting kernel, infrared completion, and cofinal refinement are separate premises
Quarks	no nonzero source-only physical mass emitted	common-scale reciprocal-ray falsification; rejected register-Clebsch candidate; exact six-scalar interface; restricted source-spread lower bound; scoped real-axis no-go	the reciprocal-ray candidate fails across the tested common scales. The register-Clebsch pairing result permits separate invariant channels, and its restricted unordered multiset is exact and runtime-target-free conditional on the declared alphabet.